SPAD.neXt Documentation

Table of contents

SPAD.neXt Manual
About SPAD.neXt
SPAD.neXt discord community
System Requirements
Supported hardware
Check Hardware
Installation
Simulation Specific Steps
SimConnect Configuration
MSFS: Enable PMDG data access
X-Plane Configuration
First Start
Configuration Wizard
Common Tasks and Issues
SPAD
Online Services
Fresh reinstall
Move SPAD.neXt to new Computer
License Issues
MSFS Specific Topics
X-Plane specific Topics
Saitek/Logitech MP/RP blinking
USB Power Management
New User Series
MSFS SimConnect
L:Vars / H:Events and B:Events
Features & Functions
Conditions / Expressions
Device Specific
Arduino / SPAD Serial
Authentikit
Cockpit Simulator CDU
Honeycomb
Joystick (Generic HID)
miniCOCKPIT
RealSimGear
Saitek/Logitech
Script Panel

Stream Deck
Thrustmaster TCA
vDECK
vFIP
Virtual Avionics
vJoy
VRinsight
WinCTRL
WINGFLEX
X-Touch MC Mode
X-Touch Midi Mode
Simulation Specific
ATR 600 (MSFS)
AS CRJ 550/700/900/1000 (MSFS)
FBW A32NX (MSFS)
FFX Citation X (MSFS 2024)
FFX/MG HJet (MSFS)
FFX Vision Jet (MSFS)
FSW C414AW
PMDG 737 (MSFS)
PMDG 777 (MSFS)
SWS Kodiak (MSFS)
WT CJ4 (MSFS)
WT G1000 NXi (MSFS)
PMDG 777 (P3D)
UI Familiarization
Home Page
Profiles Page
Expressions
Expressions: Values
Referencing Simulation Data
Expressions: Operators
Functions
Expressions: Examples
Tuner Acceleration
Event Execution Order
Virtual Devices
vFIP
vDeck
MSFS 2024

MSFS2024 PMDG Data Access
MSFS 2020
MSFS2020 PMDG Data Access
X-Plane
Pre-Requisites
Dataref Tool
Spad.neXt Gets its Type Rating
Dataref vs. Command: Know Your Copilots
Basic Flight Controls – Axis Setup
Basic Flight Controls - Button Setup
P3d and FSX
Other Simulations
USB HUB Requirements
CDU Devices
CPFlight Devices
Saitek FIP (Flight Instrument Panel)
Saitek FIP: Driver & Function Check
Loupedeck Image Devices
Midi Devices
X-Touch Mini
Serial (COM) Devices
Skalarki
VRInsight Devices
Gauges and Extensions
SPAD Gauges: GaugeSettings
C# Scripting Interface
C# Scripting: ScriptStub
C# Scripting: Provide Values
C# Scripting: PreCompile
Device Interface
Device Communication Flow
Device INIT-Phase
Device Configuration-Phase
Device options
Device COLORSET configuration
Device PROFILE configuration
Device OUTPUT Configuration
Device INPUT configuration
Device input AXIS
Device input ENCODER

Device input PUSHBUTTON
Device input ROTARY
Device input SWITCH
Device input SWITCH3
Device LABEL
Device config: Advanced topics
Device STATESCAN-Phase
Device General Commands
Device commands
Page-Control
Device SPAD.neXt Events (2)
Device Simulation Events
Device Data (Channel 5)
Device LED Update (6)
Device Display Update (7)
Device Input Updates (Channel 8)
Device Virtual Power
Device Custom UI
Device HTML UI
Device preconfigured definition
Serial V2
Command 1,RAISE
Command: 0,AUTH
demo-ino-snippet
X-Plane Datarefs & Commands
FSUIPC: Custom Offsets
RestAPI
HTML Rendering API
Other API's
General FAQ
MSFS: Lost profile aircraft assignments
Running release and beta side by side
26.7.0
0.9.24
0.9.23
0.9.21
0.9.14.0
0.9.13.39
Glossary
Old-Getting-Started-Guide

SPAD.neXt Manual

WORK IN PRORESS

(c) Ulrich Strauss https://www.spadnext.com

Just a test what happens if i edit something

https://docs.spadnext.com

About SPAD.neXt

SPAD.neXt set out as a replacement for Saitek Flight Simulation Panels (http://www.saitek.com/) drivers, but supports much more hardware today. While most manufacturers provide drivers for their hardware, these have several shortcomings, notably lack of configurability. SPAD.neXt closes this gap and provides almost unlimited options to configure your hardware not only for use with flight simulation software.

SPAD originally stands for Saitek Panel Advanced Driver, which was an earlier attempt to develop substitute drivers by Massimo de Nadal (http://fstools.weebly.com). Unfortunately, development stopped several years ago. Thus, Ulrich Strauss made a "neXt" approach. Despite similar names, SPAD.neXt is not based on the former SPAD Software but has been written from scratch.

SPAD.neXt had the reputation of being a bit hard to master initially. However, the latest versions enjoy a redesigned user interface as well as several helpful enhancements like online profiles and code snippets which should make it easier to come to terms.

Additional sources of information about SPAD.neXt are:

SPAD.neXt discord community

The SPAD.neXt Discord Community

Since there is no longer a forum for asking general question or getting a helping hand, there is a SPAD.neXt community discord for this purpose.

After the Video-Guides , the discord is the first stop ressource for any questions or problems.

> Please note: The SPAD:neXt discord community is not a service run or maintained by SPAD.neXt. It's a separate (free) product, which is used as tool.

Joining the Discord Community

To join the SPAD.next Discord community follow this link

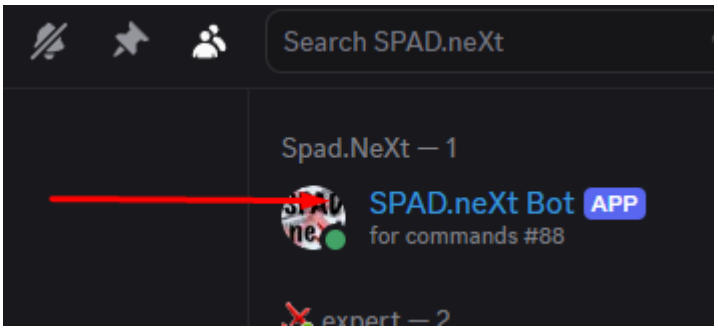
Please note that most of the channels on the SPD.neXt Community Discord are only accessible by licensed users. You will only see a very few channels, and the only channel you can write to is #trial-users.
This is the channel where you can ask general questions and our community experts will take care of it.

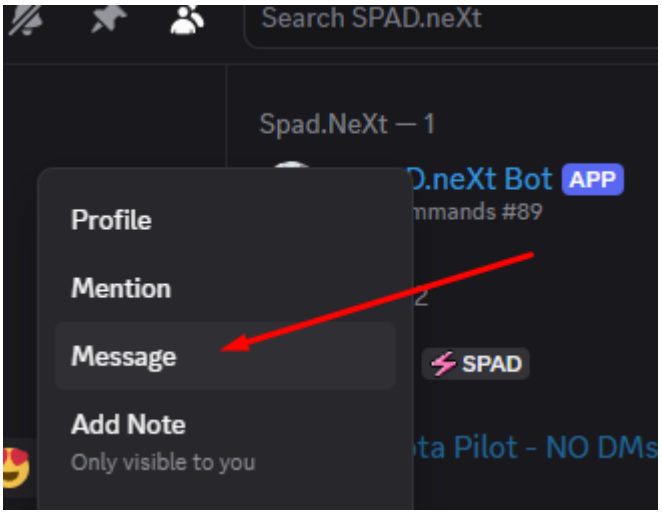
Community Guidelines

(WIP)

Linking your License

- Exit SPAD.neXt
- open the SPAD.neXt Discord

- On top of the users list (right side) you will see the "SPAD.neXt Bot"
  (If you do not see the users list open it up with the Memberlist icon right next to the search box upper right)
  

- Right-Click it and select "Message" to open a direct conversation with the bot
  

- enter `!link YOURORDERID` as message to him (Replace YOURORDERID with your actual Order/LicenseID. This is **not** the activation code! It's a 8-12 digit number)
- You will receive an email from the bot with the final command to link your license. This is to ensure a valid email is tied to the license in case we ever need to contact you. (No other data than your email and name is stored by SPAD.neXt)
- Copy the command from the email (should be something like `!link YOURORDERID-XXXX`
- Enter it in the direct conversation with the bot. The bot will tell you if it was successful
- If successful you should see all the registered-only channels in the SPAD.neXt discord now

Why linking my license?

If you link your license on the community discord, you will

- Get access to all discord channels available for your license
- Be able to open support tickets from inside SPAD.neXt

- Get access to the latest beta versions of SPAD.neXt if you are eligible (Complete Edition license required or invitation).
- Get access to the top edge alpha versions of SPAD.neXt (invite only)
- Get in touch with all the other SPAD.neXt users to exchange experiences

System Requirements

- Operating Systems: Windows 7, Windows 8 / 8.1, Windows 10, Windows 11 (**64 bit only**)
- .Net Framework 4.8 or later (pre-installed under Windows 10 ff.)
- Supported Flight Simulators:
  - Microsoft Flight Simulator 2020 (MSFS)
  - Microsoft Flight Simulator 2024 (MSFS24)
  - FS9 (FSUIPC only)
  - FS2000 (FSUIPC only)
  - FSX (Service Pack 2 or Acceleration Pack required)
  - FSX Steam Edition
  - Flight Sim World (FSW)
  - Prepar3D v2.3 and later
  - X-Plane
- Supported other Software:
  - SCS Euro Truck Simulator 2 / American Truck Simulator
  - iRacing
  - Train Sim World
  - Run8
- For some simulations a specific plugin needs to be installed.

Important:

Due to Windows security restrictions, SPAD.neXt and the simulation must be run with the same credentials. So, if you run the simulation as administrator you will have to do so as well for SPAD.neXt.

> For SPAD.neXt to operate correctly in background the Windows Game Mode must be disabled currently.

Supported hardware

Hardware supported by SPAD.neXt

> All hardware marked with (*) requires the SPAD.neXt complete edition

Directly supported with custom UI:

Arduino / Serial / Custom devices

- SPAD.neXt Serial Protocol v1
- SPAD.neXt Serial Protocol v2
- USB-HID Joystick Library (Standard Joystick UI)
- Mobiflight devices that do not require a custom mfmc configuration

Cockpit-Simulator / Cockpitmaster

> For Daisy-Chaining Cockpit-Simulator devices, the MCP/FCU must be the device connected via USB to make sure SPAD.neXt can correctly remap devices.
> The 24V power connection is required for daisy chaining

- 737 CDU Left/Right (*)
- 737 MCP (*), MCP-LITE (*)
- 737 EFIS Left/Right (*)
- A320 FCU/EFIS (*)
- v2 CDU (*)
  - 320A MCDU, 737X, 737M, 747X, 777X CDU

CoreFlightTech

- Custom firmware required.

CPFlight

- MCP
- Radio
- Efis
- ATC

Elgato

- Stream Deck Mini – 6 Buttons (*)
- Stream Deck – 15 Buttons (*)
- Stream Deck MK.2 – 15 Buttons (*)
- Stream Deck XL – 32 Buttons (*)
- Stream Deck XL Plus – 36 Buttons, 6 Encoders (*)
- Stream Deck Plus – 8 Buttons + 4 Encoders (*)
- Stream Deck Neo (*)
- Stream Deck Bulk (*)

Generic USB Joystick/HID Devices

- All USB-HID Joysticks – If you can see a device in the Windows USB-Gamedevices, SPAD.neXt will discover it and use a generic joystick UI

GoFlight

- GF-46 GF-166 GF-EFIS GF-LGT GF-LGT-II GF-MCP GF-MCP Pro GF-MESM
- GF-P8 GF-RP48 GF-SECM GF-T8 GF-TQ GF-WCP GF-45
- GF-ATC

Honeycomb

- Alpha (generic Joystick UI)
- Bravo/Bravo Lite
- Charlie (generic Joystick UI)

- Sierra (generic Joystick UI)

Loupedeck

> Loupedeck Software 6.0.3 **required**
>
> No later version of Loupedeck Software is supported.

- Live / Ct / Live S (*)
- Razer Streamdeck (*)

Midi Devices (In & Out)

- General/Standard Midi
  - Device Inputs will be discovered when it being used the first time
- Behringer X-Touch (Mackie Mode)

miniCOCKPIT

- miniFCU
- miniEFIS
- miniOVHD

Octavi

- IFR1

OpenCockpits

- MCP v3 , EFIS

Phidgets

- All Devices [WIP]

PU Air Korea

- CDU v2 (*)

RealSimGear

(Firmware >= 3.0.24 required)

- CIRRUS-SW1
- G1000MFD15
- G1000XFD / G1000XFD1 / G1000XFD2
- GMA-Addon
- GCU47X
- GCU479
- GFC500
- GFC700
- GMA350
- GNS430
- GNS530
- GTN650
- GTN750
- SW1
- Cirrus Vision Jet Cockpit

Saitek / Logitech

> Saitek has been acquired by Logitech and all devices will be renamed. This documentation uses the old common names for the devices.

- Backlight Information Panel (BIP)
- Cessna TPM
- Cessna TrimWheel
- Cessna Yoke
- Pro Flight Instrument Panel (FIP)

- Pro Flight Multi Panel
- Pro Flight Quadrant (USB)
- Pro Flight Radio Panel
- Pro Flight Switch Panel
- Pro Flight Yoke
- X.52/Pro
- X55/X56 (Standard Joystick UI)

Shakeprint Simulations

- Class Echo Desktop
- Class Echo Mobile
- Shakeprint Knob

SimInnovations

- Knobster

Skalarki

- HOMELine/Desktopline: All devices (*)
- PROLine: All devices (*)

Thrustmaster

- TCA Airbus Side Stick (Standard Joystick UI)
- TCA Airbus Quadrant (incl. addons)
- TCA Boeing Quadrant
- TCA Boeing Yoke
- TM TPR / T-Flight / Warthog etc. (Standard Joystick UI)

Turtlebeach

- Velocity One Yoke

Ultimarc

- PACLed / PACLed64 / PACNano
- PACDrive

Virtual Avionics

> Since Virtual Avionics gone out of business support for the devices is "as it is" and is no longer maintained.

- EFIS (Partial)
- MCP

Virtual-Fly

- Compass
- Switch Trims Panel

VRInsight

- Airbus MCP Combo II
- Boeing MCP Combo II
- GPS
- Overhead 737 (*)
- Overhead A320 (*)
- CDU 2/3 (SPAD.neXt driven display complete edition only)
- ProPt, uProPt

WingFlex

- A320 FCU Cube
- A320 RMP Cube
- DAP-500

WinCTRL / WinWing

> WinWing has been rebranded to WinCTRL recently

- Airbus FCU + EFIS L/R
- Airbus MCDU (*)
- Boeing PFP 3N / 4 / 7 (*)
- Boeing PAP 3 / PAP 3 MAG
- Boeing PDC 3N/3M
- PAC-32
- AGP
- ECAM
- TCAS
- RMP

Check Hardware

HowTo check if the hardware is working correctly

Before opening a support ticket because a device is not working, it is highly recommended to confirm your hardware is working correctly.

For devices, that show up in Windows USB Gamecontrollers:

- [ ] Close SPAD.neXt and any flight sim
- [ ] open browser and navigate to https://gamepad-tester.com/
- [ ] Select the device in question. (Device not listed: Hardware problem)
- [ ] Press all buttons / switches in questing, and observe if the B-values on the webpage will change.
- [ ] Values do **not** change: It's a hardware issue. Maybe USB cable not connected firmly

For Saitek Multipanel/Radiopanel/Switchpanel:

- [ ] You can download the old Saitek Testapplication here

For Midi devices:

- [ ] Check the functionality here

For other devices:

- [ ] Close any flight simulator and open SPAD.neXt.
- [ ] In SPAD.neXt enable the programming mode (the "Lego"-Icon next to the save icon on the lower left)
- [ ] Open the Event-Monitor , Undock it and start it.
- [ ] Press the switches/buttons in question
- [ ] Check the Event-Monitor sees the events.

Installation

After downloading the latest installer from the website (https://www.spadnext.com/download/download-spad-next.html) extracting it, double-clicking the downloaded installation file "SPAD.neXt.Setup.exe" starts the installation.

In case a security question pops up, confirm that the Publisher is "Ulrich Strauss" and click "Run".

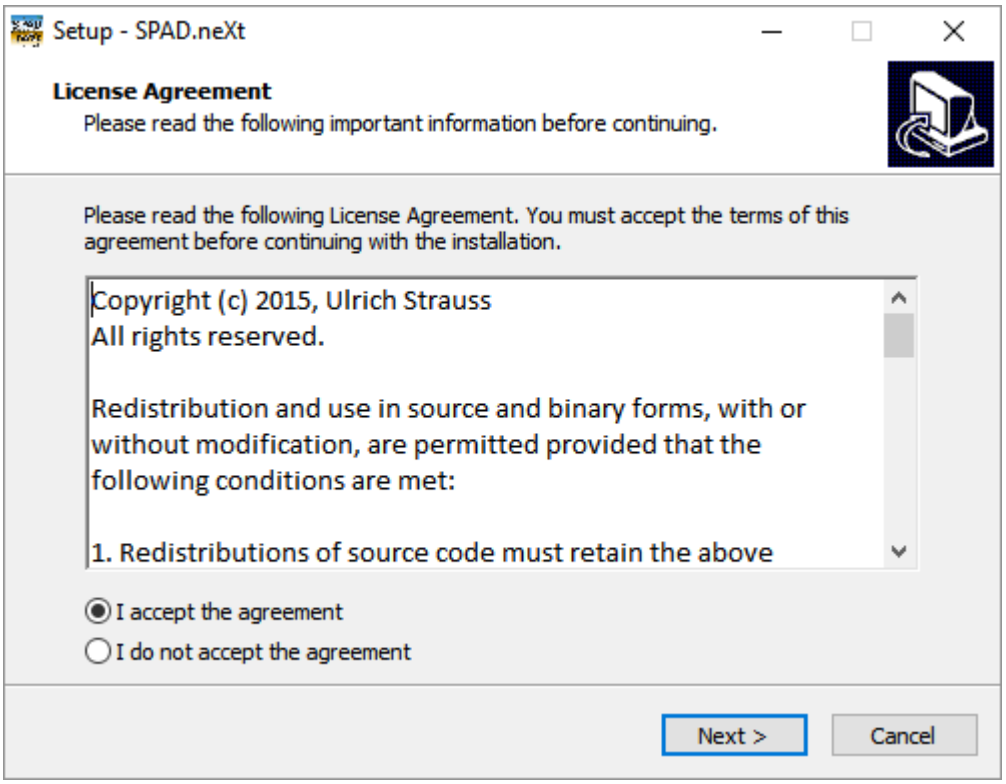


Fig. 1 License agreement

Click on Next to accept the license conditions and continue the installation.

The following screen lets you choose the location of the installation.

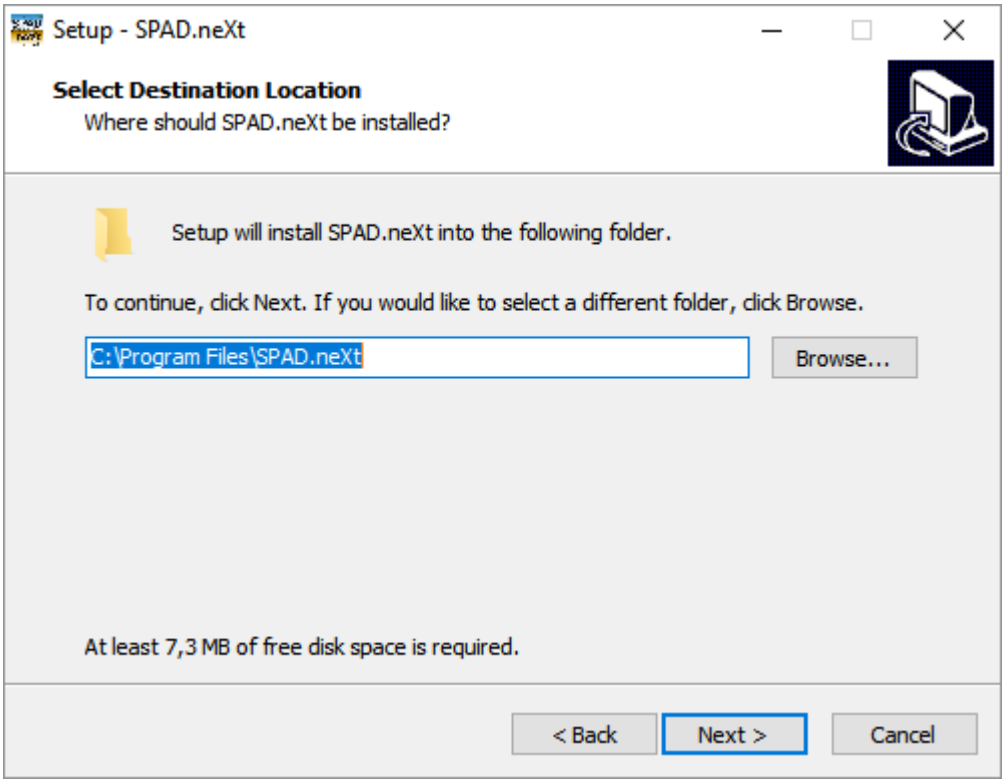


Fig. 2 Install Location

For normal operation, it is recommended to install SPAD.neXt at the default location, however if you would like to install SPAD.neXt somewhere else, you can change the location here.

Due to Windows security restrictions, do not install SPAD.neXt to the Root-Folder of your system drive (e.g. C:\SPAD.neXt)! The software will not be able to access its necessary files. If in doubt, just click next.

For review the selected location will be shown. And clicking on next will start the download process, if necessary.

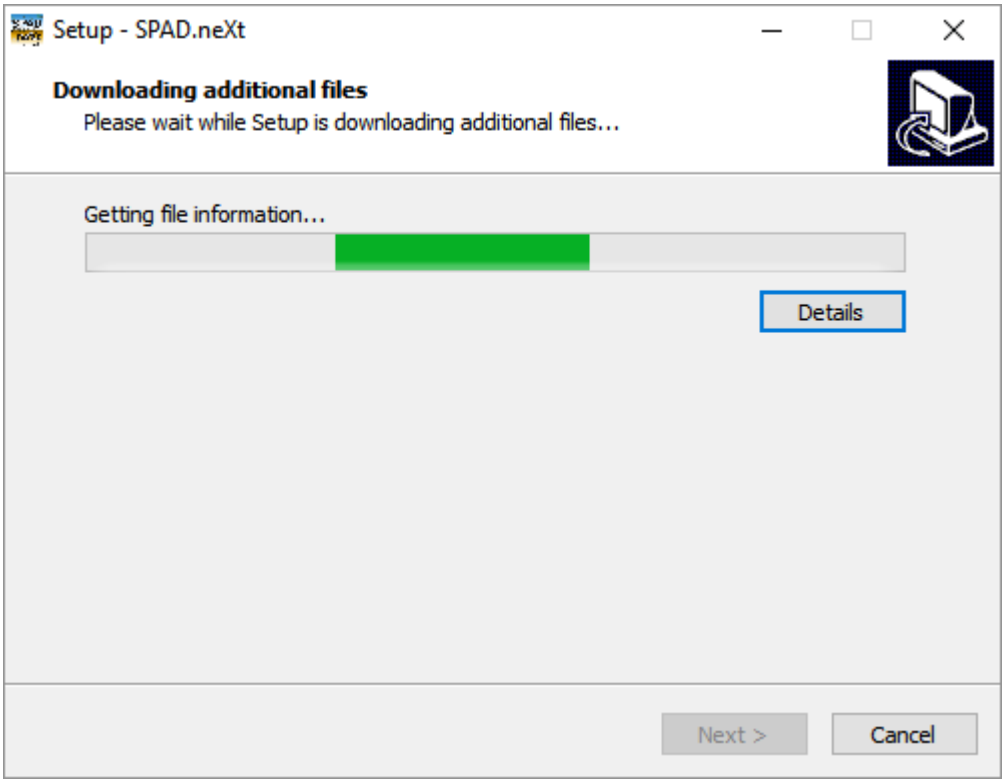


Fig. 3 Downloading installation files

After all files have been downloaded and installed the installation is complete

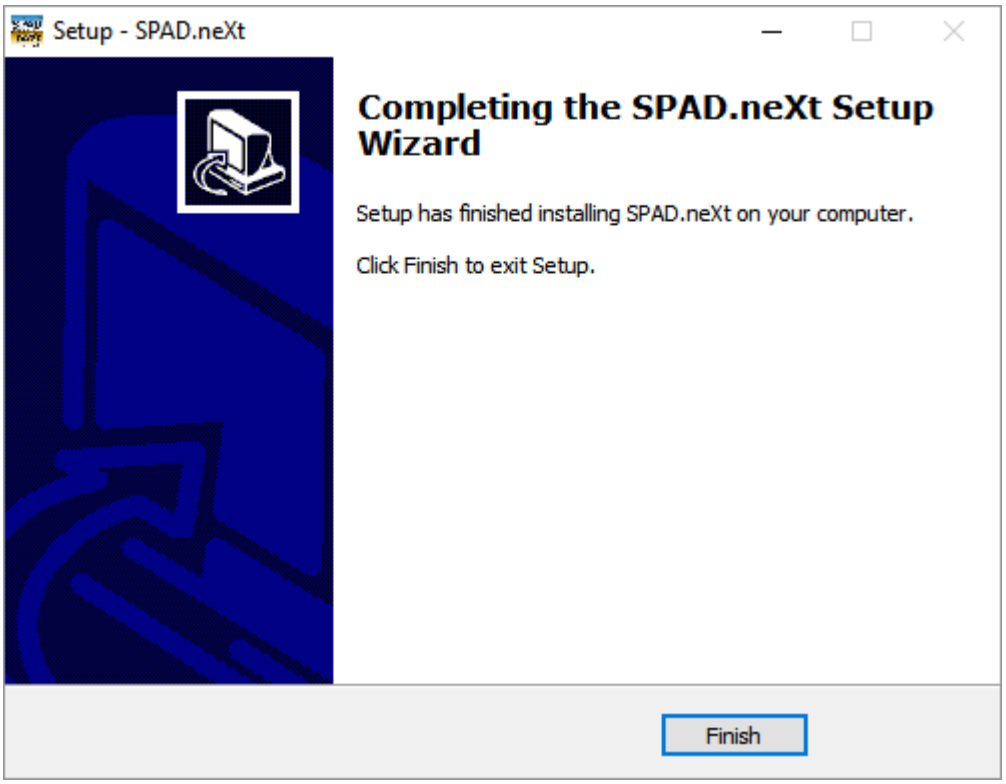


Fig. 4 Installation completed

and an Icon will be placed on your Desktop automatically.

Simulation Specific Steps

The following pages describe the steps and necessary configuration that has to be done for specfic simulations.

SimConnect Configuration

Applies to: FSX (Box and Steam edition) / P3D / MSFS

SimConnect is a synonym for a protocol Microsoft published together with the FSX Acceleration Pack/SP2. All simulations that are based on FSX (this includes Prepare3D and MS Flight Simulator 2020) support this protocol. For a complete description of SimConnect and its feature please see Microsoft's ESP Documentation.

No additional Installation is necessary on any client computer, since SPAD.neXt uses a built-in version of SimConnect. However, if you plan to user additional plugin/software that require SimConnect-Support you might need to install the SimConnect Client Packages. See the simulator documentation on details.

If you plan to run SPAD.neXt on the same computer as the simulation runs on (common case), then the simulation should have all necessary configuration created already, and you can skip this steps.
If later in the process of the configuration wizard you run into problems, get back to this pages to check the configuration.

Simconnect configuration file location

In the next paragraphs SIMCONNECT.XML will be referenced. This always references the SimConnect configuration file for the sim and can be found at the following location:

For FSX (Box Edition)

%APPDATA%\Microsoft\FSX

For FSX (Steam Edition)

either %APPDATA%\Microsoft\FSX or %APPDATA%\Microsoft\FSX-SE

For Prepar3D

%APPDATA%\Lockheed Martin\Prepar3D v2

%APPDATA%\Lockheed Martin\Prepar3D v3

%APPDATA%\Lockheed Martin\Prepar3D v4

%APPDATA%\Lockheed Martin\Prepar3D v5

depending on the installed version of Prepar3D

For MSFS (Store or XBox Gamepass Edition)

%LOCALAPPDATA%\Packages\Microsoft.FlightSimulator_8wekyb3d8bbwe\LocalCache

For MSFS (Steam Edition)

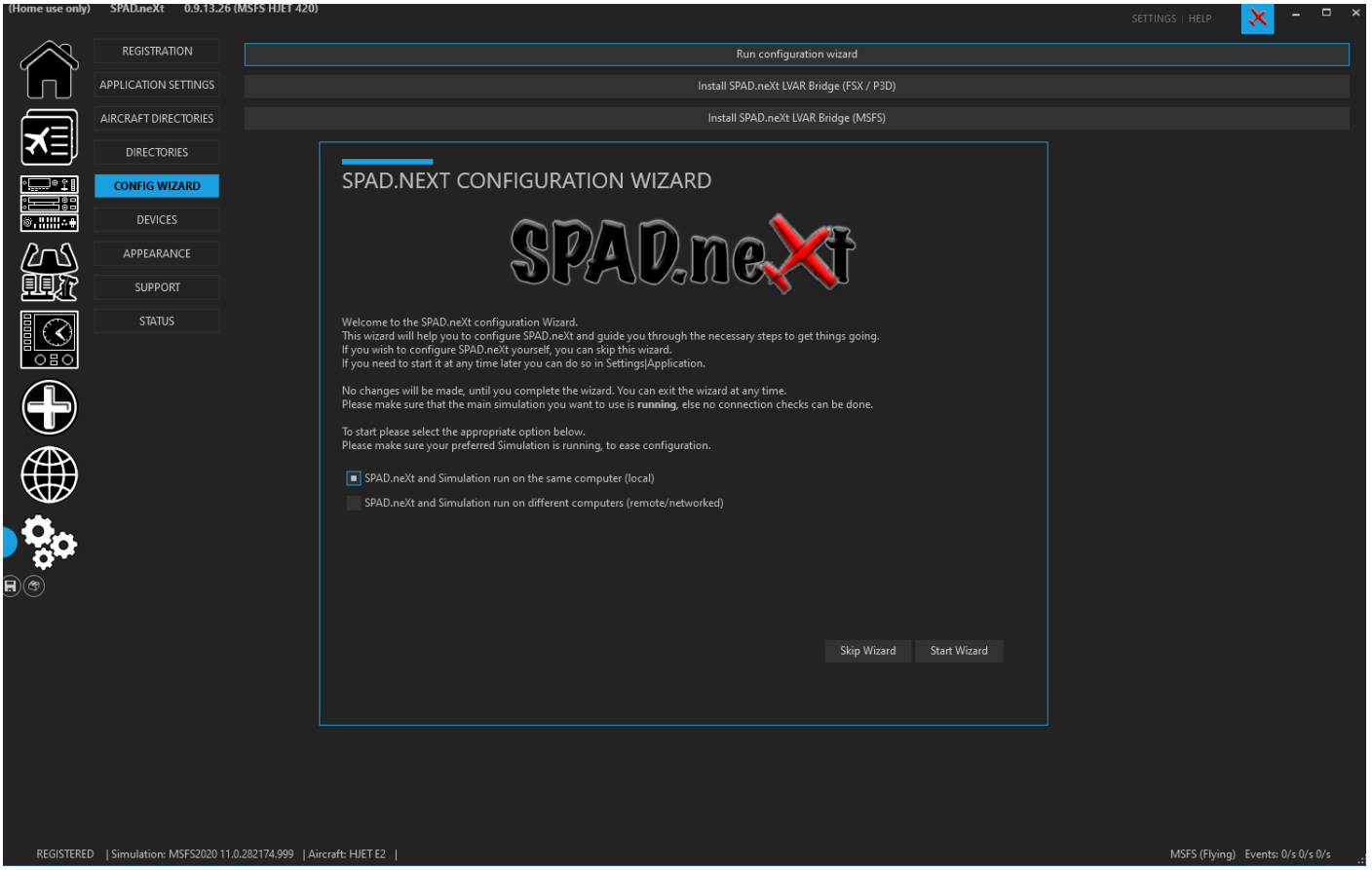
%APPDATA%\Microsoft Flight Simulator\Packages

For MSFS (Box Edition)

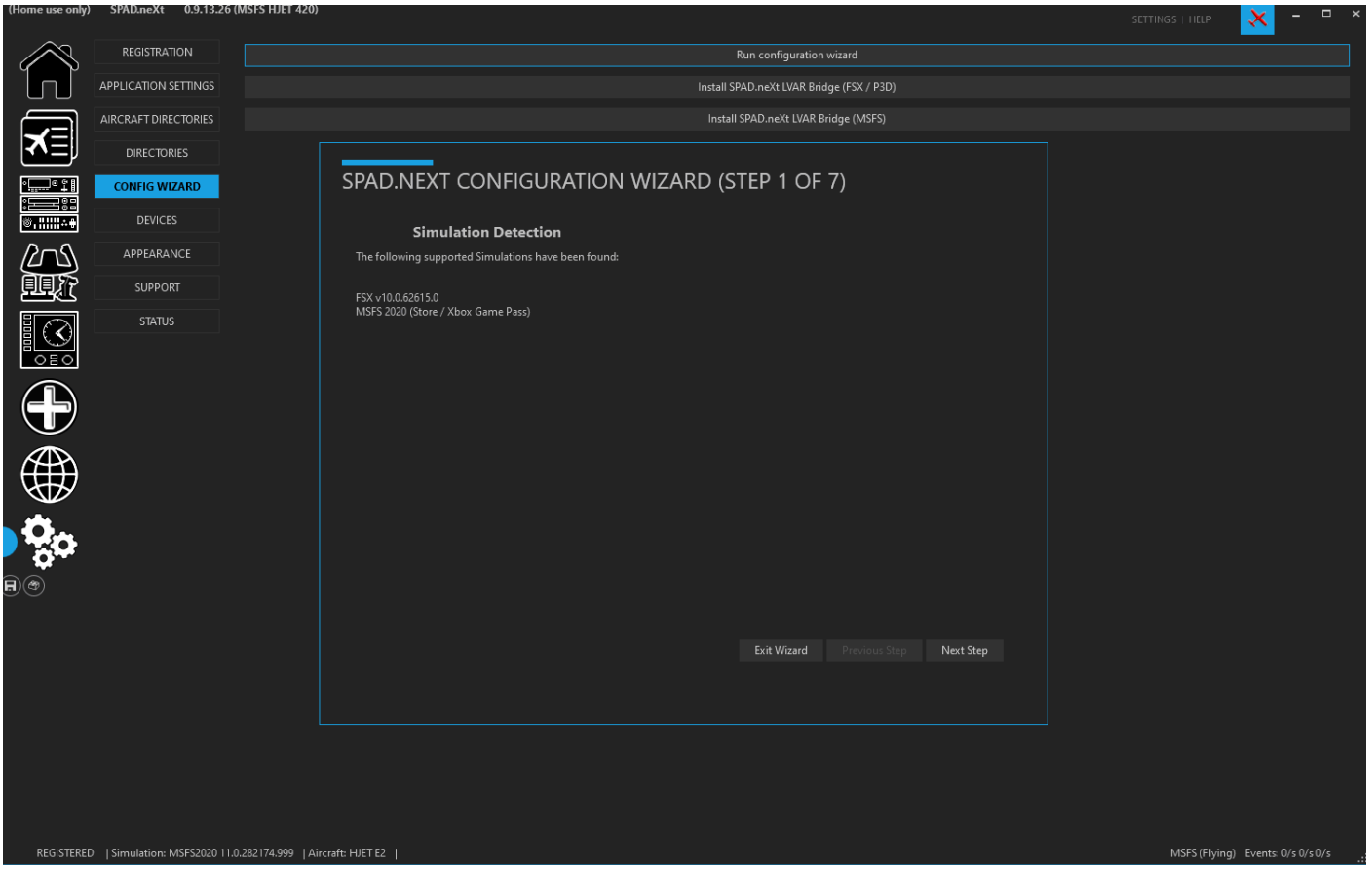
%LOCALAPPDATA%\MSFSPackages

Local Configuration (Sim and SPAD.neXt on same computer)

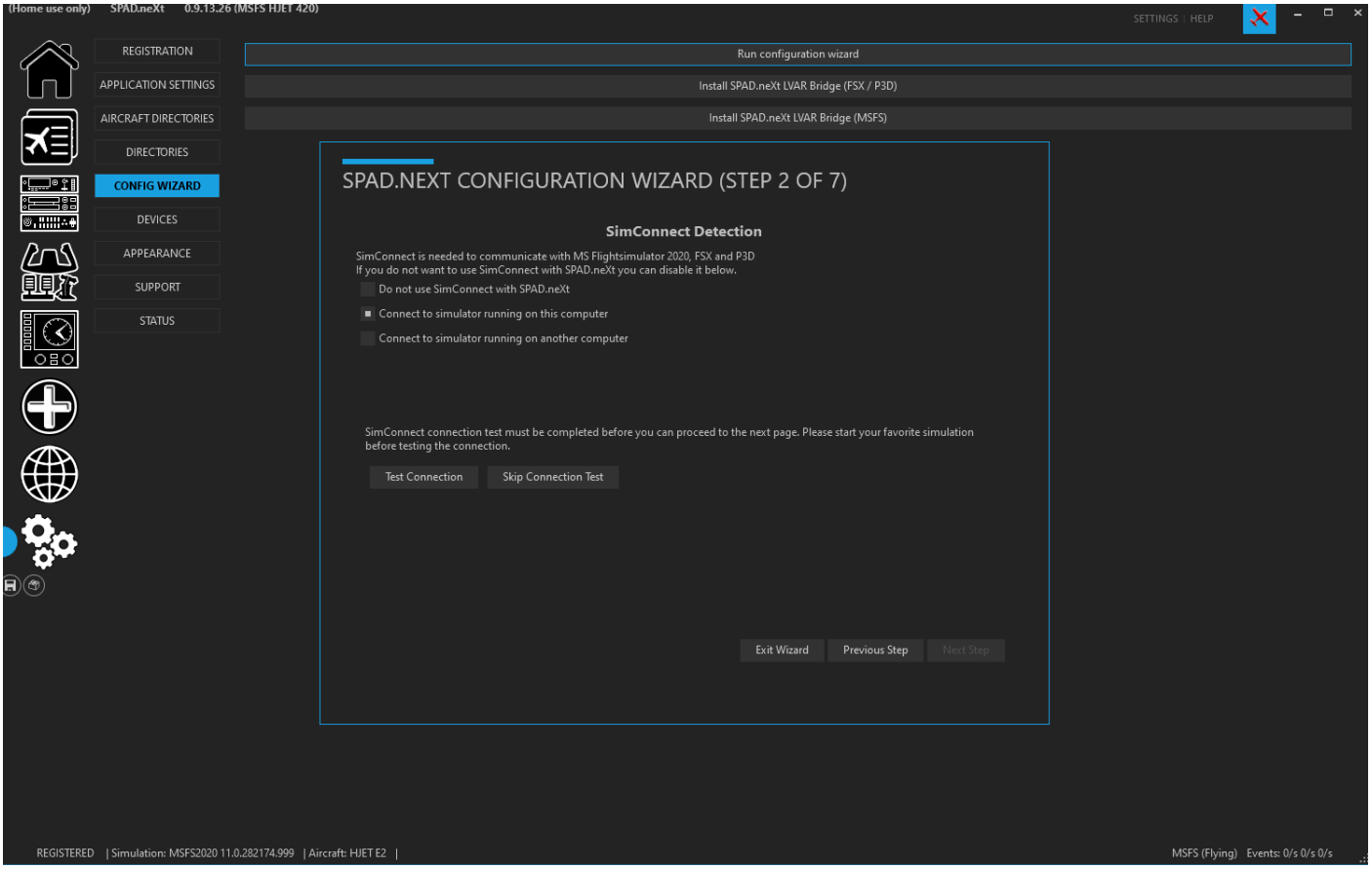
Run The Configuration Wizard and select the Local pc option. This is for running SPAD.neXt on the same PC that the simulator is running on.



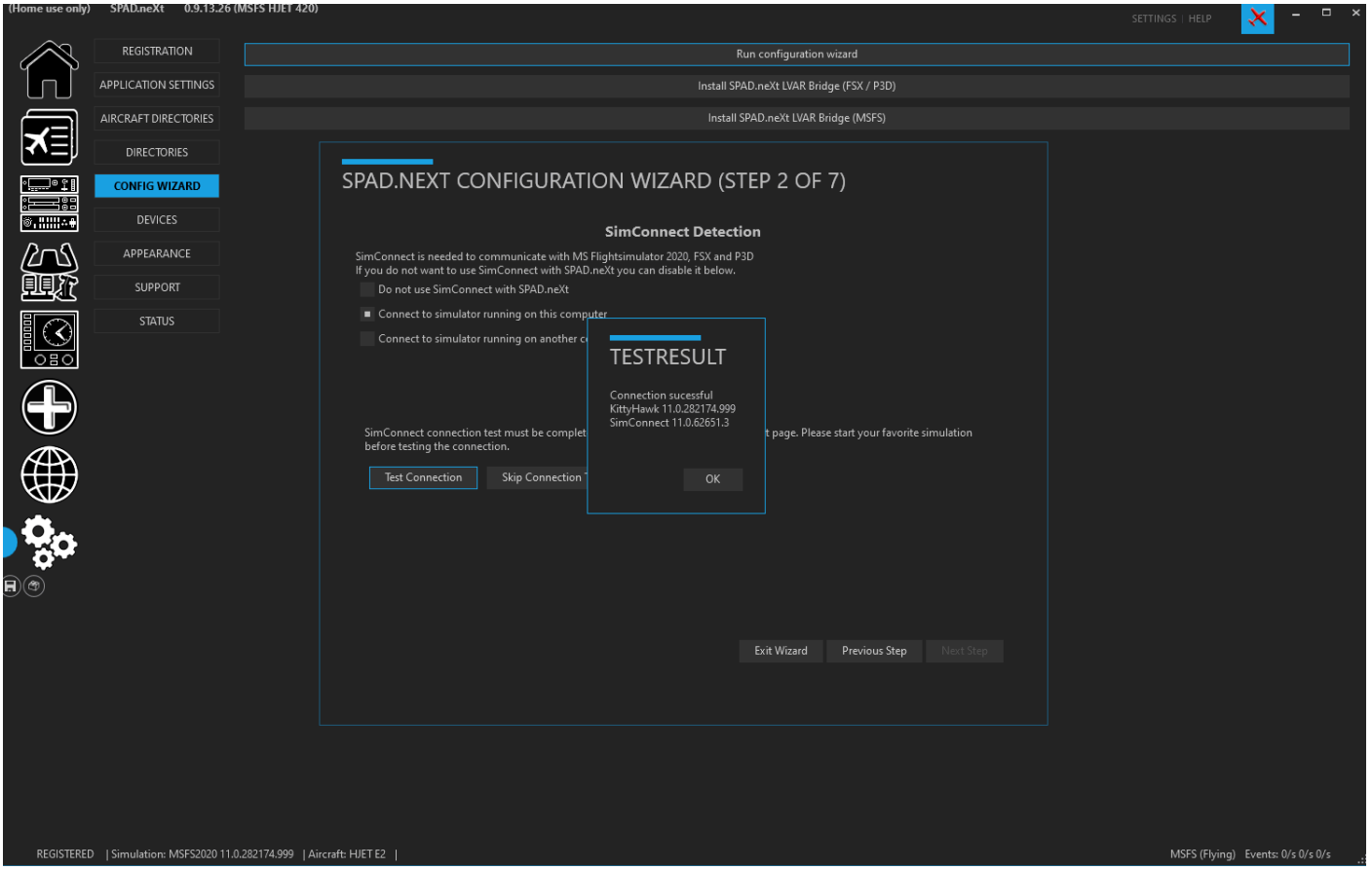
STEP 1: Detection of local simulators. if it looks correct then move to the next step. Pictured here it has found both FSX Steam and MSFS:



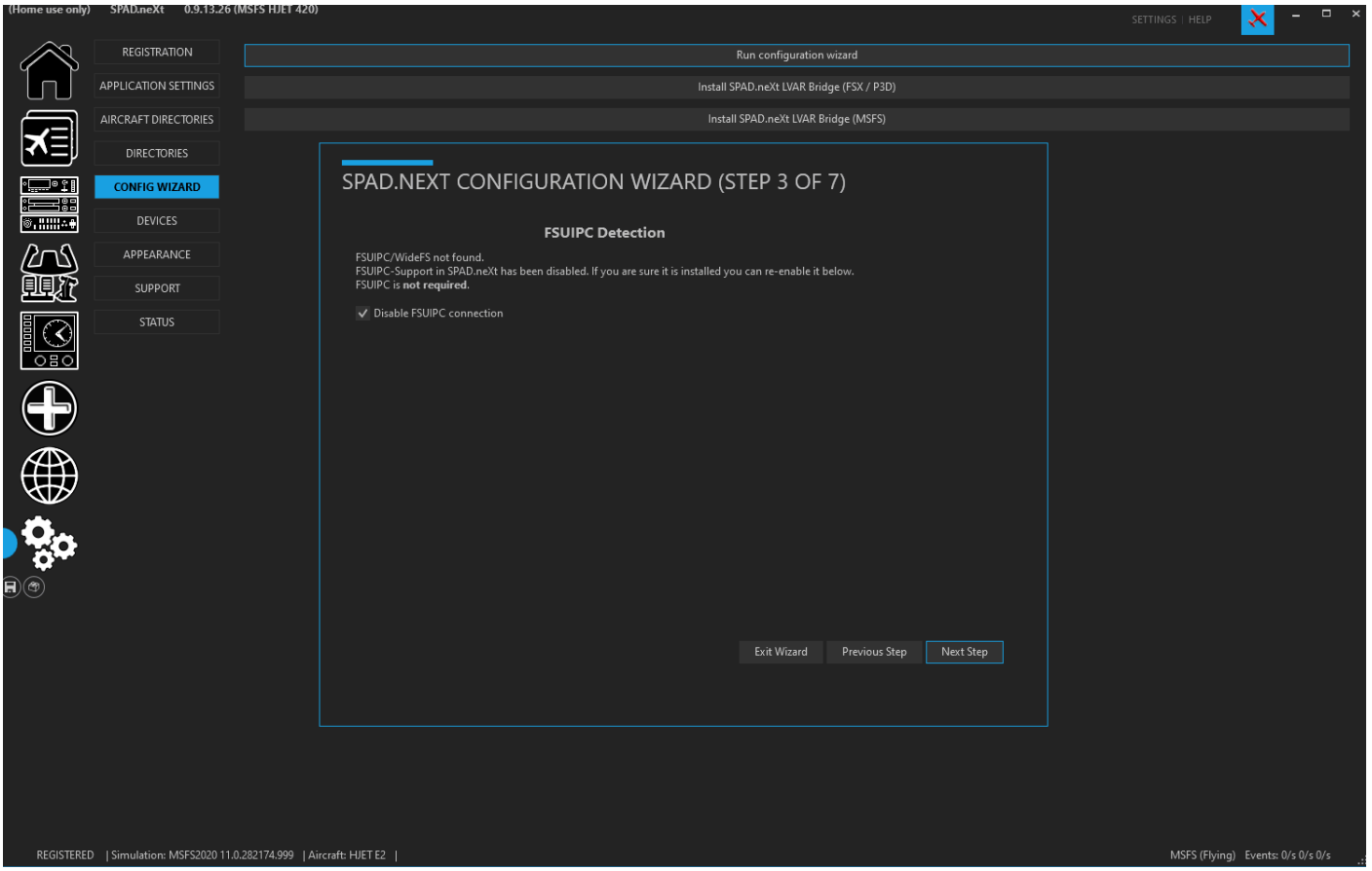
STEP 2: SimConnect Detection / Enablement. Since running locally with the sim then the selection with sim on this computer is desired.



Test Connection - you need a simulator running. In this example MSFS is running when the test is run

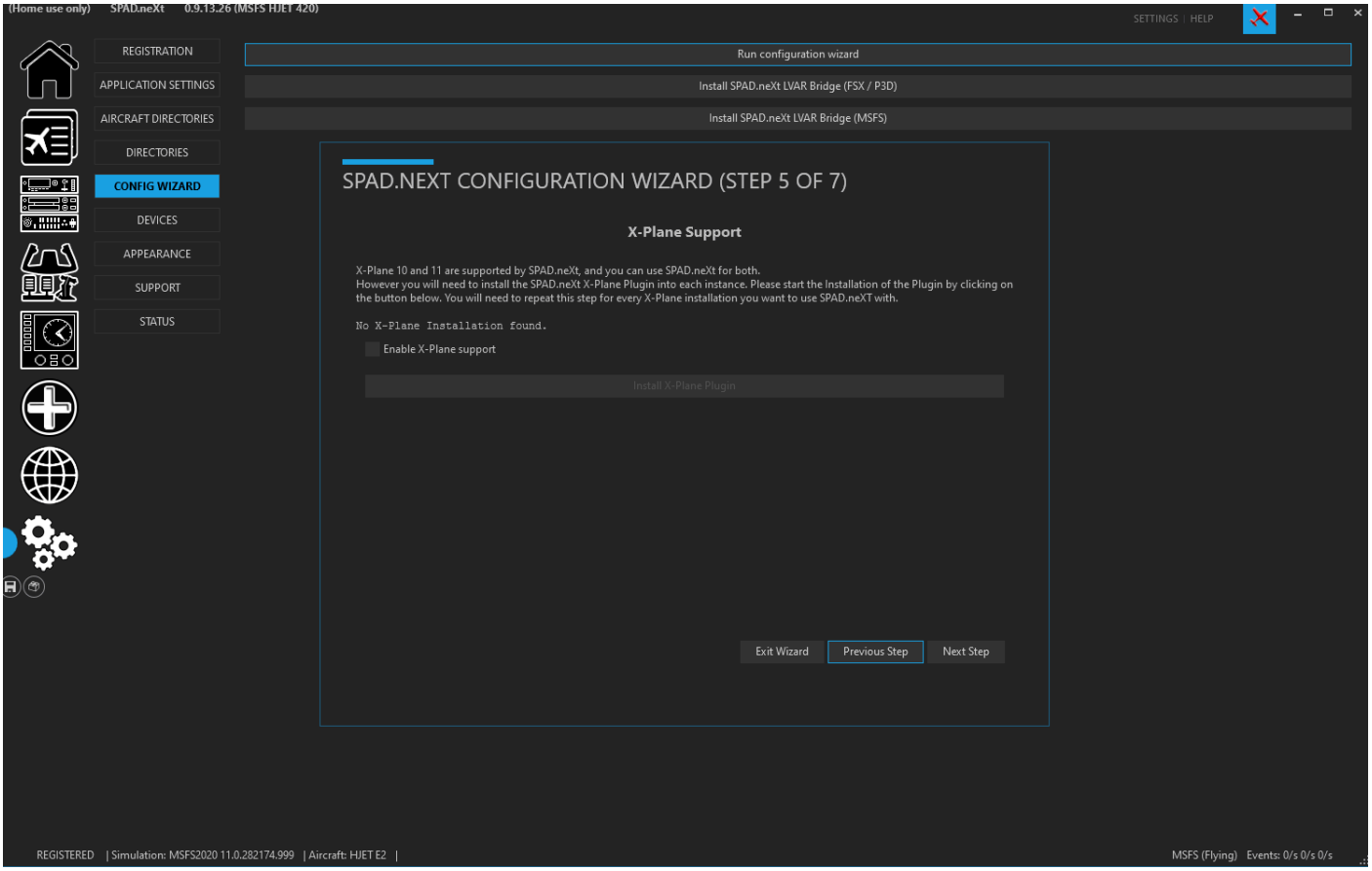


STEP 3: FSUIPC Detection. FSUIPC is not running on this PC thus disable is set. Note this can be re-enabled from the Settings menu later on if the need for FSUIPC is required.

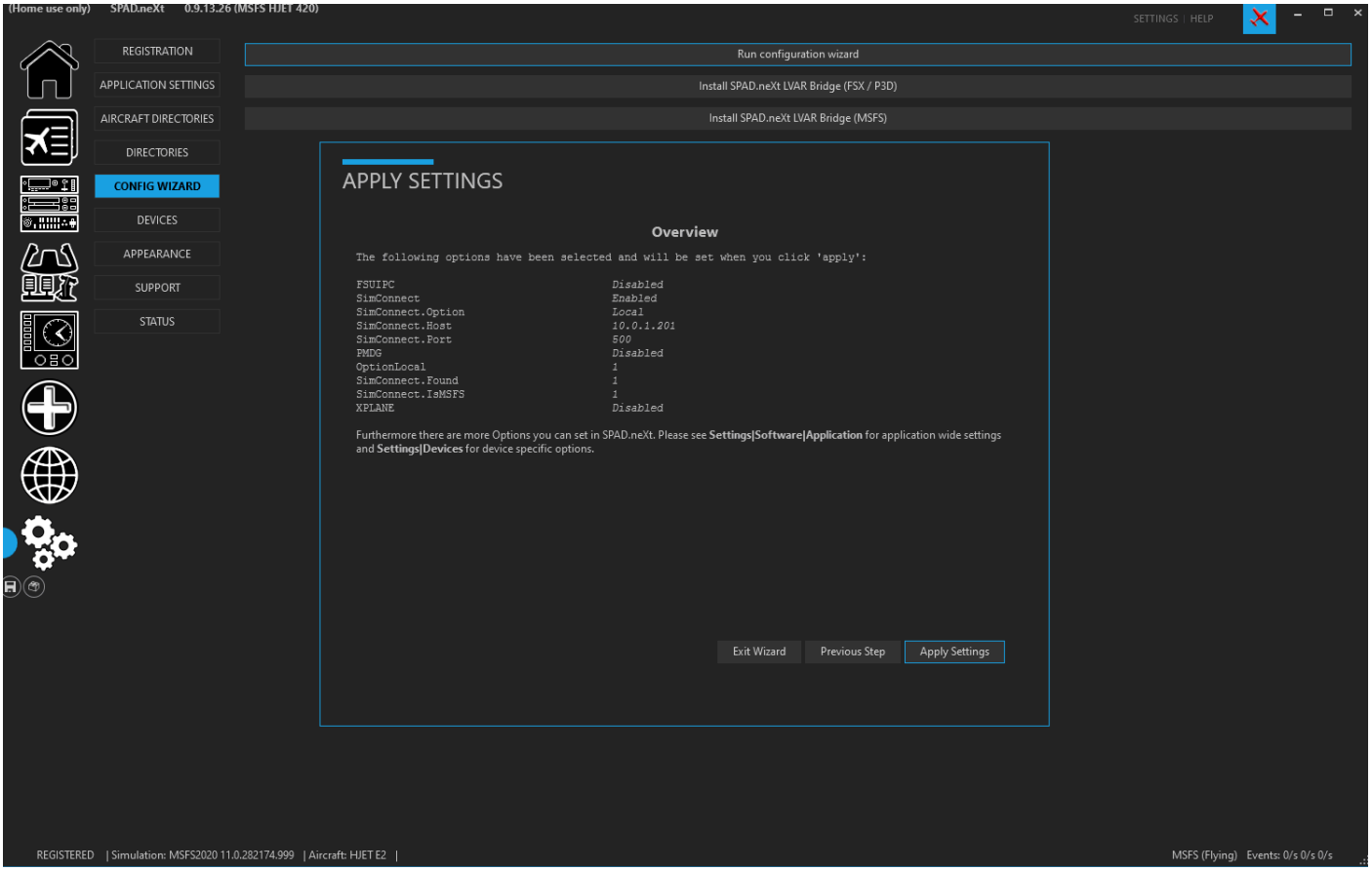


STEP 4: This is skipped for this installation.

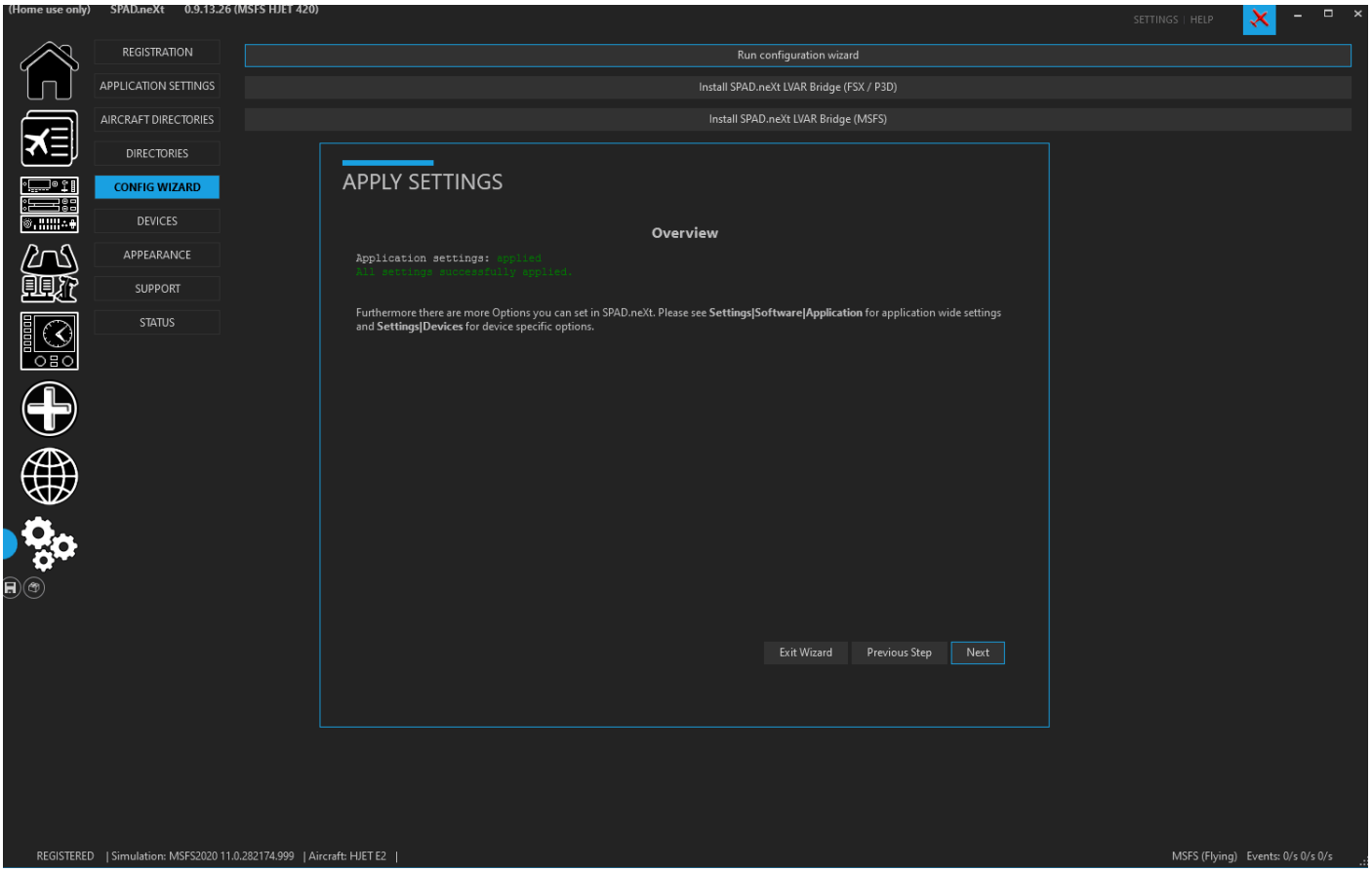
STEP 5: X-Plane Support. Enable if you have X-Plane installed but SPAD.neXt did not auto discover it.. else leave disabled.



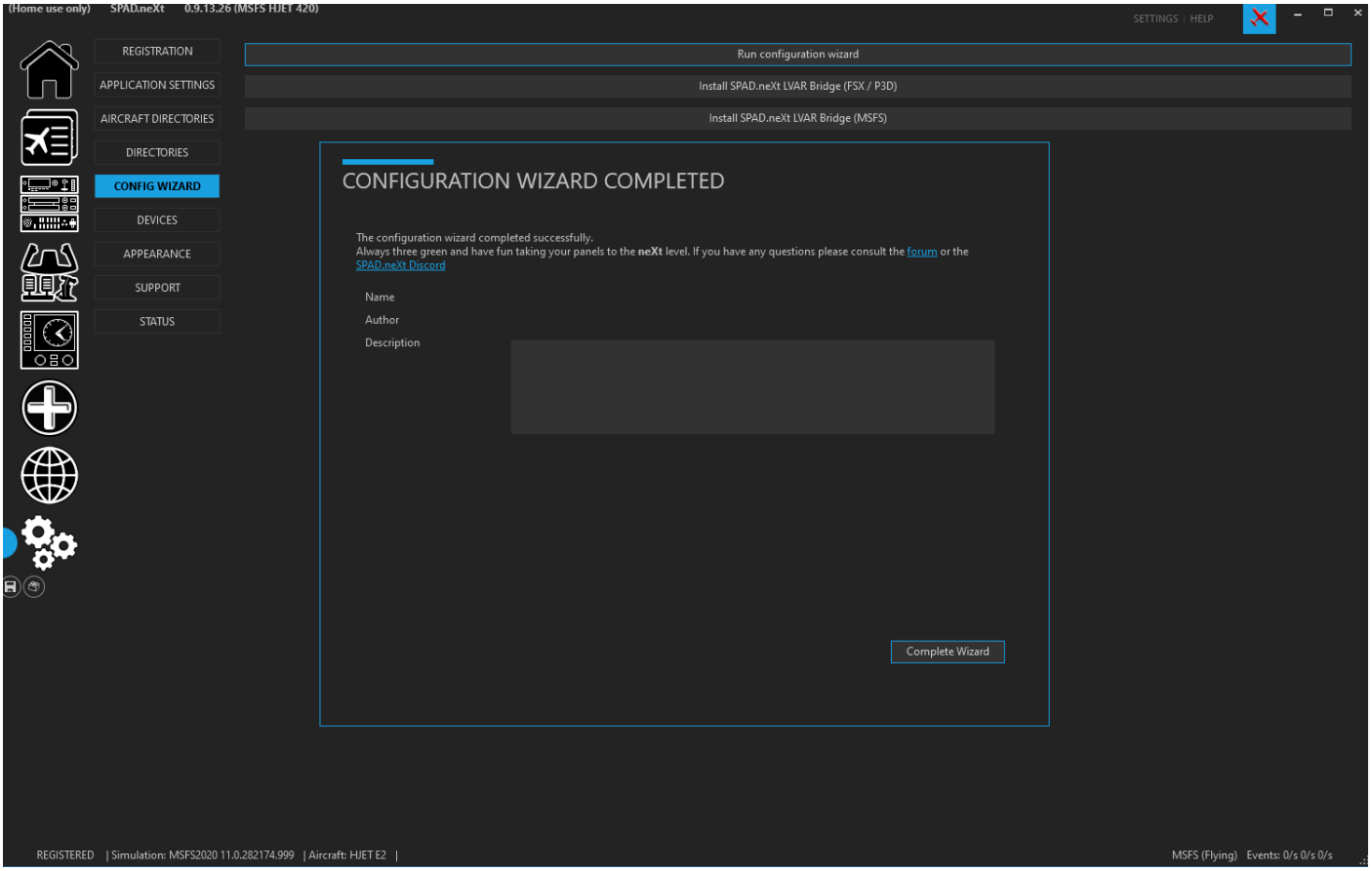
STEP 7: Review selections (note anything can be enabled later in the settings menu manually)



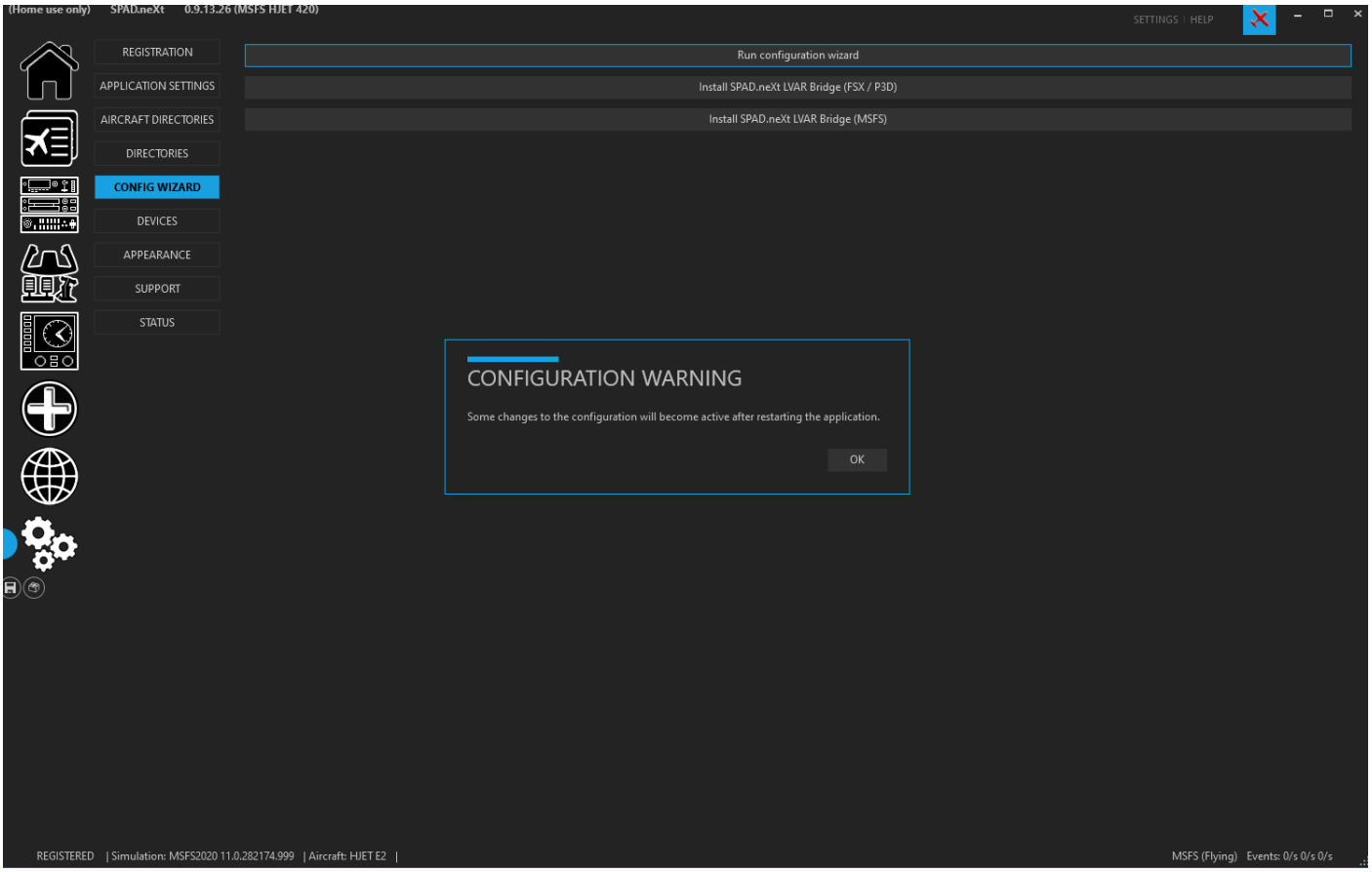
Apply Settings and Continue



Click Next



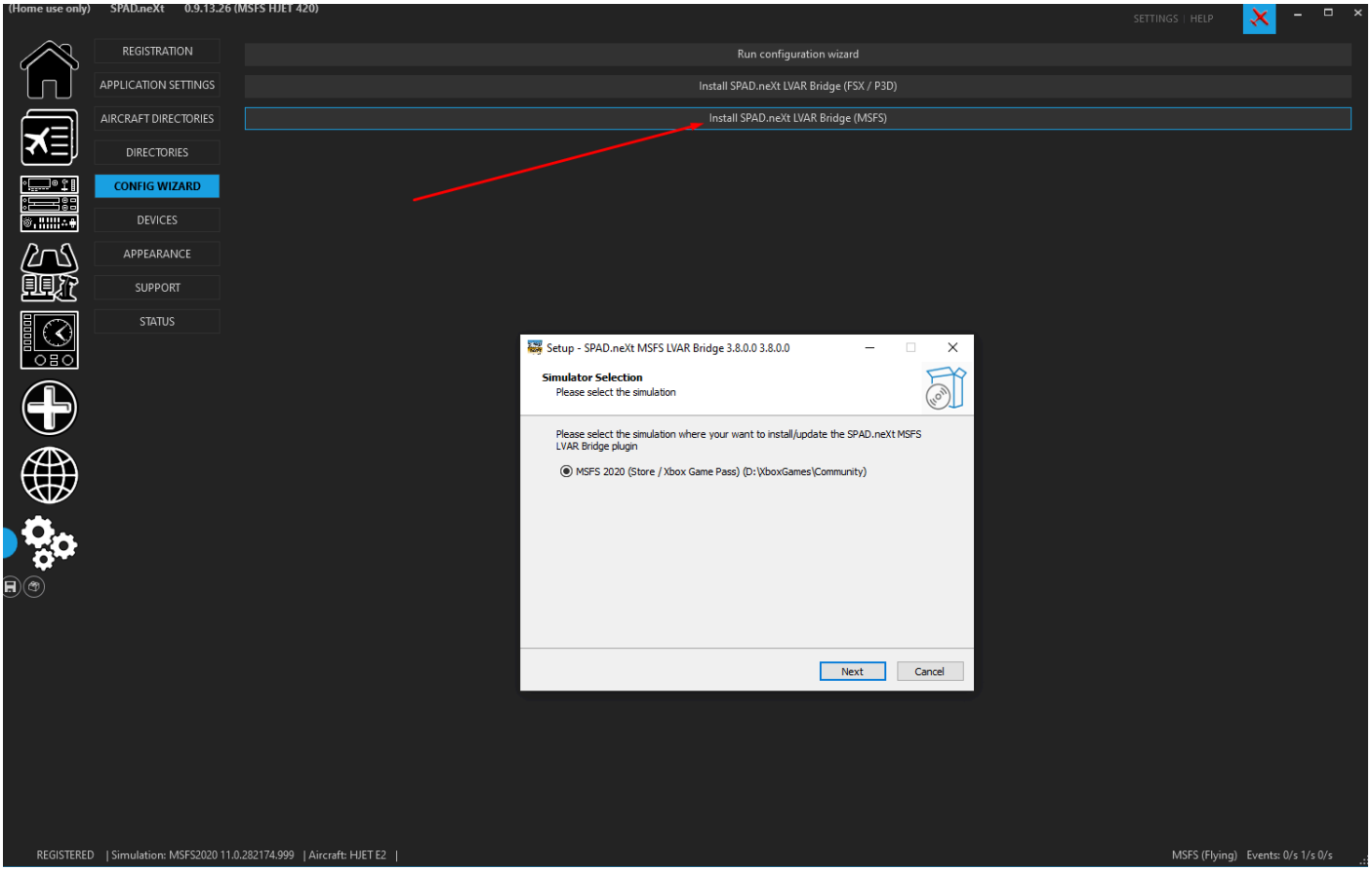
Click Complete Wizard



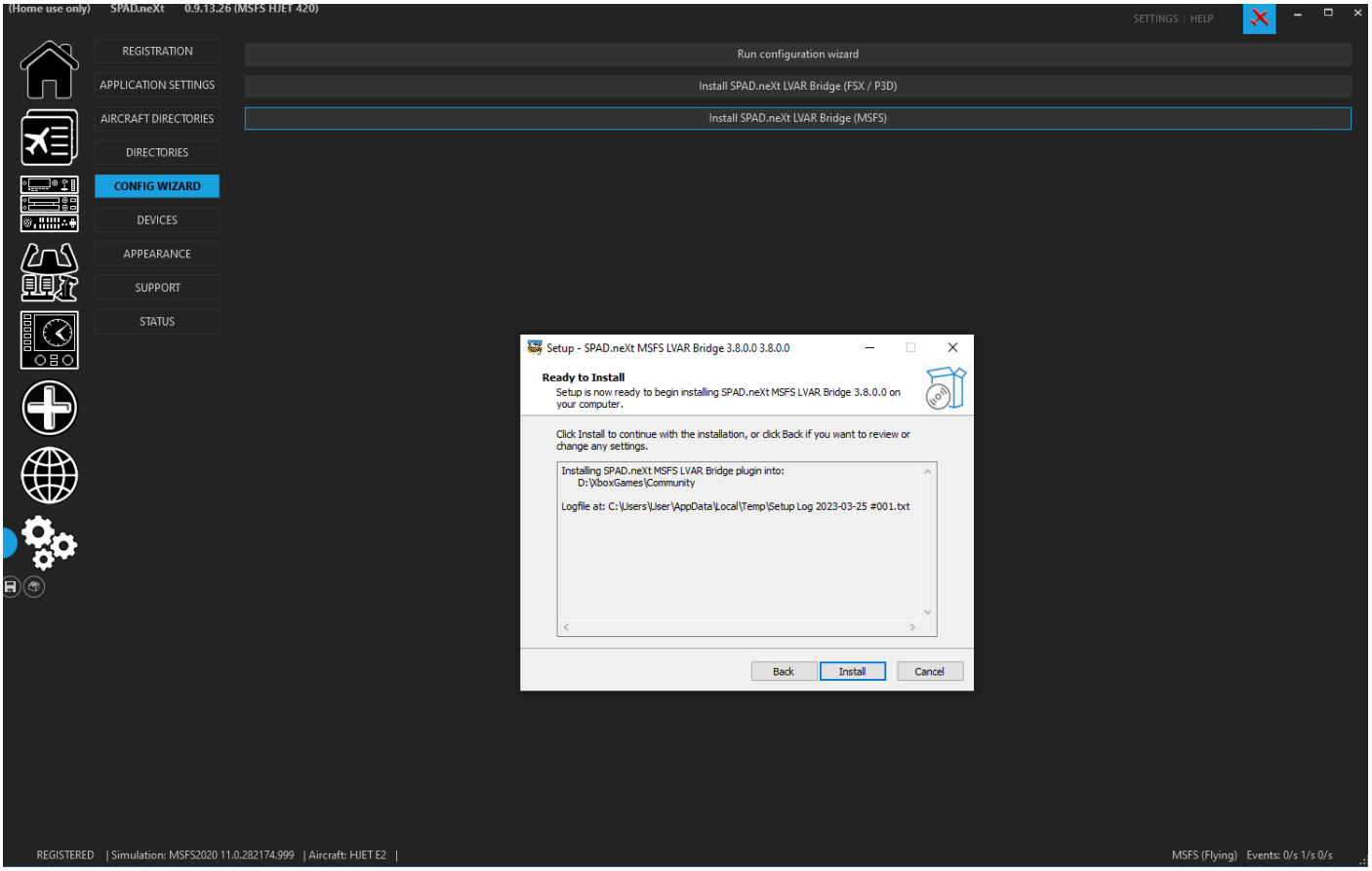
Changes Require a Restart of SPAD.neXt. HOWEVER!!! we can install the H:Event and L:Var Bridge before we do that!! Just make sure that you exit MSFS first (or restart it

after running the install. Like all Community Add Ons MSFS has to see them on initial startup otherwise they won't be loaded until a MSFS Restart!)

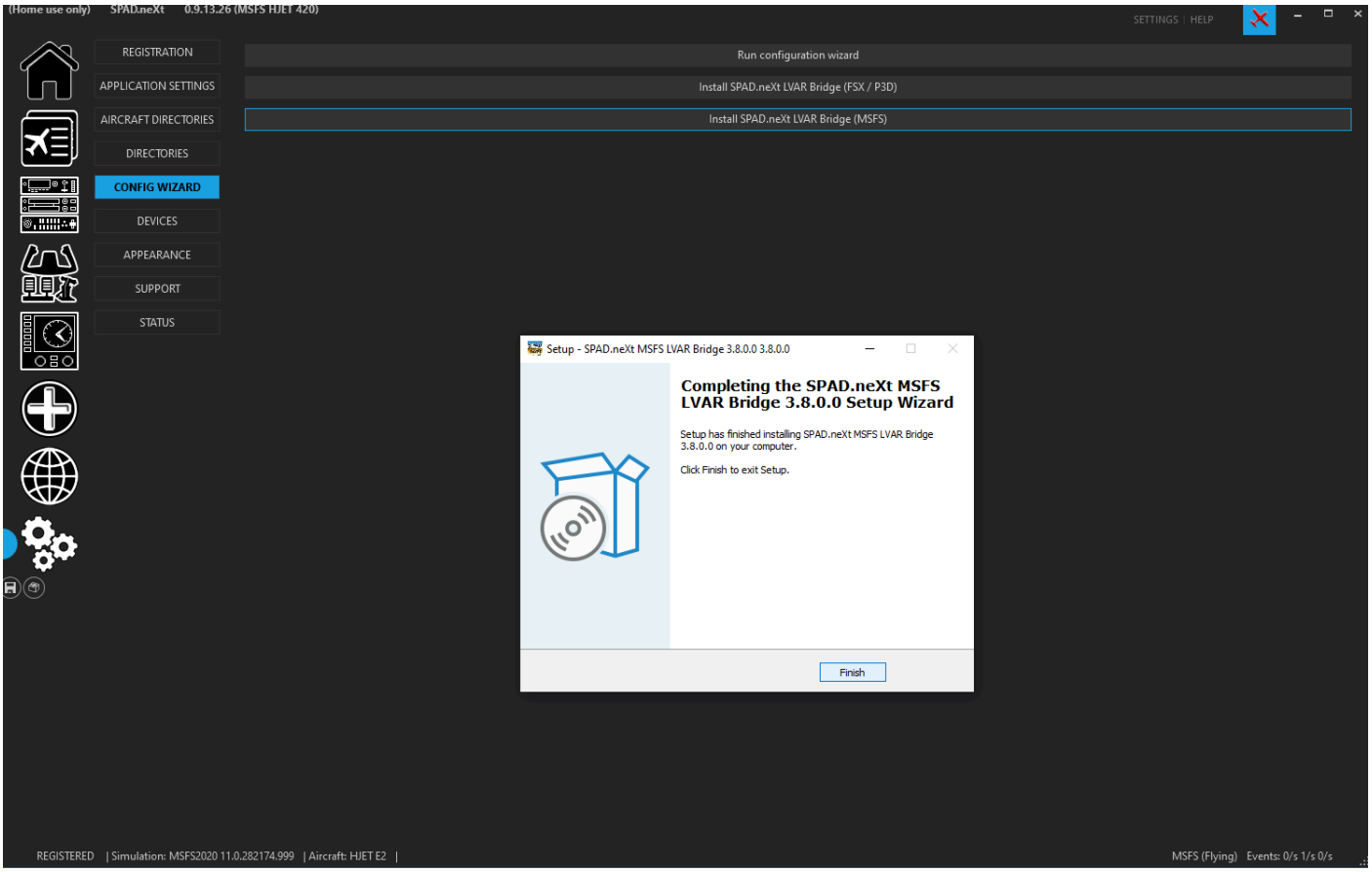
Click on Install LVAR Bridge (do this for all Sims you have installed.. example is MSFS..)



Click on the NEXT button



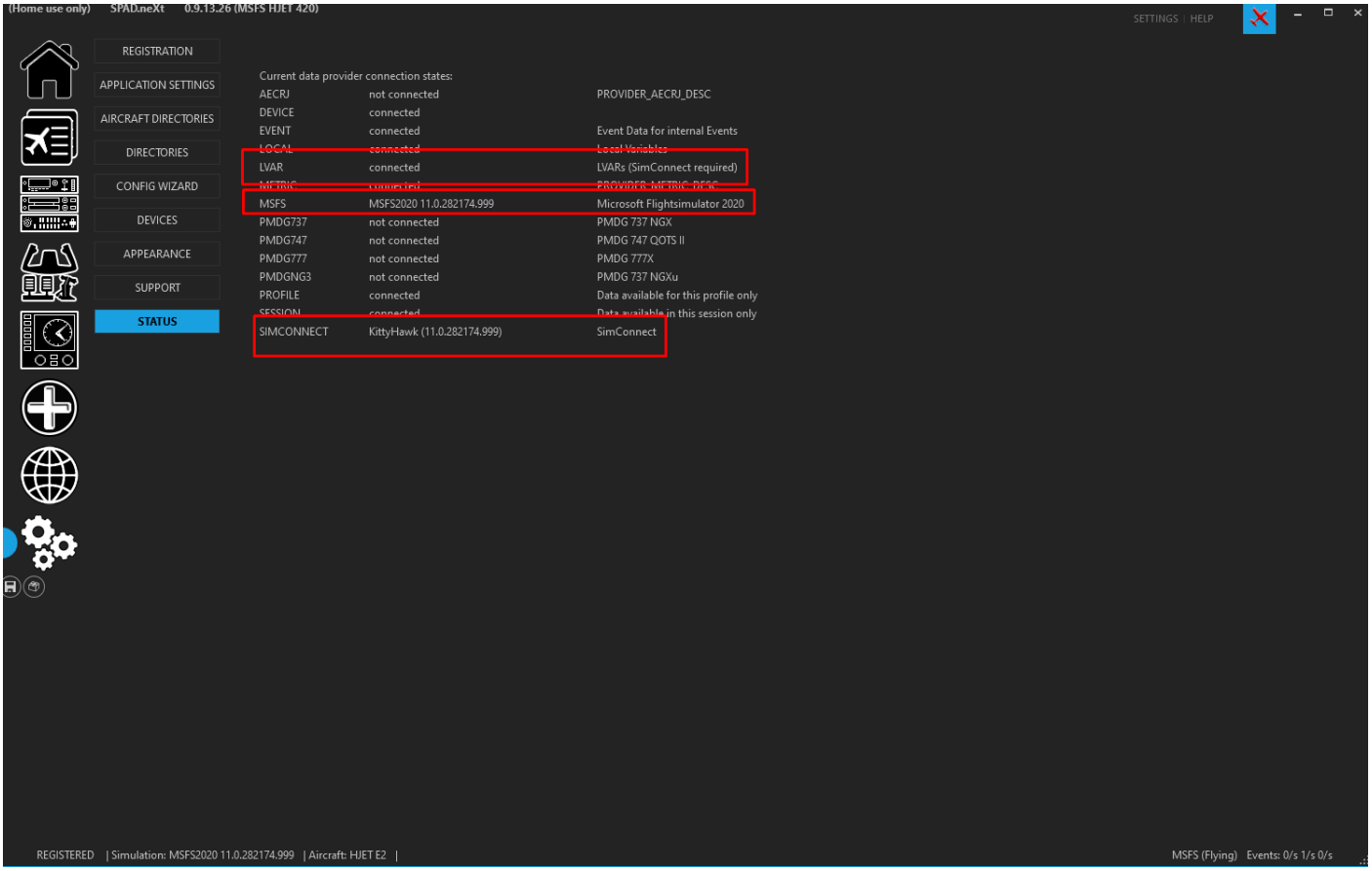
Click on Install



Click on Finish

Now Restart MSFS if it was still Running and Close and Relaunch SPAD.neXt

With MSFS back up and Running and SPAD.neXt Relaunched go to the Status Page to confirm the SimConnect and LVAR Connections:



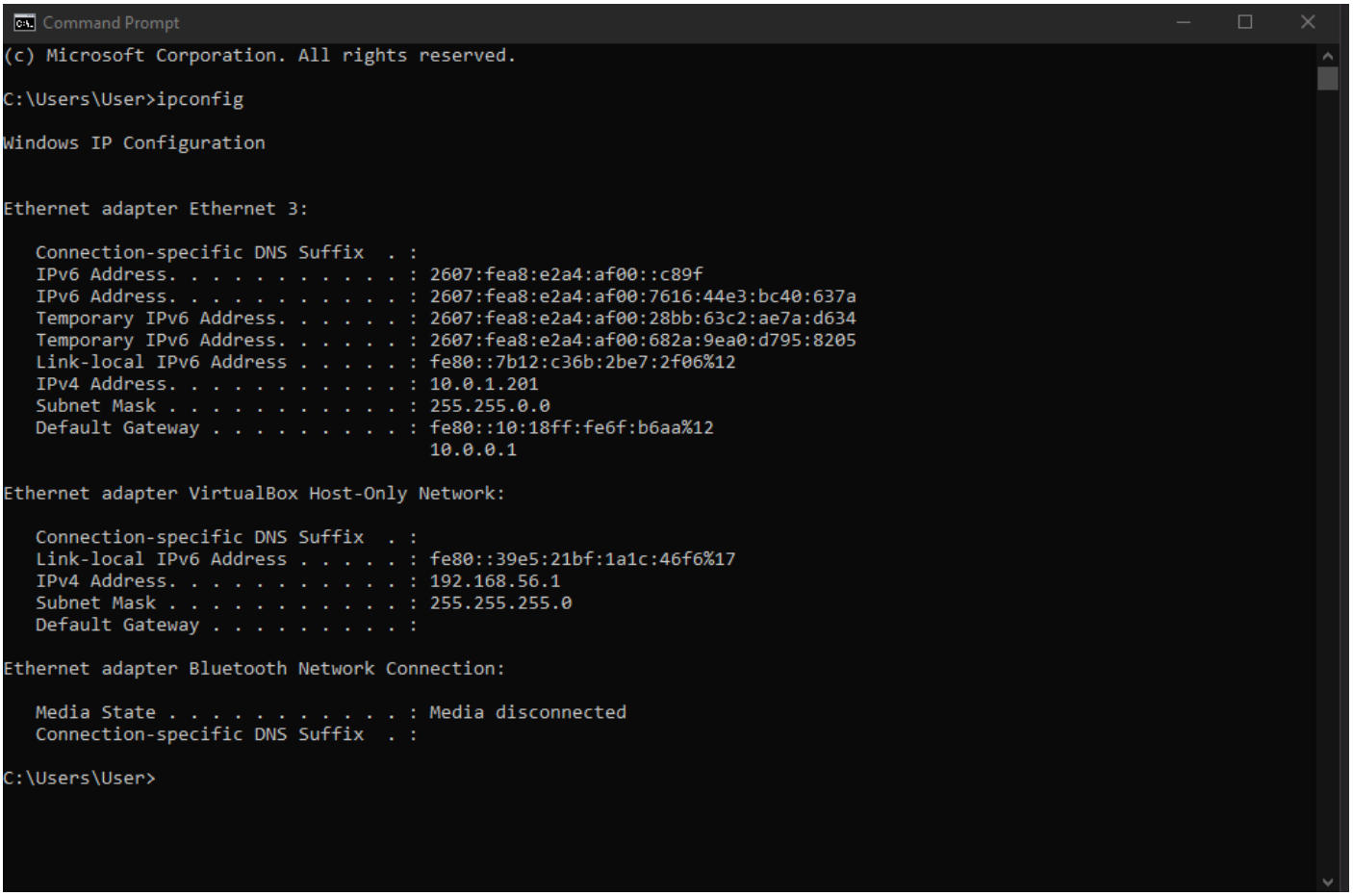
Remote Configuration (Simulator and SPAD.neXt on Different Computers)

Add The following to your XML File On the PC Running the Sim. No need to edit a file on the Remote PC. In this example the IP Address is the Address of the Sim PC.

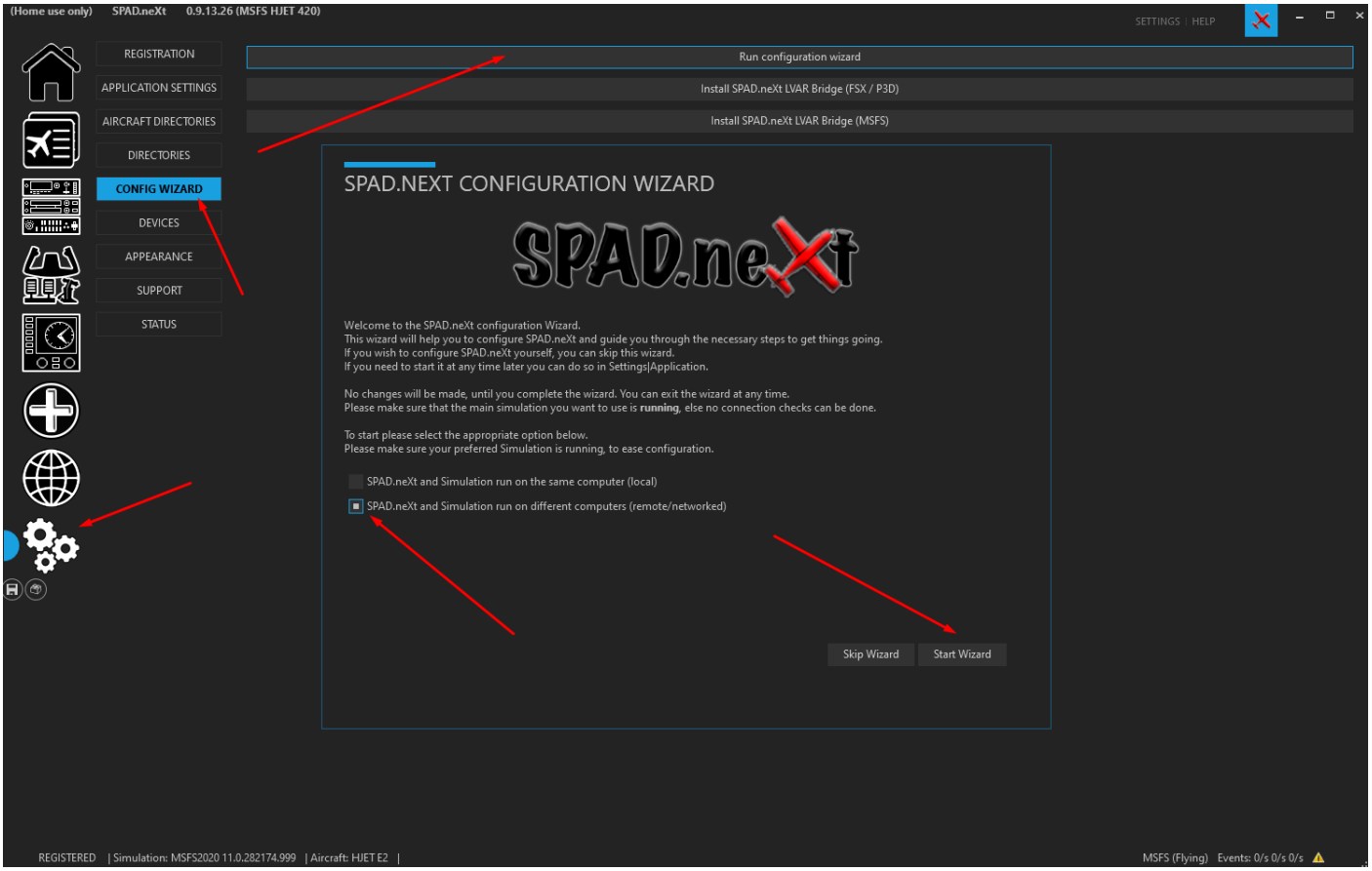
```
<SimConnect.Comm>
    <Disabled>False</Disabled>
    <Protocol>IPv4</Protocol>
    <Scope>global</Scope>
    <MaxClients>64</MaxClients>
    <Address>10.0.1.201</Address>
    <Port>500</Port>
    <Disablenagle>False</Disablenagle>
</SimConnect.Comm>
```

An easy way to figure out your ip address is to press the WINDOWS key and type in CMD then press Enter. This will launch the Command Prompt.

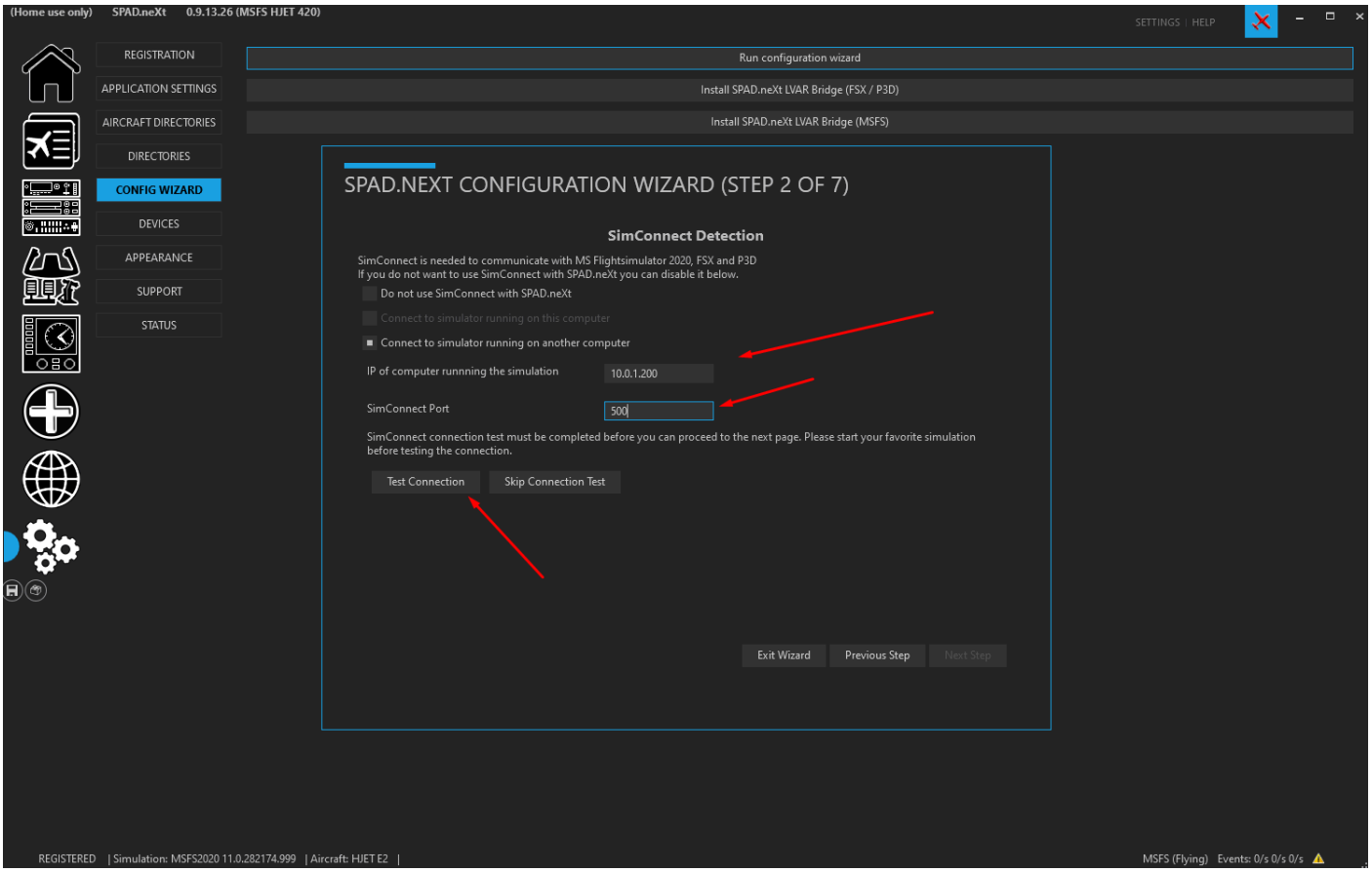
Type in "ipconfig" and press enter



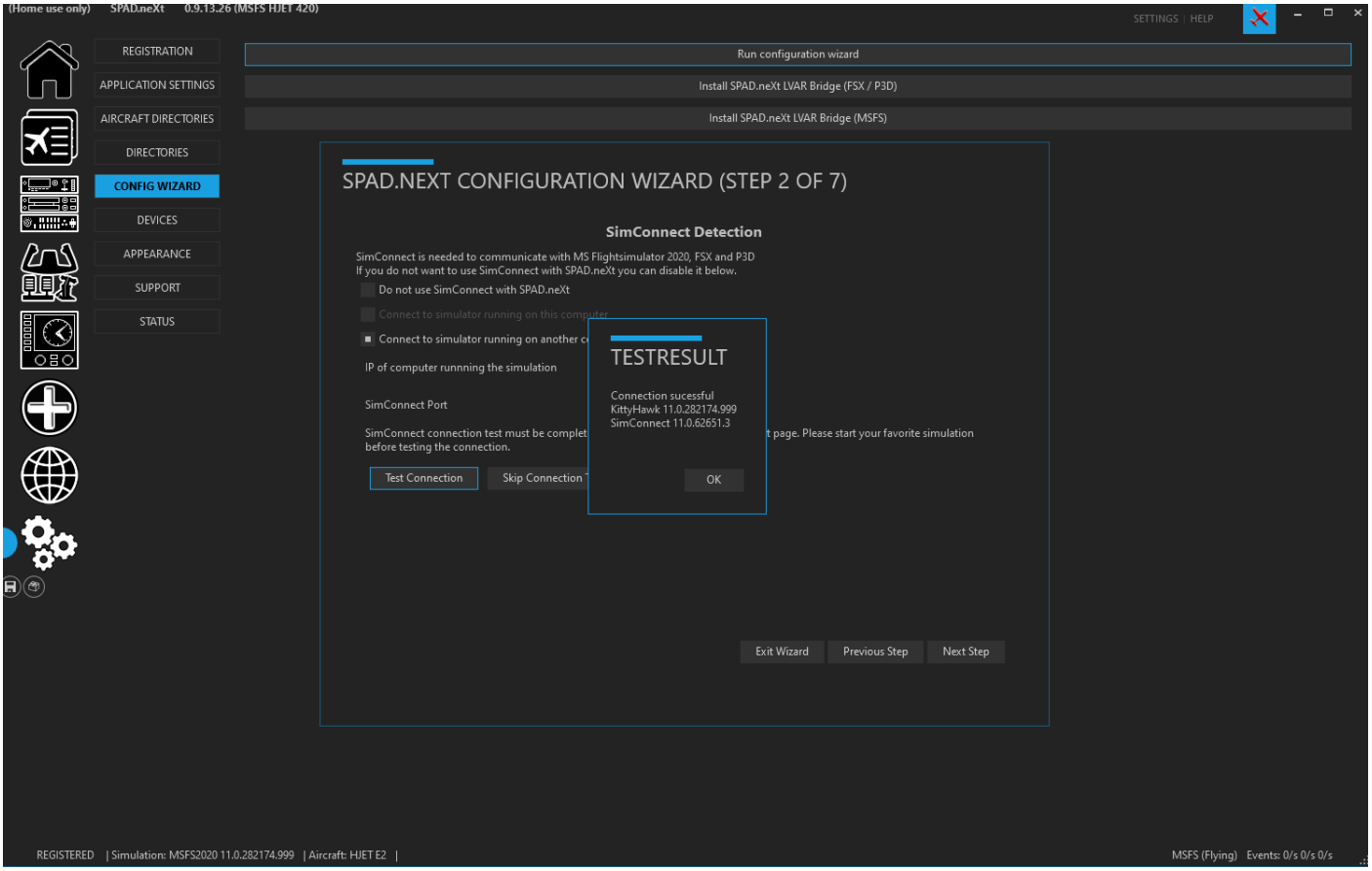
RUN The Config Wizard on the Networked PC. Make sure to select Remote/Network then Start the Wizard.



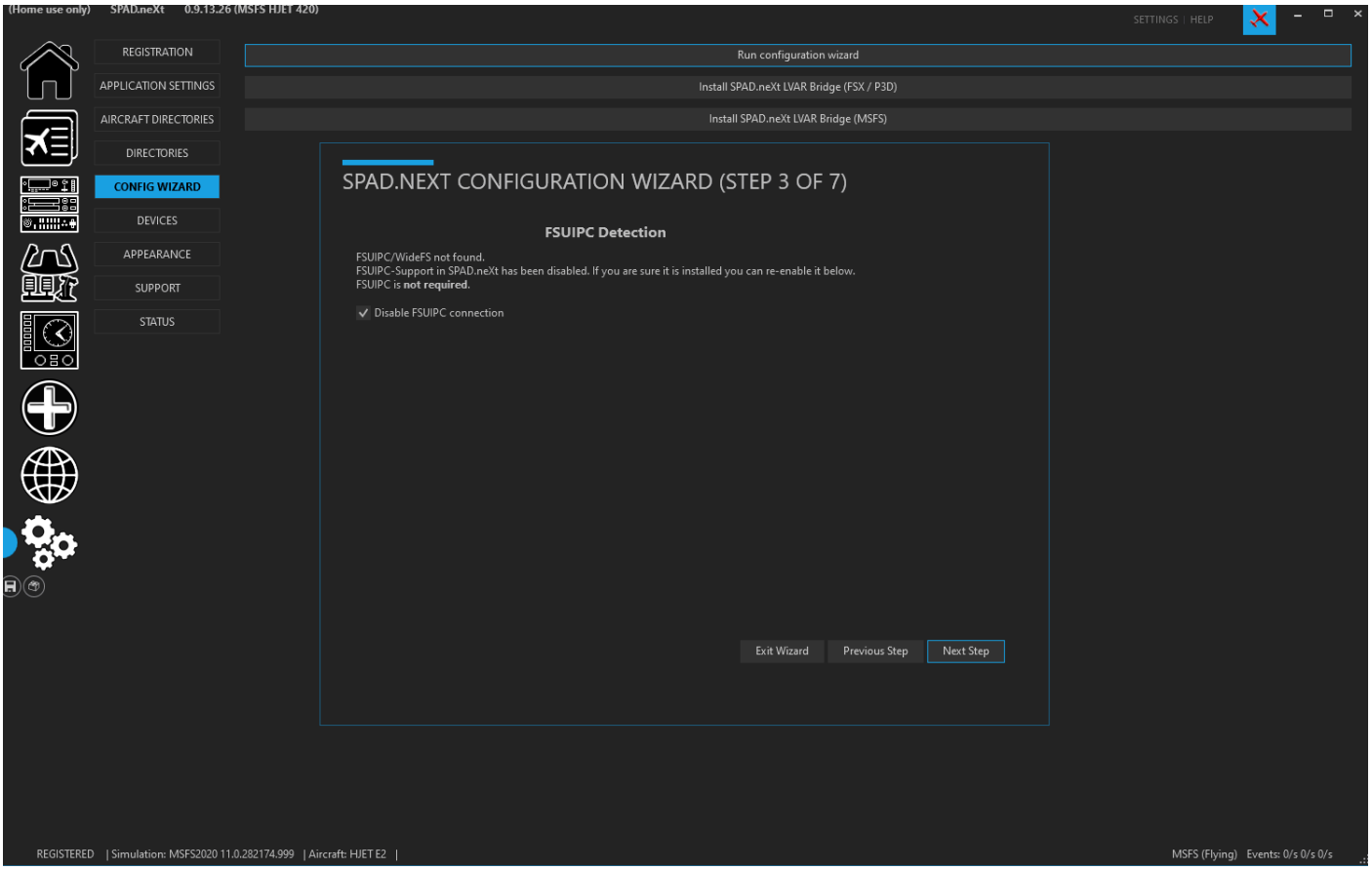
STEP 2: Add the IP Address and Port number as entered in the Sim PC XML.



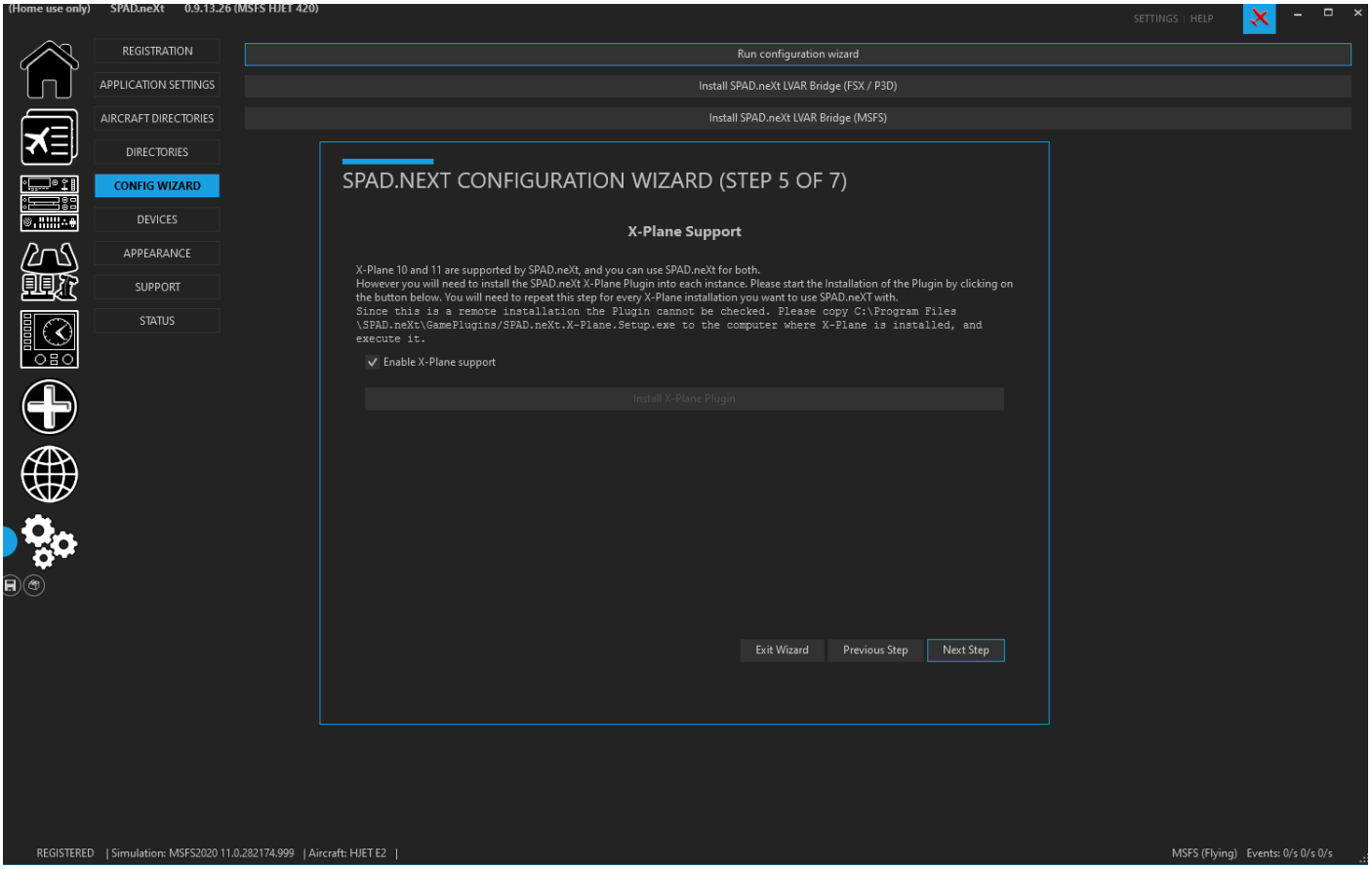
Run the Test Connection



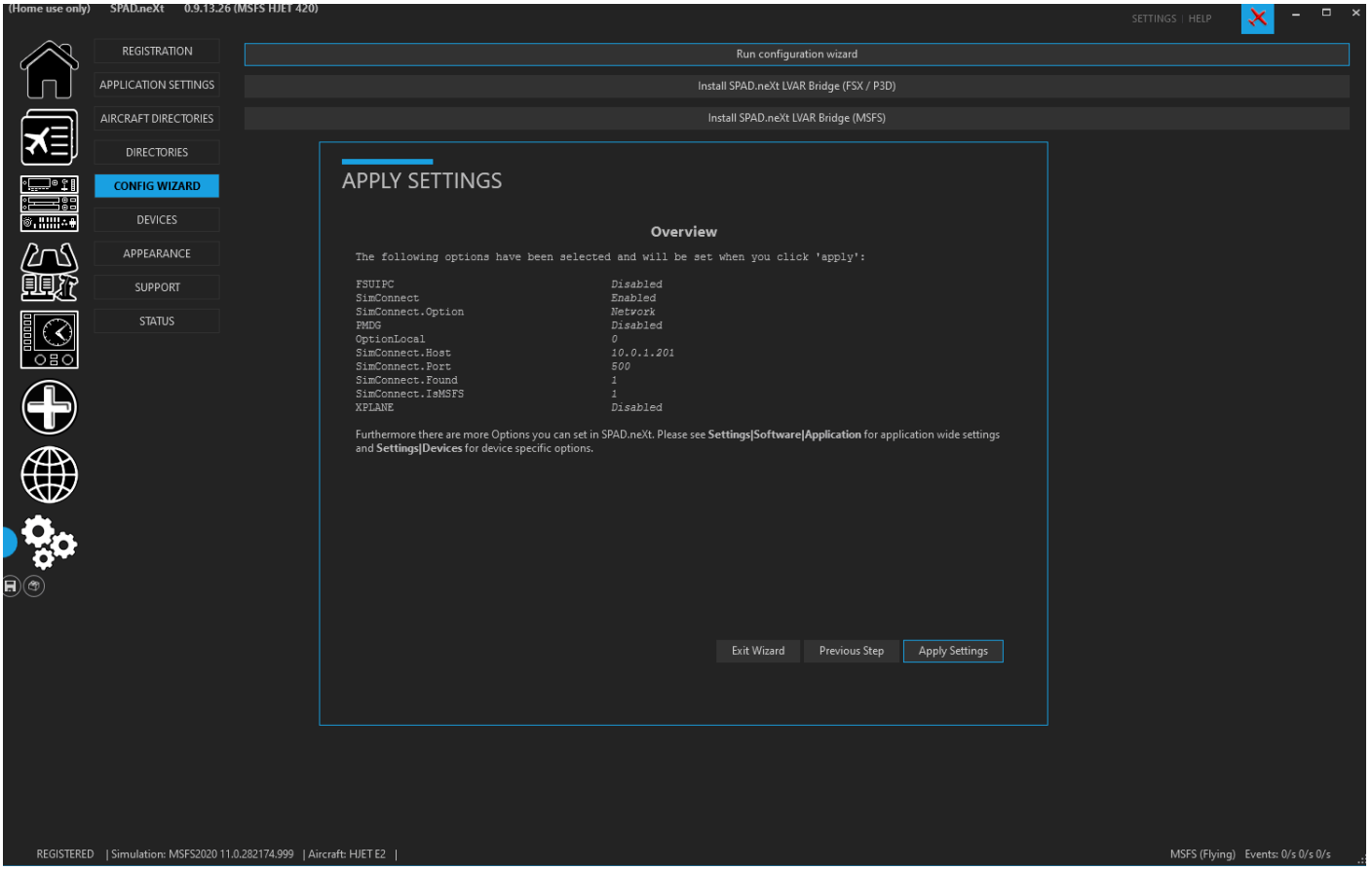
STEP 3: FSUIPC.. this is if you want to be able to link FSUIPC offsets through SPAD. For Network connection you will need WideFS running.

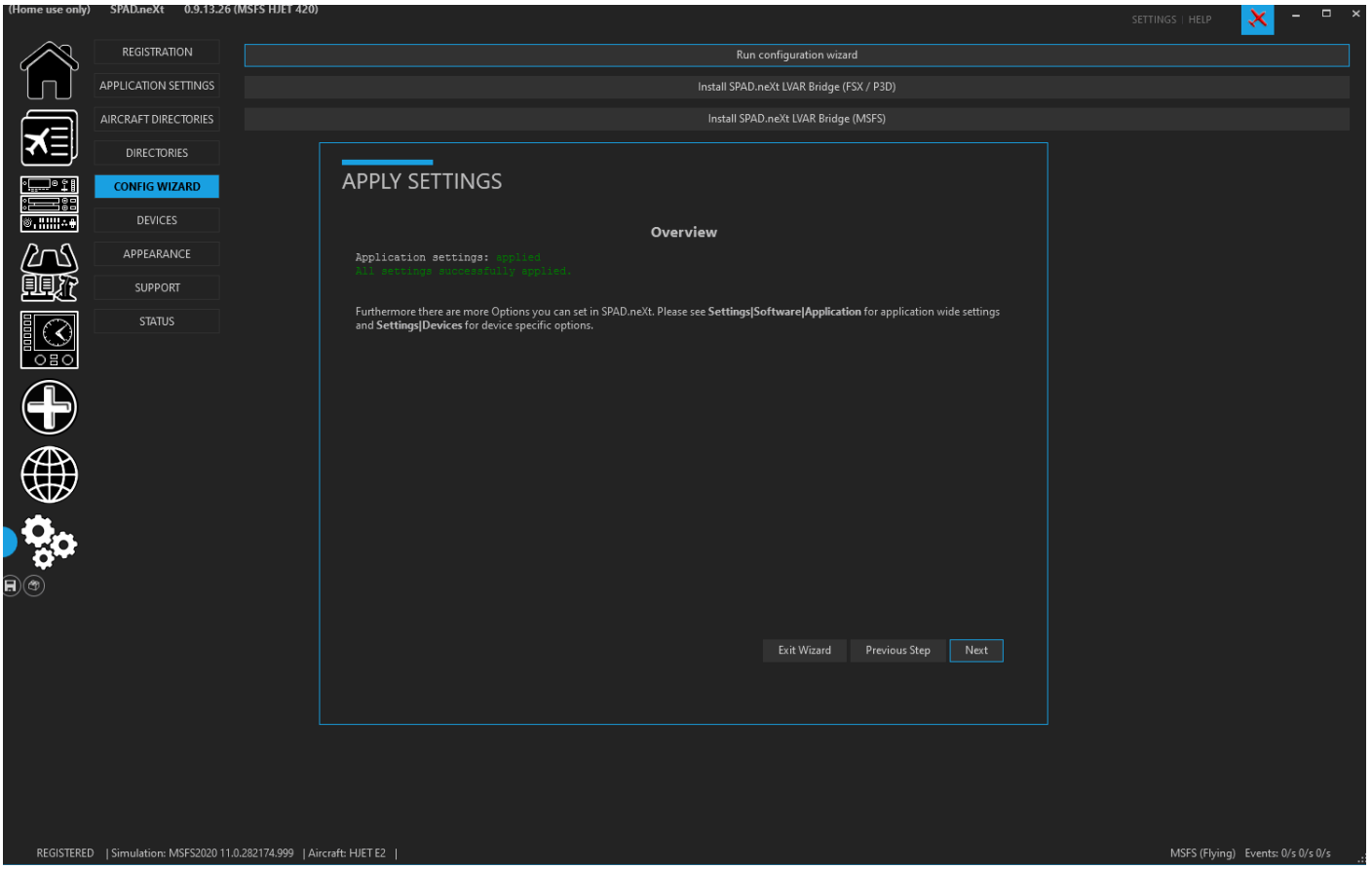


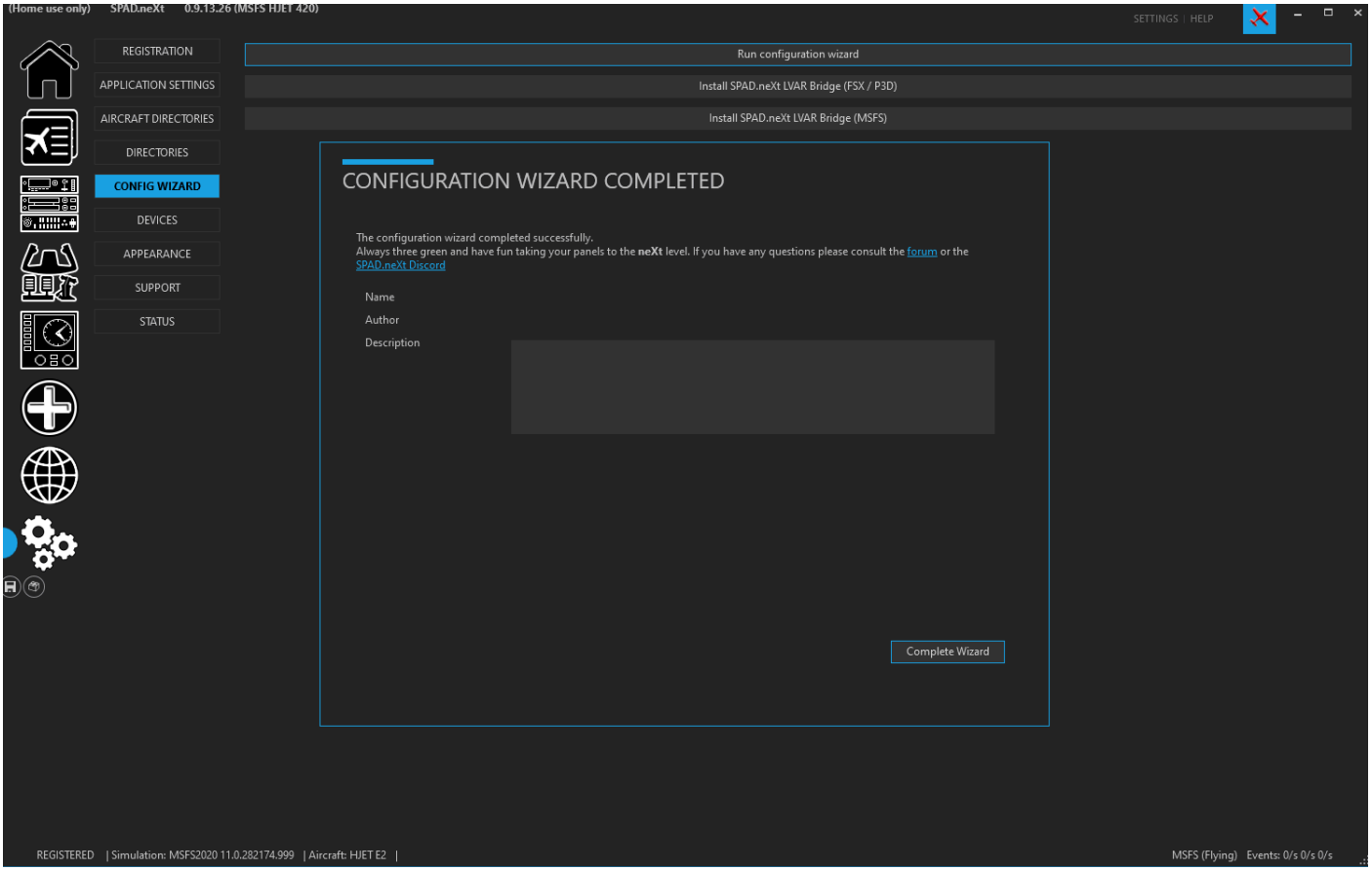
STEP 5: XPlane Support (Note Step 4 is skipped on Network Installs) enable XPlane if you have it on the Sim PC as well .. Note that you can only make 1 connection so unlike MSFS where you could have Local SPAD and Remote SPAD at the same time XPlane can only be 1.. so chose wisely...



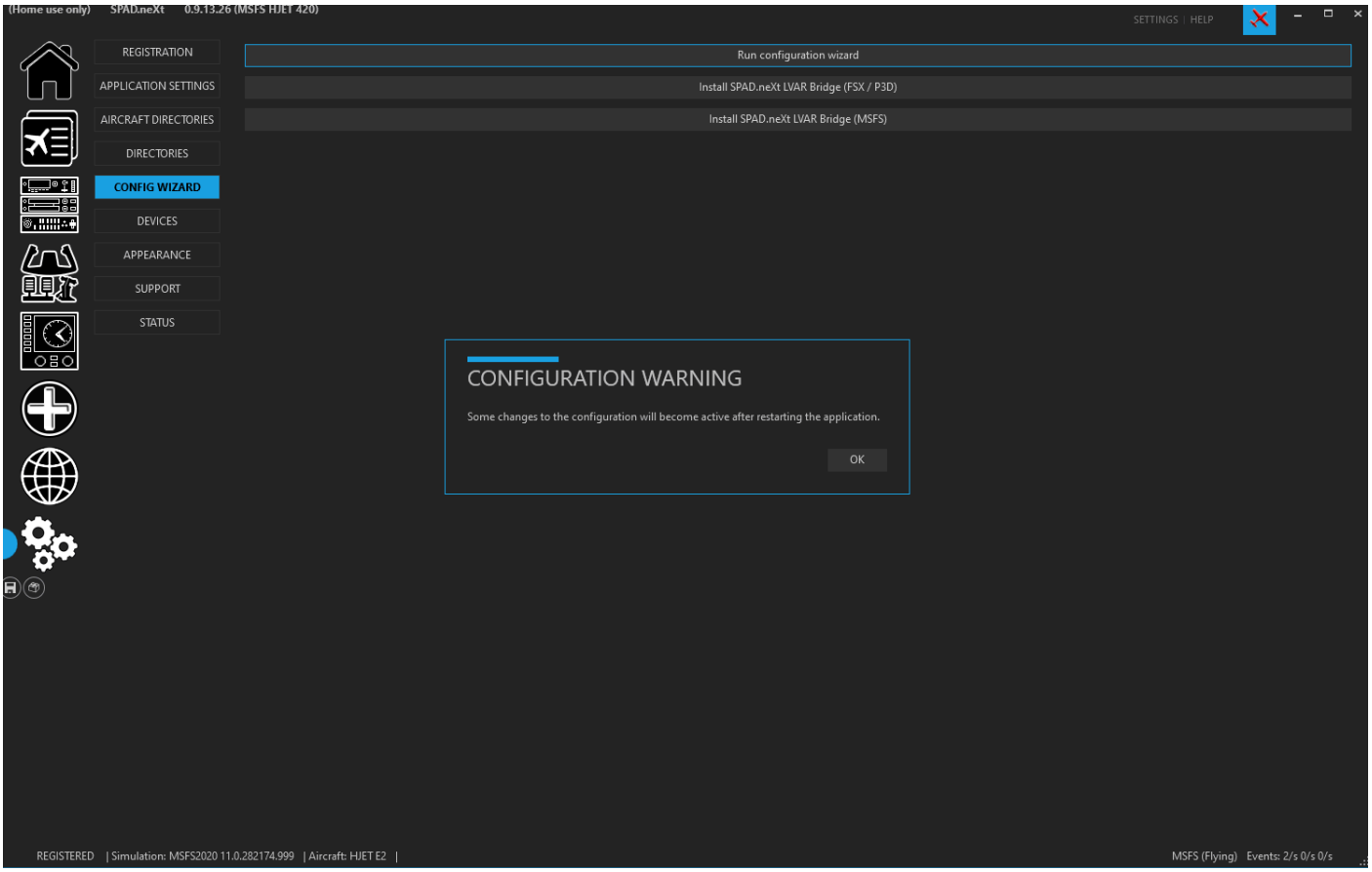
Click Apply!! Step 6 is also not required on Network Installations.


(Home use only) SPAD.neXt 0.9.13.26 (MSFS HJET 420)
SETTINGS | HELP
REGISTRATION
APPLICATION SETTINGS
AIRCRAFT DIRECTORIES
DIRECTORIES
CONFIG WIZARD
DEVICES
APPEARANCE
SUPPORT
STATUS
Run configuration wizard
Install SPAD.neXt LVAR Bridge (FSX / P3D)
Install SPAD.neXt LVAR Bridge (MSFS)
APPLY SETTINGS
Overview
The following options have been selected and will be set when you click 'apply':
FSUIPC Disabled
SimConnect Enabled
SimConnect.Option Network
PMDG Disabled
OptionLocal 0
SimConnect.Host 10.0.1.201
SimConnect.Port 500
SimConnect.Found 1
SimConnect.IsMSFS 1
XPLANE Disabled
Furthermore there are more Options you can set in SPAD.neXt. Please see Settings|Software|Application for application wide settings and Settings|Devices for device specific options.
Exit Wizard
Previous Step
Apply Settings
REGISTERED | Simulation: MSFS2020 11.0.282174.999 | Aircraft: HJET E2 |
MSFS (Flying) Events: 0/s 0/s 0/s


(Home use only) SPAD.neXt 0.9.13.26 (MSFS HJET 420)
SETTINGS | HELP
REGISTRATION
APPLICATION SETTINGS
AIRCRAFT DIRECTORIES
DIRECTORIES
CONFIG WIZARD
DEVICES
APPEARANCE
SUPPORT
STATUS
Run configuration wizard
Install SPAD.neXt LVAR Bridge (FSX / P3D)
Install SPAD.neXt LVAR Bridge (MSFS)
APPLY SETTINGS
Overview
Application settings: applied
All settings successfully applied.
Furthermore there are more Options you can set in SPAD.neXt. Please see Settings|Software|Application for application wide settings and Settings|Devices for device specific options.
Exit Wizard
Previous Step
Next
REGISTERED | Simulation: MSFS2020 11.0.282174.999 | Aircraft: HJET E2 |
MSFS (Flying) Events: 0/s 0/s 0/s



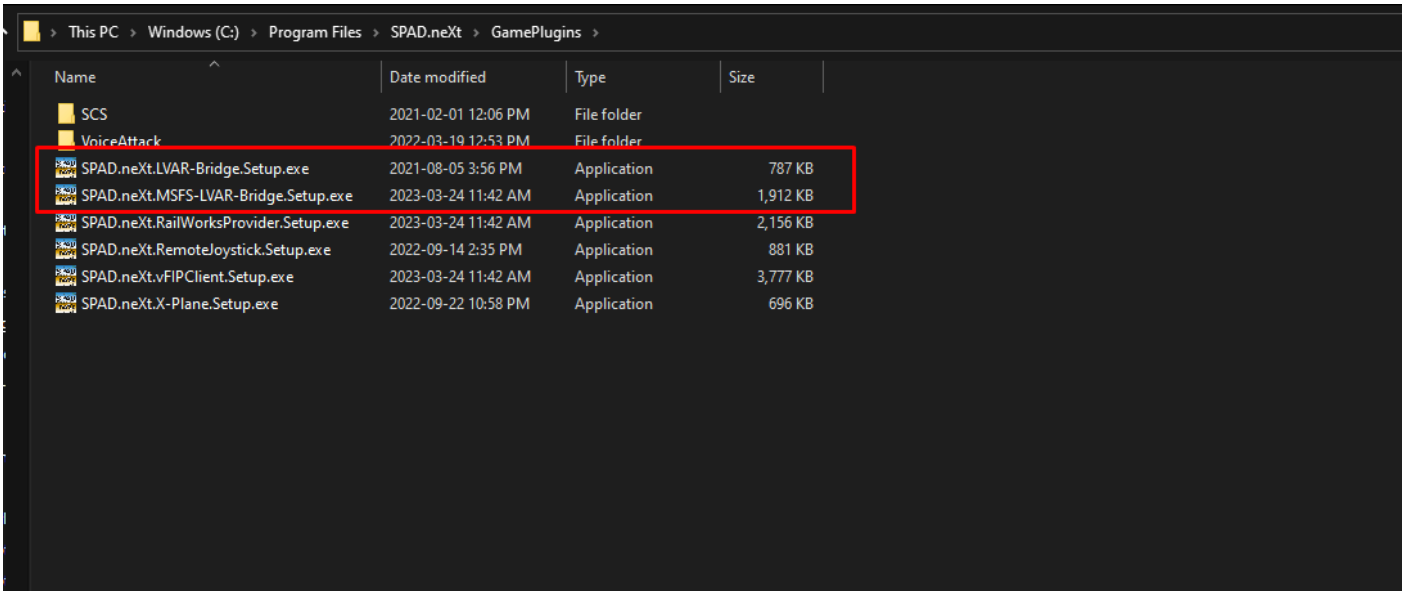
Now you will need to restart!



Close SPAD.neXt and then Restart it for all the settings to take effect.

IMPORTANT!!
With a Network Connection where no SPAD.neXt was installed on the Local Sim PC the LVAR Bridge will need to be installed Manually:

Find the LVAR Bridge Installers on the Network PC.
Default install location -- C:\Program Files\SPAD.neXt\GamePlugins



Copy the Bridge Installers to the Sim PC and run them on it.

These are needed for Complex Add On that use L:Vars (Local Variables defined by the simobjects when loaded.. ie when you load up a plane..) And also with MSFS where the new Instruments are JS/HTML and require the "H:Events" which must pass through the WASM Module and not directly Via SimConnect.

MSFS: Enable PMDG data access

Please see following pages

for MSFS 2020

for MSFS 2024

X-Plane Configuration

X-Plane detection

Wenn running the SPAD.neXXt Configuration Wizard, or installing the SPAD.neXt X-Plane plugin, you will automatically be presented all found X-Plane installations on the computer.

However if you installed instances of X-Plane manually (e.g. by copying), those cannot be detected automatically and you will have to tell SPAD.neXt where to find them:

- [ ] Open Windows File-Explorer and navigate to `%LOCALAPPDATA%`
- [ ] There should be a file named `x-plane_install_[versionnumber].txt` (e.g. x-plane_install_12.txt) for each x-plane version (10/11/12) you have installed/using
- [ ] If that file is missing create it
- [ ] edit the file for your version and add all paths where you have installed that X-Plane version

eg.:
`W:\X-Plane 12/`

- [ ] Now SPAD.neXt and the Plugin-Installer should recognize all X-Plane installations.

X-Plane Plugin Configuration

If SPAD.next and X-Plane are running on different computers, you will have to tell the plugin where SPAD.neXt is running.

First start X-Plane once, and exit it again.

Now head to your X-Plane installation folder and navigate to Output\Preferences

Here you will find a file called "**SPADneXtConnector.sdm**". Open it in your favorite text editor (e.g. notepad)

The file will look like something this (do not delete any additional entries!):

```
{
"ServerHost": "127.0.0.1",
```

```
"ServerPort": 8181
}
```

replace "127.0.0.1" by the ip (IPv4 or IPv6) of your SPAD.neXt-PC.

The port should be ok, however if the port 8181 on your SPAD.neXt-PC is not available SPAD.neXt will automatically select another one. To find out which one, head to Settings->Application->Expert in SPAD.neXt. You will find an option "Remote Services Port" there and the Port that is being used.

Save the file, and make sure the firewall on the SPAD.neXt-PC has port 8181 open.

First Start

Starting SPAD.neXt the first time

Configuration Wizard

Common Tasks and Issues

This section and sub-pages describe common tasks and solutions for common problems

Thrustmaster TCA Speedbrake / Flaps are reversed

Make sure the Speedbrake and Flaps are calibrated following the Thrustmaster Guide: https://ts.thrustmaster.com/download/accessories/manuals/TCA_Quadrant/TCA_Quadrant-Throttle-addons_Calibration.pdf

SPAD

Reset SPAD.neXt license

Delete the file `%APPDATA%\SPAD.neXt\SPAD.neXt.lic`\ SPAD.neXt will start as trial and you can reenter your license key.

How to properly open a Support Ticket

1. Start SPAD.neXt
2. (reproduce problem)
3. in SPAD.neXt click in Settings->Support
4. and either Select **an exiting** ticket from the list and click on UPDATE TICKET or click on Create new ticket
5. Add any info needed and hit submit.

Note: if for reproducing the problem no simulation connection is needed please do so from a fresh SPAD.neXt start without Simulator running.
If simulator is needed do it after a new start not after a over the pond flight
It's no joy to gravel through megabytes of unnecessary logfiles if you update the ticket after a 5 hour flight. Processing of such tickets is declined.

If you cannot open the ticket from inside SPAD.neXt send an email to support@spadnext.com and attach the file `%APPDATA%\SPAD.neXt\logs\SPAD.log`

Available SPAD.neXt update channels

There are 3 update channels available for SPAD.neXt

- **Release**: This is the public release of SPAD.neXt. Your SPAD.neXt installation will automatically update to it, if your update subscription is valid for it
- **Beta**: The upcoming pre-release, available only for "Complete Edition"-License users , or when assigned for special tests.

- **Alpha**: The current development version of SPAD.neXt. This update channel is invite only for early adopters. To apply for participation, open a ticket from within SPAD.neXt. Development focus of current Alpha version can be reviewed in the SPAD.neXt-Discord in the appropriate #release-faq thread

To change your update channel, head to Settings->Application->Expert in SPAD.neXt.

All update channels available to you will be selectable.

> Note: All update-channels other than Release **require to join the SPAD.neXt community discord and link your SPAD.neXt license** to your discord user properly. All communication for non-release updates is done via discord exclusively.

Windows Defender claiming "Trojan:Script/Wacatac.B!ml" Infection

All releases of SPAD.neXt are virus scanned against the 3 most common scan-engines, and all executables are digitally signed. However sometimes Windows Defender will complain when trying to run e.g. a Plugin-Installer and claim it being infected with "Trojan:Script/Wacatac.B!ml".
This is a known Problem of how Windows Defender handles compressed files sometimes and usually is a false-positive.
First countercheck that the digital signature of the file is intact:

- Right click the file and select "Properties"
- Select the tab "digital signature" and confirm it states "signature is valid"

If the signature is intact, you need to update you Windows Defender virus database and clear the detectioncache:

1. Open command prompt as administrator and change directory to c:\Program Files\Windows Defender
2. Run “MpCmdRun.exe -removedefinitions -dynamicsignatures”
3. Run "MpCmdRun.exe -SignatureUpdate"

If the signature is **not** intact. Delete the file immediately and report to support@spadnext.com
If in doubt, you can always submit the file to Microsoft for analysis.

Bitdefender claiming Infection

From time to time Bitdefender changes its detection startegies and might flag some required components of SPAD.neXt as infected and refuses to load them, which might result in SPAD.neXt crashing or malfunctioning.

It will take some time until those components are whitelisted by Bitdefender again.

Only known workaround, apart from submitting a ticket to Bitdefender and report a false-positive and wait for them to react, is to define a Bitdefender exclusion for SPAD.neXt installation directory.

Online Services

Online Services Requirements

To be able to access the SPAD.neXt online services, the following requirements must be met:

- The time of your computer must not differ from World-Reference-Time.
- SPAD.neXt must be able to communicate with "https://update.spadnext.net". You can check if communication is working, by opening https://update.spadnext.net/checkpage.html in a browser. It should just state "Works"

Online Services Restrictions

If your SPAD.neXt update subscription is expired, you can only access snippets and profiles that have been published before the expiration date. You not be able to publish any snippets/profiles or update your existing ones.

"NOT_AUTHORIZED" Error when publish a snippet/profile

Most likely your update subcription has expired or you are using an outdated version of SPAD.neXt which is no longer supported.

"Online Services not available" when accessing Online Snippets

Please check the time synchronization on your computer.
Due to heavy attacks to our services we had to increase the security, and requests with an invalid/outdated timestamp will be rejected.
Your computer time most likely reports to differ from World Reference Time by more than 3 minutes.
(Right click Windows Start icon -> Settings -> enter "time" -> Change Date/Time and make sure Synchronization is on)

Fresh reinstall

Completely reset SPAD.neXt installation

To reinstall SPAD.neXt completely, proceed as follows:

1. Make a backup copy of "Documents/SPAD.neXt" (It contains your profiles)
2. Uninstall SPAD.neXt
3. Delete the Folder %APPDATA%\SPAD.neXt
4. Delete the Folder %LOCALAPPDATA%\SPAD.neXt
5. restart computer
6. Reinstall SPAD.neXt using the latest installer.

Move SPAD.neXt to new Computer

To move/copy SPAD.neXt to a new computer, please complete the following steps:

On OLD computer

- [ ] Start SPAD.neXt, head to Settings->Registration and deactivate the license
- [ ] Make a backup of Documents\SPAD.neXt (it contains your profiles etc)

On new Computer

- [ ] Install SPAD.neXt and the LVAR Bridge
- [ ] Start SPAD.neXt , head to Settings->Registration and activate your license
- [ ] Check your Documents Settings. IF your Documents are located on OneDrive it is highly recommended not to store any SPAD.neXt configuration there.
To move your SPAD.neXt data outside the OneDrive, create a Folder on a hard-drive, the in SPAD.neXt at Settings->Directories change the location of the Documents to it.
- [ ] Restore previously saved backup of Documents\SPAD.neXt to either Documents or the new location chosen above

License Issues

Lost license information

- For licenses bought **after 12.11.2016** via Avagate/2Checkout, you can log in to your customer account at https://secure.2co.com/myaccount/ to see your order history and license information.

- For licences bought **before 12.11.2016** via SimMarket , simply log in to your simmarket account at https://secure.simmarket.com . Your license information is listed there.

- **If you have linked your license on the SPAD.neXt community Discord, you can ask the bot about your license details using the "!licinfo" command.**

  If this does not help, open a ticket from within SPAD.neXt to request your license information and attach a proof of purchase for the license. (Important: Mark it as license issue!)

  If you cannot start SPAD anymore, open a ticket via email (support at spadnext.com) from the email you used to purchase SPAD.neXt. Providing your order number and a proof of purchase will speed up the process.

  If you do not have access anymore to the email address you purchased SPAD.neXt from, a proof of purchase is mandatory to have it re-assigned.

"There is a Problem with your license, please contact support"

If your SPAD.neXt installation all of the sudden complains about a license problem, this is normally caused by a Windows update changing vital information.

- Delete the file %APPDATA%\SPAD.neXt\SPAD.neXt.lic
- SPAD.neXt will start in trial mode and allows you to re-enter your license information
- if it still complains contact support@spadnext.com

MSFS Specific Topics

LVAR Bridge Outdated (MSFS 2020)

If SPAD.neXt keeps reporting that the LVAR Bridge is outdated:

1. Download the latest MSFS LVAR Bridge Installer at https://www.spadnext.com/download/download-spad-next.html
2. Proceed as follows
3. Exit MSFS 2020 and SPAD.
4. Delete any folders named "spad-bridge-module" in:

   MSFS Store/Xbox Version

   ```
   %LOCALAPPDATA%\Packages\Microsoft.FlightSimulator_8wekyb3d8bbwe\LocalSt
   ```

   MSFS Steam Version

   ```
   %APPDATA%\Microsoft Flight Simulator\Packages
   ```

5. Delete any folders named "spad-bridge-module" in your Community folder. Default locations:

   MSFS Store/Xbox Version

   ```
   %LOCALAPPDATA%\Packages\Microsoft.FlightSimulator_8wekyb3d8bbwe\LocalCa
   ```

   MSFS Steam Version

   ```
   %APPDATA%\Microsoft Flight Simulator\Packages\Community
   ```

6. Install the latest version of the MSFS LVAR Bridge and select the MSFS 2020 location.

7. Start MSFS 2020 and wait until it reaches the main menu.

8. Start SPAD.neXt.

LVAR Bridge Outdated (MSFS 2024)

If SPAD.neXt keeps reporting that the LVAR Bridge is outdated:

1. Download the latest MSFS LVAR Bridge Installer at https://www.spadnext.com/download/download-spad-next.html

2. Exit MSFS 2024 and SPAD.

3. Delete any folders named "spad-bridge-module" in subdirectories of:

   MSFS Store/Xbox Version

   ```
   %LOCALAPPDATA%\Packages\Microsoft.Limitless_8wekyb3d8bbwe\LocalState\WA
   ```

   MSFS Steam Version

   ```
   %APPDATA%\Microsoft Flight Simulator\Packages\WASM
   ```

4. Delete any folders named "spad-bridge-module" in your Community folder. Default locations:

   MSFS Store/Xbox Version

   ```
   %LOCALAPPDATA%\Packages\Microsoft.Limitless_8wekyb3d8bbwe\LocalCache\Pa
   ```

   MSFS Steam Version

   ```
   %APPDATA%\Microsoft Flight Simulator\Packages\Community
   ```

5. Install the latest version of the MSFS LVAR Bridge and select the MSFS 2024 location.

6. Start MSFS 2024 and wait until it reaches the main menu.

7. Start SPAD.neXt.

LAG/FPS spikes with SPAD LVAR Bridge installed

During the first five minutes after starting a flight (after advancing beyond the Ready to Fly screen), the SPAD.neXt LVAR Bridge regularly scans for new LVARs. Depending on your system, you may notice lag spikes while it scans.
You can adjust the scan parameters in SPAD.neXt under Settings → Application → MSFS.

Name	Purpose	
MaxLVarSearchTime	Time in minutes the bridge will scan for LVAR after flight is started	Increase it to look for LVAR longer
MaxDigBatch	Number of LVARs scanned every frame	Decrease the value to reduce the load on the simulator.
MaxDigDepth	Maximum number of LVARs to search for	

X-Plane specific Topics

Known incompatible X-Plane Plugins

```
- SAM - Scenery Animation Manager (stairportsceneries.plugin.aos)
```

This xplane-plugin can cause the SPAD plugin to crash and then crash X-Plane. Since that plugin is outdated and known to be incompatible to xp12 and is no longer maintained for a long time already, it is recommended to uninstall it

X-Plane crashes with custom datarefs.txt

If X-Plane crashes after the aircraft is loaded and you have a custom datarefs.txt in place, check the X-Plane logfile for the last "AddDataref"-Entry from SPAD. This is normally the dataref that causes a problem, and most likely you will see a deprecated warning. Removing that dataref (and all datarefs in the same group) from your datarefs.txt should resolve this issue.

X-Plane error: Discover port blocked

Most likely you have something running like e.g. Hyper-V which reserved the ports that are used by X-Plane for external communication. You will need to tell it not to reserve those ports.
You can check with this commands:

```
netsh int ipv4 show excludedportrange protocol=tcp
netsh int ipv4 show excludedportrange protocol=udp
```

None of the lists should include port **49707**
You can change that reserved range as described here

https://dandini.wordpress.com/2019/07/15/administered-port-exclusions-blocking-high-ports/

Saitek/Logitech MP/RP blinking

If your Saitek/Logitech devices are constantly activating for a short time even when SPAD.neXt is not running, whenever you are activating a window or open the Start-Menu, this is caused by a Windows Service called "GameInput Service".
Usually it gets installed and activated when you install MSFS via Steam, to enable the Steam-Remoteplay features.
If you do not need this feature, you can get rid off that blinking:

- [ ] Start Services manager (Windows key "services")
- [ ] For the Services "GameInput Service"
  - [ ] Stop the service if it is running
  - [ ] Set the Start-Type to "disabled"
- [ ] reboot computer
- [ ] It might be necessary to repeat this steps after a windows update or if the Steam-GameInput Redistributables are reinstalled by a game installation

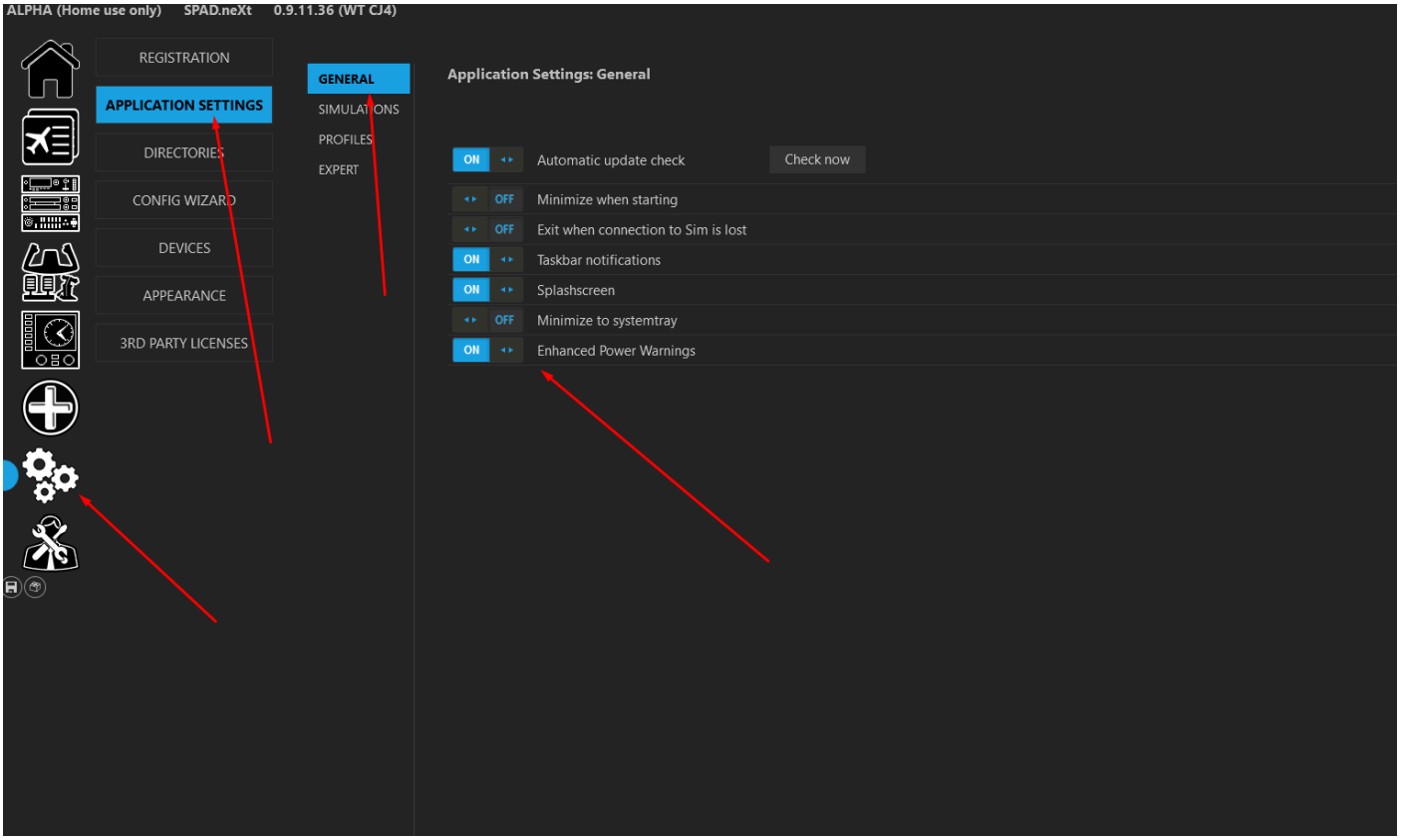
USB Power Management

Devices stop reacting / Displays stay blank after a while

Windows is putting the devices to sleep... So after you start spad and it runs and sees the devices... eventually they go to sleep when there is no interaction with them for a while...

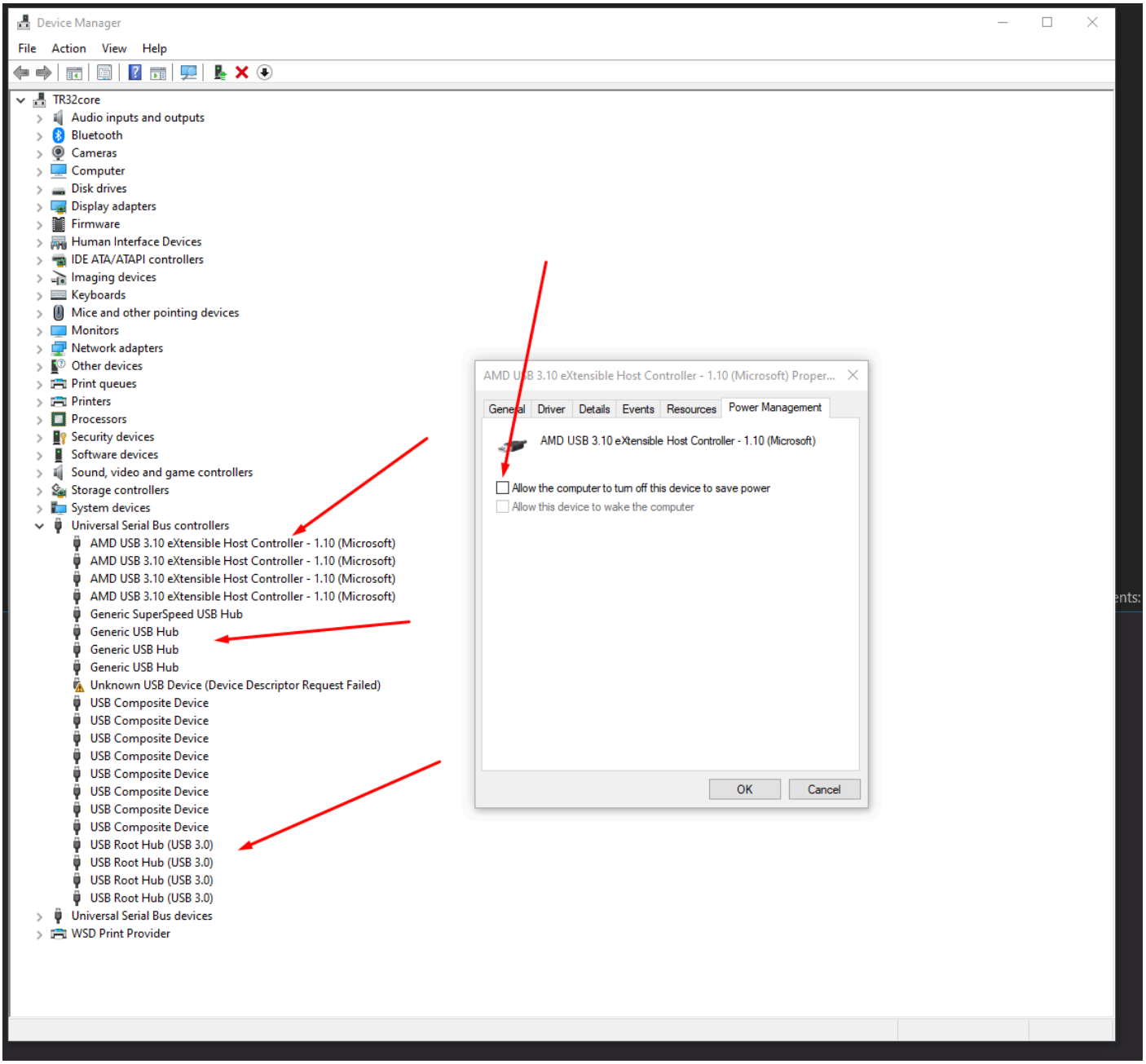
Please also see this article about the common requirements for connecting USB devices to flight sim computers: USB HUB Requirements

> Windows updates likes to turn power management back on.[1]

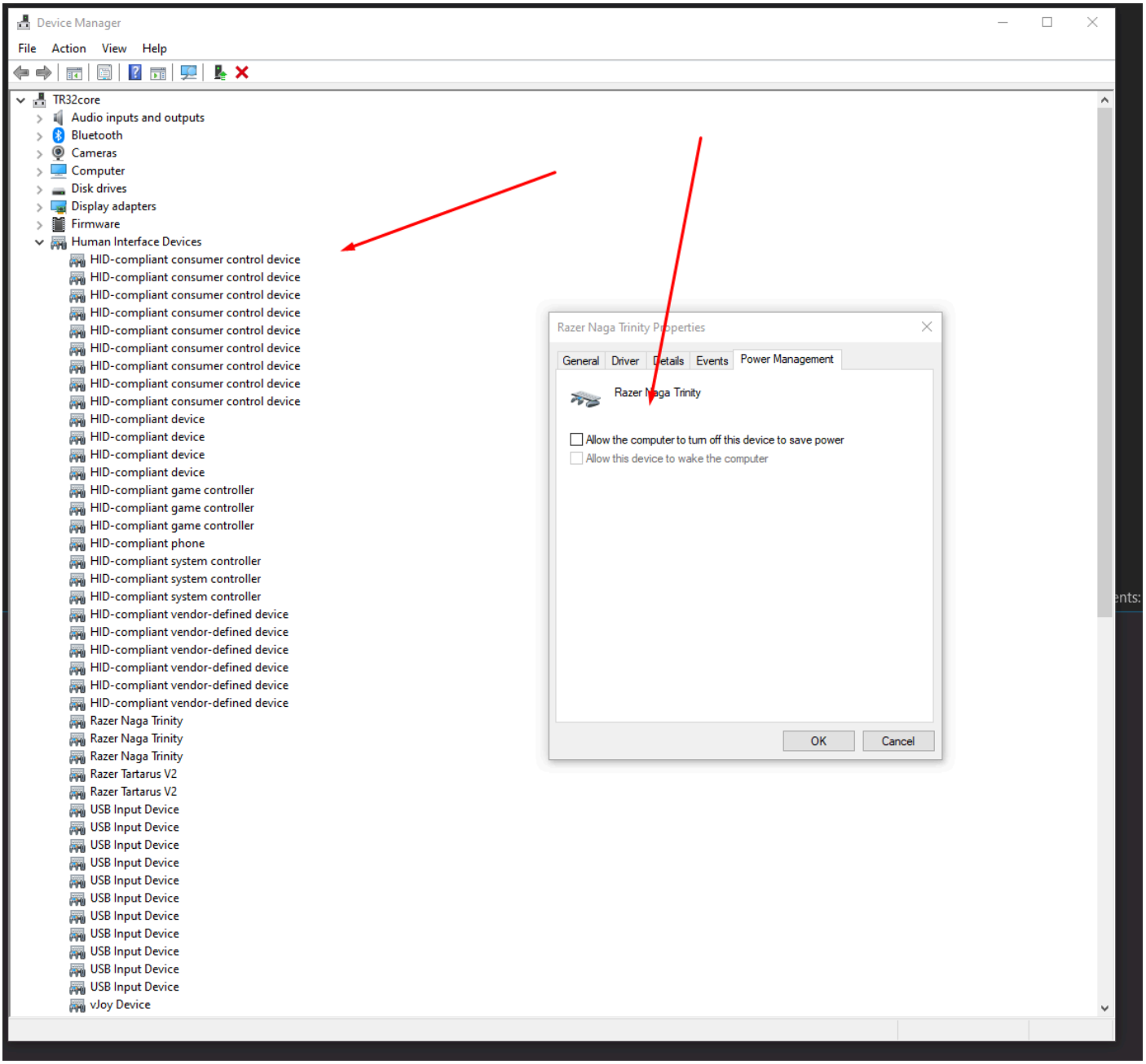


You Need to disable the power management at all levels!

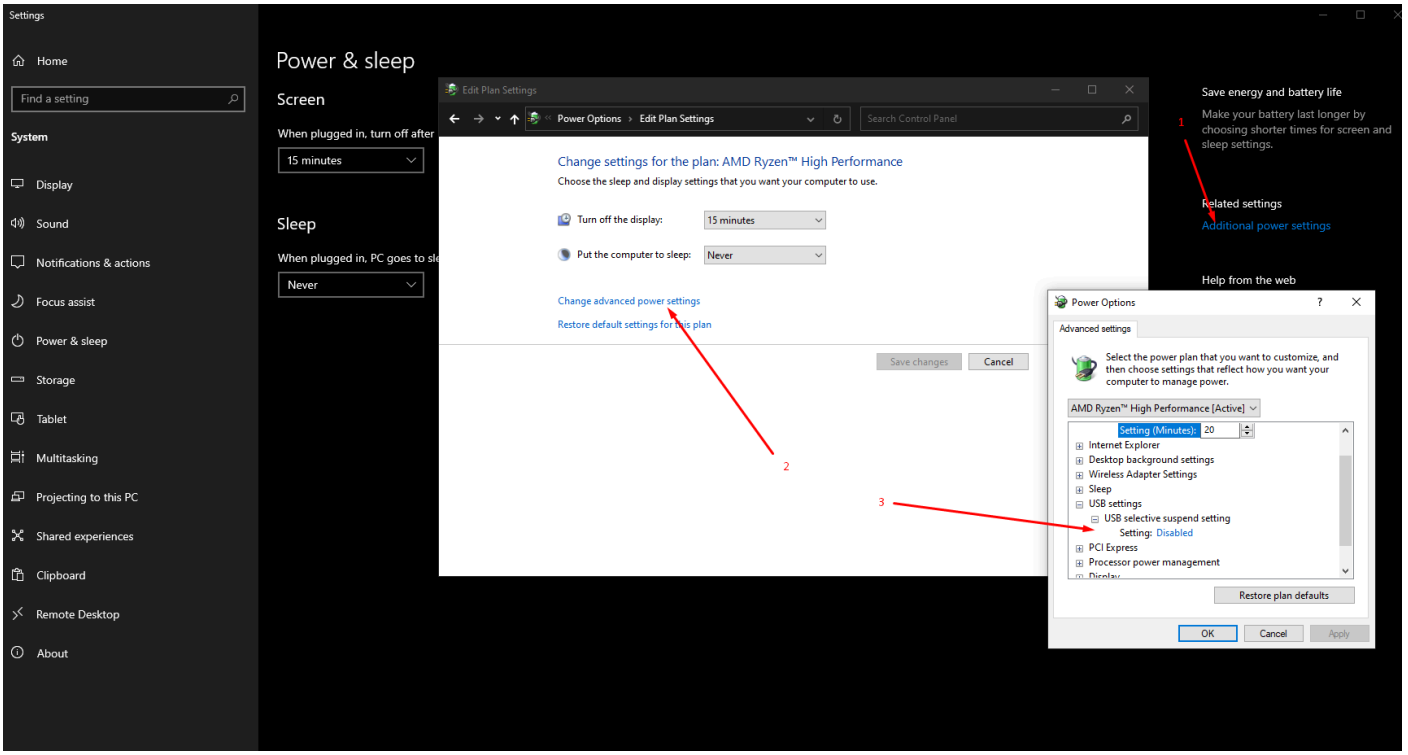
Controllers, Hubs, Root Hubs...



Also check all the devices as well



Also make sure in your system energy options you have also disabled the selective suspend


Settings
Home
Find a setting
System
Display
Sound
Notifications & actions
Focus assist
Power & sleep
Storage
Tablet
Multitasking
Projecting to this PC
Shared experiences
Clipboard
Remote Desktop
About
Power & sleep
Screen
When plugged in, turn off after
15 minutes
Sleep
When plugged in, PC goes to sl
Never
Edit Plan Settings
Power Options › Edit Plan Settings
Search Control Panel
Change settings for the plan: AMD Ryzen™ High Performance
Choose the sleep and display settings that you want your computer to use.
Turn off the display:
15 minutes
Put the computer to sleep:
Never
Change advanced power settings
Restore default settings for this plan
Save changes
Cancel
Save energy and battery life
Make your battery last longer by choosing shorter times for screen and sleep settings.
Related settings
Additional power settings
Help from the web
1
2
3
Power Options
Advanced settings
Select the power plan that you want to customize, and then choose settings that reflect how you want your computer to manage power.
AMD Ryzen™ High Performance [Active]
Setting (Minutes): 20
Internet Explorer
Desktop background settings
Wireless Adapter Settings
Sleep
USB settings
USB selective suspend setting
Setting: Disabled
PCI Express
Processor power management
Restore plan defaults
OK
Cancel
Apply

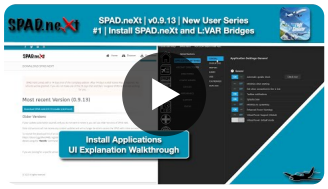
Footnotes

1. ↩

New User Series

Videos targeted at "new users" with updates to Features that can apply to even seasoned SPAD.neXt Users!

Install SPAD.neXt and Take a Tour of the User Interface to get Familiarized.



New to SPAD.neXt then Start here. Install Applications and UI Walkthrough

Binding Buttons and Axis in SPAD.neXt does not replace the sim bindings. Make sure that you create blank MSFS Profiles or you will end up with "doubling of events".

Delete in MSFS FIRST!!!

Before we start configuring our Hardware in SPAD.neXt we need to remove the Bindings from MSFS 2024 on the devices we plan to use. If we leave behind assignments in MSFS 2024 then those events will still fire! So let's Clean it up quickly!

New User 1.2 - Remove Bindings in MSFS 2024 First!

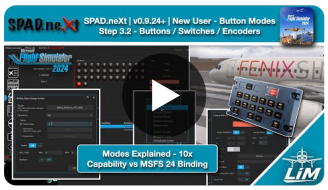
MSFS 2020 and MSFS 2024 Require the L:VAR Bridge to function properly. This is also the H:Event Gateway as those have to be done via the WASM Module as well. In FSX/P3D it is not "required" unless you are using advanced planes.. which well makes it required. Also same Wizard area for installing the XPlane Plugin.
Bonus: How to Relocate your community folder!

Install the L:VAR Bridge and Move 2024 Community Folder

The 2nd installment in the New User series. Profiles.... Let's make sure our Profile settings are correct and then get into what Profiles are and what the Profile Tab interface does for us!

What is a Profile?

3rd Installment is all about understanding Buttons, Encoders, Rotary Position Switches, Toggle Switches and even Rotary Dial modes for Joysticks.... "Generic" Joysticks can apply different Modes and Event Types to the "HID Button". SPAD.neXt supercharges the capabilities of what you can do with the assignments vs using default sim binding system. We will use the Fenix and show how to find the Variables and map examples of each!



https://youtu.be/9ZjXxk6yWu0

4th Installment - Axis Assignments. What is the difference between Standard and Custom Axis along with how to use the features in the Custom Axis.. For Version 0.9.12 and previous .. See Next Video for Axis Capability in version 0.9.13

Standard and Custom Axis... When to use which?

Axis in Version 0.9.13.39 and Newer!! Big Changes to the Axis UI and Capabilities. The assignments are more aligned with Buttons and Switches! Big Feature enhancement with the Addition of the ability to Map Conditions to Axis!! Easy to create a FWD and REV thrust now on the same Axis!

Axis UI has Improved!! v0.9.13.39+ must see!

Connecting the Logitech/Saitek Panels in SPAD.neXt usually means that there is some software to remove and some power management to deal with. Follow these steps before we move on to pulling in configurations for devices.

Step 5: Connecting Logitech/Saitek Panels

Take a Multi panel from Zero to Hero in the first 5 Minutes! How to easily download the configurations from the "ONLINE SNIPPET" Database. This one we use the Example of the WT CJ4 and after the first 5 minutes we break down the configuration and assignments!

Step 6: Download a Device Snippet for the Multi Panel + Learn about the WT CJ4 example!

Logitech Radio Panel - selecting a configuration from the online snippet database!

Get a Radio Panel up and Running in less than 2 Minutes!

Logitech Switch Panel - Download a SNIPPET for the C172 and then we walk through the "HOW" we figure out the events. This is done by leveraging the Event Monitor which allows us to interact in the cockpit and see what events that is setting off in the sim!!

Setup the Logitech Switch Panel and Learn about the Event Monitor!

Updated for Version 0.9.24.23+. How to use the Data Monitor Tool in SPAD.neXt to find and view Variables including L:VARs. Video also leverages the MSFS Dev Mode Behavior tool to quickly identify Syntax and Variables to then Add to the Data Monitor. Learn how to Update LVARs and Undock the monitor to make short work of your bindings. Fenix A320 FCU/EFIS examples in the video.

https://youtu.be/c1EQBE3663A

Event Monitor and walk through some examples in the TBM. How to figure out which events are sent when buttons and switches are moved.

How to use the Event Monitor

Stream Deck Basics for version 0.9.12 and Before. Version 0.9.13 Introduced New "Mini Gauge" capabilities. See the Stream Deck Section for newer examples.

Getting the Stream Deck up and running!! Plus grab a fantastic Snippet for the WT CJ4!

Custom Axis are powerful and understanding the control parameters and how to use them is important! With MSFS now using the _EX1 extended axis there are planes that won't support or work when it is a "Standard" simconnect axis. This video will explain the Advanced capabilities that you can achieve with ease in SPAD.neXt !!

Custom Axis Control Parameters Explained

Programming LEDs! Extensive power and capability for lighting and Flashing LEDs was always available. With this though meant "simple" LED assignment required a little more complexity like having to assign at least TWO change types for even a basic ON / OFF pairing.

With v0.9.15 the NEW LED system (also now common across most devices!) offers a few options on how to Link the LEDs! The New system is really cool and offers quicker mapping and direct copy paste between devices!!

New User #13 - Program LEDs on Any Device!

MSFS SimConnect

This Section includes Content to explain how SimConnect works and how Events/Data work within MSFS.

At FSExpo 2023 I was able to give a talk on how SimConnect works from the "users" point of view. The goal was to breakdown the way in which 3rd party planes and interfaces work. Also targeting the how and Why things work the way they do. Note that this was in 2023 and most of the information is still accurate with the exception that we DO have access now to Send B:Events.

How SimConnect Works - From The User Perspective

During Flight Sim Expo 2025 I had the opportunity to deliver another presentation regarding how users can leverage the SDK to understand better configuring hardware!

With the release SU15 in MSFS 2020, then further in 2024 Release many updates and capabilities were exposed. These Improvements open even more possibilities for the user! MSFS 2024 DEV Mode is even more powerful for quickly identifying controls to be bound!

Thank you again to the FSA team that puts together these great Events! Especially to all the people that volunteer at the event to help make it happen for the community!

https://youtu.be/V3Ari9iieVo

L:Vars / H:Events and B:Events

This Section is focused on L:VARs, H and B Events ... These are the more customized items per plane and delve deeper into the programming of custom events and finding them.

L:VARs and H:Events are additional Variables and Sim Control events that are not part of the "Standard" events in Sim connect. This requires an Additional Bridge be installed.

In both Prepar3d and MSFS L:VARs are required so we can work with the Extended capabilities of Add On Aircraft. This is how all of the complex aircraft will interact data wise for their custom systems implementations. The most notable being Auto Pilot Data. Add On Planes need to run their own systems to simulate the modes and functions of the simulation. Then they "under the hood" take control of the plane and fly it.... Thus we need to gain access to View and Control that Data!

The H:Event is the new JS/HTML Gauge Events found in MSFS which also replaced many of the older simconnect events like the GPS controls commands. MODs like the Working Title CJ4 and the Fly By Wire A320 have also required the separation of the Standard Autopilot and Data to allow them to implement their controls. This meant modifying the commands and Data. We are able via the Bridge to also send these Commands into the sim. Additional names in use for this will be WASM Bridge or WASM Module.

If you are interested in understanding all of the different data types the SDK docs are valuable source of info.
https://docs.flightsimulator.com/html/index.htm#t=Additional_Information%2FReverse_Polish_Notation.htm%23Types

MSFS 2020 and MSFS 2024 Require the L:VAR Bridge to function properly. This is also the H:Event Gateway as those have to be done via the WASM Module as well. In FSX/P3D it is not "required" unless you are using advanced planes.. which well makes it required. Also same Wizard area for installing the XPlane Plugin. Bonus: How to Relocate your community folder!

Install the L:VAR Bridge and Move 2024 Community Folder

One of the early examples of working with L:VARs and Add On Aircraft

Getting Started with L:Vars using the CRJ

Client Events. How to Add custom events that are not currently in the SPAD.neXt Event List.

Add H:Events or K:Events Manually

L:VAR control to work around some conflicts of WT GNS and TDS GTN integrations.

Disable WT GNS LVARS to Run TDS External Mode

Working with Another Add On Airplane example. The MS ATR and L:VARs to take control of the entire plane!

Find The Control L:VARs via the Data Monitor

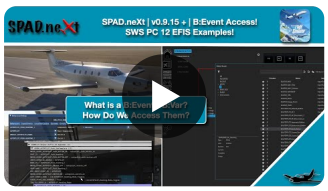
Dev Mode in MSFS can help you discover the Events/Vars used by each plane. If the plane is encrypted and the XML locked out.. just use Dev Mode!

XML or Dev Mode - Find H:Events

L:VAR example to create a button to "rebuild" the Engine on the SWS Kodiak!

Kodiak Engine Reset - LVars and Events

B:Events are finally Accessible (sure we are limited to _SET for now due to MSFS limitations..) and this video is the first covering of this new feature!



B:Events Now Available -- How To Find and Use Them!

In MSFS 2024 we can utilize the Behaviors tab inside of Dev Mode to quickly identify the events and variables that are being used with the Virtual Cockpit Interactions. This video will show you how to interpret and verify in Dev Mode then assign that to a button or switch in SPAD.neXt.

MSFS 2024 - Use Dev Mode and Quickly Find Events and Variables

B:Events have many "custom" extensions. There are some common extensions like like _ SET / INC / DEC / ON / OFF / TOGGLE that can easily be added with a Right Click now!! Even Better is that we can easily Add any custom extension and get all of the control we need for these interfaces.

It is Finally Possible to say we can control it all now!! well I hope :) I am sure something will come up and break it again though. Use Dev mode - Identify the Event and then Add it in SPAD.

MSFS 2024 : Quickly Add Custom B:Events Into SPAD.neXt

Features & Functions

We will use this section for videos where we are focused on SPAD.neXt Features or Functionality to leverage to complete our tasks

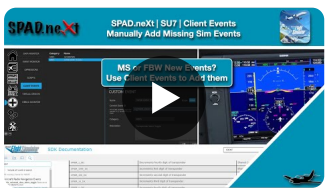
Conditional Events and Local Variables are powerful for creating intelligent actions. We may want to have multiple things on a single Button or Encoder. We can use conditions to allow them to figure out which event to send on that click! In this Video we use Multiple Conditions and a Local Variable to create our own Acceleration Events!!

Use a Local Variable and Multiple Conditions to create our own Encoder Acceleration!!

How to Remove the Warning about Trim Percent and Replace those actions using Trim Position instead! This video shows you how to do that and explain why this warning is now appearing. Also great example using the Event Monitor and Data Monitor Features!!

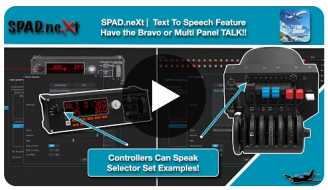
Resolve Trim PCT Warning with Trim Position!

From time to time the SDK can be updated or other developers like Fly By Wire may add their own sim events that SPAD.neXt is not aware of yet. In those situations you can leverage the Client Events Tab in the Menu to add in any of these events that are missing..



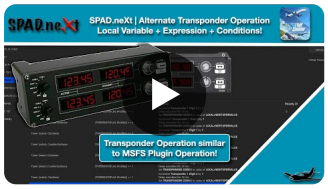
Client Event Tab: Add Missing Sim Events to SPAD.neXt

SPAD.neXt has a very useful Text to Speech Feature. In this short Feature video we use it to setup button events and script events to have the Bravo and Multipanel tell us what their selectors are assigned to!



Text to Speech: Devices Can Talk to you!

Another application where we needed to use Local Variables, Expressions and Conditions to make the Radio Panel operate in a different mode of operation for the transponder. The operational result is a way to visualize how to leverage these SPAD.neXt features to enhance your sim experience!!



Local Variable + Expression + Conditions = Alternate Operation!!

Use Local Variables and Conditional Events to put Multiple Functions onto a single Knob. Here we use the example of toggling a Variable on the X-Touch by pushing the Knob to flip between Heading and Speed Controls. We also setup a second Local Variable to make the Second Knob flip between Altitude adjustment in 100 Foot increments to 1000 Foot Increments.

https://youtu.be/_hV32_FKfLA

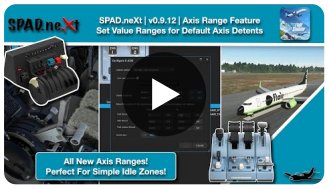
This Video focuses on using a Local Variable that we Create to Allow us to assign conditions to what the Radio Panel will Display. This example focuses on the Radio Panel and switching between displaying Baro in in/HG or MB but could be applied to anything!!

Local Variable and Conditions = Multiple Display Options!

Using the "Import Device" feature. Now you can Import a device from one of your other local profiles. No longer need to switch profiles.. copy the device... switch back to Paste it!!

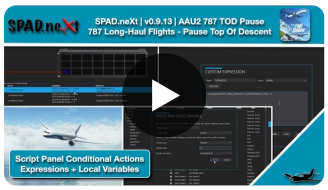
Import Device Configuration From Another Local Profile

v0.9.12 introduced the Axis Range Feature to Standard Axis Configuration. Now it is possible to create "zones" for the Full / Idle and Reverse ranges. This will be really helpful for High resolution devices like the TCA throttles or Low Resolution where the end stops can sometimes mis a value or two..



Axis Ranges added in v0.9.12

Using an L:Var, an Expression and a Local Variable we will create a Script Panel Conditional Action to Pause the AAU2 WT 787 at the Top of Descent!

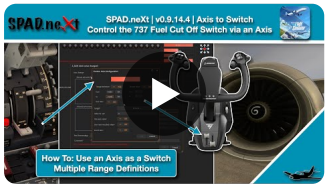


Pause the AAU2 WT 787 at Top of Descent

Big Changes with Axis GUI and Capabilities!! Much Desired Ability to bind Conditions with Axis is now here!! v0.9.13.39 is released and we can finally directly manipulate Axis mappings without the need for scripts!

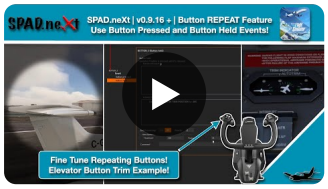
Apply Conditions To Axis Now. Example Toggle Thrust Reverse Mode

How to use an Axis to control a Switch. Using Custom Axis with Range Definitions we can apply Axis Value Ranges that will Trigger Sending Sim Events with Fixed Parameters. The Example we use is the PMDG Fuel Cut Off Switches controlled by the Thrustmaster Boeing Yoke lower "throttle" axis.



Control the Fuel Cutoff Switches with the Boeing Yoke Lower Throttles

How to Best use the Button Held Mode along with a Button Pressed Mode to implement Repeat Events. Adding a copy of the Button Pressed Event, Changing to HELD mode and Editing to get the best result for triggering repeat Events when the Button is Held.



How To Repeat Events with Button HELD Mode! - Still use a Press though :smile:

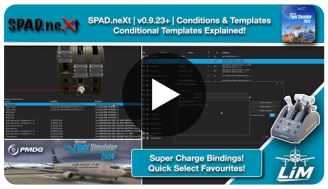
Conditions / Expressions

This section contains guides covering topics on: Conditions, Conditional Templates and Expressions.

Conditions enable a user to apply "rules or approvals" as to when an Assignment to any Button, Axis, Switch, Encoder — etc .. are allowed to be sent. This Enables users to bind numerous sets of assignments with independent logic for when they will be enabled.

Conditional Templates allow creating a single or series of conditions that can then be wrapped up into a "human readable" Label which can then be quickly selected from a drop down list. This will Super Charge Assignments of complex Condition Sequences and speed up device configuration!

Examples are built in this Guide for the PMDG 737-800 in MSFS 2024 using a TCA Boeing Throttle but the premise can be applied to any number of devices as it covers: Throttles, Rev Switches, AP Mode Selectors and Encoders + Buttons with Multiple Assignments based on Modes.



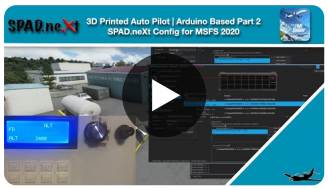
https://youtu.be/NJ0U5ugV0XY

Device Specific

The Sub Channels here are broken out by Vendor

Arduino / SPAD Serial

Step 1 to building your own Arduino Autopilot!



Step 2 assigning the Events and Config of Arduino AP in SPAD.neXt

Let's take a Dual Concentric Encoder + an Arduino Pro Micro and make a "SPADster". Using the Stream Deck we will control the Local Variable to select which Function/Knob/Wheel or Events the Encoders will be controlling.. Great extension to your Stream Deck!

Dual Encoder + Arduino = SPADster :) ... Great Extension for the Stream Deck!!

Authentikit

For further Information about AuthentiKit see https://authentikit.org

AuthentiKit + MSFS + SPAD.neXt

https://www.youtube.com/watch?v=Zuy56KggmDU

Cockpit Simulator CDU

USB Hardware 737 CDU

First Look Video. SPAD.neXt driving the Display and control of the Keys. Follow up videos on configuration and CDU layouts to come... Just wanted to get something from an early Alpha build out there so people could see just how good this interface is!!

Quick Look at the Cockpit Simulator CDU in MSFS with the CRJ!

Here we go with the Early Alpha Implementation and configuration.. This will probably be obsolete in the next week or two but for now it works great!!

Configuration and Mapping CS CDU with CRJ L:VARs

Support for WT CJ4 Mod .. When you have another Mod :) which provides similar implementation as the FBW A32NX. In this video we setup the CM CDU with the CJ4 and get a basic Flight plan with Departure and Arrival entered

Cockpit Simulator CDU Configuration for the WT CJ4

Next Up we have the Configuration for working with the PMDG 737 in MSFS!! This implementation is just Amazing !!!

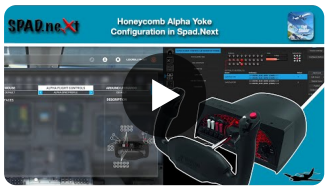
PMDG 737 (MSFS) and the Cockpit Simulator CDU

The Cockpit Simulator CDU can also be used with the CJ4 and NEW AAU2 updates.. Split Left and Right operation with complete snippets posted for download.

Also learn how to add your own Data and Expressions to custom display pages or just the SPAD Menu.. Examples on Hardware Display and Key Brightness Controls given.

Configure the Cockpit Simulator CDU for the CJ4 AAU2 Update!

Honeycomb

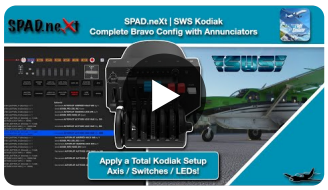


Starting Point for Setup of the Alpha

Configuration of the Alpha for a "Single Engine" MSFS C172

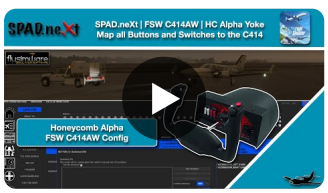
Alpha and Bravo Config for the Aerosoft CRJ 550/700/900/1000

New Features and Capabilities for the Honeycomb Bravo



Setup All Axis, Switches and Annunciators for the SWS Kodiak

Bravo / FBW A32NX Axis Configuration



Alpha Setup - FSW C414AW

Bravo Config and Snippet for the PMDG 777 (MSFS)

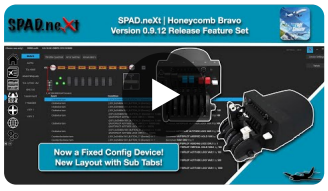
FFX/MG HJet Honeycomb Bravo Configuration

FFX/MG HJet Honeycomb Alpha Configuration

Re-Configure the Alpha Yoke to use the Default Configs!

This Video is a General Update to How the Bravo Layout and UI is handled in the Release version of SPAD.neXt v0.9.12!

There was a video update early in the Alpha phase but near the end of testing the final layout including sub tabs was introduced.



Honeycomb Bravo v0.9.12 Explained! New Config and New Layout!

Guide to the Bravo configured with the MS ATR 42/72-600. Includes the Download Snippet of the Configuration and the walk through of all of the controls. Note you must follow the steps to Calibrate and configure the ATR EFB. Look in the Chapters section.

HC Bravo and the MS ATR 600

Big Changes with Axis GUI and Capabilities!! Much Desired Ability to bind Conditions with Axis is now here!! v0.9.13.39 is released and we can finally directly manipulate Axis mappings without the need for scripts!

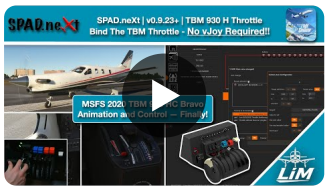
Conditional Axis - Toggle Between FWD Thrust and REV Thrust

Dash 7 Bravo Configuration. This includes setting up Custom Condition Lever Axis with Multiple Ranges and all Starter Switches required for successful Starts. Decision was made to go with 2 Axis For Throttles and 4 for the Prop/Condition Levers. AP and Annunciators fully functional.

Dash 7 Bravo Config Including Snippet

How to Bind the TBM 930 Throttle in SPAD.neXt to have Control and Animation Working!!!

vJoy method is no longer needed. We can use B Events and K Events to tie it all together!!



https://youtu.be/4a9ZQ3aXV6I

Joystick (Generic HID)

Using a Leo Bodnar Board with "Switches".. Also some Expressions are used!!

miniCOCKPIT

Videos for the miniCockpit Products

miniCOCKPIT's miniFCU is fully supported in SPAD.neXt v0.9.15.38 and Newer. Upgrading the Firmware to the Custom SPAD.neXt firmware (written by miniCOCKPIT) enables the additional event abilities like Press Short, Press Long, Press, Release and Held for Buttons. LEDs / DOTs and Labels are also now independent from the "Airbus" Modes for additional Customization.

In Release Version 0.9.23.31 - SPAD.neXt now Uploads and Controls the Firmware for miniCOCKPIT Devices. Additionally this version added the support for the miniEFIS as well. The previous Video is now removed as there is no longer a need to manually change the firmware via external Software.

https://youtu.be/483gVFGi8tQ

RealSimGear

Adding a GNS or GTN Device to SPAD.neXt

Configure the GTN 750 to work with the TDS GTNXi Add On

First Steps to add a GMA 350 for use in SPAD.neXt

Configure the GMA 350 to work with Standard Aircraft

Add RSG GNS430 and Download the 430 Setup

Add RSG GNS530 and Download the 530 Snippet

Connect and Apply Config for the RSG GMA-ADDon

Saitek/Logitech

Looking to get started with Saitek / Logitech Panels? Well first thing to do is make sure you have the Software and power management cleaned up!

First Steps to take in getting your Saitek/Logitech Panels into SPAD.neXt

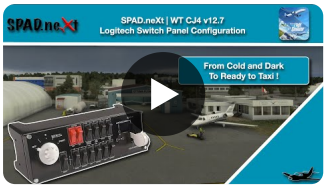
Working Title CJ4 Configuration Download and Assignments walk through

Advanced Radio Panel Configuration Walk Through... Works with VATSIM!!

Saitek MultiPanel and the G1000 NXi from WT

Re-Purpose the Radio Knob on the DME Selector for the NXi PFD FMS Knob!

More Features added to the NXi == New Snippet Update to the MultiPanel!



WT CJ4 v12.7 Switch Panel Configuration with Cold and Dar Start Up!

Configure the Multi Panel for the WT CJ4 and Fly Fully Controlled Auto Pilot Modes!

Setup the Radio Panel with the WT CJ4. Use the Knobs to control the PFD and the MFD.

Logitech / Saitek BIP Panel Basics

Alternate Transponder Mode Operation

Multipanel Config for the PMDG 737 in MSFS

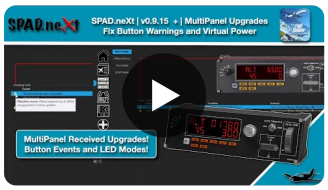
FFX/MG HJet Logitech Multipanel Config

FFX/MG HJet v1.0.4 Logitech MultiPanel Config

This Video focuses on using a Local Variable that we Create to Allow us to assign conditions to what the Radio Panel will Display. This example focuses on the Radio Panel and switching between displaying Baro in in/HG or MB but could be applied to anything!!

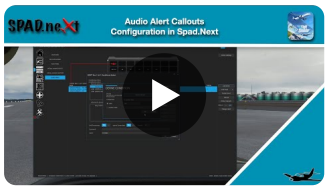
Local Variable and Conditions = Multiple Display Options!

The MultiPanel in version 0.9.15+ has been upgraded to the Button and LED event systems. Users will find warning messages on older configurations for the Buttons. This video explains how to quickly edit your configurations and update to more efficient LED settings. Virtual Power is also affected by this so make sure you updates those -- tip on using a specific circuit in MSFS to better link it to avionics bus instead of a switch.



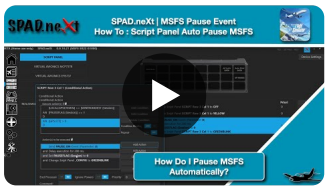
Fix MultiPanel Warnings and Obsolete Events! v0.9.15+

Script Panel



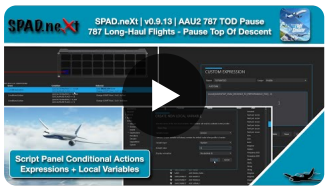
Let's Make some Audio Callouts and Play Audio Files... Text to Speech would also Work

Use the Script Panel to track L:Vars and write Sim Vars for vPilot



Let's play with the Script Panel to Pause the Sim For us!

Using an L:Var, an Expression and a Local Variable we will create a Script Panel Conditional Action to Pause the AAU2 WT 787 at the Top of Descent!



Pause the AAU2 WT 787 at Top of Descent

Stream Deck

Original Stream Deck Implementation. If you are running version 0.9.13 or newer then you should skip down to the v0.9.13 video with the new implementation.

Getting Started with the Stream Deck and SPAD.neXt

How to make a Dual Concentric Encoder with an Arduino.. Tie it to a StreamDeck and you now have a "Spadster".

Version 0.9.13 Implemented the New Stream Deck interface with Layers and "Mini Gauge" capability. This Video reviews the new Implementation and how to use the layers.

v0.9.13 New UI for Button Images and Text

Control the LVARs of the FFX Vision Jet to implement the VAMS iPad on a Stream Deck!

Vision Jet v2 VAMS Stream Deck Mapping

I have been wanting to cover the "Quadster" I have been working on for some time... Thanks again to Shakeprint for working with me to make this custom mount.

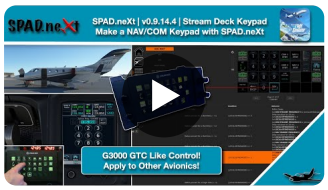
With the Vision Jet v2.1 on the path to release I felt updating the Stream Deck Device Snippet would be a great time to tie in the Quadster and show it off!! There is a lot to unpack with 22 pages of StreamDeck Control tied to 4 Dual Concentric Encoders!!

Stream Deck XL + Quadster = Vision Control Overload!

The SWS Kodiak has Engine life and Stress!! Nothing worse then getting Setup and then accidentally putting it over the edge. Here is how we can implement ramp "Engine Overhaul" to get going again!

Kodiak Engine Reset - LVars and Events

Create Custom Made Keypads for controlling Sim Variables via Stream Deck. In this video we use the example of the HJet and G3000 GTC touch control of the Com and Nav Frequency entry. This can be applied to other variables where Data Entry is desired.



Stream Deck Keypad Entry using Local Variables

Change Pages on Any Stream Deck from Any Other Device! Assign Multiple Page Changes on any connected Stream Deck or Page Enabled Devices from a single Button!!

Change Pages on a Stream Deck from Another Stream Deck or any Controller!

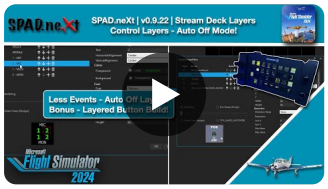
Time to Jump Into the Stream Deck Plus and it's features. This video includes downloading a Base Page Snippet to understand how to manipulate the displays and interactions. This is one versatile product!

Stream Deck Plus Features Explained

Quick Video covering the new "Right Click Functions Menu" to help speed up the editing and creation process of your Stream Deck or Loupedeck configs!

Stream Deck Right Click Function Menu

Video Covering how to use the New Auto Layer Mode and Increase efficiency in Button Gauge Designs. Bonus regarding leveraging Layers with other switches which have Transparency and reduce the need for the same switch over and over generated with backgrounds and text.



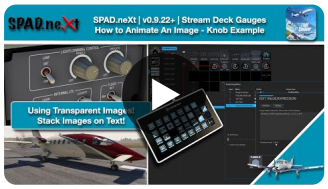
Stream Deck Layer Mode - Auto Off Feature

Stream Deck Image Animations

In this video we show how to use Data to dynamically Animate Images. We use Transparent Images to layer over Text and a Variable to drive it.

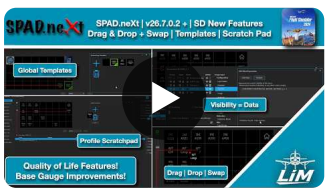
Expressions in SPAD.neXt are extremely powerful and the Rescale Function is used to easily remap the L:Var to the Rotation value required!

Learn more about available Functions here:
https://docs.spadnext.com/features/expressions/functions



Animate Images - Button Gauges - Rescale Expression Example

This video we cover many new Stream Deck Features added to SPAD.neXt. We also cover how some of the basics in the Gauge Layer have Improved!! Including Linking much of the base layer directly to data to shorten build times!!



https://youtu.be/1McZjPWjgQg

Thrustmaster TCA

Setup the TCA TQ and use Multiple Axis with the A320

Setting up the TCA TQ plus Add On to work with the AS CRJ

TCA TQ with Addon - FlyByWire A32NX

FFX/MG HJet TCA TQ + Addon Configuration

TCA TQ+Addon New Layout Overview

In this Video we delve into the Thrustmaster TCA Airbus 320 TQ with Add ONs and configure a complete setup for the PMDG 737 Series! Don't worry the Snippet is available now so you can download and fly in just a couple of minutes!!

PMDG 737 TCA TQ + Addons

Plenty of Jets have Throttle FADEC Detents! Let's use Multiple Custom Axis to map these into custom ranges on the Throttle Axis! This method would work for any Throttle since it does not rely on the TCA TQ's Virtual Detent Buttons!

Throttle FADEC Detents via Multiple Custom Axis

ATR - 600 Full TCA TQ with Addons

For the First Device Tutorial Regarding the PMDG 777 I decided to use an Airbus Setup :). The TCA Airbus Captains Pack! This is just what happens to be at my desk vs in the main simulator.

Thrustmaster TCA Airbus Captain Pack and the PMDG 777

Use a vJoy to download any Device Snippet ! Using vJoy as a "clipboard" we can download a Device snippet to the vJoy then copy events to a different device type!

Example in this video we will Download a Bravo Setup to load onto a TCA Boeing Throttle!!

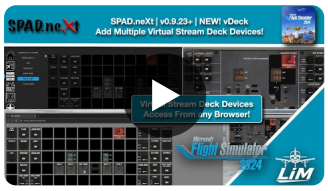
https://youtu.be/oHp8IPEFu1I

vDECK

New Feature in 0.9.23.31 Release! Virtual Stream Deck Devices!! accessible from any network connected web browser!! Use Tablets and Touchscreens to create multiple custom user interfaces!!

Add Virtual Stream Decks to your Setup!! Access over the network via a Tablet or second PC via a web browser!

Download Snippets from Stream Decks and vDecks for access to all kinds of premade setups!!



https://youtu.be/wCRNGLec8-Q

vFIP

Section regarding the Virtual FIP Operation and Configuration

This first Video in the section covers the Basics of what a vFIP is (a virtual representation of the Logitech Flight Instrument Panels...) and how to add them to your Profile. Covers also how to access the available Gauges and how to add local folders for your Purchased or Community downloaded products.

vFIP Configuration and Operation. Add Gauges to a second screen!

Virtual Avionics

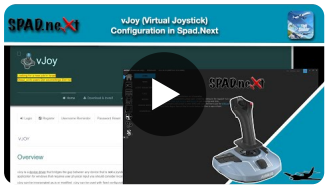
Adding a VA 737 MCP with P3D and the PMDG 777

Working with the VA 737 MCP in MSFS Aircraft

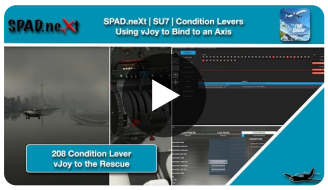
Configure the VA MCP to work with PMDG 737 in MSFS

vJoy

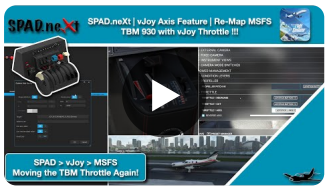
Download vJoy here... https://github.com/shauleiz/vJoy/releases



This was to work around the View / Camera System Issues



Use vJoy to bridge the gap to the Condition Levers after SU7 adding them in.



vJoy Axis - Re Map the Throttle in MSFS

Use a vJoy to download any Device Snippet ! Using vJoy as a "clipboard" we can download a Device snippet to the vJoy then copy events to a different device type!

Example in this video we will Download a Bravo Setup to load onto a TCA Boeing Throttle!!

https://youtu.be/oHp8IPEFu1I

VRinsight

How to Add a VRi Panel to Control the Sim

WinCTRL

WinWing Devices

How to Add WinWing MCDU and CDU Devices to SPAD.neXt. This video will also explain the Global Device Setting and the Profile specific Settings. Basic Walk through of downloading a Snippet and some of the UI functions — like using CTRL Key to show mappable Buttons and Events.

https://youtu.be/OX5Wj0sOQPw

The FFX C750 CitationX is able via the Community Mod to deliver the CDU / FMC Data for display on the SPAD supported CDU devices. This video focuses on the WW MCDU however ANY support CDU/MCDU will work in the same way.

https://youtu.be/CnXtIN-aIGc

Add the WinWing FCU and EFIS Devices to SPAD.neXt.
We can quickly Add the hardware and then Download Published Device Snippets which get us going with our profiles!

https://youtu.be/wsfjr2aq0M0

Maximize your PAP3 MCP usage! Using SPAD.neXt we can control the Autopilot systems of any airplane! Let's use the FFX Citation X as an example and fire up MSFS 2024 to get it working!

https://youtu.be/COh2dEzkHX4

PMDG 737 introduces a new FANS2/B layout configuration. In this video we will show you how to find the PMDG Switch Numbers and Turn them into Rotor Brake Events.

Just like other events (cabin door or the Headset Sound) that have not been added to the PMDG SDK with release we can use the Rotor Brake Events directly. With PMDG — Send Events and Read Variables!! We use the WinCTRL PAP7 as the example hardware but this will work with any supported CDU

https://youtu.be/D3GEHB1VoM0

This Video Guide covers setting up the PAC32 with Addons in SPAD.neXt. How to download a Device Snippet for it — the example is using the Fenix A320 and even calibration of the axis in the MCDU. Get the throttles setup and flying in less than 5 minutes!

https://youtu.be/PpmNlBhaGWM

In this video we get the MSFS WT747v2 Avionics exporting the CDU DATA.

We use SPAD to configure our hardware and display the WebSocket data on the Screens.

https://youtu.be/Tc4zQvg8sd4

Hook up your Hardware CDU or Run the Bonus WEB CDU included with the Plugin. Download Plugin and install it in the MSFS 2024 Community to control the UNS-1 found in the WT 707 and the SAAB 340!

Download is linked in the Chapter Marker!

For a List of Supported Hardware CDU Devices in SPAD.neXt:
https://docs.spadnext.com/getting-started/supported-hardware

For the List of Supported Planes in SPAD.neXt for CDU Screen Support:
https://docs.spadnext.com/hardware-specific/cdu-devices

https://youtu.be/67NAeFTEzG4

WINGFLEX

Video Guides for the WINGFLEX Brand Controller Devices

This video covers setting up the DAP-500 from WINGFLEX. This is a GFC-500 replica device. Complete Device Snippets are available for download and will cover many Planes. This video example we show jumping into the HJet from FFX/MG download a K event and A Var based snippet to fully configured in just a couple minutes!

https://youtu.be/uRwivg4Ot5s

X-Touch MC Mode

v0.9.20+ X-Touch Now uses Mackie Control Mode Only!

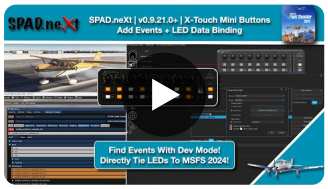
The SPAD.neXt GUI for the X-Touch Mini is ALL NEW!! Version 0.9.20 + brings direct MC or Mackie Control Mode Integration! We are no longer having to deal with any kind of device side LED or page workarounds! MC mode enables the creation of Unlimited Pages with direct LED control. We can now set this up just like other devices with Full Direct LED and Paging control!!

X-Touch MC Mode Setup - All New UI Interface!

This Video covers the Page System when in MC Mode for the X-Touch Mini. How to Add, Duplicate, Re-Order and Navigate the paging system. We also cover Assigning Page Navigation Events and LEDs to track which Page the X-Touch is currently On.

X-Touch MC Mode Page Management

This Video will cover working with the Buttons of an X-Touch Mini in MC (Mackie Control Mode). This includes using Dev Mode to Find the Button Event you are trying to emulate. Performing that Assignment and then Working with LEDs to tie them to Data Values (Sim Vars).



X-Touch MC Mode Buttons and LEDs

This XTouch Video Covers the Encoders and examples using the new Trim and Fan Modes for the Encoder Ring LEDS

X-Touch Encoders and Ring LEDs

X-Touch Midi Mode

Configuration of Midi Devices

> Starting with version 0.9.21 SPAD.neXt no longer supports the X-Touch mini midi mode. Please see the MC Mode section.

Setting up a Midi Device using the X-Touch Mini. Covers application configuration and UI to Hardware explanation. Example walkthrough for configuration Events to Encoder, Buttons and the LEDs to conditional events.

Function	RX CH	RX Command	RX value
Operation Mode Select	GLOBAL CH	CC 127	0 = Standard Mode, MC Mode LED off 1 = MC Mode, MC Mode LED on 2-127 = ignored
Preset Layer Change	GLOBAL CH	Program Change	ONLY in Standard Mode (see above): 0 = Layer A, A-button LED on 1 = Layer B, B-button LED on 2-127 = ignored
LED Ring Behavior Change	GLOBAL CH	Encoders 1-8: CC 1 – CC 8	0 = Single 1 = Pan 2 = Fan 3 = Spread 4 = Trim 5-127 = ignored
LED Ring Remote Control (Value Change)	GLOBAL CH	Encoders 1-8: CC 9 – CC 16	0 = all LEDs off 1-13 = LEDs 1 (left) – 13 (right) on 14-26 = LEDs 1 (left) – 13 (right) blinking 27 = all LEDs on 28 = all LEDs blinking 29-127 = ignored
Buttons LED Remote Control	GLOBAL CH	Upper Row 1-8: Note 0 – Note 7 Lower Row 9-16: Note 8 – Note 15 Layer A, B LEDs: Not assignable; only 1 of 2 LEDs lights if Standard mode selected (see above "Preset Layer Change")	Note off or Note on with Velocity 0: Button LED off Note on with Velocity 1: Button LED on Note on with Velocity 2: Button LED blinking Note on with Velocity 3-127: ignored

X-Touch Mini Initial Setup and Config Examples

Moving on from the Basics to controlling Axis with Control Change Values. We use the Slider/Fader of the XTouch to control some Overhead Lights and some Lvars!! We also throw in some review of using the Data Monitor and Event Monitor to figure out which variables or events we need to control.

X-Touch Mini Slider/Fader Configuration

Use Local Variables and Conditional Events to put Multiple Functions onto a single Knob. Here we use the example of toggling a Variable on the X-Touch by pushing the Knob to flip between Heading and Speed Controls. We also setup a second Local Variable to make the Second Knob flip between Altitude adjustment in 100 Foot increments to 1000 Foot Increments.

Local Variables, Conditional Events = Multiple Controls per Knob

In this video we map the XTouch Mini to work with the WT G1000 NXi and the SWS Kodiak! Note this layout and config should work with any G1000NXi equipped plane! Full Autopilot control, PFD, MFD and even COM/NAV Radios and Audio Support.. Perfect for VATSIM users!

XTouch Setup for G1000 NXi

MS ATR 600 X-Touch Autopilot Configuration

Simulation Specific

The Sub Channels here are broken out by Simulation. Platforms as well as Add On Aircraft where multiple Videos are specific to that Add On or MOD. Some of these may also be listed under device specific.

ATR 600 (MSFS)

This List is for videos specific to the ATR 42-600 and 72-600 in MSFS

Full TCA-TQ Configuration. All ATR TQ controls fully supported. Plus ability to control the Starters and Starter Mode. Bonus for the Power Management selector!

ATR - 600 Full TCA TQ with Addons

Find The Control L:VARs via the Data Monitor

MS ATR 600 X-Touch Autopilot Configuration

Guide to the Bravo configured with the MS ATR 42/72-600. Includes the Download Snippet of the Configuration and the walk through of all of the controls. Note you must follow the steps to Calibrate and configure the ATR EFB. Look in the Chapters section.

HC Bravo and the MS ATR 600

AS CRJ 550/700/900/1000 (MSFS)

Step 1... You need to Get the L:VAR Bridge Installed... v0.9.11.5 Release and Newer do not require the Alpha or Beta Release!

Install the L:Var and H:Event Bridge. Alpha or Beta No Longer Required.

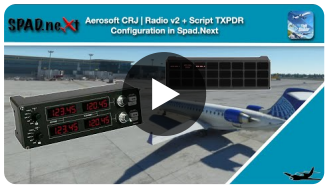
Step 2... We need to Learn how to Find the Data using the Data Monitor. Also how we can then interact with that Data to make changes. We can use the Switch Panel for this one!

How to Find the L:VARS and how to make Data Changes to them

Step 3... Let's now take what we learned in the previous video and get the Multi Panel working with the AS CRJ FCP. In this one we are going to learn about how some items need additional commands if we want to also have the Knobs and Buttons Animate and make Sounds in the cockpit!

Setting up the Logitech Multi Panel to control the CRJ FCP

Step 4... Onwards to the Radio Panel configuration! Also since not all of the data has L:Vars (Nav Radio STBY) or initially the CRJ was not updating Transponder State we will use Data and Script Panels to work around them.



Configuring the Logitech Radio Panel and Using Data Tricks with the Script Panel

Cockpit Master CDU configuration .. early Alpha implementation.

AS CRJ early Alpha Implementation

Let's get our Alpha and Bravo Controls Fully Configured for use with the Aerosoft CRJ Series!! Plus a bonus walk through the cockpit with each assignment then we do a Departure from CYYZ to put it through the paces!!

Aerosoft CRJ specific Configurations!

FBW A32NX (MSFS)

Videos on this page may also be found in other Sections. Grouped here for convenience

Configure the Bravo Axis for the FBW A32NX

TQ + Addon Full Device Configuration + FlyPad config

FFX Citation X (MSFS 2024)

Videos containing Content for the FFX Citation X

Using the Community Mod to export the FMC Data to SPAD. All of the H events were previously imported when the WEB FMC was implemented.

https://youtu.be/CnXtIN-aIGc

FFX/MG HJet (MSFS)

Videos specific to configurations for the HJet

First video is on the Logitech MultiPanel Configuration!! Plus we do a demo flight of the configuration and using the features like Climb VNAV, FLC, VS and show how the very unique CSC "cruise control" mode works as well!

FFX/MG HJet MultiPanel Configuration

The Second HJet video focuses on the Honeycomb Bravo and our Arrival and Approach into CYTZ in this video

FFX/MG HJet Honeycomb Bravo Configuration

Setup and Download the Device Snippet for the Honeycomb Alpha!!

We walk through the configuration of all switches and also setup the H:Events under Client Events!!

After Setup we jump into the Cockpit and demo all the controls including the Custom Checklist Control!! Then we walk through doing the basics of Cold and Dark using the Switches to Turn on the Electrics, Follow the Checklist and then Start up the Engines!!

FFX/MG HJet Honeycomb Alpha Configuration

We move on to the TCA TQ with the Addon Package! Full Configuration and even do an in cockpit demo with Startup using the our work!!

FFX/MG HJet TCA TQ + Addon Configuration

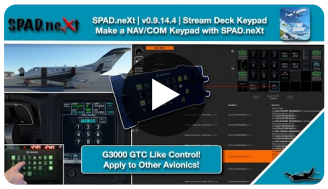
Using an X-Touch Midi Controller to Configure the Overhead Lights Control

X-Touch Mini Slider/Fader Configuration - Overhead Lights

v1.0.4 of the HJet is here!! Which Means the Autopilot Bindings have all be reworked to use the standard Variables and Events!!... Well Let's get back at doing the HJet again..

HJet v1.0.4 New MultiPanel Bindings

Make a Nav and Com Keypad Interface with a Stream Deck! This implementation is similar to that of the GTC Com and Nav Pages!



Stream Deck Keypad Entry using Local Variables

FFX Vision Jet (MSFS)

Videos focused on the FFX Vision Jet

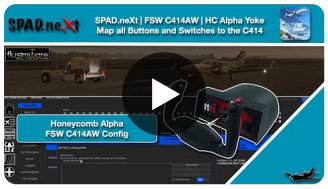
Vision Jet v2 VAMS Stream Deck Mapping

Vision Jet v2 GTC Operation and Bezel Control Mapping

Stream Deck XL + Quadster = Vision Control Overload!

FSW C414AW

Content Focused on the FSW Cessna 414AW



Setting up the Buttons / Switches and Mags for the C414AW

PMDG 737 (MSFS)

Videos Focused on the PMDG 737 for MSFS

PMDG SDK must be enabled for each PMDG aircraft in MSFS. See here for details.

Please remember PMDG Simulation Events work **exactly** like the cockpit.
Use the "PMDG Special Actions" to do with the event what you would do in cockpit with the mouse! Events from the Category **PMDG 737 NGXu** must be used.

Video Guides

In the first Video we step through the requirements for Configuring the PMDG SDK and the SPAD.neXt Simulation Settings. We also show how the Custom Interactive SDK interface to configure Data Vars and use the Event System unique to the PMDGs

First Step - Configure PMDG SDK and SPAD.neXt

In The Second Video for the PMDG 737 in MSFS we delve into adding the Cockpit Master CDU to SPAD.neXt and downloading the default Snippet to get full control over the the in sim CDUs.

PMDG 737 (MSFS) and the Cockpit Master CDU

The Third Video for the 737 PMDG focuses on Programming the MCP (mode control support). We delve into how the SDK works regarding Buttons / Switches / Knobs and Encoders with regards to the PMDG Interaction Events. Also how to find and Map LEDs to Annunciators and Displays to Values!!

Configure 737 MCP Devices to Control the PMDG 737 MCP

Continuing on in the 4th PMDG 737 Video we configure the Honeycomb Bravo for use with the MCP (AP functions) as well as all other Axis and Switches!! Full implementation of the LEDs and Annunciators -- don't forget you need to be on the at least 0.9.12.37 for this!!

Configure the Bravo TQ for the PMDG 737

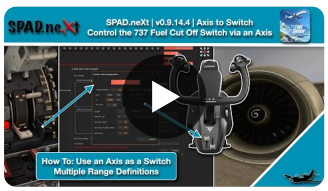
Sticking with Autopilot Configurations we move on to the Logitech/Saitek Multi Panel device. Download the Snippet or use the walkthrough to configure it yourself.

Configure the MultiPanel for the PMDG 737(MSFS)

In this Video we delve into the Thrustmaster TCA Airbus 320 TQ with Add ONs and configure a complete setup for the PMDG 737 Series! Don't worry the Snippet is available now so you can download and fly in just a couple of minutes!!

PMDG 737 TCA TQ + Addons

How to use an Axis to control a Switch. Using Custom Axis with Range Definitions we can apply Axis Value Ranges that will Trigger Sending Sim Events with Fixed Parameters. The Example we use is the PMDG Fuel Cut Off Switches controlled by the Thrustmaster Boeing Yoke lower "throttle" axis.



Control the Fuel Cutoff Switches with the Boeing Yoke Lower Throttles

Big Changes with Axis GUI and Capabilities!! Much Desired Ability to bind Conditions with Axis is now here!! v0.9.13.39 is released and we can finally directly manipulate Axis mappings without the need for scripts!

Conditional Axis - Toggle Between FWD Thrust and REV Thrust

PMDG 777 (MSFS)

Videos Focused on the PMDG 777 for MSFS

First thing that needs to be done is enabling the SDK. This will need to be done for every variant and sometimes even after a PMDG upgrade if it resets the files.

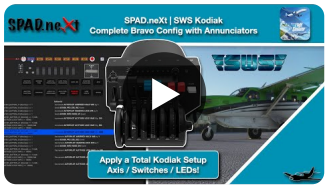
PMDG Needs the SDK Enabled!! Do This First for all 777 Variants!

For the First Device Tutorial Regarding the PMDG 777 I decided to use an Airbus Setup :). The TCA Airbus Captains Pack! This is just what happens to be at my desk vs in the main simulator.

Thrustmaster TCA Airbus Captain Pack and the PMDG 777

SWS Kodiak (MSFS)

Kodiak Configs and Examples



Full Bravo Setup of the Kodiak

XTouch Setup for G1000 NXi

HC Alpha for the SWS Kodiak

Kodiak Engine Reset - LVars and Events

WT CJ4 (MSFS)

Working Title CJ4 Updates and specific content here

Working Title has heavily modified the CJ4 and with that required introducing L:Vars and H:Events. The Custom Autopilot system they built uses these Variables and Events to create the shim between the Default Autopilot system and their custom FMS and FMC changes to the Proline 21 system. Specific videos to configuring devices for the WT CJ4 Mod. Some videos may be found in other sections as well.

With the Switch Panel we will configure it so that we can perform a Full Cold and Dark Startup of the CJ4, We will also add some conditions to the Avionics so that we can go into Dispatch mode when the GENs are off or go into Full Mode when they are On! the full electrics panel is mapped and so is the Left side of the Lights panel.

Configure the Switch Panel for the WT CJ4 and do a Cold and Dark Startup!

Take the CJ4 from Working Title v0.12.7 and configure the Logitech Multi Panel to have full control over the FCP (auto pilot). Using the H:Events and L:Vars to have complete control over this really well done Plane!

Configure the Multi Panel for the WT CJ4 and Fly Fully Controlled Auto Pilot Modes!

Now we move on to configure the Honeycomb Alpha and Bravo to take control of our CJ4 in the quest to limit the amount of mouse clicking we need to do :)

Setup the Alpha and Bravo for the CJ4

The WT CJ4 implements some events that make it impossible to control the Nav Radio Frequency changes. It intentionally over writes any changes made to the Active

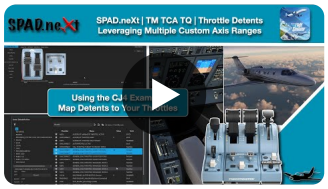
frequency. With that in Mind we move ahead to configure the Radio Panel to also control the PFD Menu Knobs and the LWR Panel knobs for use with the MFD Lower Menu and the Charts control!

Setup The Radio Panel with the CJ4 v0.12.7 and use the NAV Knobs to control the PFD and MFD Knobs

Stream Deck Snippet and the WT CJ4

Hardware CDU Implementation for the CJ4

Map the CJ4 FADEC Detents! Let's use Multiple Custom Axis to map these into custom ranges on the Throttle Axis! This method would work for any Throttle since it does not rely on the TCA TQ's Virtual Detent Buttons!



Throttle FADEC Detents via Multiple Custom Axis

WT G1000 NXi (MSFS)

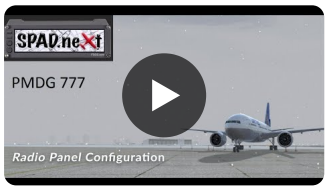
While the NXi will be the default it currently is not. We will add related things here as well.

NXi v0.7.1 Multi Panel Configuration and Snippet Download

PMDG 777 (P3D)

Configuring SPAD.neXt and PMDG 777

Making the Logitech Multi Panel work with the PMDG 777



Making the Logitech Radio Panel work with the PMDG 777

Making the Logitech Switch Panel work with the PMDG 777

Making the VA 737 MCP work with the PMDG 777

UI Familiarization

This Section will introduce you to the User Interface and what each of the sections contain

When SPAD.neXt Launches and after initialization you are greeted with the following Home Page as seen in Fig 1

Fig 1

The UI Contains Navigational Elements On the Left Hand side. In addition there are two "quick selects" found in the Upper right side to Jump to the Settings and Support sections of the program. Seen in Fig 2 with the red arrow pointing to the location

Fig 2

The Left Most Column of the User Interface is the Navigation Column. The Navigation Column is always visible while the second column to the right is the sub navigation or "Tabs" for the UI component that this selected in the Navigation Column. Fig 3 points to these elements

Fig 3

The SPAD.neXt Version in operation can be found at the Top of the UI on the Left Side. If running on an Alpha or Beta channel (selectable in Settings) that will also be displayed as seen in Fig 4. Next to the version number the current "Profile" is displayed in brackets.

Fig 4

The Lower Left will display text regarding the current state of connection to the simulator and even what version you are on. Side note if something happens to your license (like say you upgrade your network card to a 2.5Gb) and it is reset this will be a good indication as it will revert from a Registered License to a Trial and easy to find here. The Simulation connected and even aircraft loaded will also appear here at the bottom as seen in Fig 5

Fig 5

The Lower Right is displaying the events traffic that SPAD.neXt is Interacting with the simulation as seen in Fig 6.

Fig 6

Home Page

This Topic covers the UI Elements found on the Home Page Menu Section

When the Home Page is selected (Fig 1) we see the Half Circle next to the Image of the House. Also the First Sub Tab will always select when switching the UI to a different Navigation Column. This Menu has Page "Home" has one very important part .. the Link to join the discord. You will find plenty of Help from many users in the discord group.

Fig 1

The Second Tab is selected "News" the content section will be filled with the SPAD.neXt news. Even though there is a lot going on with SPAD.neXt the news does not actually get updated that often ... It takes time to do that and the dev tends to spend more time developing the product and sharing new stuff in the Discord group... Fig 2 has an example of the UI when on the News Tab.

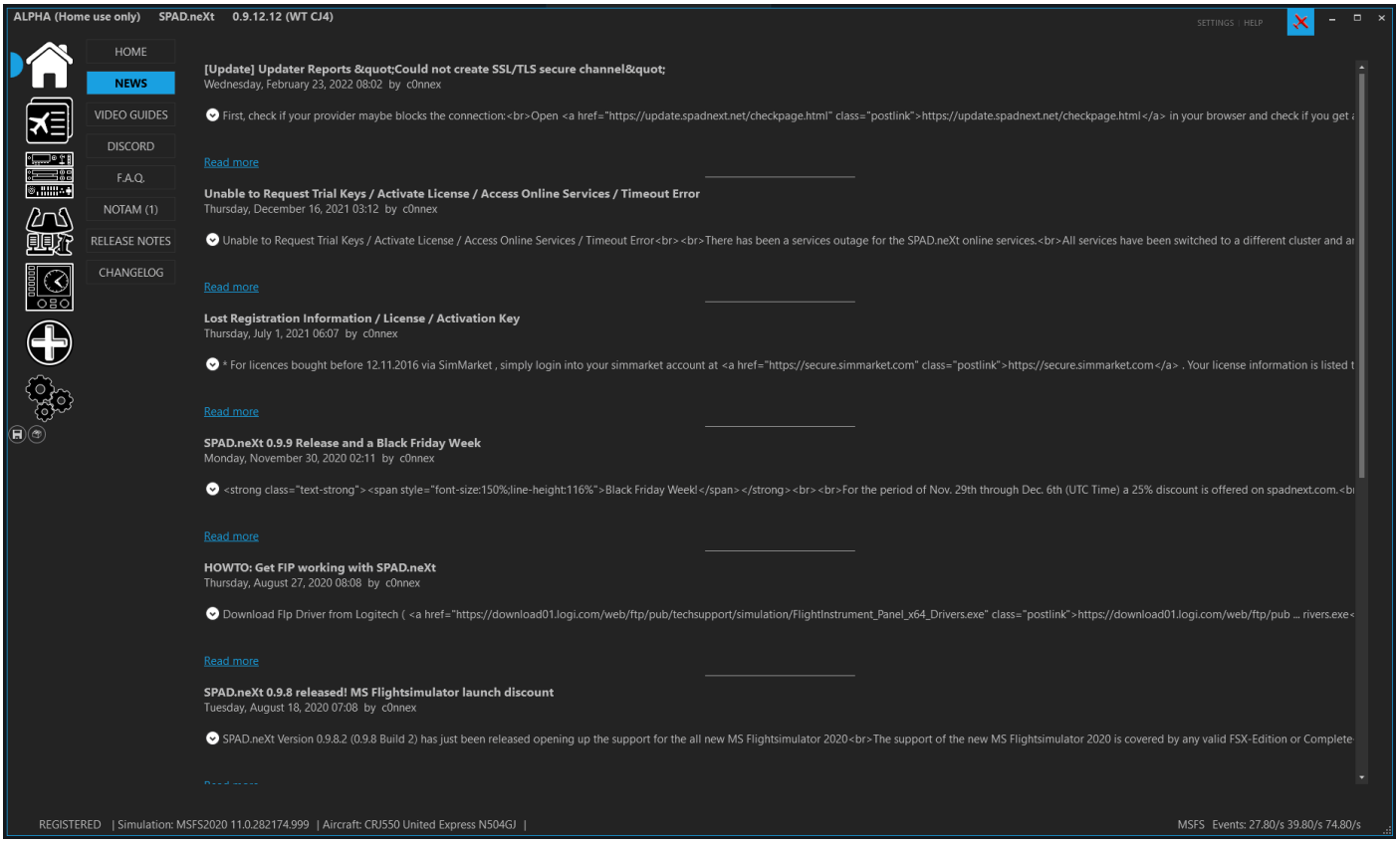


Fig 2

The Third Tab will display the "Video Guides" menu. A third Column with subject sorting is displayed that will then display the video content to the right (as seen in Fig 3). This content is a contribution of Video based How To Guides for many topics. There are Series like the "New User" series which can take you through many aspects when you first get going with SPAD.neXt. Just click on the Video Icon and it will take you to the YouTube Video.

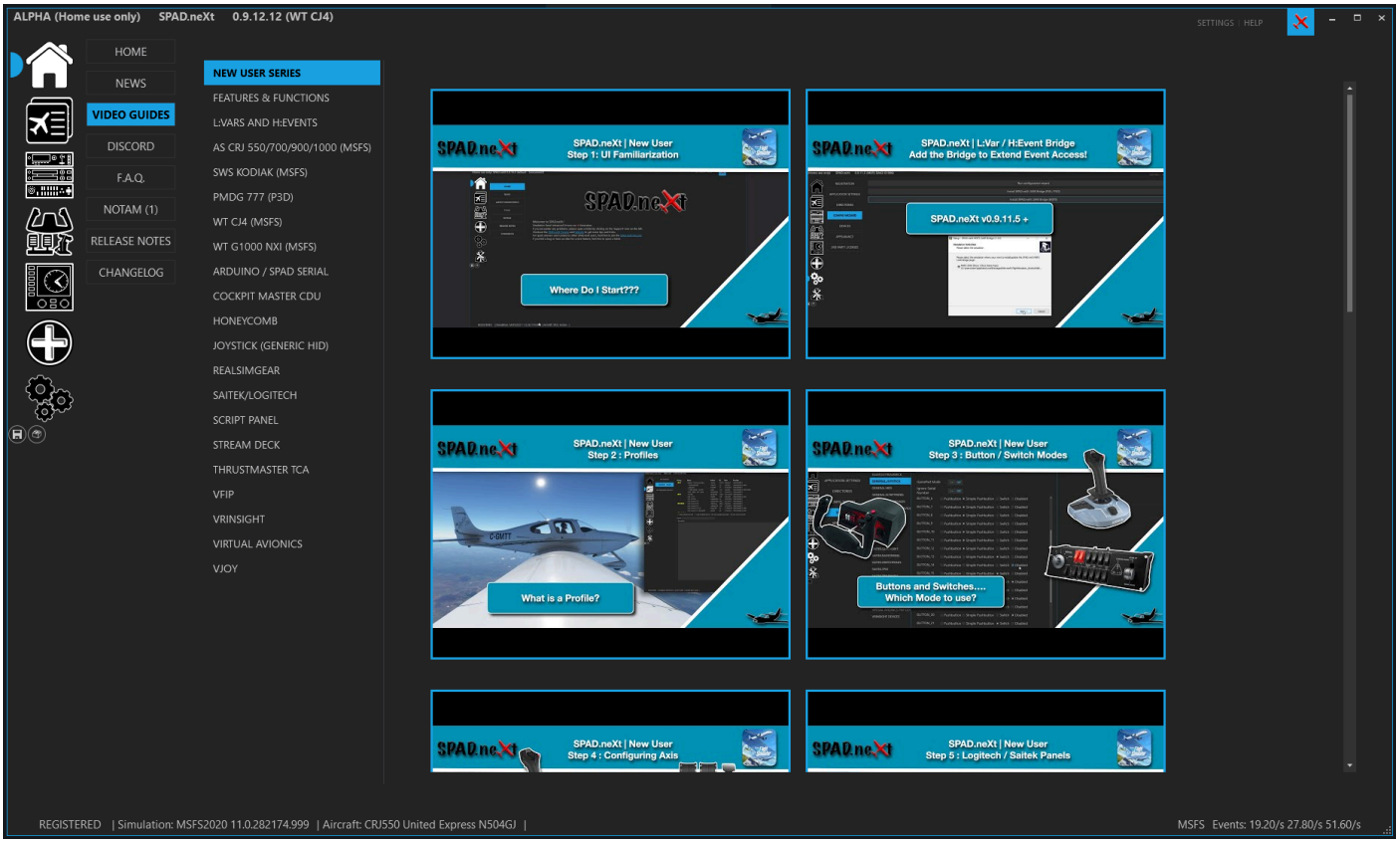


Fig 3

The Fourth Tab is for the "Discord" (Fig 4). This is a tremendous resource with real time interaction from many of the users and even the developer. It is different than a Forum but has much more real time interaction and results usually in some great back and forth. There is a search and there are "channels" to use for different types of information.

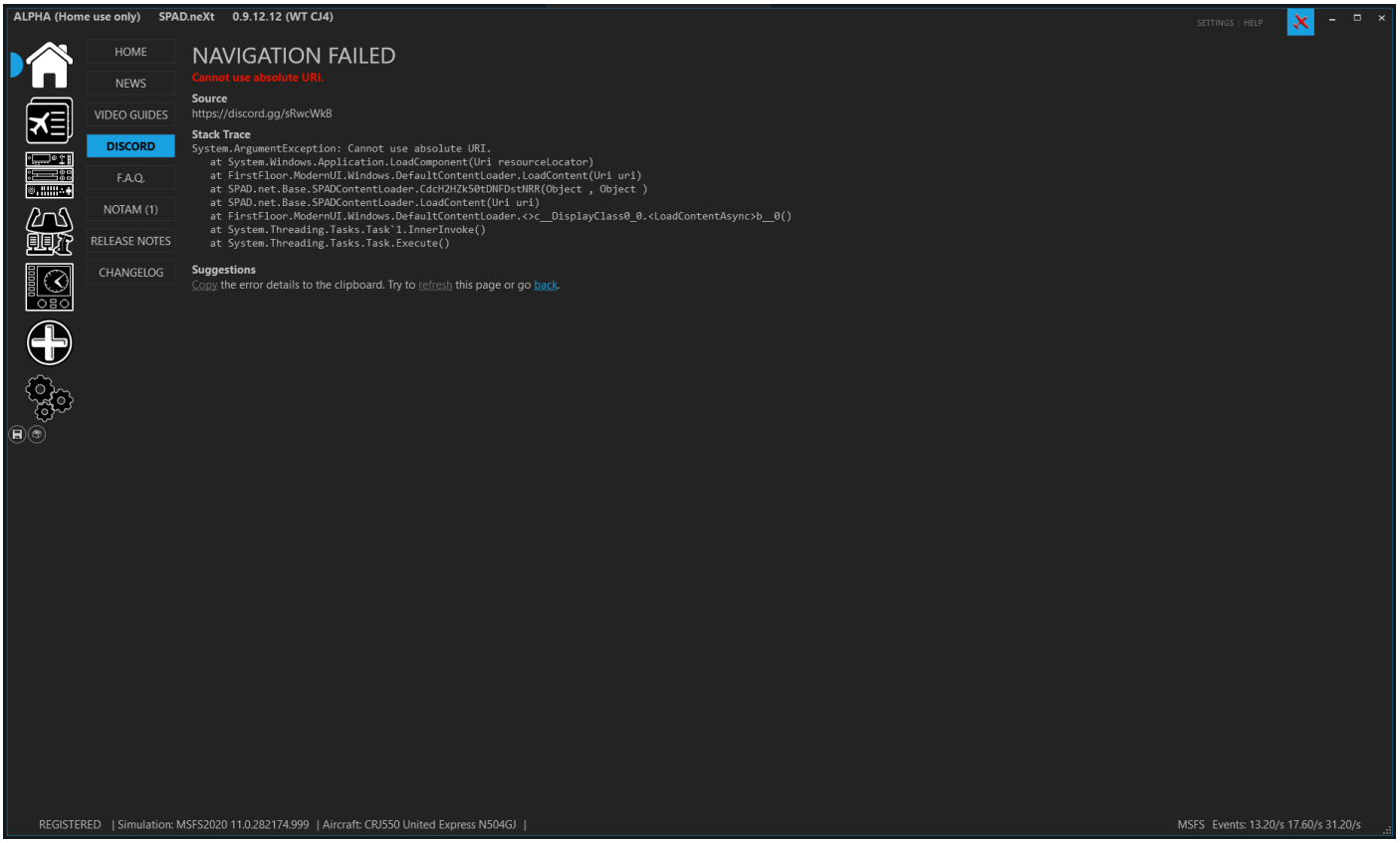


Fig 4

The Fifth Tab contains the "FAQ" ... Yes many of the questions are already directly in the UI and if you use this page you might find what you are looking for. Fig 5 shows a recent update with plenty of information and especially one example at the top that many could find the answer right here in the UI!!

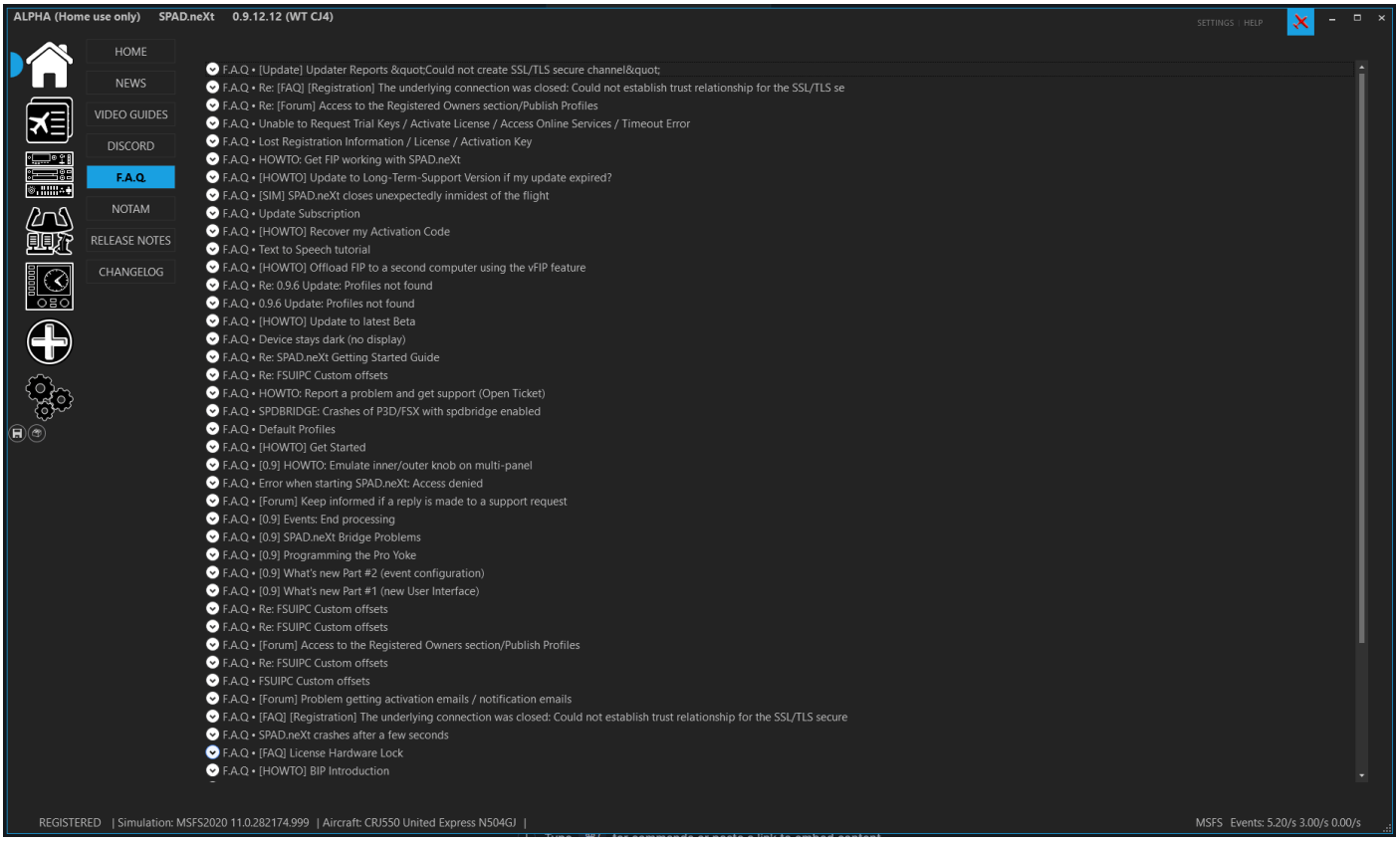


Fig 5

The Sixth Tab contains the "NOTAMS" menu (Fig 6) which will house some important information. Unlike News which is a what is going one or what is new.. Notams is a NOTICE to Air Missions (FAA just changed it to this...) which has temporary information that when resolved would no longer be part of this section and would either move to News, Release notes or FAQ.

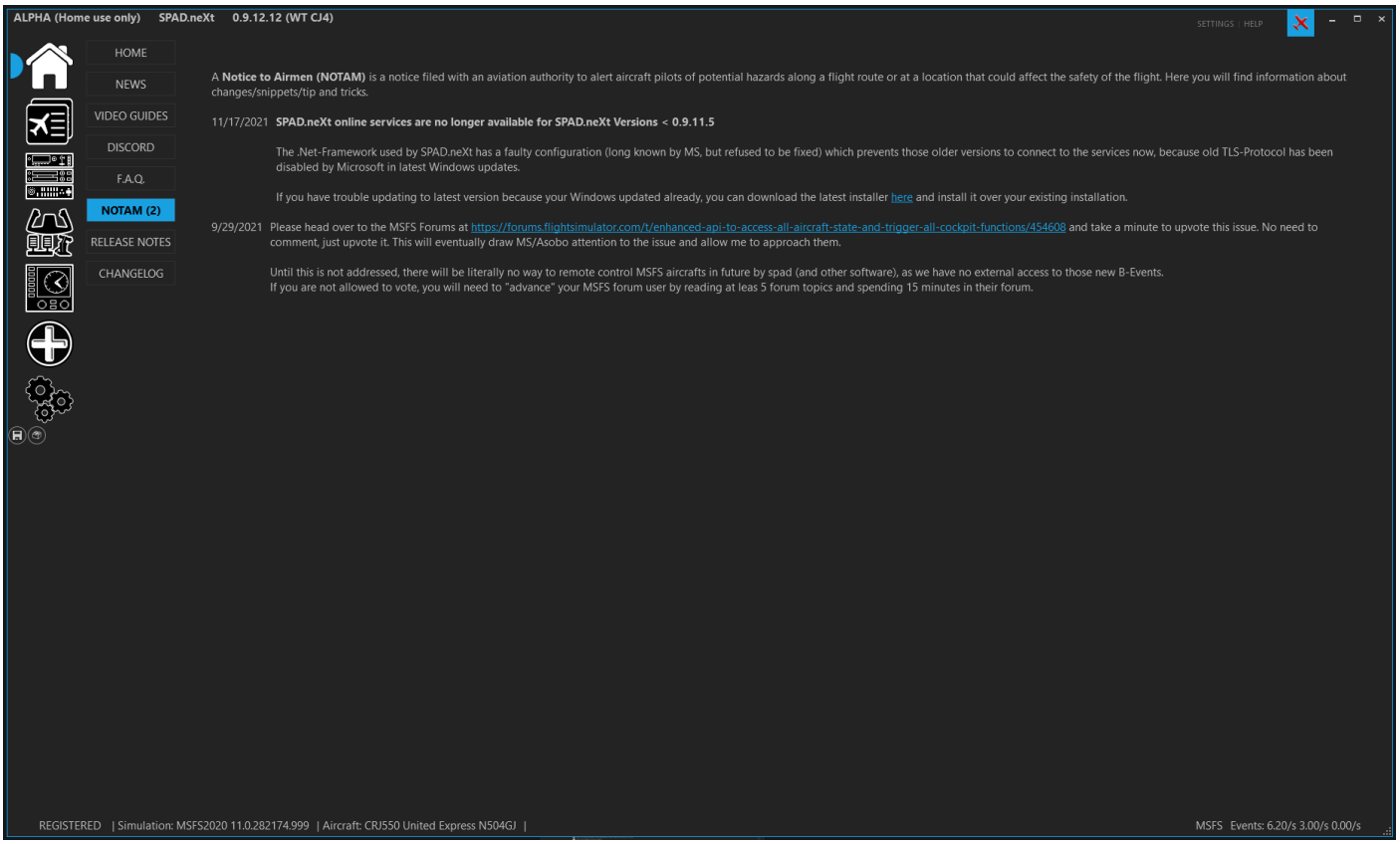


Fig 6

The Seventh Tab is the "Release Notes" Menu. This section (Fig 7) contains all of the Released Version numbers and what has been included with the product in these releases. This is NOT information about Alpha or Beta channels. This is about the released product versions. This is also helpful information as you will know what your license is entitled too if you do not pay for the update subscription after the first year of included updates expires.

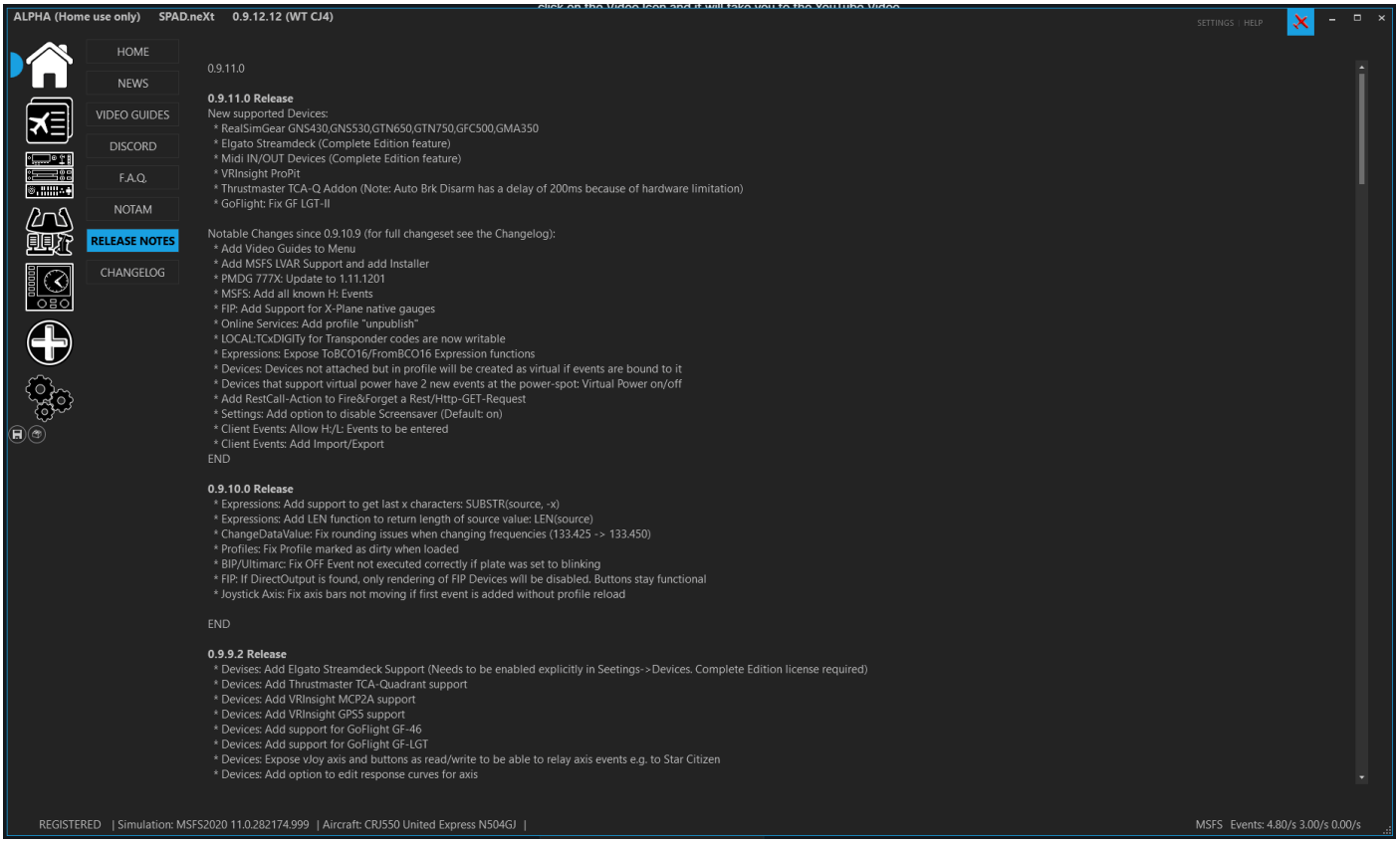


Fig 7

The Eighth Tab contains the "Change Log" Information. Fig 8 shows an example of the content displayed in this section. There can be frequent updates to the Alpha builds and this information will document all of the builds and what was done in each one. Note that when a "release" is performed this information will all Roll Up into the the Release Notes Section with the official release build and number.

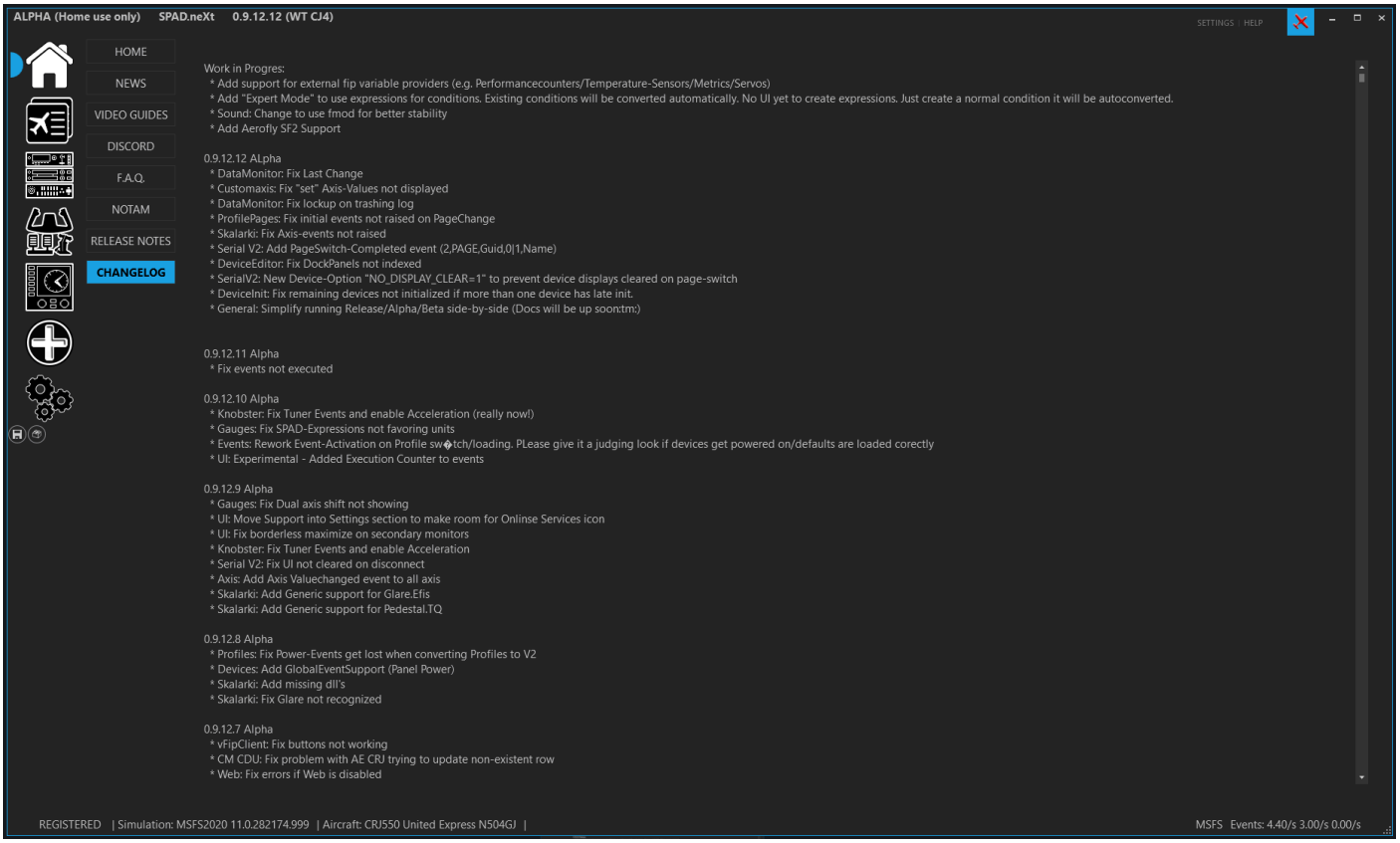

ALPHA (Home use only) SPAD.neXt 0.9.12.12 (WT CJ4)
SETTINGS | HELP
HOME
NEWS
VIDEO GUIDES
DISCORD
F.A.Q.
NOTAM
RELEASE NOTES
CHANGELOG
Work in Progres:
* Add support for external fip variable providers (e.g. Performancecounters/Temperature-Sensors/Metrics/Servos)
* Add "Expert Mode" to use expressions for conditions. Existing conditions will be converted automatically. No UI yet to create expressions. Just create a normal condition it will be autoconverted.
* Sound: Change to use fmod for better stability
* Add Aerofly SF2 Support
0.9.12.12 ALpha
* DataMonitor: Fix Last Change
* Customaxis: Fix "set" Axis-Values not displayed
* DataMonitor: Fix lockup on trashing log
* ProfilePages: Fix initial events not raised on PageChange
* Skalarki: Fix Axis-events not raised
* Serial V2: Add PageSwitch-Completed event (2,PAGE,Guid,0|1,Name)
* DeviceEditor: Fix DockPanels not indexed
* SerialV2: New Device-Option "NO_DISPLAY_CLEAR=1" to prevent device displays cleared on page-switch
* DeviceInit: Fix remaining devices not initialized if more than one device has late init.
* General: Simplify running Release/Alpha/Beta side-by-side (Docs will be up soon:tm:)
0.9.12.11 Alpha
* Fix events not executed
0.9.12.10 Alpha
* Knobster: Fix Tuner Events and enable Acceleration (really now!)
* Gauges: Fix SPAD-Expressions not favoring units
* Events: Rework Event-Activation on Profile sw?tch/loading. PLease give it a judging look if devices get powered on/defaults are loaded corectly
* UI: Experimental - Added Execution Counter to events
0.9.12.9 Alpha
* Gauges: Fix Dual axis shift not showing
* UI: Move Support into Settings section to make room for Onlinse Services icon
* UI: Fix borderless maximize on secondary monitors
* Knobster: Fix Tuner Events and enable Acceleration
* Serial V2: Fix UI not cleared on disconnect
* Axis: Add Axis Valuechanged event to all axis
* Skalarki: Add Generic support for Glare.Efis
* Skalarki: Add Generic support for Pedestal.TQ
0.9.12.8 Alpha
* Profiles: Fix Power-Events get lost when converting Profiles to V2
* Devices: Add GlobalEventSupport (Panel Power)
* Skalarki: Add missing dll's
* Skalarki: Fix Glare not recognized
0.9.12.7 Alpha
* vFipClient: Fix buttons not working
* CM CDU: Fix problem with AE CRJ trying to update non-existent row
* Web: Fix errors if Web is disabled
REGISTERED | Simulation: MSFS2020 11.0.282174.999 | Aircraft: CRJ550 United Express N504GJ |
MSFS Events: 4.40/s 3.00/s 0.00/s

Fig 8

Profiles Page

This section describes the layout and use of the Profiles Menu Page

When the Profiles Menu Page is selected on the Left Side Navigation Column the UI will look similar to Fig 1. This is where you create configuration profiles that you can then Link to Aircraft in your simulations. The current Active Profile will already be selected and you will see the Red SPAD.neXt plane to the left as well. If you select another Profile the Highlight will change but the Red Plane will stay next to the Active Profile.

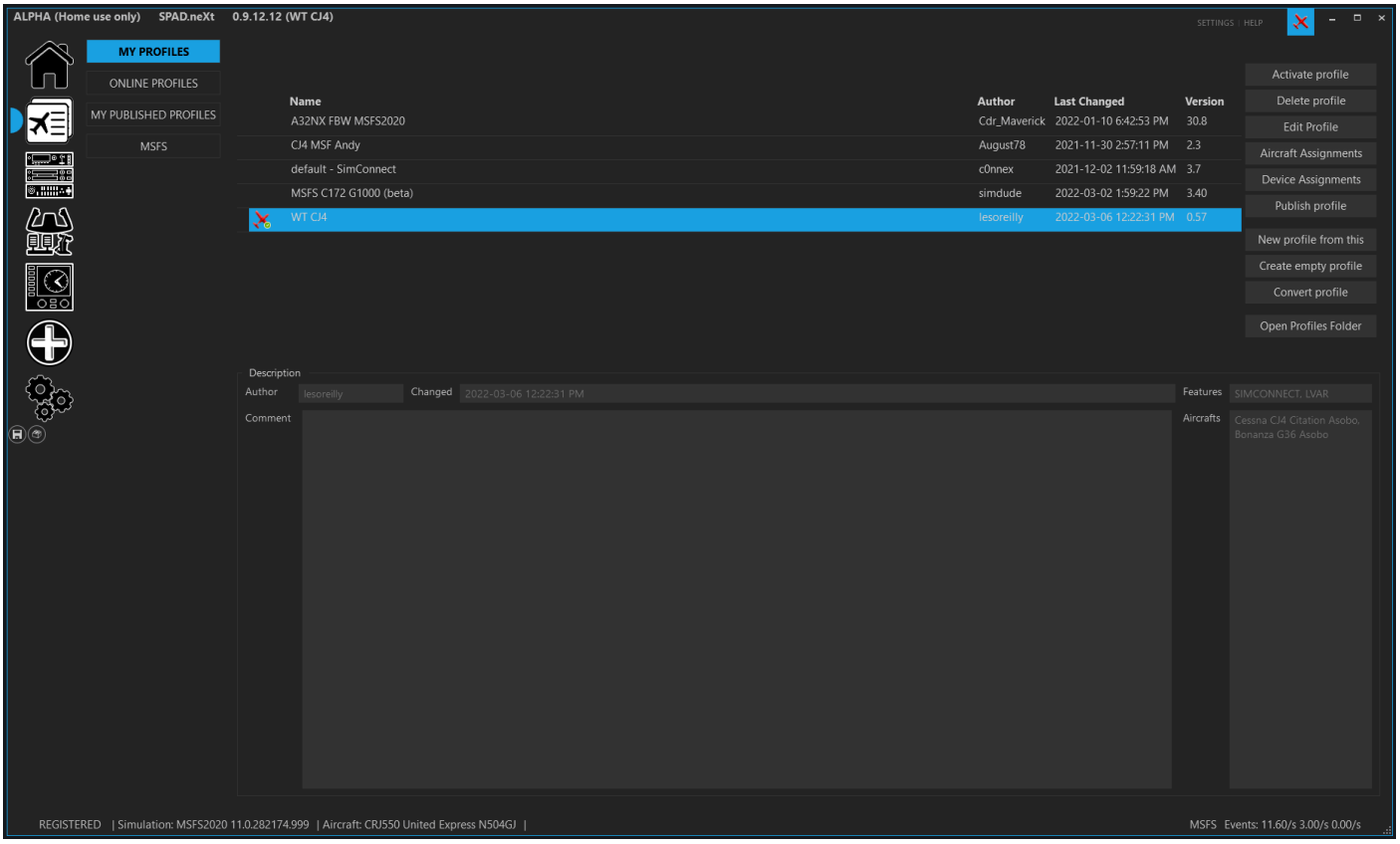


Fig 1

The Sub Tab Navigation is on the Left side like other Menus. The First Three Menus (called out in Fig 2) will be present. Additional Navigation will be available when you create Profiles that have a "\" in the name to instruct SPAD.neXt to create sub directories for sorting them. As seen here there is a MSFS directory that was created (this will be further explained as you continue to read)

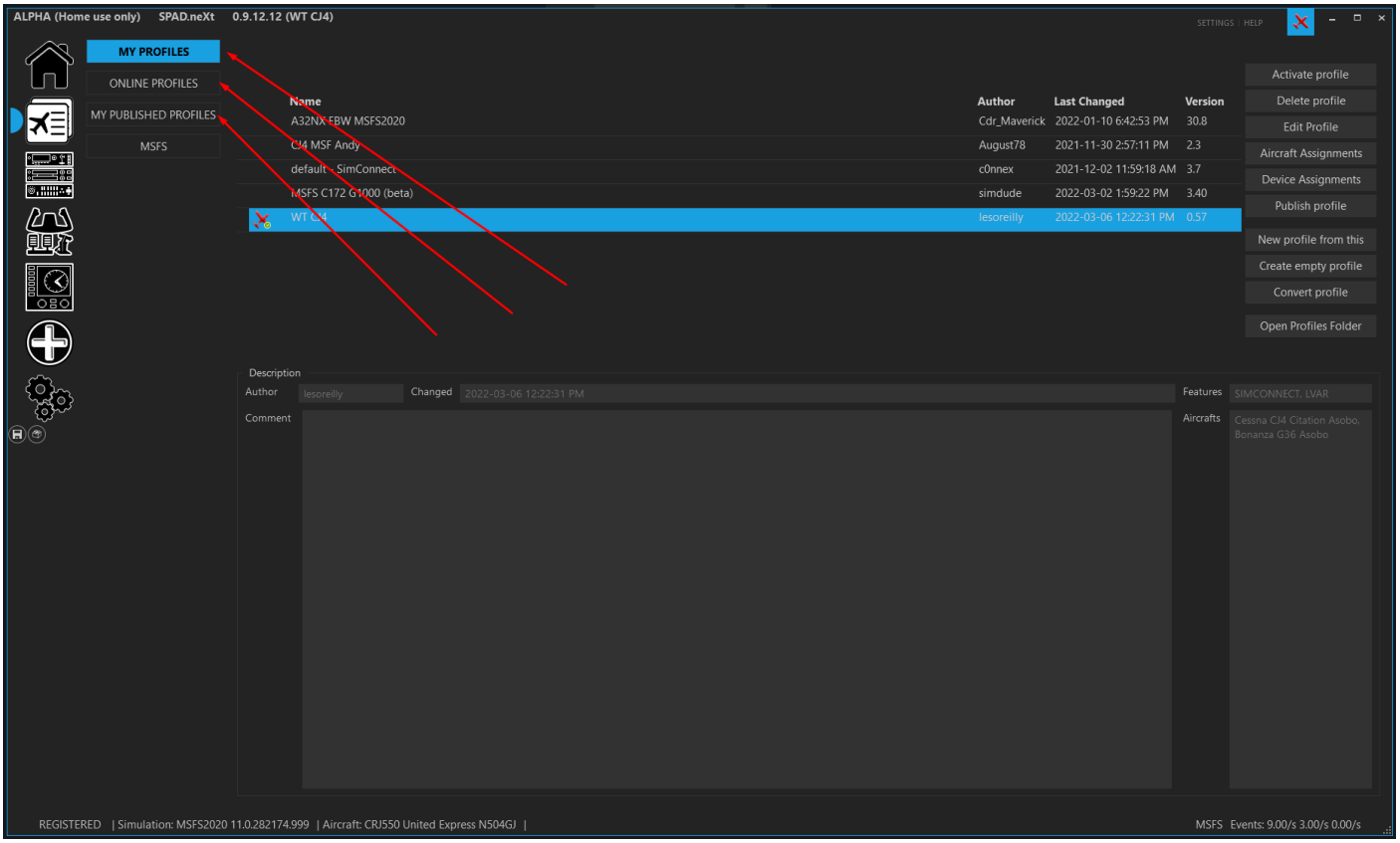


Fig 2

The Three Sub Navigation sections are My Profiles (kind of like the "root directory" of profile storage that is found in your Documents example C:\Users\User\Documents\SPAD.neXt\profiles). If no sub folders are created by the user then all Profile XML files will be located there.

Online Profiles (Fig 3) will display all of the available Online Profiles that have been published and your license has entitlements for (note that any published profile AFTER the expiry of an Update Subscription will not be available). Called out in Fig 3 is the filtering criteria and ability to search for Profiles as well.

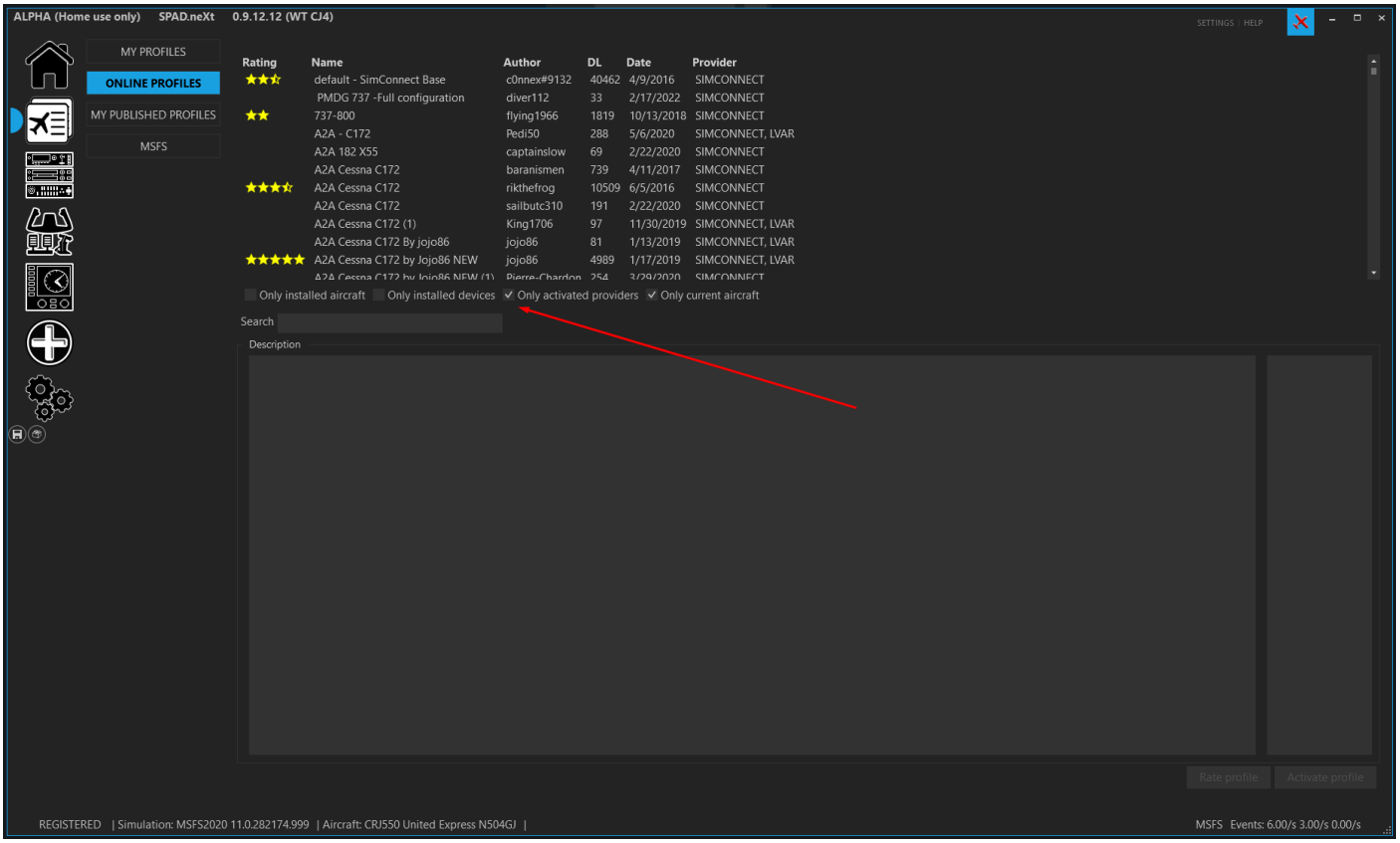


Fig 3

My Published Profiles Will display the Profiles that you have published to the Online System and have shared with the community. Fig 4 displays an example of what you published menu could look like. Note that again the Filters at the bottom can also be used.

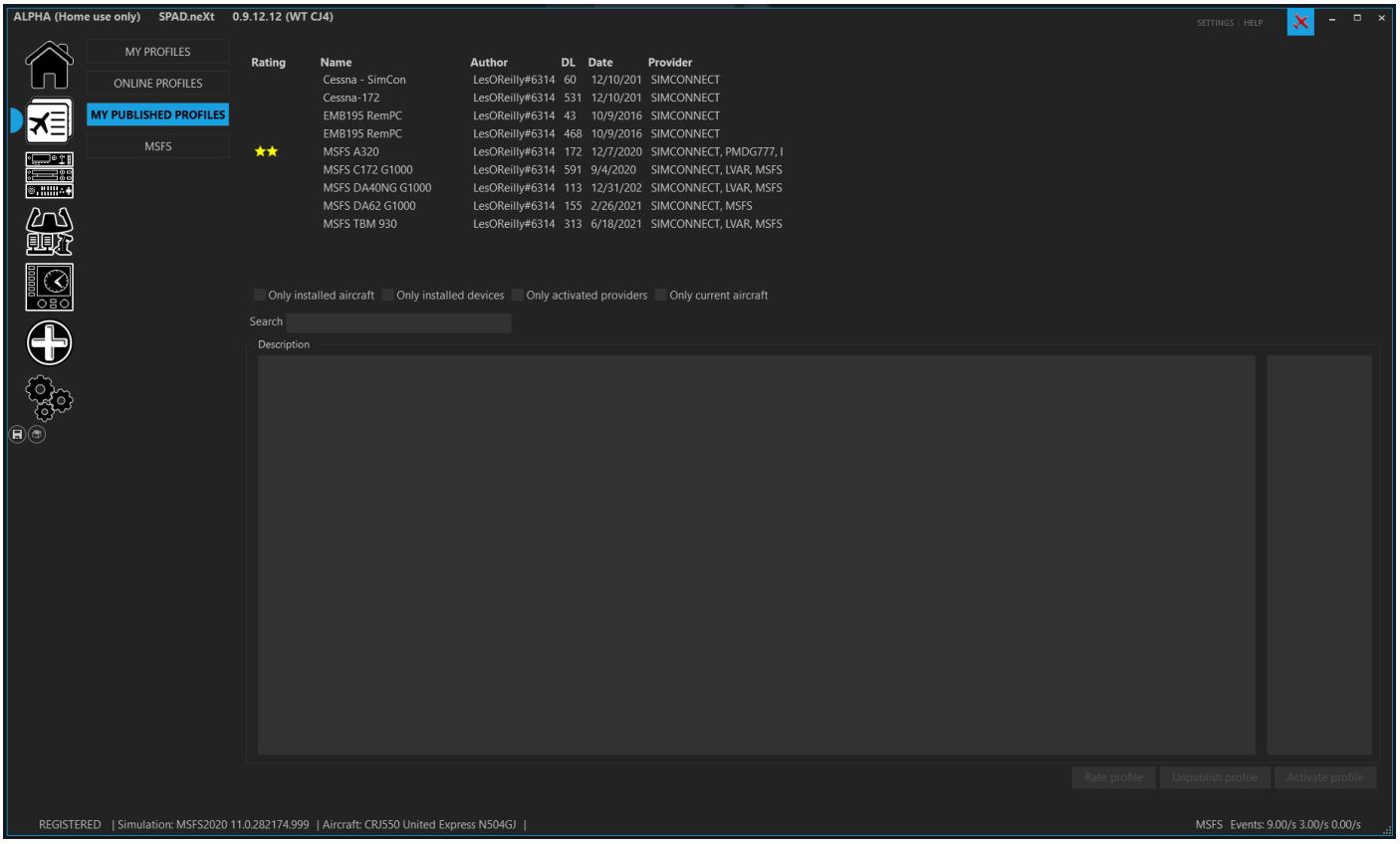


Fig 4

In the case of Fig 5 where we have the example of a sub directory for MSFS it can be seen that the name of the Profile was edited to have MSFS\ added to the front of it. This instructs SPAD.neXt to create a subfolder of that name in your profiles directory and then place the profile in that location. Note: You may need to restart SPAD.neXt after creating a new directory that was not present before launch.

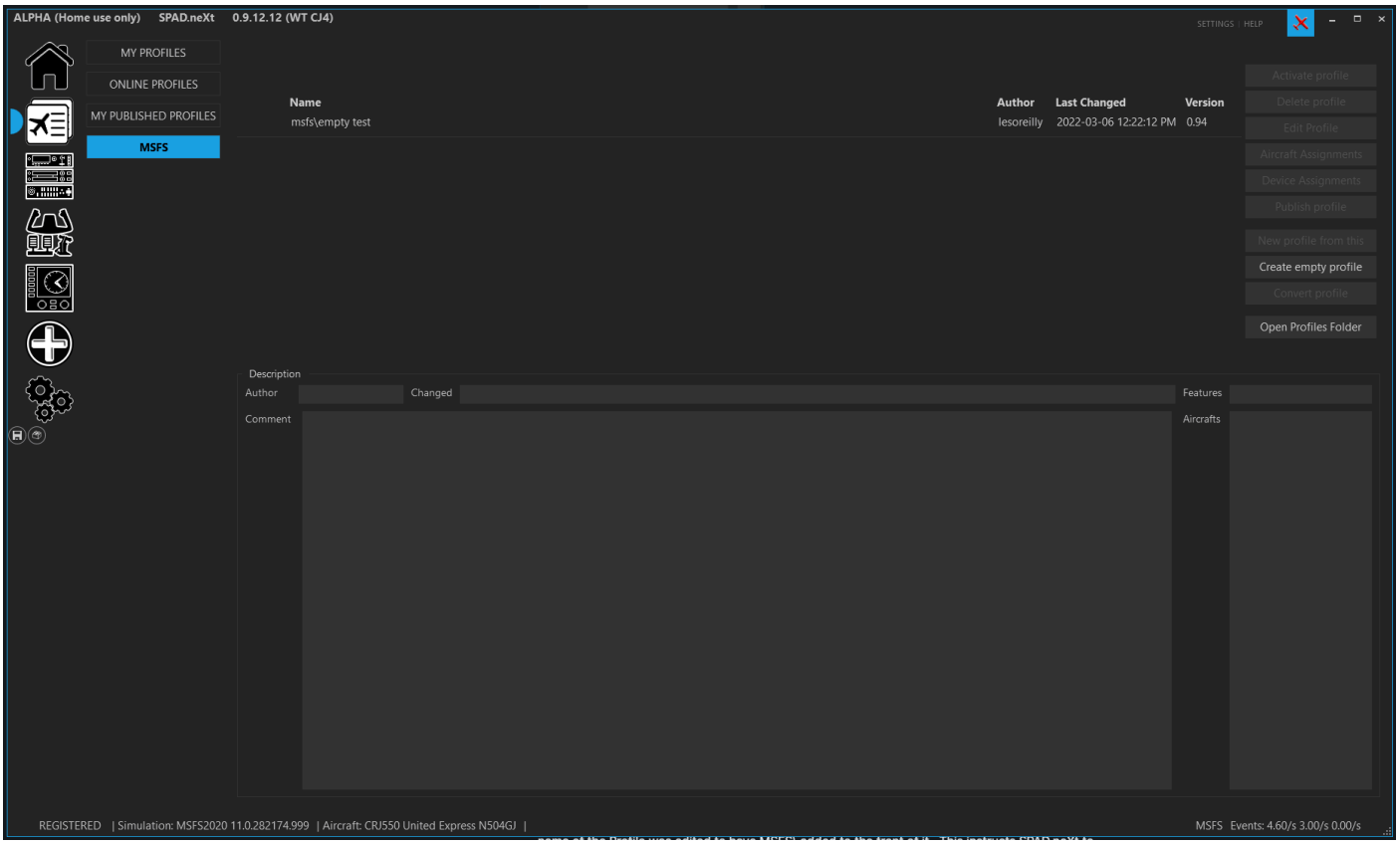


Fig 5

When we look at Fig 6 and it's Callout you will see that the Right Side of the UI is actions regarding the Profiles. When Actions are not available they will be "greyed" out and can not be selected. In Fig 6 we see that since no profile is Highlighted in the MSFS subdirectory that we are viewing then it is not possible to perform any actions other than "Create Empty Profile" which will start a totally new and blank profile with nothing assigned to your devices. The other Action is "Open Profiles Folder" which will launch Explorer and take you to the Folder or "Sub Folder" that you are currently viewing.

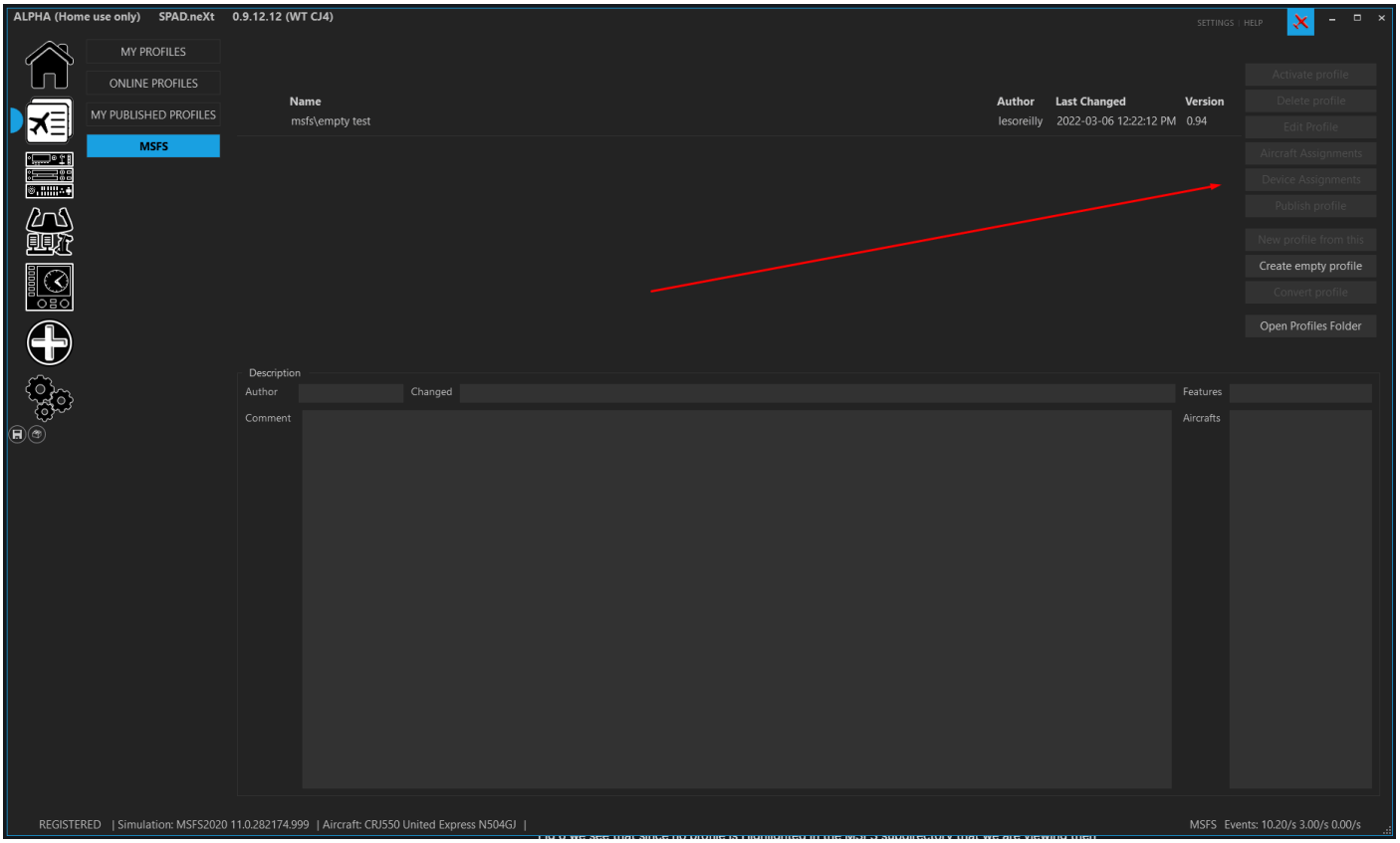


Fig 6

If we Highlight the Profile in the MSFS directory that we are currently viewing we can see that all of the right hand side Profiles actions become selectable (see Fig 7). The Top two actions are straight forward.

Activate Profile will Make the Highlighted Profile load it's configuration of devices. Note if you had another Profile open that was edited it should prompt you to save... However it is always good to manually press the Save icon first to be safe

Delete Profile is the other pretty straight forward action that will allow you to delete the Profile entirely ... Hey maybe you made a new version from an old profile and now that you are happy with your changes you are ready to clean up the old stuff...

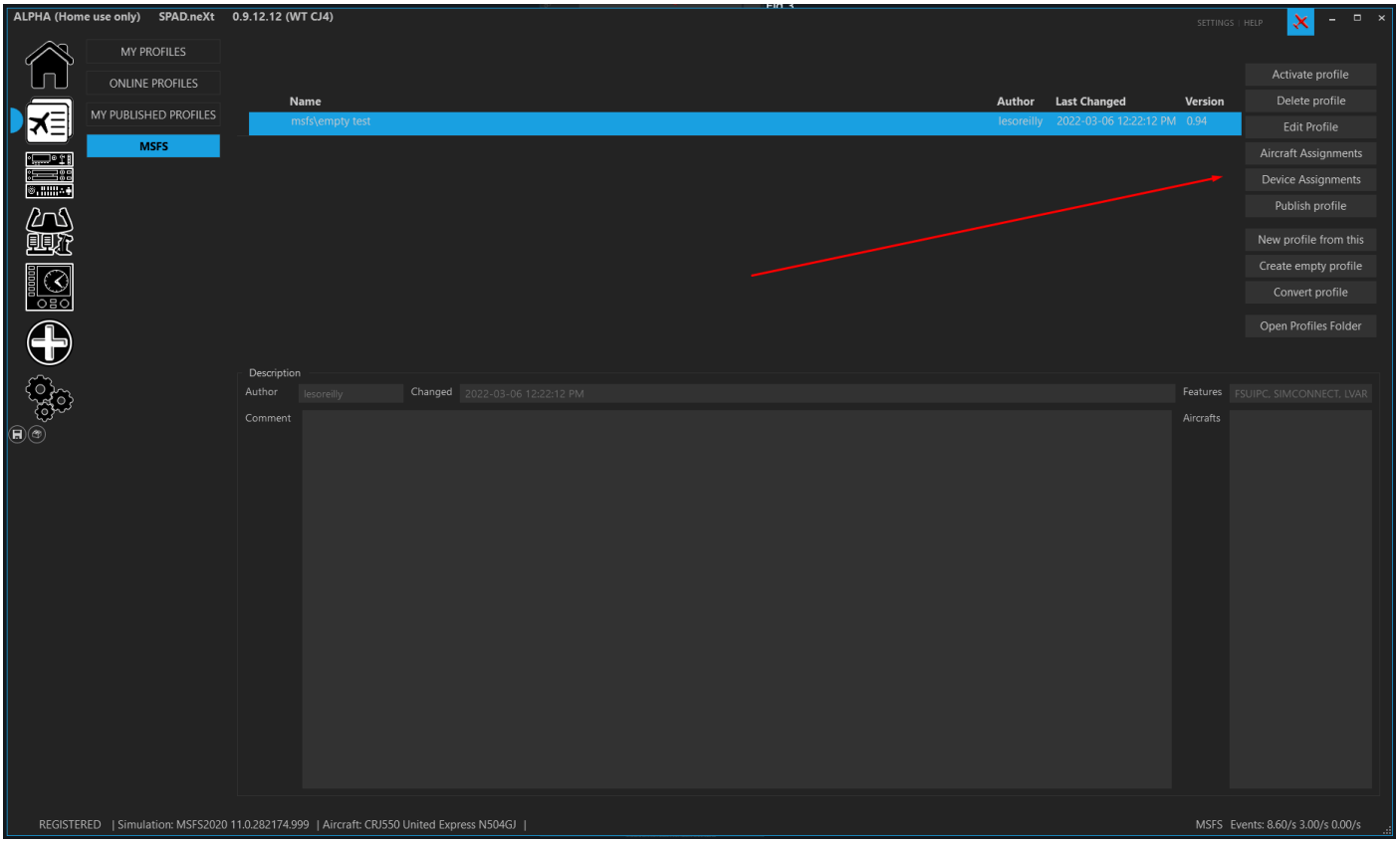


Fig 7

Moving on to the Next Action Item "Edit Profile" in Fig 8. This will open an additional Dialog Box that allows you to modify or "Edit" the selected Profile. Keep in mind that you do not need to activate a profile to use this editor. This is not about Editing the Devices but rather the Properties of this Profile.

When the edit dialog opens up you are now able to Modify the Profile name (here you see how it is in a sub folder and removing that MSFS\ would move it back to the root profiles folder).

Version numbers can be edited manually. As you make changes and edit the profile the versions will automatically increment. here you can change them yourself.

Description section allows you to write up all the details and information that you would like about this profile. The more you put into this the more others can read about it if you are to publish this to the online system.

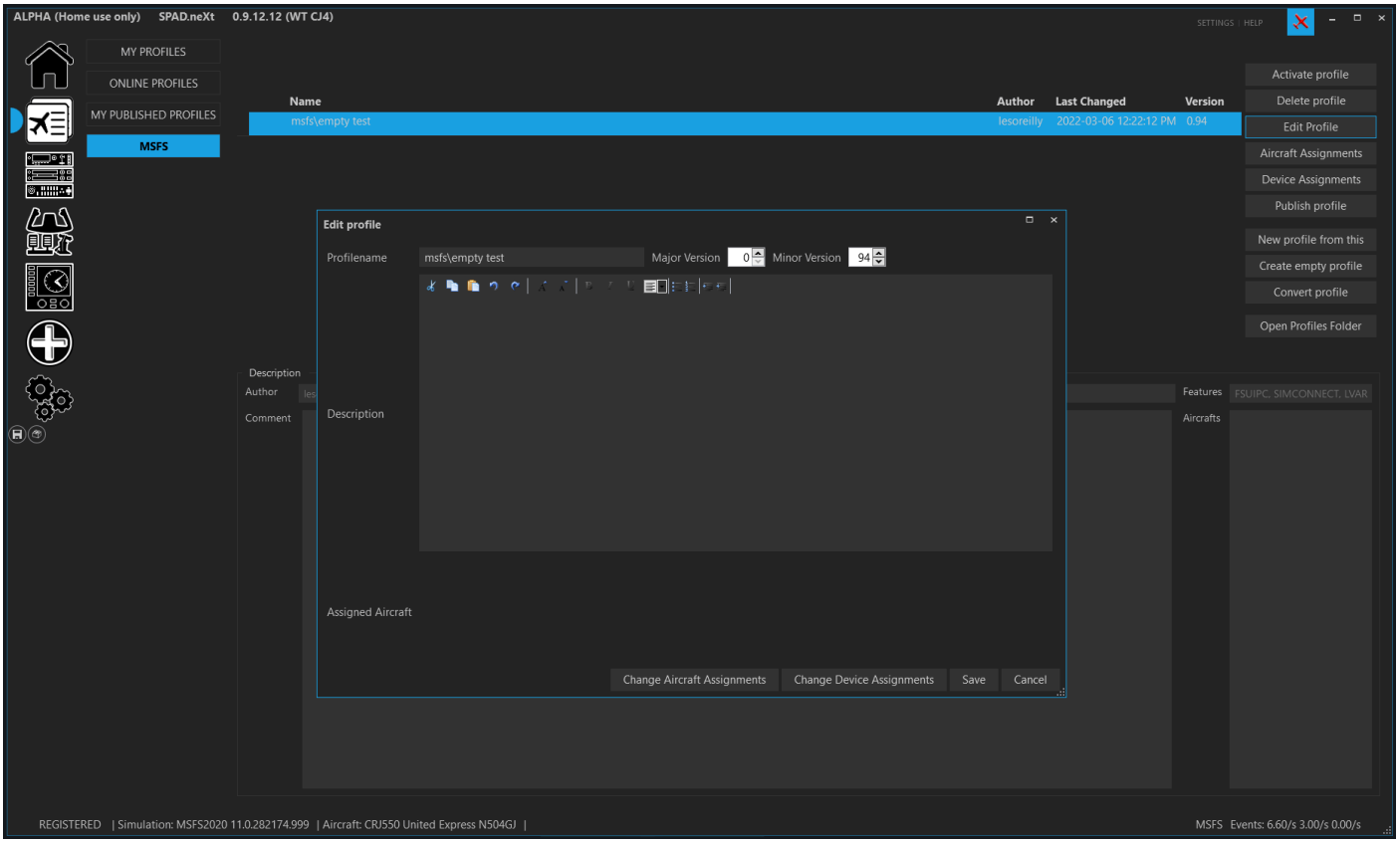


Fig 8

The next selection in the "Edit Profile" Action is found on the Bottom and will open up another dialog that will allow you to edit the Aircraft that you want assigned to this profile. Note that in Fig 9 this instance of SPAD.neXt is operating on the "networked" mode and is only able to see the list of the current active plane. Local installations that are run on the same machine will see additional Aircraft.

Select the Aircraft on the Left Side that you want to link to this profile and then press the Arrow to "transfer" them to the assigned aircraft on the Right hand side. If you want to remove an aircraft you can select them on the Right Side and use the Left Arrow to remove them from the assignments Field. Note: If you were editing a Profile and add an aircraft that is already assigned in another profile you will be warned that making the assignment will remove it from the other Profile. Once finished press Save to confirm the assignments.

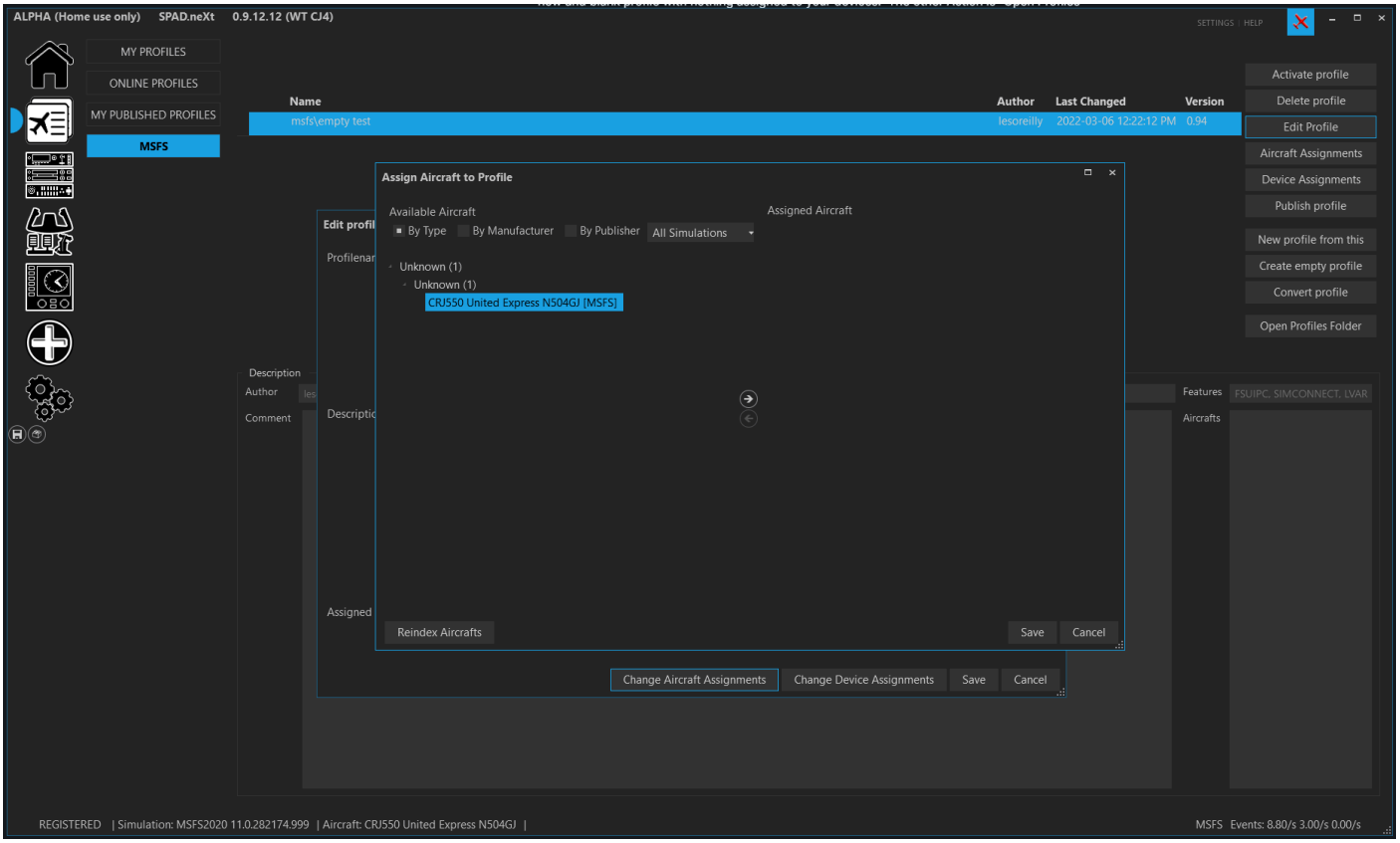


Fig 9

Continuing on with the Edit Actions is the Device Assignments operation (Fig 10) which will launch another dialog that lists all of the device configurations that were saved with this profile and which physically connected devices they are assigned too. This can also help with re-mapping in case a duplicate device that only get instantiated by windows was out of order... Once complete click OK

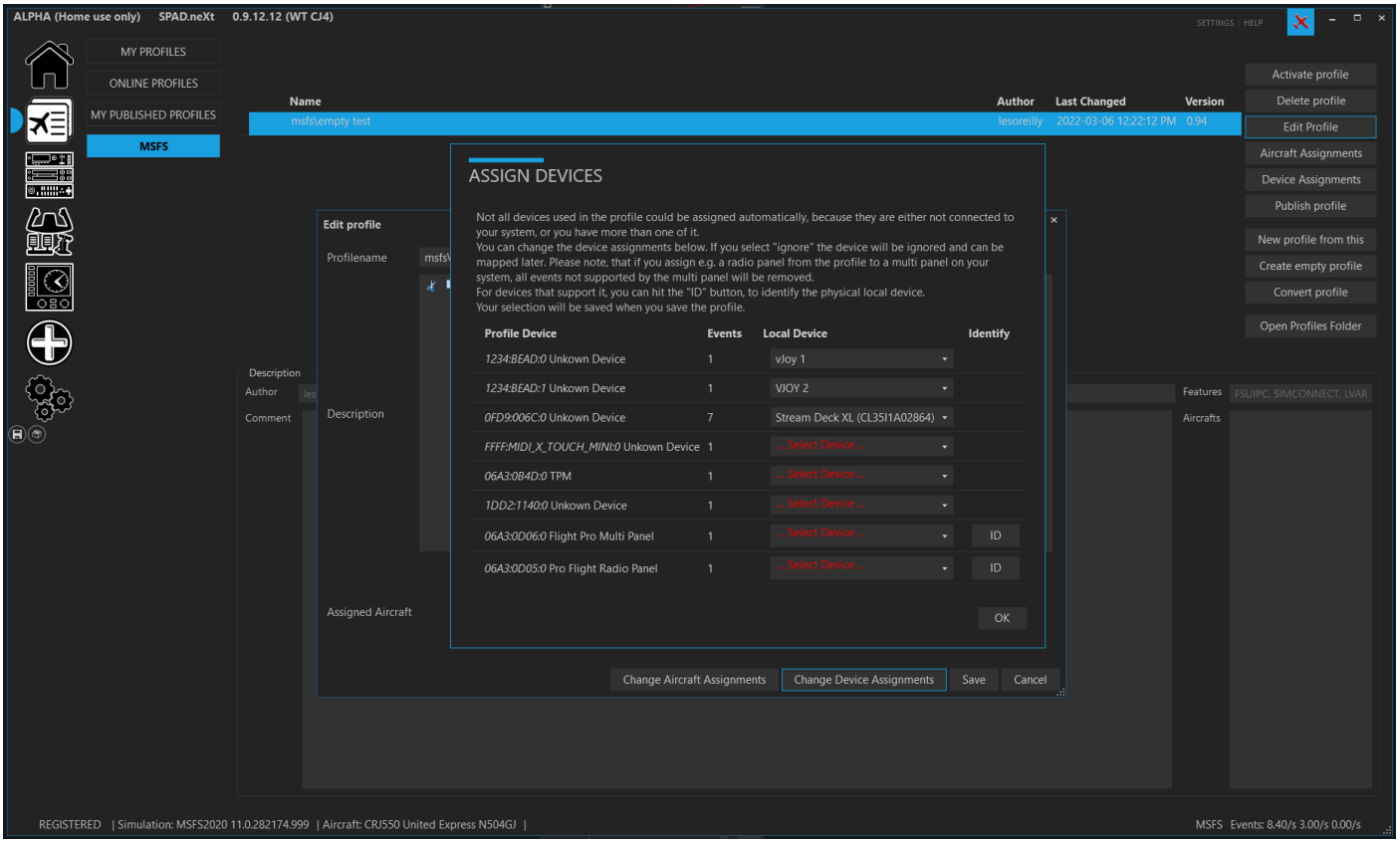


Fig 10

Moving on to the next Action item "Aircraft Assignments" in Fig 11 we see that there are quick selects that effectively take you to the same editor for the Aircraft Assignments as we saw in the Edit Profile section. This is no different in operation however saves the button click of going into Edit Profile first when you just need to link a new Aircraft

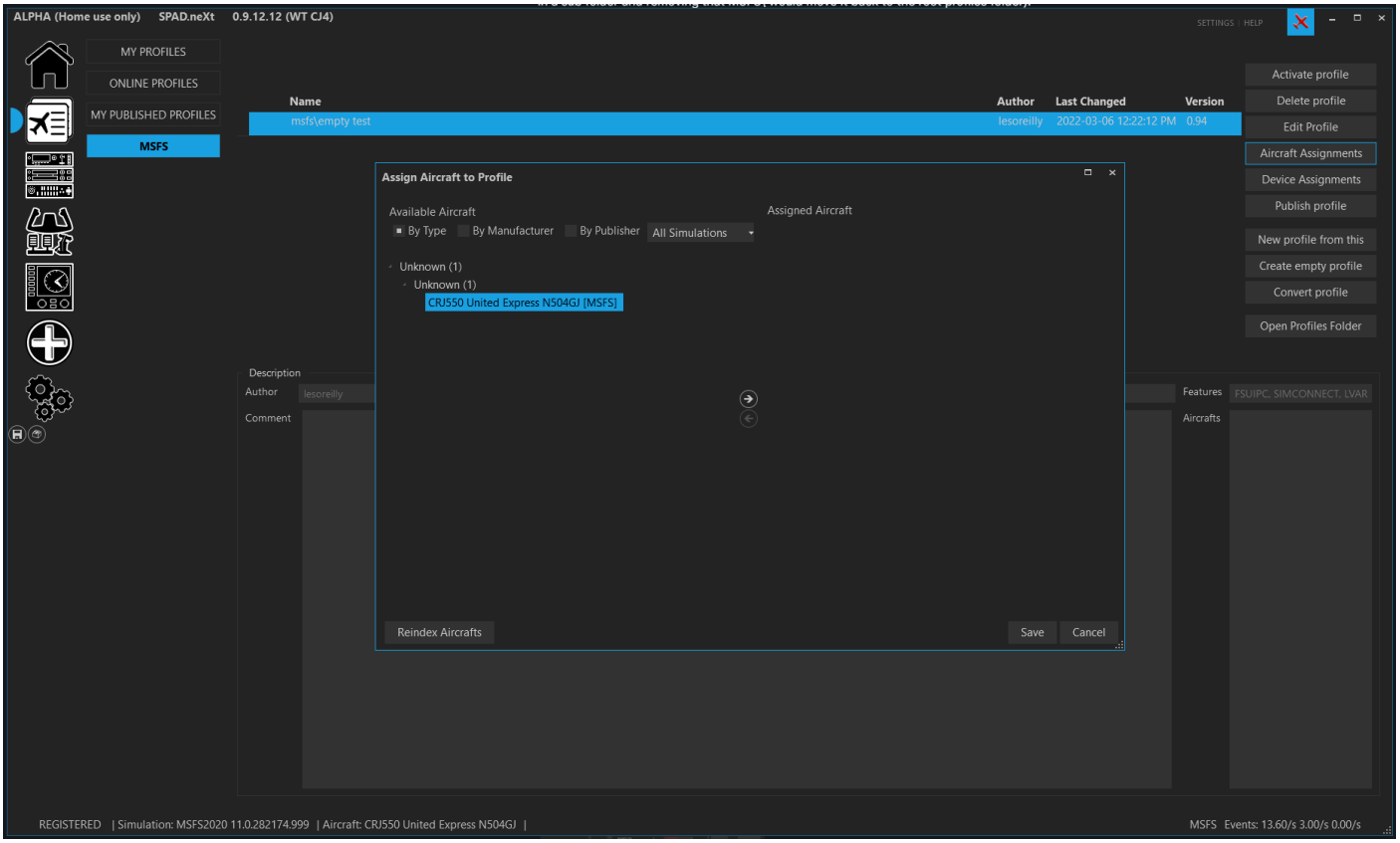


Fig 11

"Device Assignments" (Fig 12) is an action item which also has the quick navigation and does not required editing the Profile to bring up the dialog editor.

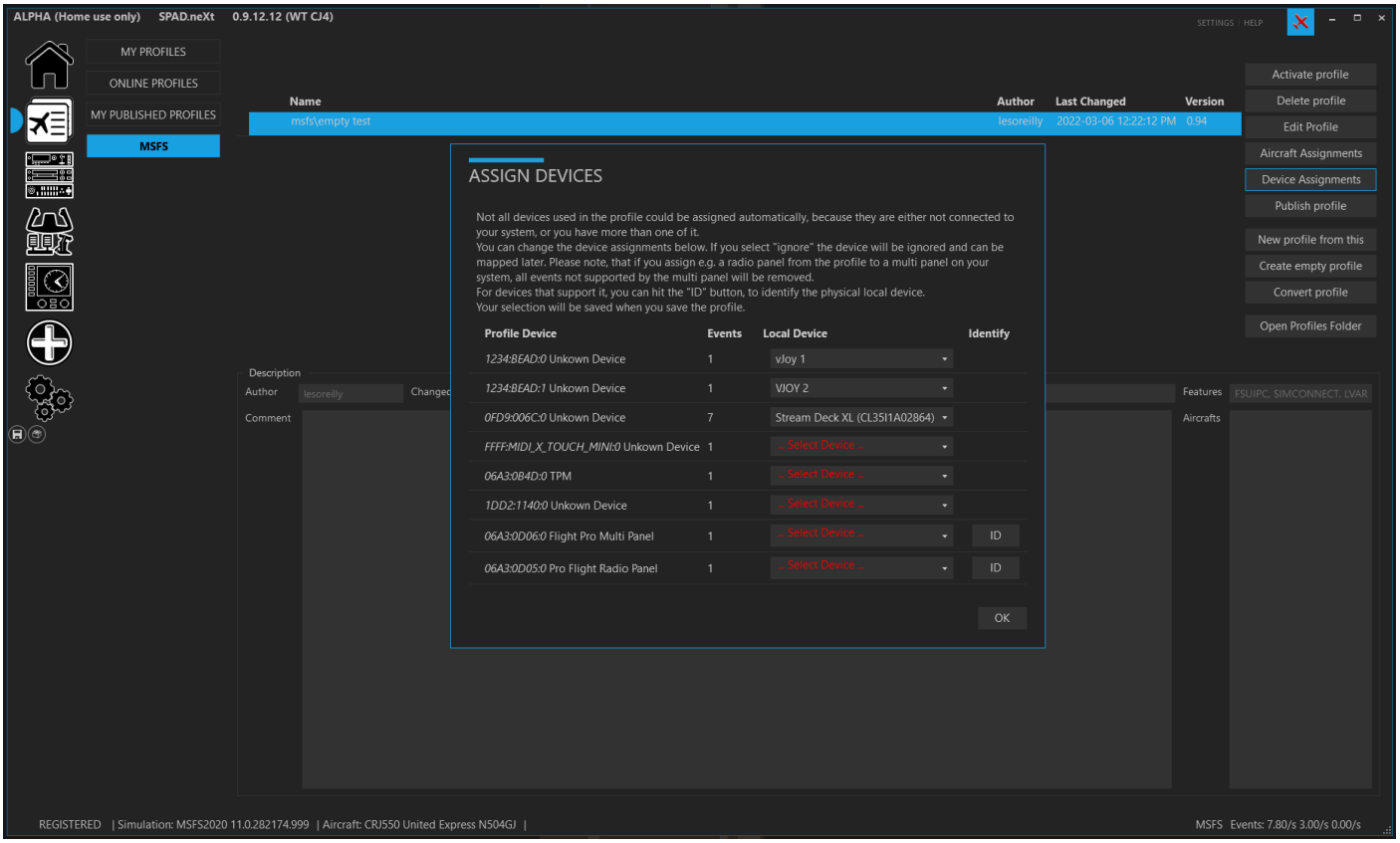


Fig 12

"Publish Profile" action will bring up the Dialog to allow you to share the entire Profile to the online community. In Fig 13 we see what this context menu looks like and that it allows you to pick which devices are included with the publish and which aircraft if you want to help filter it for very specific single or multiple different planes.

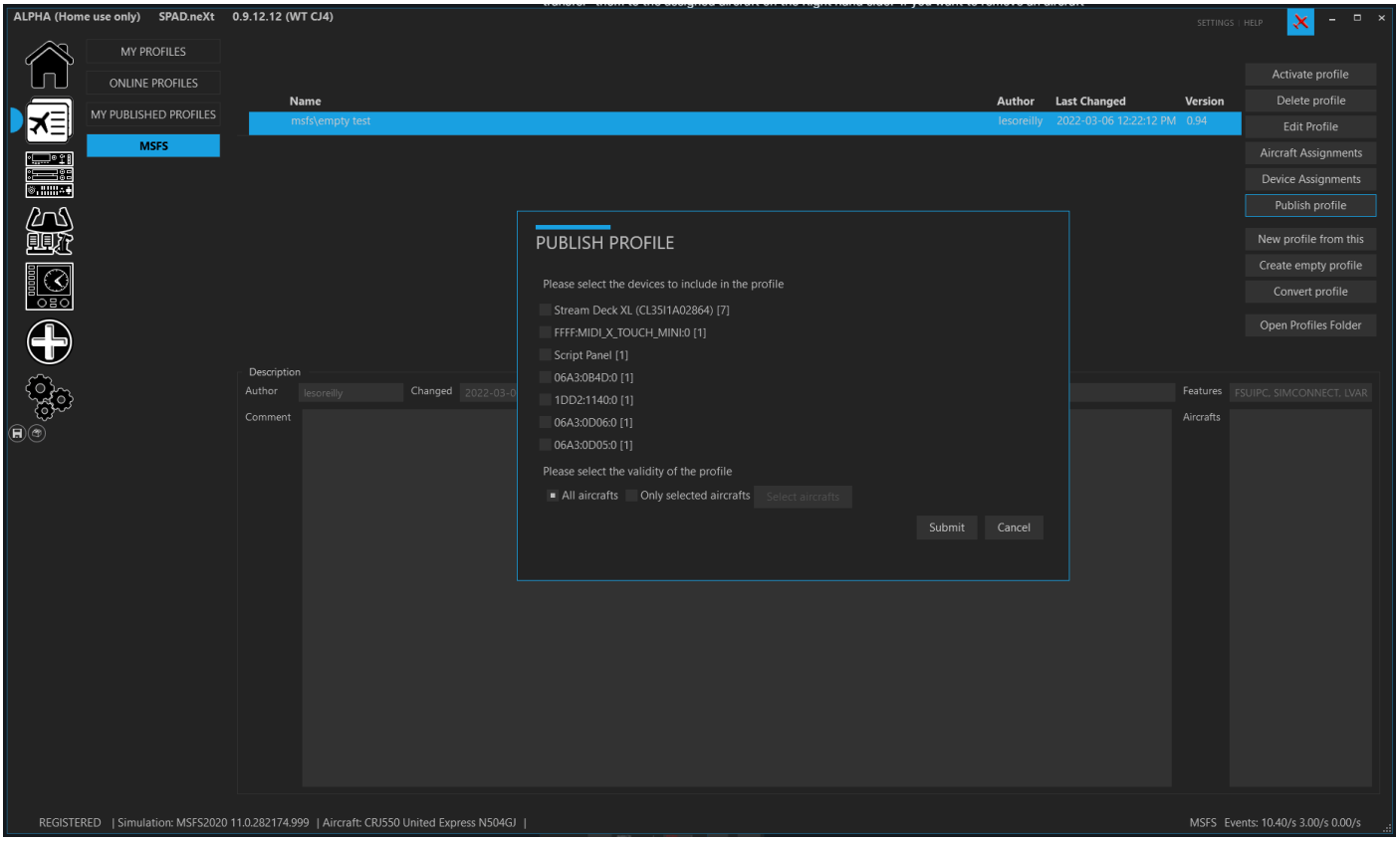


Fig 13

The 7th Action Item is the "New Profile from this" which will allow you to create a new profile but based on all of the device configurations that you have created in this profile. Effectively another term would be Cloning the profile. As seen in Fig 14 when this is selected you are prompted with having to enter a new Name for this profile. This is a helpful way to take a profile that is setup perfect for one plane but needs a couple of tweaks to work well with another.... ie the C172 G1000 just needs some different switch assignments than say the DA-40 with the G1000.

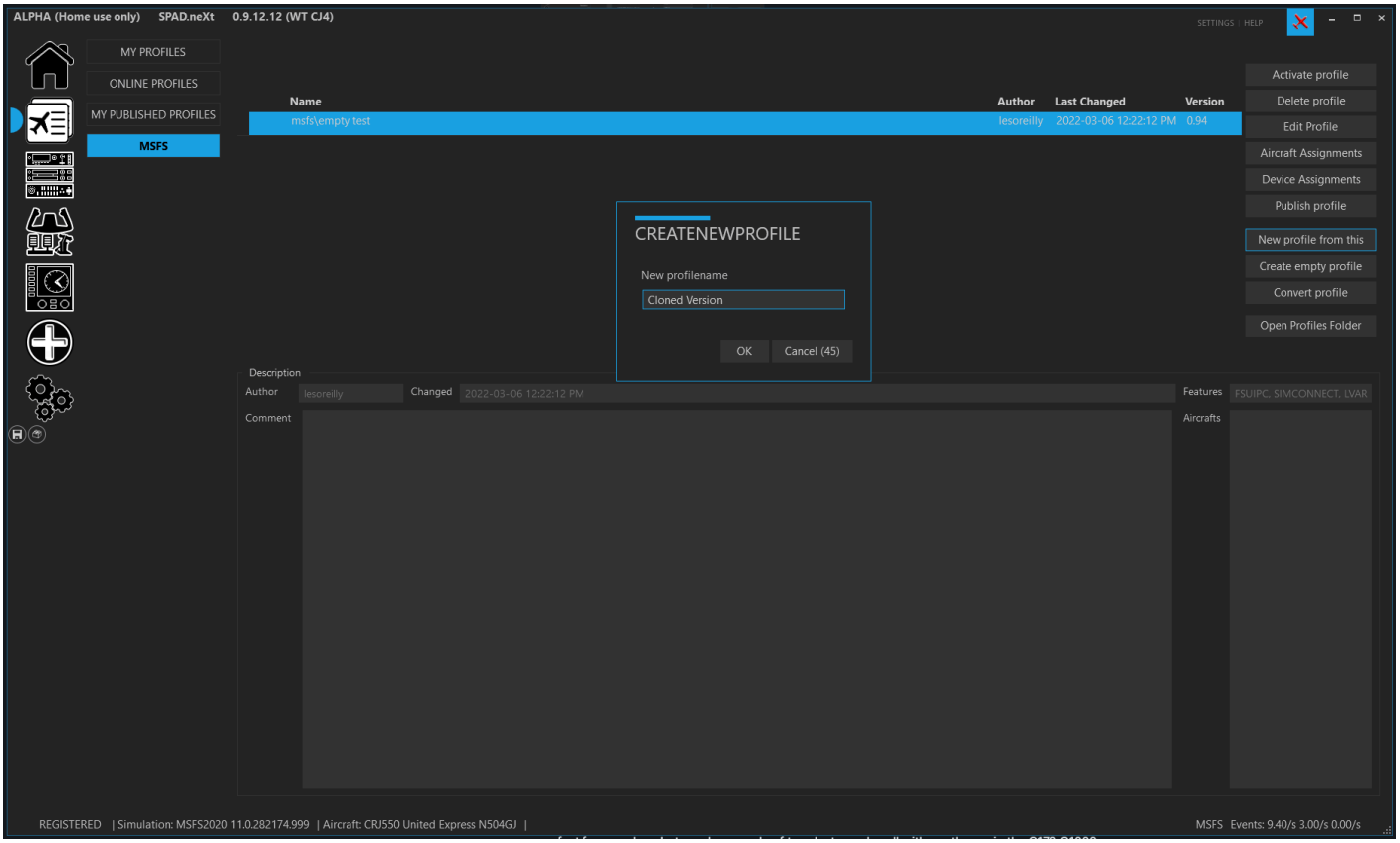


Fig 14

The Last Action is "Convert Profile". This will take older version of the profile an convert them to the new version. This will result in a backup being created in your directory. Note that if you activate an Old Profile that needs to be converted you will be prompted to do that anyway.

Expressions

SPAD.neXt contains a powerful expression evaluation system, which can be used throughout all events and actions.

If not stated differently in the event/action the result of the expression is expected to be a number.

By default all data references will be interpreted as numbers. You might need to apply conversion (see Functions) or preface your expression with `''+` (two single quotes) to enforce the expression system to interpret (and automatically convert) everything as a string.

The underlying expression system (ncalc) is Open-Source and can be found here

Expressions: Values

Values

A value is a token representing a concrete element. This can be:

- an integer
- a floating point number
- a date time
- a boolean
- a string
- a function
- a parameter

Integers

They are represented using numbers.

```
123456
```

They are evaluated as **Int32**.

Floating point numbers

Use the dot to define the decimal part.

```
123.456
.123
```

They are evaluated as **Decimal**.

Scientific notation

You can use the **e** to define power of ten (10^).

```
1.22e1
1e2
1e+2
1e-2
.1e-2
1e10
```

They are evaluated as Double

Dates and Times

Must be enclosed between sharps.

```
#2008/01/31# // Note it's US format!
```

The are evaluated as **DateTime**. Expression always use invariant culture (en-us)

Booleans

Booleans can be either **true** or **false**.

```
true
```

Strings

Any character between single quotes "'" are evaluated as **String**.

```
'hello'
```

You can escape special characters using ****, **\'**, **\n**, **\r**, **\t**.

To enter special characters (e.g. a degree symbol) you can either hold the Alt-Key and enter the asciicode (https://theasciicode.com.ar/extended-ascii-code/degree-symbol-ascii-code-248.html) via the numpad (degree = Alt 248) or add the unicode representation \u00B0 (unicode for degree symbol)

Function

A function is made of a name followed by braces `()` , containing optionally any value as arguments.

```
Abs(1), doSomething(1, 'dummy')
```

Please read the functions page for details.

Simulation Data

You can reference any data available by enclosing the datareference in `[]`

```
2 + [x]
```

Please read the Referencing-Simulation-Data page for details

Referencing Simulation Data

To reference any simulation data in an expression, you can add the reference of the data to an expression by enclosing it in []

```
Round(  [SIMCONNECT:AUTOPILOT ALTITUDE LOCK VAR]  / 1000 )
```

The result will be the current value of SIMCONNECT:AUTOPILOT ALTITUDE LOCK VAR (e.g. 10000 (feet)) divided by 1000 and then rounded to a full number = 10

Expressions: Operators

Expressions can be combined using operators. Each operator as a precedence priority. Here is the list of those expression's priority.

Logical

These operators can do some logical comparison between other expressions:

- or, ||
- and, &&

> The **and** operator has more priority than the **or**, thus in the example above, **false and true** is evaluated first.

Relational

- =, ==, !=, <, >
- <, <=, >, >=

Additive

- +, -

Multiplicative

- *, /, %

Bitwise

- & (bitwise and), | (bitwise or), ^(bitwise xor), << (left shift), >>(right shift)

  > 2 >> 3

  Unary

- !, not, -, ~ (bitwise not)

Primary

- (,)

$$2 * (3 +2)$$

Functions

Name	Description	Us
Name	Description	Usage
Abs	Returns the absolute value of a specified number.	Abs(-1)
Acos	Returns the angle whose cosine is the specified number.	Acos(1)
Asin	Returns the angle whose sine is the specified number.	Asin(0)
Atan	Returns the angle whose tangent is the specified number.	Atan(0)
Ceiling	Returns the smallest integer greater than or equal to the specified number.	Ceiling(1.5)
Cos	Returns the cosine of the specified angle.	Cos(0)
Exp	Returns e raised to the specified power.	Exp(0)
Floor	Returns the largest integer less than or equal to the specified number.	Floor(1.5)
IEEERemainder	Returns the remainder resulting from the division of a specified number by another specified number.	IEEERemainder(3, 2)
Log	Returns the logarithm of a specified number.	Log(1, 10)

Name	Description	Us
Log10	Returns the base 10 logarithm of a specified number.	Log10(1)
Max	Returns the larger of two specified numbers.	Max(1, 2)
Min	Returns the smaller of two numbers.	Min(1, 2)
Pow	Returns a specified number raised to the specified power.	Pow(3, 2)
Round	Rounds a value to the nearest integer or specified number of decimal places. The mid number behaviour can be changed by using EvaluateOption.RoundAwayFromZero during construction of the Expression object.	Round(3.222, 2)
Sign	Returns a value indicating the sign of a number.	Sign(-10)
Sin	Returns the sine of the specified angle.	Sin(0)
Sqrt	Returns the square root of a specified number.	Sqrt(4)
Tan	Returns the tangent of the specified angle.	Tan(0)
Truncate	Calculates the integral part of a number.	Truncate(1.7)
Timespan	Takes a number and interprets it as a timespan in milliseconds	Timespan(1000)

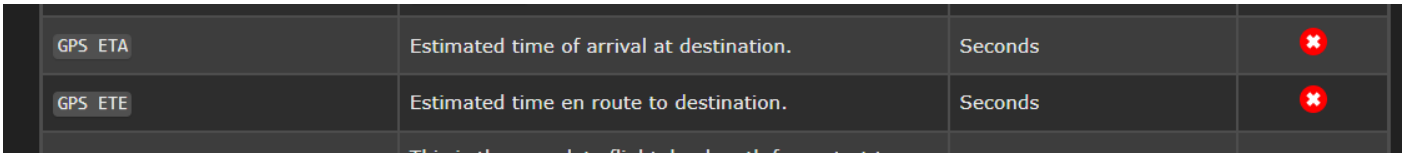
Name	Description	Us
ToNumber	Converts the argument into a number (e.g. if it's a string)	ToNumber('100') ToNumber([SIMCONNECT:
ToString	Converts the argument into a string	
ToChar	Convert argument to a char	ToChar(int)
GetValue	Get the Value of the Reference the argument is pointing to.	GetValue('SIMCONNECT:A
Format	Formats argument-1 using the formatting rule argument-2	Format(123.2356 , '0.00'
SubStr	Extracts a substring SubStr(value, startIndex, [length]) Startindex can be negative to start at end	
Len	Gets the Length of the argument as string	
Concat	Concat a list of values to a string	Concat(val1,val2,val2,val
ToBCO16	Convert argument to BCO16	
FromBCO16	Convert argument from a number to BCO16	
ToBCD / FromBCD	Convert to/from BCD (Com frequencies) format	
Select	Select a indexed value (index) from argumentlist (haystack)	Select([SIMCONNECT:SOM
Rescale	Rescale a value linear from inMin...inMax to outMin..outMax	Rescale(value,inMin,inMa

Name	Description	Us
Count	Counts the number of occurrences of a character in a string	Count(value,char)
Random	Create a Random number	Random() Random(max) Random(min,max)
In	Is a value in a list?	In(value,v1,v2,v3...)
Replace (case-sensitive)	Replace all occurrences of a string	Replace(input,old,new)
Replace (case insensitive)	Replace all occurrences of a string (case insensitive)	Replace(input,old,new,1)
IsMatch	Does string match a regular expression?	IsMatch(input,regex)
IndexOf (case-sensitive)	Find first occurrence of a string (0 based, -1 if not found)	IndexOf(stringWhat,string
IndexOf (case-insensitive)	Find first occurrence of a string (0 based, -1 if not found)	IndexOf(stringWhat,string
IsEmpty	Check if string is empty	IsEmpty('') IsEmpty([LOCAL:PAGEID]
IsNotEmpty	Check if string is not empty	IsNotEmpty('') IsNotEmpty([LOCAL:PAGE

Expressions: Examples

Updated when interesting Examples have been Given in Discord

Converting Time from Sim Vars:



GPS ETA	Estimated time of arrival at destination.	Seconds	
GPS ETE	Estimated time en route to destination.	Seconds	

Example involves using the GPS ETA or ETE and getting conversion to a clean Hours : Minutes : Seconds. The Expression can use a combination of two functions to convert the Variable.

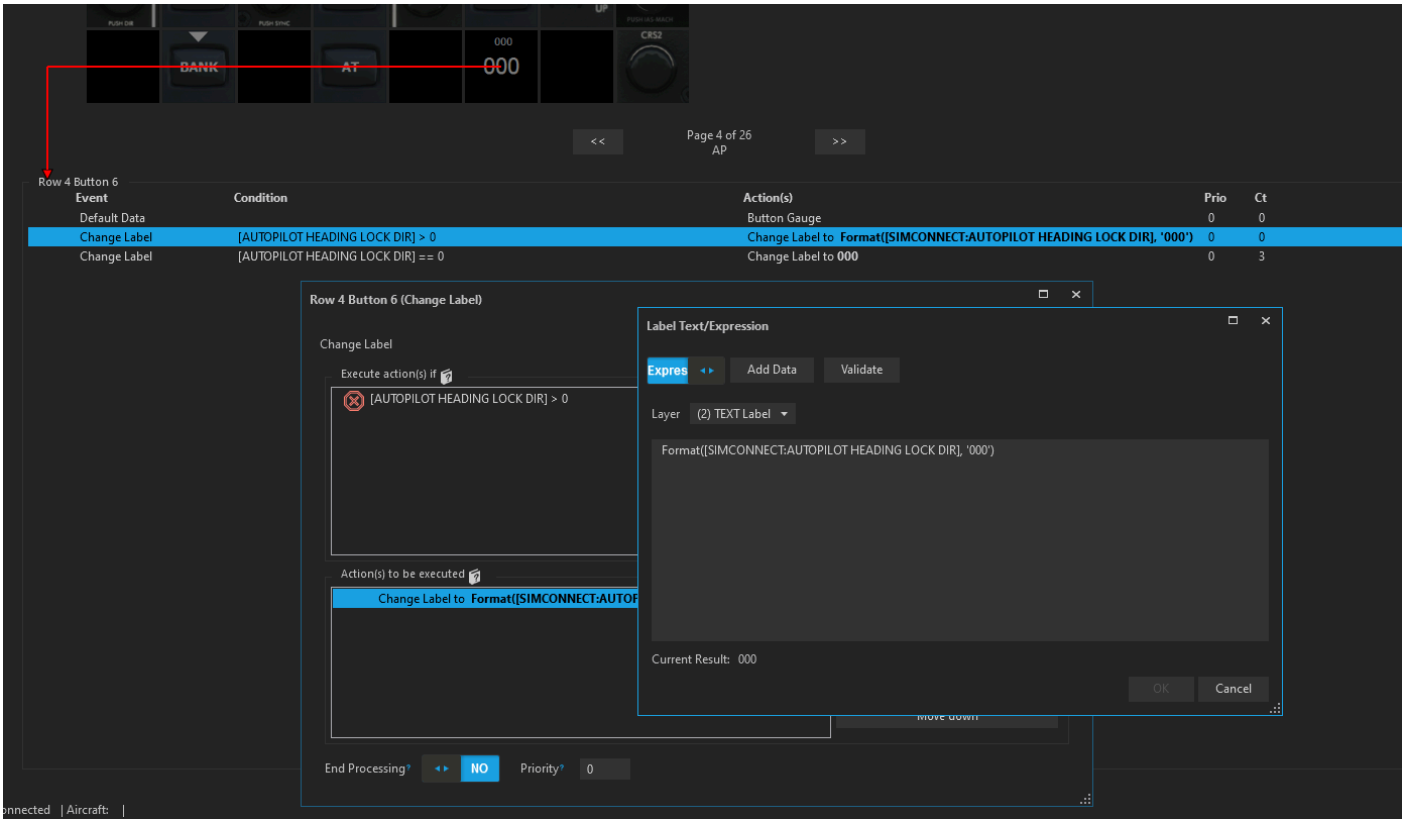
```
// Format and Timespan

Format(TimeSpan([SIMCONNECT:GPS ETE]*1000),'hh\\:mm\\:ss')
```

IF Statement Examples

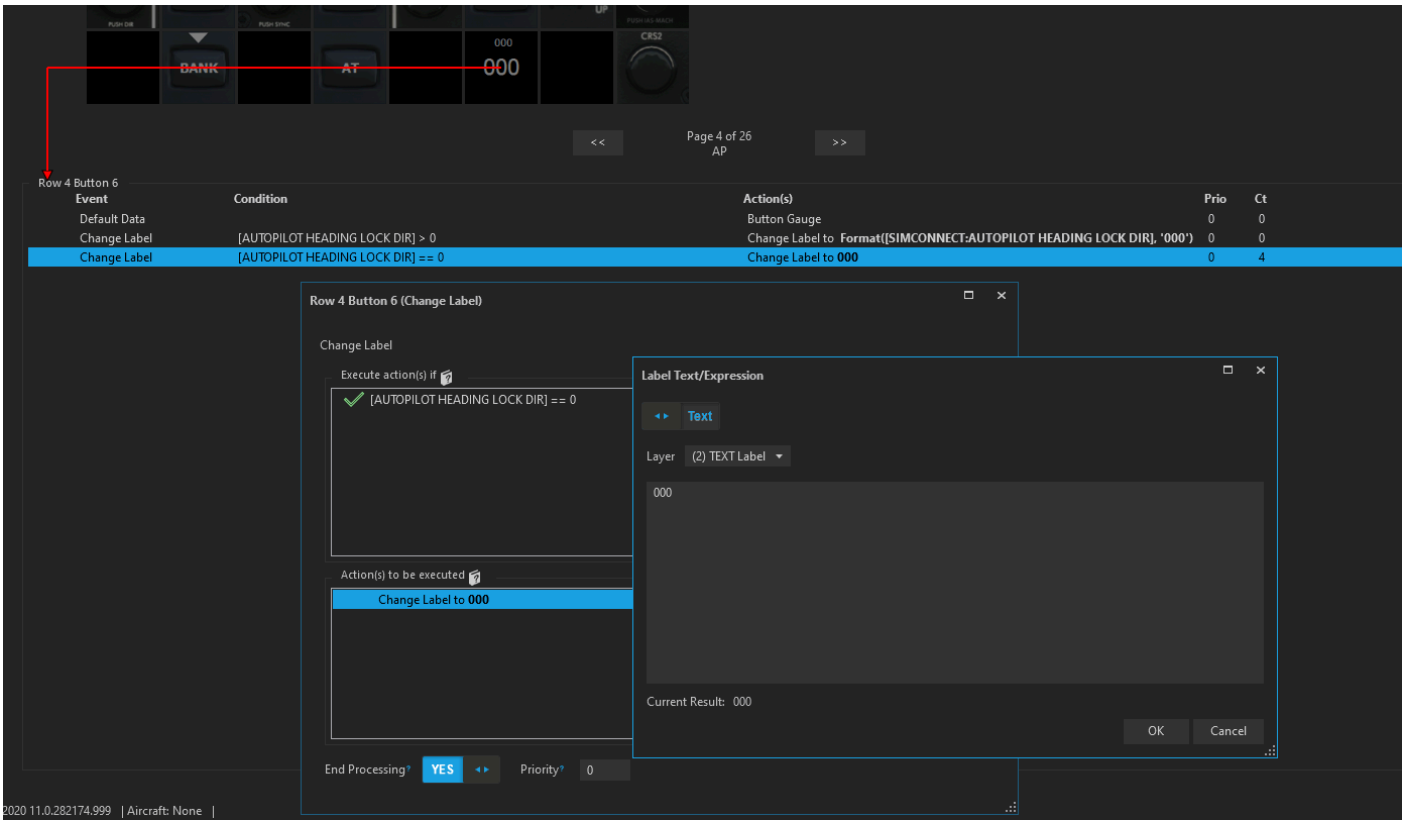
Simple IF statements can be added to Expressions. This can reduce the overall number of "scripts/conditional display events" required to be added..

Example Using "multiple" Change Label Actions on a Stream Deck display. Example is to have the display show the Heading Value of 1-360 however 0 to show as 360 to match how the Garmin PFDs display the Heading Variable Values of 0 as 360.

The First Event will display the value of the VAR exactly as it is (format to always show 3 digits) while the Second Event will set it to a static text of 360 when the Variable value is equal to 0.



Condition of AP Heading Variable "GREATER" than Zero.



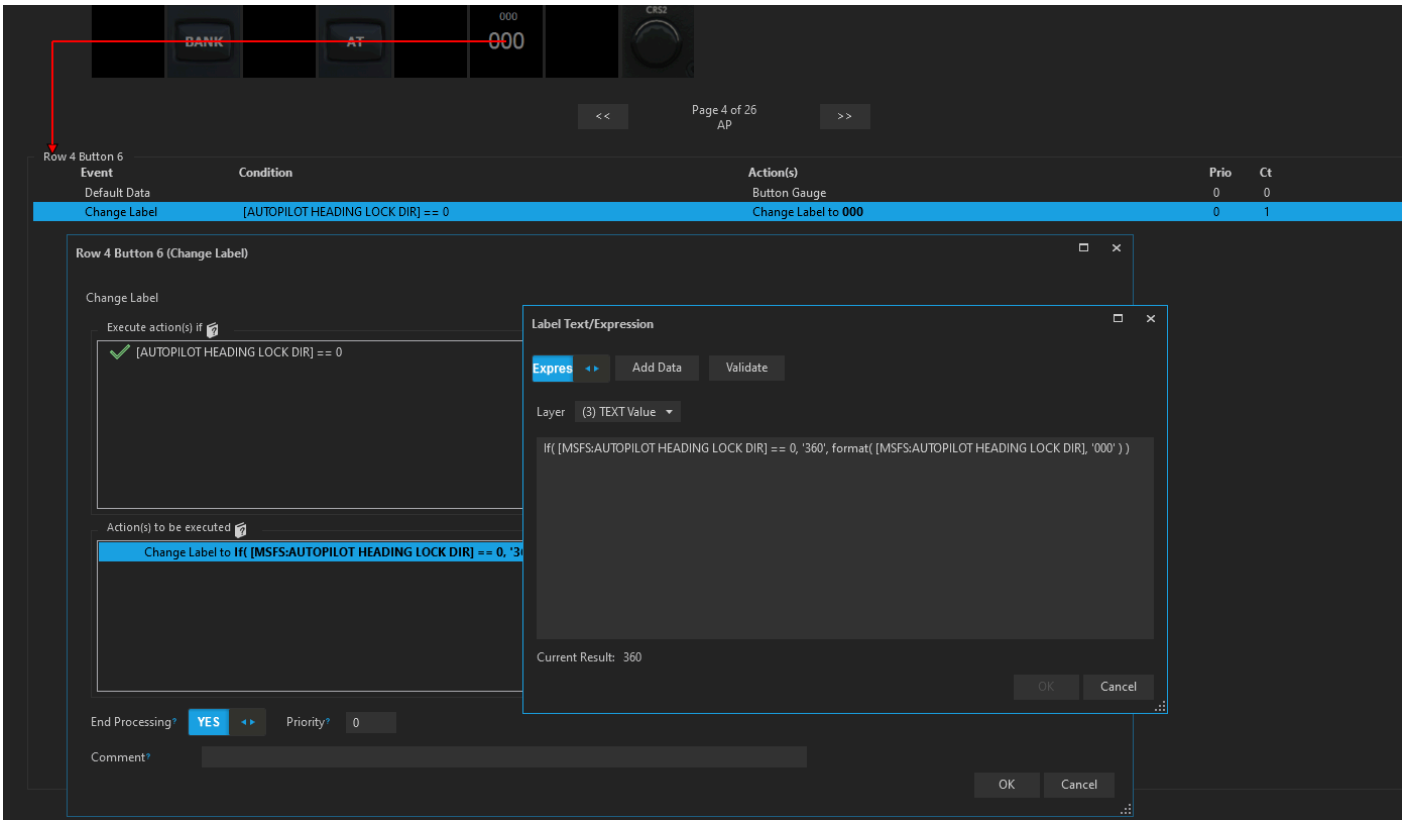
Condition of AP Heading Variable "EQUAL" to Zero.

Instead using an if statement in the first Change Label we can achieve this same result

```
// Expression If Example
```

```
If( [MSFS:AUTOPILOT HEADING LOCK DIR] == 0, '360', format(
[MSFS:AUTOPILOT HEADING LOCK DIR], '000' ) )
```

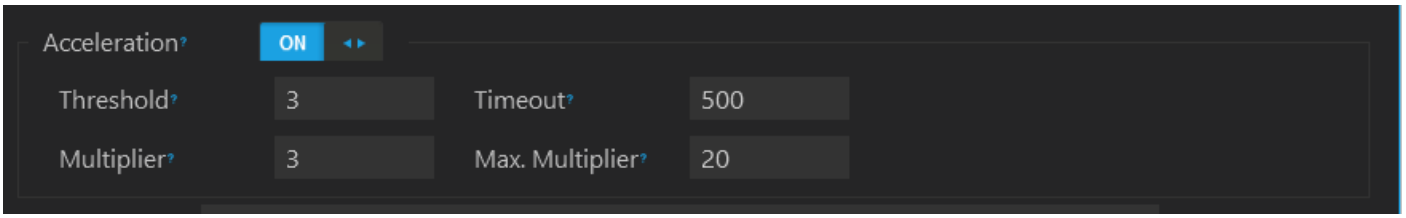
This will now insert into the Expression box and supply the same result but only requiring a single Action. The "condition" is now the AP Heading Var on "ANY CHANGE" of value to trigger the update of the display which will run the Expression to display 360 if the value is 0 else it will display the Variable .. but using Format of 3 digits to make sure the display will always have leading Zeros.



Tuner Acceleration

Accelerate tuner clockwise/counter-clockwise changes

Turning a tuner clockwise or counterclockwise can be modified so SPAD.neXt will accelerate changes in the sim (Turning the tuner faster)



Timeout: Time in milliseconds without any tuner-turn to reset the acceleration mutiplier

Threshold: Number of tuner-movements into same direction before acceleration will be started

Multiplier: The multiplier that will be applied to Increment/Decrement value for each stage when accelerated

Max Multiplier: the maximum multiplier applied

With default values this means:

if your turn the tuner clockwise like 5 clicks within 500ms with an increment value of "1" set following increments will sent to the sim:

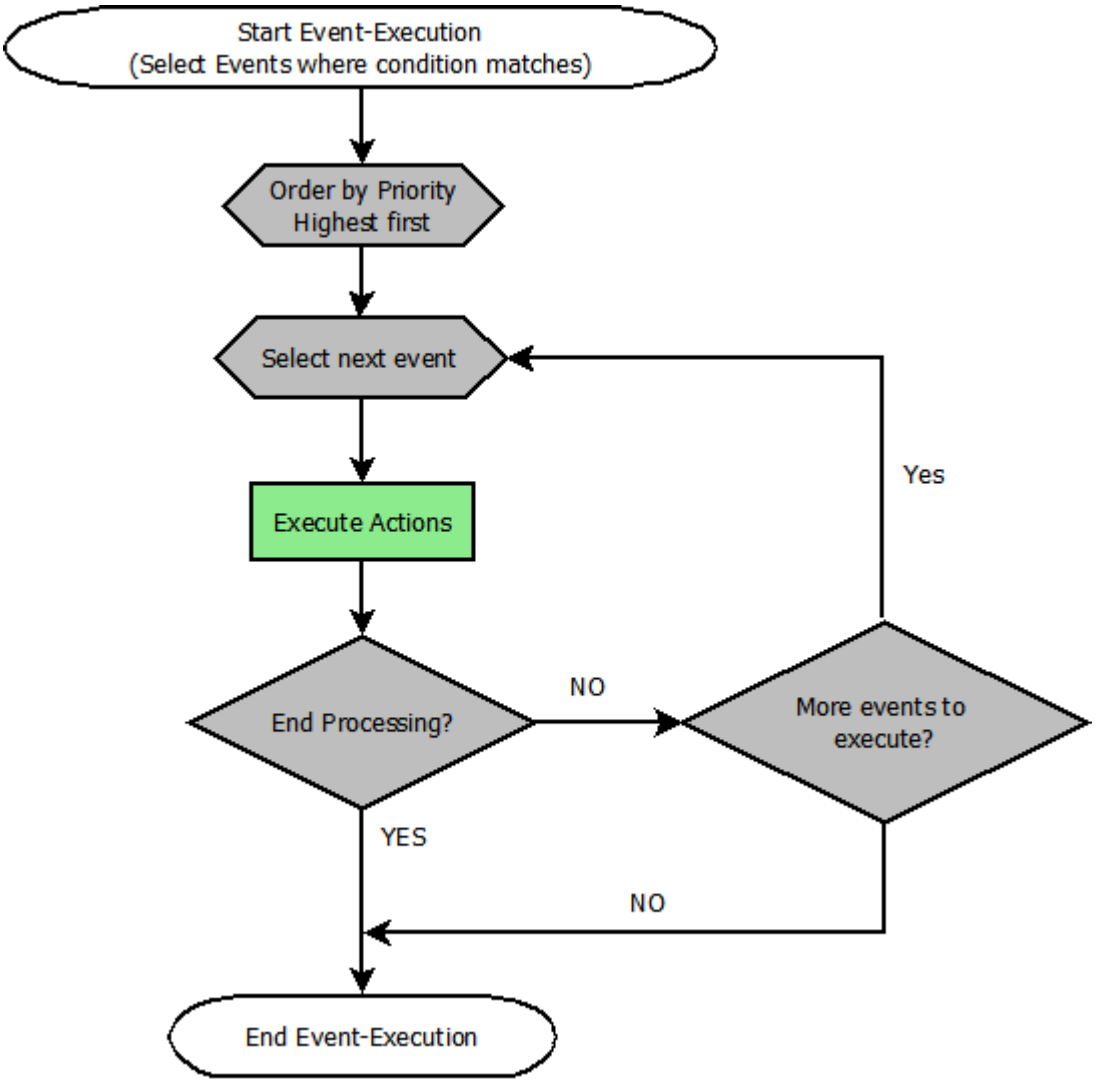
1. click: 1
2. click: 1
3. click: 1
4. click: 3 (Threshold met: (clicks - threshold) *multiplier = (4-3)* 3)
5. click: 6 (Threshold met: (clicks - threshold) *multiplier = (5-3)* 3)

Increasing timeout => Acceleration will be active for a longer time (you can turn the tuner slower) Decrease timeout => You will need to turn the tuner really fast, but be aware wacky saitek hardware eats a lot of the (counter-)clockwise clicks even that you can feel it.

Increase threshold: Acceleration will start later
Decrease threshold: Acceleration will start earlier, but might make it harder to achieve one/two-click changes

Event Execution Order

Order of Execution / EndProcessing / priority



End Processing comes into account only if you have more than one event that would be executed, but want to stop processing after a certain event was processed.

This is more or less only handy for the BIP.
Lets assume you have 2 events configured for a BIP plate:

```
a) Condition: VALUE equals 0
   Action: turn plate OFF
   Priority: 1
   End Processing: OFF

b) Condition: VALUE less than 10
   Action: turn plate RED
```

```
    Priority: 0
    End Processing: OFF
```

Now, if VALUE is *5* , all is fine, only b) will be executed, as only the condition `VALUE < 10` matches and the plate turns red

Now, if VALUE is *0* , you have a problem since first a) and then b) will be executed, as both conditions are true. This results in the plate turning red, where it should be off. If you set “End Processing” on a) to ON , b) will no longer be executed if a) already matched. So in this test case the plate will be turned off, as expected.

Virtual Devices

SPAD.neXt offers some virtual devices, which can be opened in a web browser. Those devices behave exactly like the physical devices, just that you have to "press" buttons using the mouse or touchpad.

> Virtual Devices is an extended feature and separately licensed. Either the complete edition of SPAD.neXt or the vFIP Addon is needed.

vFIP - virtual FIP

The vFIP is a virtual Flight Instrument Panel, which behaves exactly as the hardware device. You can detach it from SPAD.neXt and move it freely around.

vDeck - virtual Streamdeck

The vDeck is a virtual image-button device, similar to an Elgato Stream Deck or a Loupedeck.

vFIP

vFIP - virtual FIP

(Work in progress. Check back later)

vDeck

vDeck - virtual Streamdeck

> Note: SPAD.neXt uses the Windows built-in web-access methods to expose the vdeck to your network. To be able to access a virtual device from another computer in your network, you need to open TCP port 28001 in your firewall for your local network. (Not just SPAD.neXt.exe! The port must be open!)

(Work in Progress)

Add a vDeck in Settings->Devices->Virtual Devices

- It behaves exactly like a streamdeck
- Click on the detach button to open it as a web device
- For remote access open "http://`ipofSPAD`:28001/VDCK0001.html" in a browser
- Note: `ipofSPAD` is the IP address of the computer running SPAD.neXt. It is not `127.0.0.1`, the localhost address, which is only accessible to applications running on that computer.

How to Find PC IP - Command Prompt

1. Press **Windows key + R**
2. Type `cmd` and press **Enter**
3. In the black window, type `ipconfig` and press **Enter**
4. Look for:
    - **IPv4 Address** → This is your computer's local IP address (Example: `192.168.1.15`)

Customization

(Current vDeck version: 6)

- You can customize the web interface by editing `Documents/SPAD.neXt/html/VDCK0001.html`.

- If you customize the HTML or global CSS (`vdck.css`) and want to prevent SPAD.neXt from overwriting your changes, create a file named `VDCK0001.html.disableUpdates` or `vdck.css.disableUpdates`. Be aware that this may break functionality in future updates.
- Each vDeck has its own additional `VDECKXXXX.css` file for customization. See the corresponding file for details.
- If the HTML contains an element with the ID `spad-pagename`, its content will be set to the current page name.
- If the HTML contains an element with the ID `spad-profilename`, its content will be set to the current profile name.
- The page title is automatically set to `Profilename / Pagename`.

Double events on older touch devices

Some older touch devices send click and touch events. If you experience events being executed twice you can disable either mouse or touch events in the settings of the vDeck.

MSFS 2024

MSFS2024 PMDG Data Access

How to enable PMDG data access in MSFS 2024

> These steps are for the PMDG 737. For other PMDG aircraft, follow the same steps using the aircraft-specific INI file.

To access MSFS2024 PMDG Data from SPAD.neXt the following steps have to be completed:

- [ ] In MSFS2024 the PMDG 737 must have been loaded beyond "ready to fly" at least once and exited again.
- [ ] Exit MSFS2024
- [ ] in Windows File-Explorer open
  - [ ] Store/Gamepass Version %LOCALAPPDATA%\Packages\Microsoft.Limitless_8wekyb3d8bbwe\LocalState\WASM\MSFS2024`\pmdg-aircraft-737\work`
  - [ ] Steam Version %APPDATA%\Microsoft Flight Simulator 2024\WASM\MSFS2024\pmdg-aircraft-738\work\
- [ ] Add to end of 737_Options.ini

```
[SDK]
EnableDataBroadcast=1
EnableCDUBroadcast.0=1
EnableCDUBroadcast.1=1
```

> The EnableCDUBroadcast entries are only needed if you have a CDU Device or want to access CDU Data within SPAD.neXt

> Add `EnableCDUBroadcast.2=1` additionally to get access to the center cdu (e.g. 777)

> Make sure there is a newline/empty line after the last entry in the ini file and after each block, else it will get deleted again by the PMDG

- [ ] start MSFS2024 and load 737, start flight
- [ ] exit MSFS2024
- [ ] start MSFS2024 and load 737, start flight
- [ ] start SPAD.neXt
- [ ] Make sure in SPAD.neXt

Settings->Status
PMDGNG3 Connection shows as connected.
If it's not connected check the ini file, also make sure that you restarted MSFS at went beyond the "ready to fly" screen after making any changes to the pmdg ini files.

- [ ] Always 3 green!

All SDK Data and events are referenced in SPAD.neXt in the "PMDG 737 NGXu" (PMDGNG3) category and is **read-only**. To make any changes use the PMDG Events.

> Please note PMDG Simulation Events work exactly like the cockpit. Use the "PMDG Special Actions" to do with the event what you would do in cockpit with the mouse!
> In MSFS 2024, PMDG made some changes, and some functions can now only be triggered via LVARs.

MSFS 2020

MSFS2020 PMDG Data Access

How to enable PMDG data access in MSFS 2020

> These steps are for the PMDG 737. For other PMDG aircraft, follow the same steps using the aircraft-specific INI file.

To access MSFS PMDG Data from SPAD.neXt the following steps have to be completed:

- [ ] In MSFS the PMDG 737 must have been loaded beyond "ready to fly" at least once and exited again.

- [ ] Exit MSFS

- [ ] in Windows File-Explorer open

* Store/Gamepass Version
`%LOCALAPPDATA%\Packages\Microsoft.FlightSimulator_8wekyb3d8bbwe\LocalState\packages\pmdg-aircraft-737\work`
* Steam Version
`%APPDATA%\Microsoft Flight Simulator\Packages\pmdg-aircraft-737\work`

- [ ] Add to end of 737_Options.ini

```
[SDK]
EnableDataBroadcast=1
EnableCDUBroadcast.0=1
EnableCDUBroadcast.1=1
```

> The EnableCDUBroadcast entries are only needed if you have a CDU Device or want to access CDU Data within SPAD.neXt
>
> Add `EnableCDUBroadcast.2=1` additionally to get access to the center cdu (e.g. 777)

> Make sure there is a newline/empty line after the last entry in the ini file and after each block, else it will get deleted again by the pmdg

- [ ] start MSFS and load 737, start flight
- [ ] exit MSFS
- [ ] start MSFS and load 737, start flight
- [ ] start SPAD.neXt
- [ ] Make sure in SPAD.neXt

Settings->Status
PMDG Connection shows as connected.
If it's not connected check the ini file, also make sure that you restarted MSFS at went beyond the "ready to fly" screen after making any changes to the pmdg ini files.

- [ ] Always 3 green!

All SDK Data and events are referenced in SPAD.neXt in the "PMDG 737 NGXu" (PMDGNG3) category and is **read-only**. To make any changes use the PMDG Events.

> Please note PMDG Simulation Events work exactly like the cockpit. Use the "PMDG Special Actions" to do with the event what you would do in cockpit with the mouse!

> Please note PMDG Simulation Events work exactly like the cockpit. Use the "PMDG Special Actions" to do with the event what you would do in cockpit with the mouse!

X-Plane

✈ Welcome to the X-Plane Hangar

Alright, Captain—strap in, because we’re about to taxi into the world of **X-Plane + SPAD.neXt integration**. If you’ve been flying SPAD.neXt in the Microsoft Flight Sim skies, welcome to the slightly weirder, wonderfully flexible (and occasionally stubborn) world of X-Plane.

This part of the manual is your pre-flight briefing for everything SPAD.neXt does once it’s cleared into X-Plane airspace. From finding the right **datarefs**, to working with **custom aircraft commands**, to making your cockpit hardware talk to your sim like they’ve been best friends since flight school—this is where the magic happens.

But don’t worry—we’re not going to throw you into the left seat without a checklist. We’ll guide you step-by-step, with just enough humor to keep things from feeling like you're stuck in a holding pattern over the tech manual.

What to Expect:

- Hooking SPAD.neXt into X-Plane like a seasoned avionics tech
- Making sense of datarefs (without needing a PhD in sim-ology)
- Creating button magic with custom commands
- Troubleshooting things when X-Plane throws a tantrum

So grab a coffee, fire up your sim, and let’s turn that cockpit of yours into a finely-tuned flying machine.

Pre-Requisites

a.k.a Pre-Flight Checklist

Before we start flipping switches and soaring through the skies, let’s make sure you’ve got everything you need ready to go. Just like in real aviation, skipping your pre-flight checks is a fast track to a crash landing (metaphorically speaking, of course).
Here’s what you’ll need on-board:

✈ X-Plane Installed & Ready - https://www.x-plane.com/kb/digital-download-install/
Make sure X-Plane is installed, updated, and functioning properly. If it’s not, we’ll be grounded before we even taxi.

 SPAD.neXt Installed (Latest Version) - https://docs.spadnext.com/getting-started/untitled
Download and install the latest version of SPAD.neXt. Trust us—old versions are like using a paper map in an FMS world.

SPAD.neXt X-Plane plugin installed into X-Plane - https://www.spadnext.com/discover/how-to/connect-to-x-plane.html

 DataRef Tool Installed in X-Plane - https://docs.spadnext.com/simulations/x-plane/dataref-tool
This one’s important. The DataRef Tool is like your black box for figuring out what’s going on under the hood in X-Plane. It helps you find and test the datarefs you’ll use to connect your hardware to in-sim functions. Without it, you're basically flying blind when it comes to custom configurations.

 Supported Hardware - https://docs.spadnext.com/getting-started/supported-hardware
Whether you’re using Logitech/Saitek panels, custom switches, or a Franken-rig you built in your garage, make sure your gear is connected and recognized by Windows.

 Basic Understanding of SPAD.neXt Interface - https://docs.spadnext.com/guides-and-videos/new-user-series
You don’t need to be a systems engineer, but knowing your way around the SPAD.neXt interface will help a lot. Don’t worry—we’ll keep it smooth and turbulence-free. Using the in app video guides will keep you heading on the right course.

 Network (Optional for Remote Panels/Devices)

If you're running SPAD.neXt across multiple PCs or using remote panels, make sure your network setup is good to go. No one wants a cockpit that drops off the radar mid-flight.

Once you’ve checked all this off, you're cleared to push back and start diving into the fun stuff!

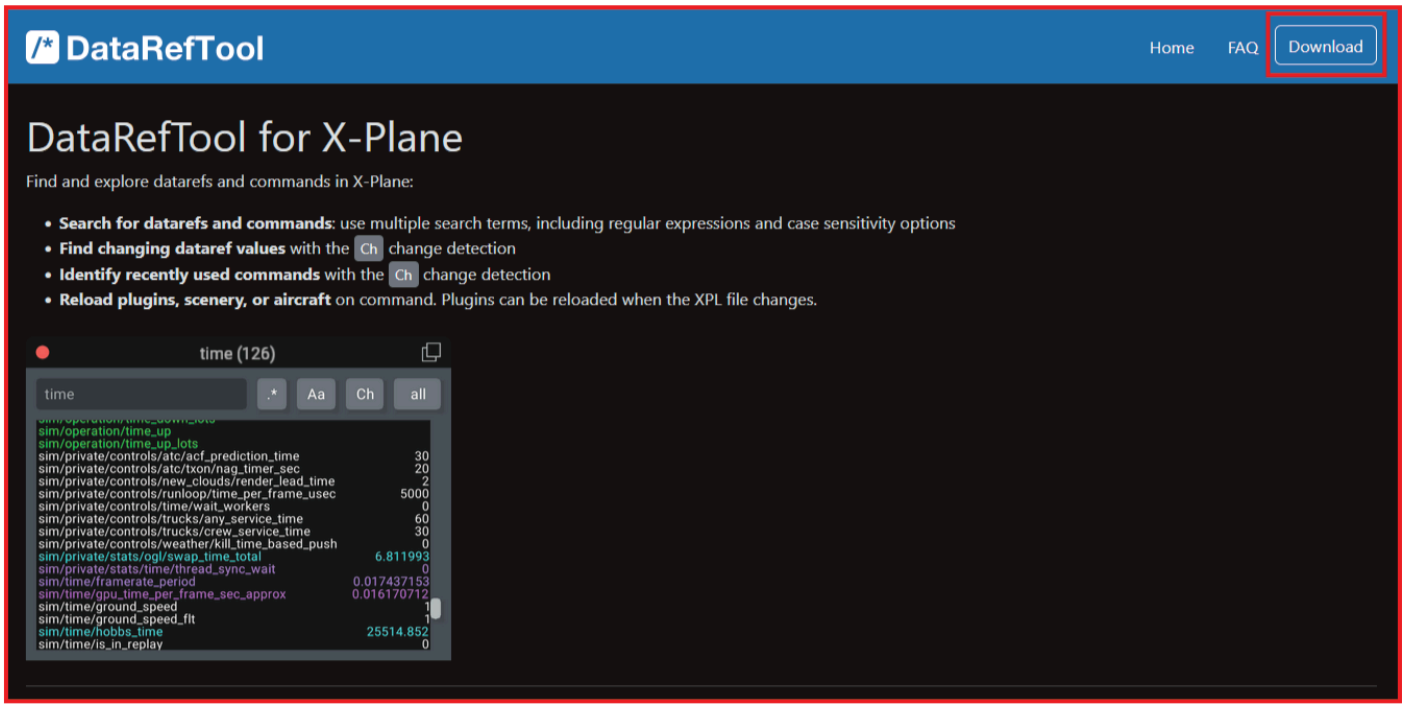
Dataref Tool

Installing The Datareftool

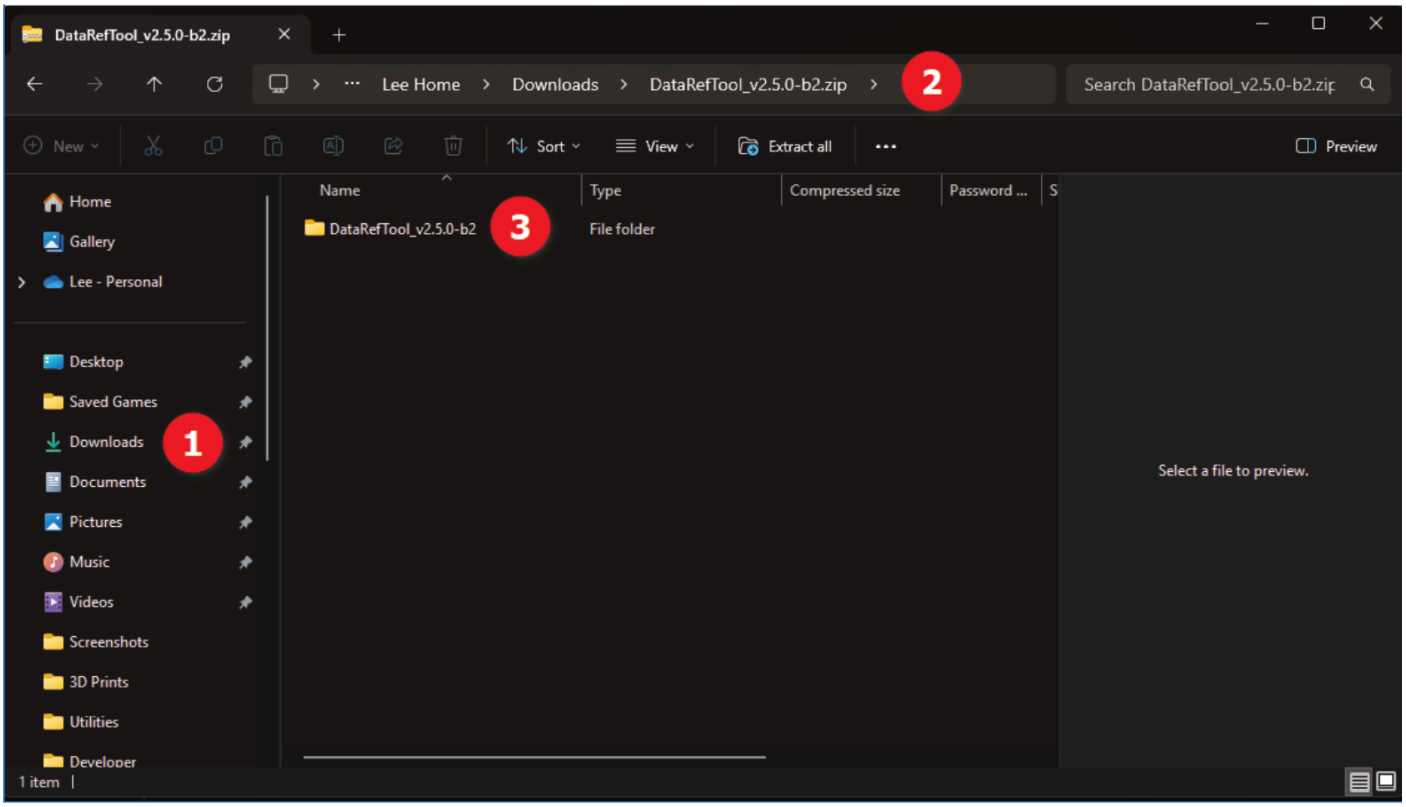
Now that we've taxied past the basics, it's time to install the DataRef Tool—your new best friend when it comes to connecting SPAD.neXt with the deep inner workings of X-Plane. Think of it as your aircraft's diagnostic scanner, letting you peek behind the curtain and see exactly what's going on with all those mysterious datarefs.

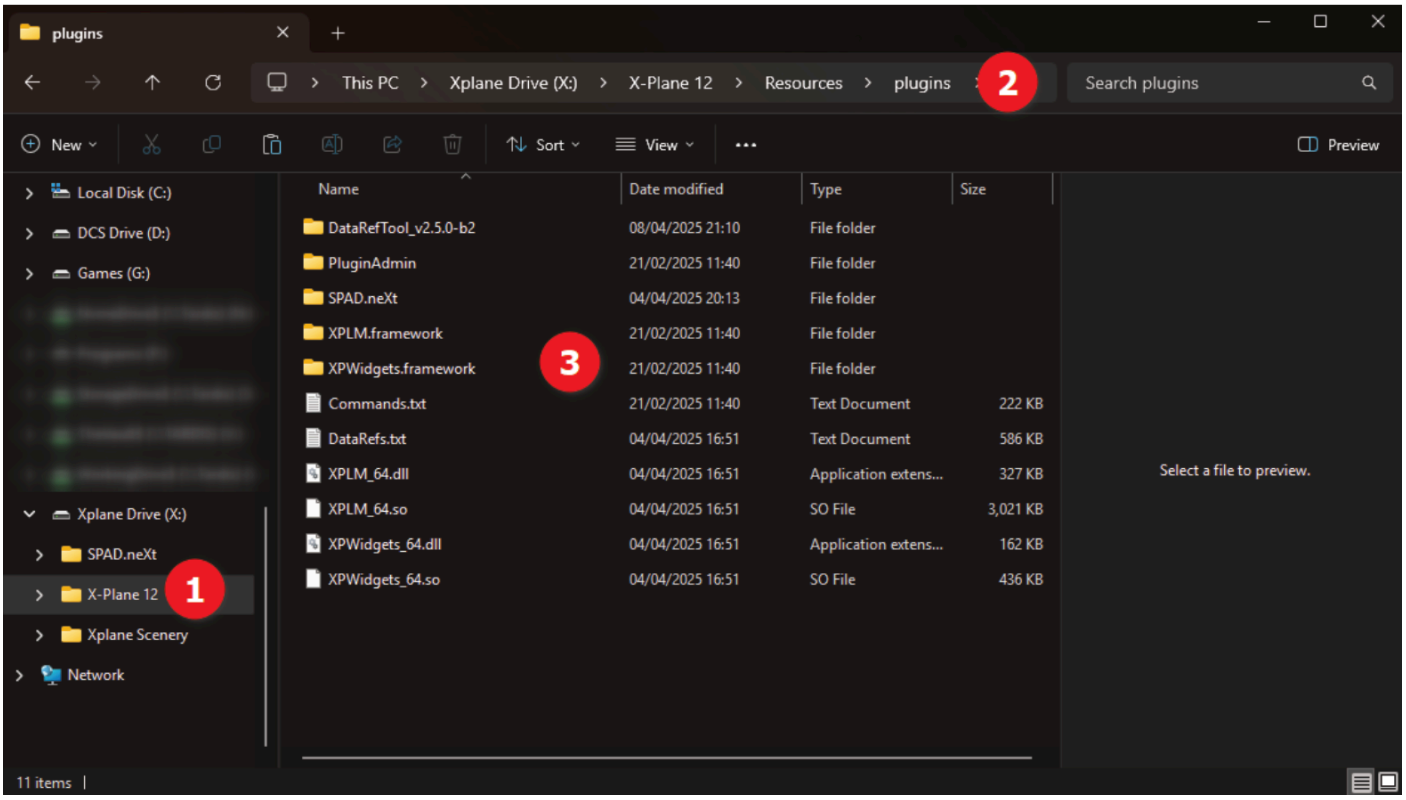
Step 1: Download the DataRef Tool
You can grab the latest version from a trusted X-Plane community site like the X-Plane.org forums. Just search for "DataRefTool" or "DataRef Editor" (note: not the same as DataRef Viewer, but they can work together). Make sure it's compatible with your version of X-Plane.

https://datareftool.com/



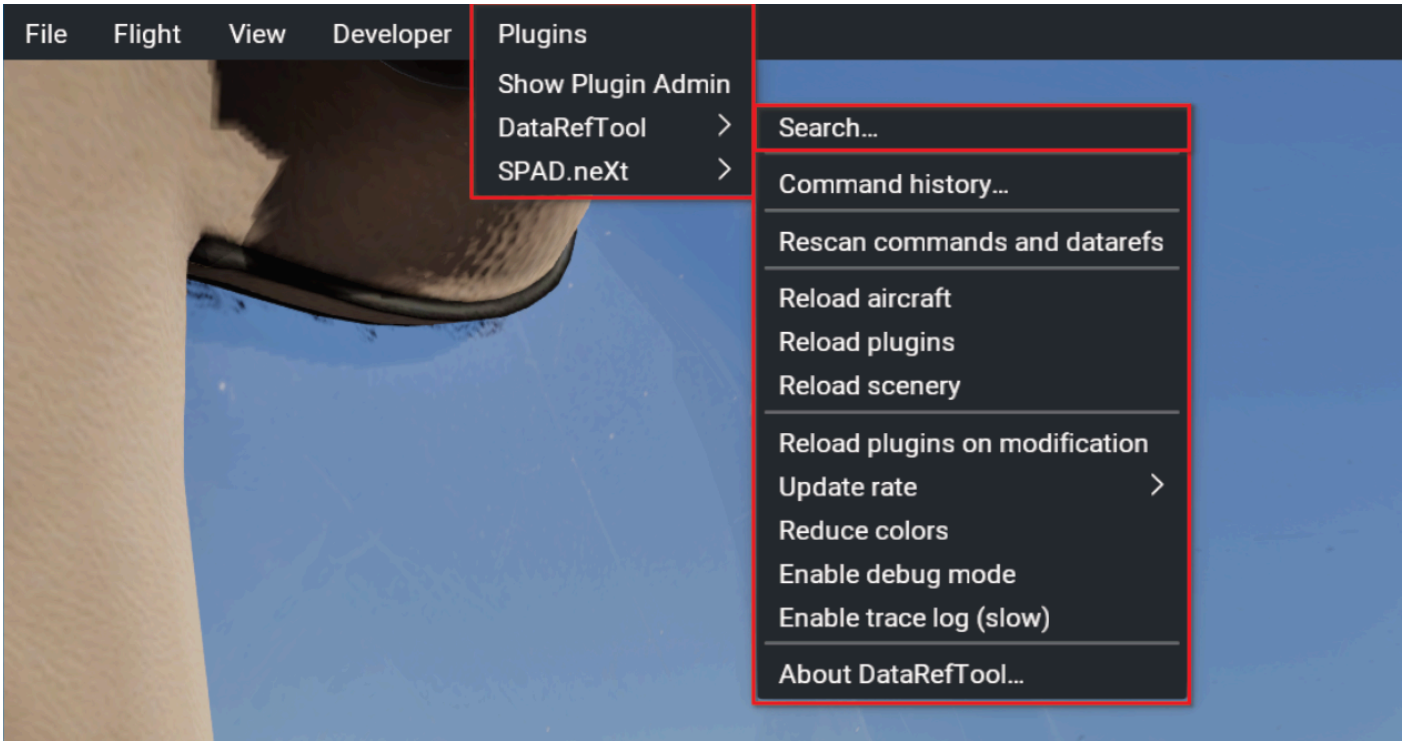
Step 2: Unzip and Install
Once downloaded, unzip the folder. Inside you'll find a plugin folder—usually named something like DataRefTool with the version number. Note the latest version is **NOT compatible with Xplane 11**
Drag and drop that folder into your X-Plane > Resources > plugins directory. That's it! No complicated installer, no bloat—just good old-fashioned drag-and-drop.





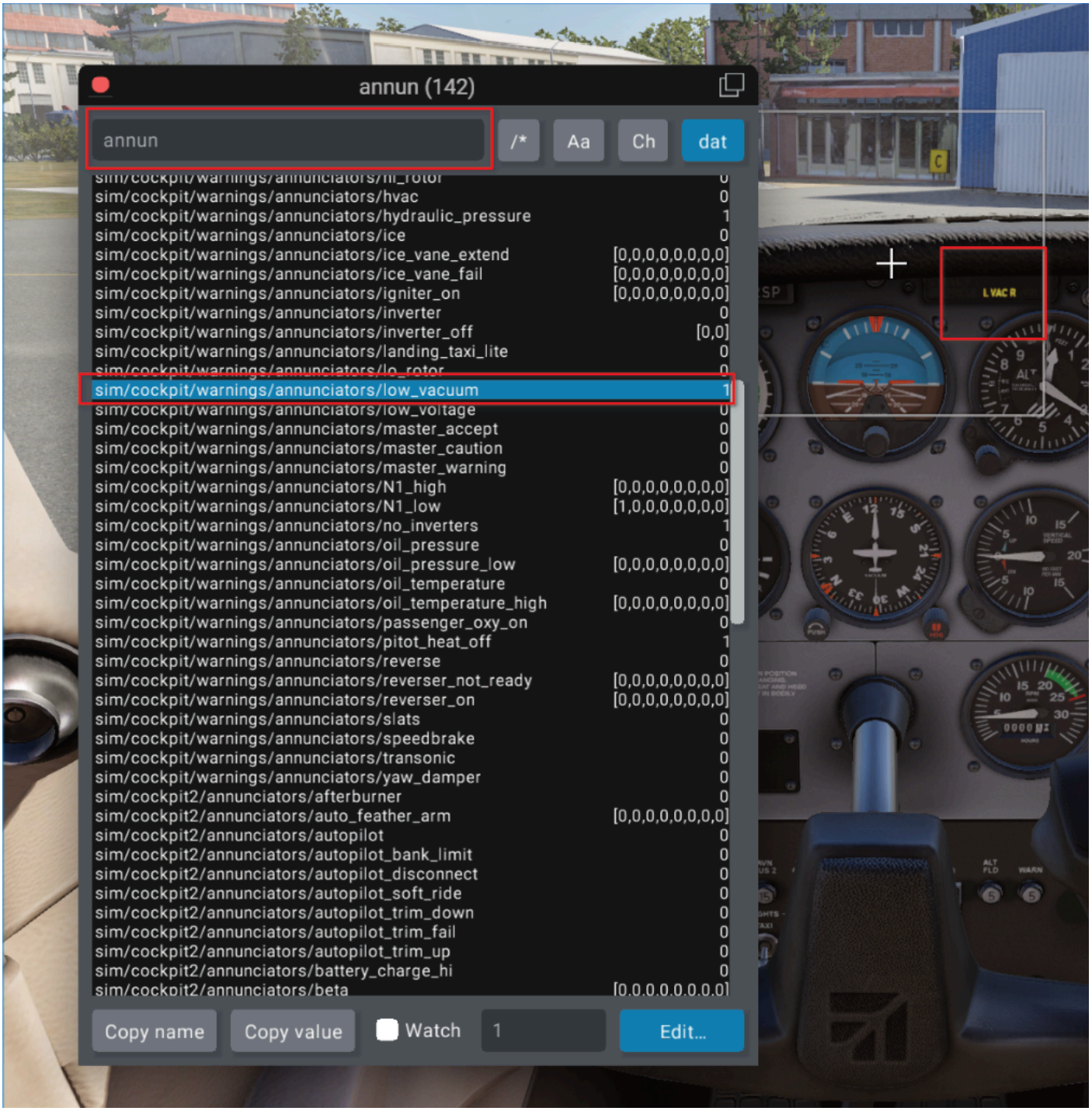
Step 3: Fire Up X-Plane

Launch X-Plane, load into any aircraft (even the default Cessna if you're just testing things out), and then check the Plugins menu at the top. You should see DataRef Tool listed there, ready for duty.



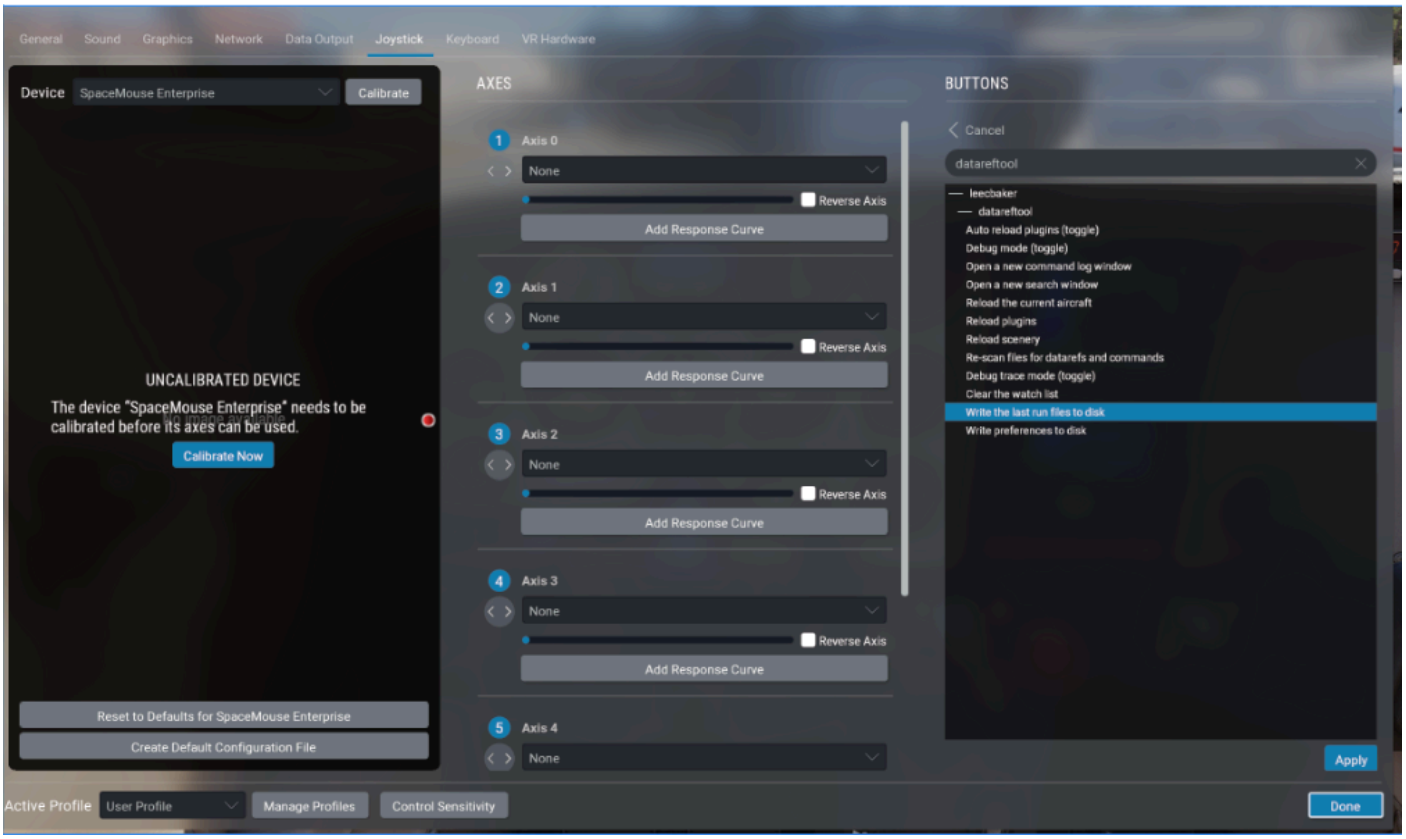
Step 4: Verify It's Working

Click on the plugin, and a search window should pop up. From here, you can start typing in keywords (like lights, battery, gear, etc.) to see live values and interact with them in real time. If you’re seeing live data and clickable checkboxes, congrats—you’re in business!

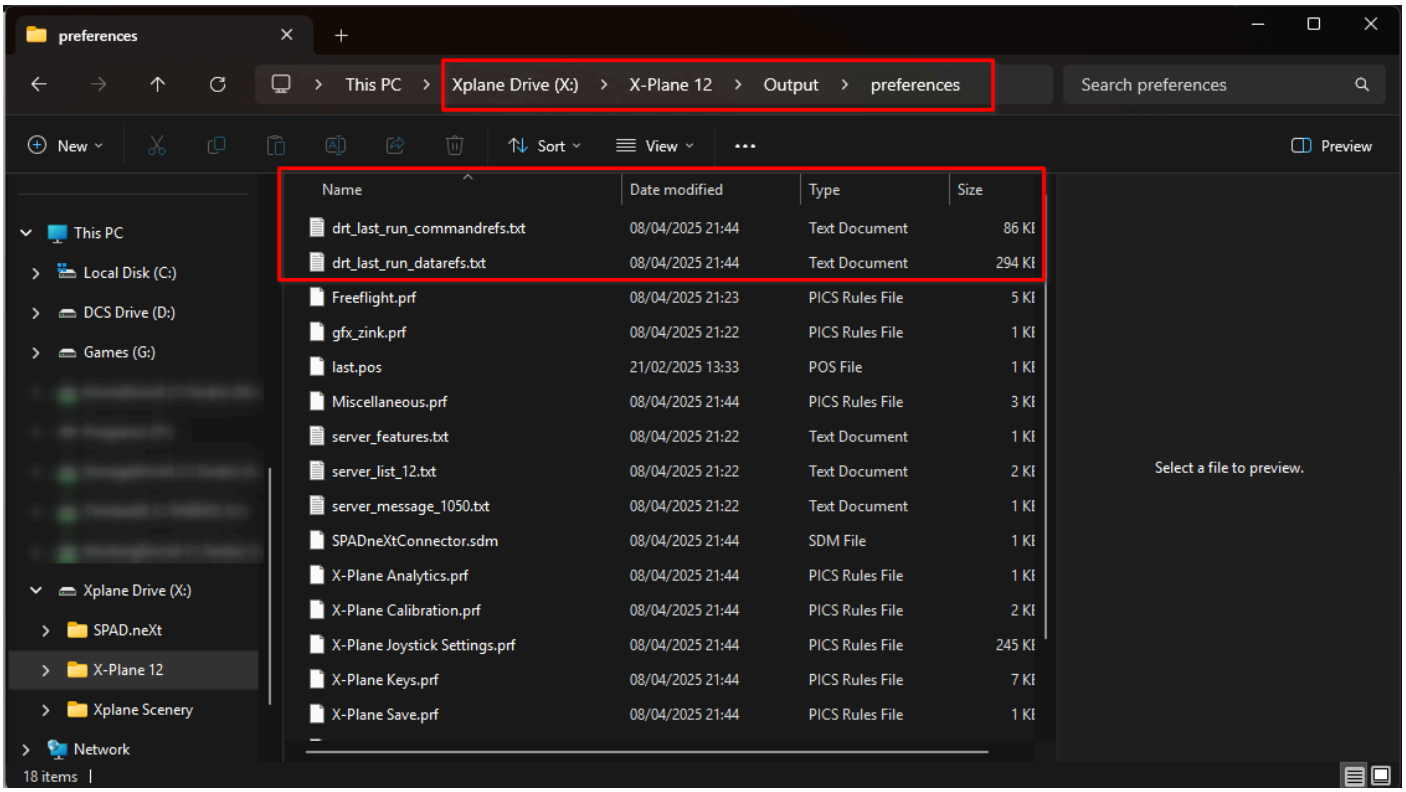


Bonus Tip:

Assign 1 button within X-plane to write the the drt_last_run_commandrefs.txt & drt_last_run_datarefs.txt files this will save you a lot of time. This will overwrite the existing files whilst in the simulator & Instead of rebooting the sim you can make your edits and reboot Spad.Next.



Pressing this button will then write 2 files to the following location.
\X-Plane 12\Output\preferences



Spad.neXt Gets its Type Rating

Datarefs & Commands

Good news, The latest version of **SPAD.neXt** no longer needs you to play file-renaming ground crew. It now automatically discovers **all X-Plane datarefs and commands** on its own.

That’s right. No more generating last-run files.
No more renaming text files like you’re hacking NORAD.
No more dragging mysterious `.txt` files into aircraft folders hoping you didn’t break something.

SPAD.neXt now scans the aircraft and sim environment directly and builds its own database like a well-trained avionics suite.

In aviation terms?
We’ve upgraded from paper charts to full glass cockpit.

Let’s be crystal clear:

 You no longer need:

- `datarefs.txt`
- `commands.txt`

 You now:

- Open SPAD.neXt
- Select your aircraft
- Search for any dataref or command directly inside SPAD.neXt
- Assign it
- Fly happily ever after

No pre-flight file rituals required.

Dataref vs. Command: Know Your Copilots

Okay, aviator—before we start wiring buttons to things like flaps, lights, ejector seats (hey, no judgment), we need to understand the two main languages X-Plane speaks: **Datarefs** and **Commands**.

Think of these like your cockpit crew. One tells you what's going on. The other *makes* things happen.

🧠 Datarefs: The Sim Whisperers

Datarefs are like the gauges and readouts in your cockpit. They **tell you the current state** of something in the sim. They're *informational*. Want to know if the landing gear is down? There's a dataref for that. Want to see your fuel level? Yep—dataref.

They're like saying:
🗣️ *"Hey sim, what's the current altitude of this bird?"*
And the sim replies:
🗣️ *"You're at 2,700 feet, boss."*

You **read** datarefs to get info. Some can be **written to** (we'll get to that), but they don't *do* anything on their own.

✍️ *Example:*
`sim/cockpit2/switches/landing_lights_on = 1`
`(This tells you the landing lights are on. Maybe. Sometimes. Depending on the aircraft. Yay, X-Plane!)`

🎮 Commands: The Doers

Commands are the sim's action-takers. They **make something happen**—like flipping a switch, toggling a light, or yelling "CLEAR PROP!" before engine start (okay, maybe not that last one… yet).

They're like saying:
🗣️ *"Hey sim, turn on the landing lights."*

And the sim *does it*—no questions asked.

You don’t get a value back, you just send the command and hope the sim complies. (Most of the time, it does. Sometimes it acts like a teenager.)

✈``` *Example:*
`sim/lights/landing_lights_toggle`
`(This tells the sim: “Toggle the landing lights!”)`

So... Which One Do I Use?

- **Use a command** when you want to make something happen. (Buttons, switches, momentary events)
- **Use a dataref** when you want to check the state of something. (LEDs, displays, conditions)
- Sometimes, use **both** if you want to change something *and* confirm it worked.

Bonus Tip: Not All Aircraft Play Nice

Some aircraft (especially custom ones) love using their own secret datarefs and commands. So if your button doesn’t seem to be doing anything, you may need to do a little detective work using **DataRefTool** to figure out what’s *actually* going on under the hood.

That's it! You've just passed the "Datarefs vs. Commands" ground school exam. No written test required, but feel free to celebrate with a smug throttle push.

Basic Flight Controls – Axis Setup

(Where We Tame the Yoke, Wrangle the Rudder, and Throttle Like a Pro)

Now it’s time to tackle the real beasts of the flight deck—**your flight control axes.** Because if your yoke is limp, your rudder’s confused, and your throttle’s spiky… you're flying a flying brick with cool lights.

✈ What Is an Axis?

Think of axes (plural of axis, not something you swing at your PC) as the analog controls—smooth, continuous inputs like:

- Yoke pitch & roll (your classic “push and pull” and “left and right”)
- Rudder pedals (tail wagging)
- Throttle(s) (moar power!)
- Mixture, Prop pitch, toe brakes, spoilers, flaps, coffee temperature knobs (okay maybe not that last one… yet)

If it moves smoothly and *isn’t* a button, it’s probably an axis.

X-plane Default Axes + Datarefs + Value Ranges

Axis Role	**What It Controls**	**Default DataRef**	**Value Range**
`Pitch`	Elevator (yoke push/pull)	`sim/joystick/yoke_pitch_ratio`	`-1.0` (nose down) `1.0` (n up)
`Roll`	Ailerons (yoke	`sim/joystick/yoke_roll_ratio`	`-1.0` (to `1.0`

	left/right)		(right)
`Yaw`	Rudder (pedals or stick twist)	`sim/joystick/yoke_heading_ratio`	`-1.0` (to `1.0` (right)
`Throttle 1`	Engine power (engine 1)	`sim/flightmodel/engine/ENGN_thro[0]`	`0.0` (ic to `1.0` (full power)
`Throttle 2`	Throttle (engine 2)	`sim/flightmodel/engine/ENGN_thro[1]`	`0.0` to `1.0`
`Throttle 3`, `4`	More throttles	`ENGN_thro[2]`, `[3]`	`0.0` to `1.0`
`Prop 1`	Prop RPM (engine 1)	`sim/flightmodel/engine/ENGN_prop[0]`	`0.0` (lo RPM) t `1.0` (h RPM)
`Prop 2`	Prop RPM (engine 2)	`ENGN_prop[1]`	`0.0` to `1.0`
`Mixture 1`	Fuel/air mix (engine 1)	`sim/flightmodel/engine/ENGN_mixt[0]`	`0.0` (cutoff `1.0` (fu rich)
`Mixture 2`	Mixture (engine 2)	`ENGN_mixt[1]`	`0.0` to `1.0`
`Speedbrake`	Spoilers / airbrakes	`sim/flightmodel/controls/speedbrake_ratio`	`0.0` (retrac

			to `1.0` (full ou
`Flaps`	Flap lever	`sim/flightmodel/controls/flap_ratio`	`0.0` (u to `1.0` (full fla
`Reverse Thrust`	Reversers active (per engine)	`sim/flightmodel/engine/ENGN_thro_use_rev`	`0` (off) `1` (on)
`Brakes`	Main brakes (left/right combined or split)	`sim/flightmodel/controls/parkbrake` or `l_brake_add` / `r_brake_add`	`0.0` (releas to `1.0` (full brake)
`Nosewheel Steering`	Nose gear steering	`sim/flightmodel/controls/nwheel_steer`	`-1.0` (to `1.0` (right)
`Landing Gear`	Gear handle (if used as axis)	`sim/flightmodel/controls/gear_handle_down`	`0` (up) `1` (dow

🎮 Step 1: Axis Configuration

(Or: Teaching Your Controls Which Way Is Up)

Alright, time to get your hands on the flight controls—literally.

Head over to the **left panel in SPAD.neXt** and select the device you want to configure. This could be your yoke, throttle quadrant, rudder pedals, or that mysterious lever you *swear* you'll use one day.

Now click into the **Axis section**.

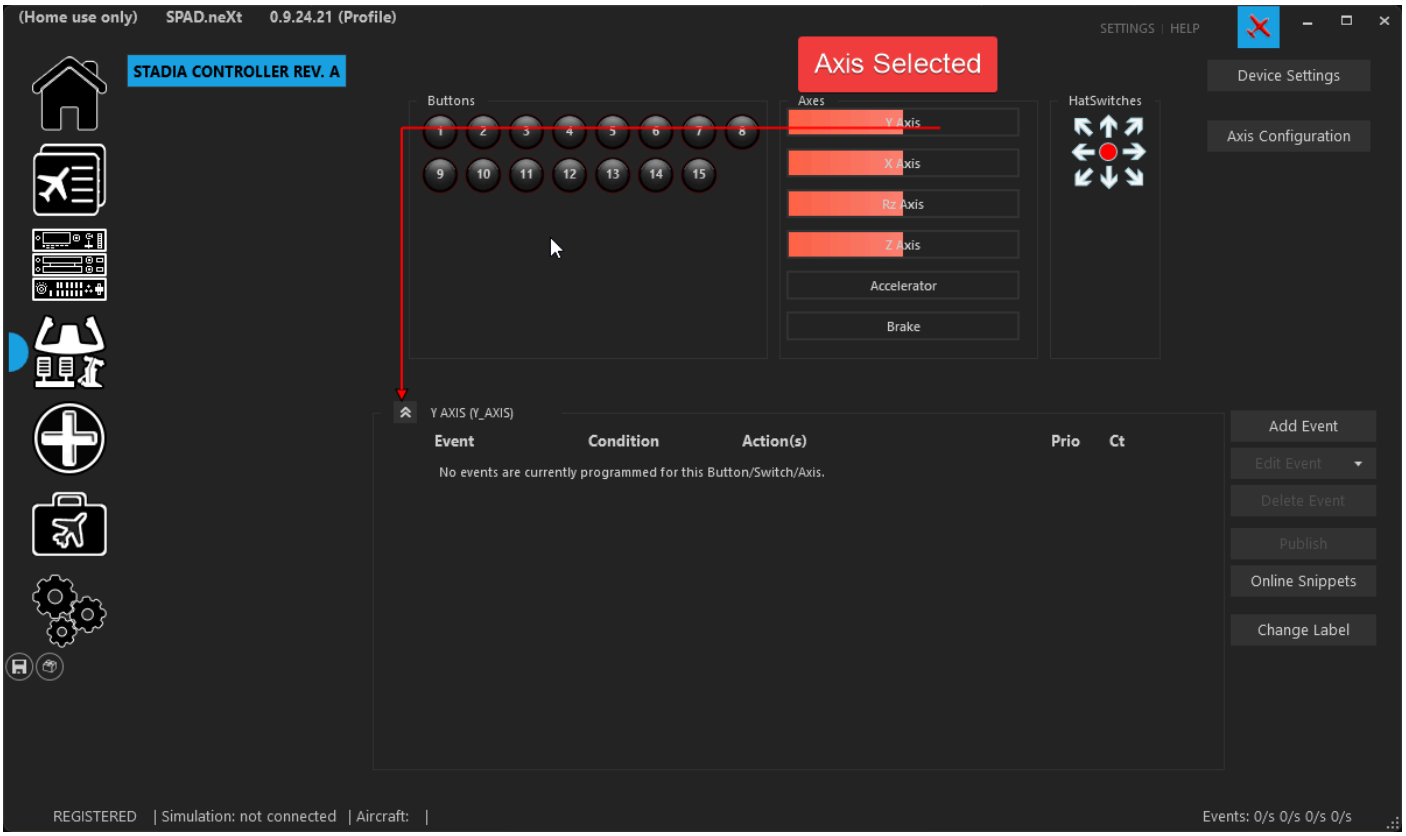
If it moves, wiggles, slides, or generally refuses to stay still—it'll show up here. Think of this as your aircraft's roll call.

🎯 Identify Your Axis

Now give your control a wiggle—nice and easy, no need to wrestle it like a crosswind landing.

SPAD.neXt will highlight the axis that's moving. Once you've found the right one, **click it to select it**.

> 🎯 *"Ah yes... Axis 2. The chosen one."*

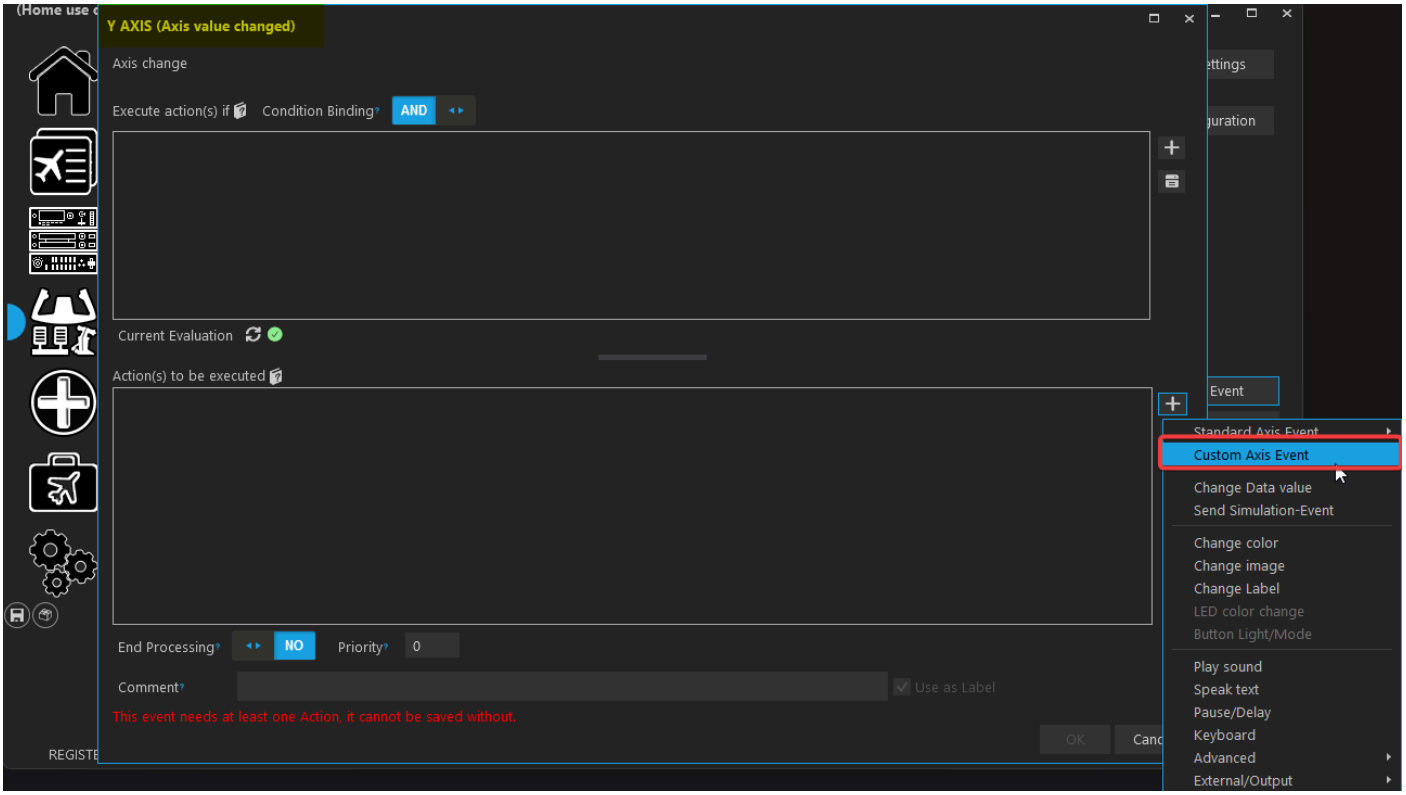


- 🎯 Create the Event
  Click "Add Event"
  Select "Axis Value Changed"

  This tells SPAD.neXt:
  🎯 "Oi, every time this thing moves—pay attention!"

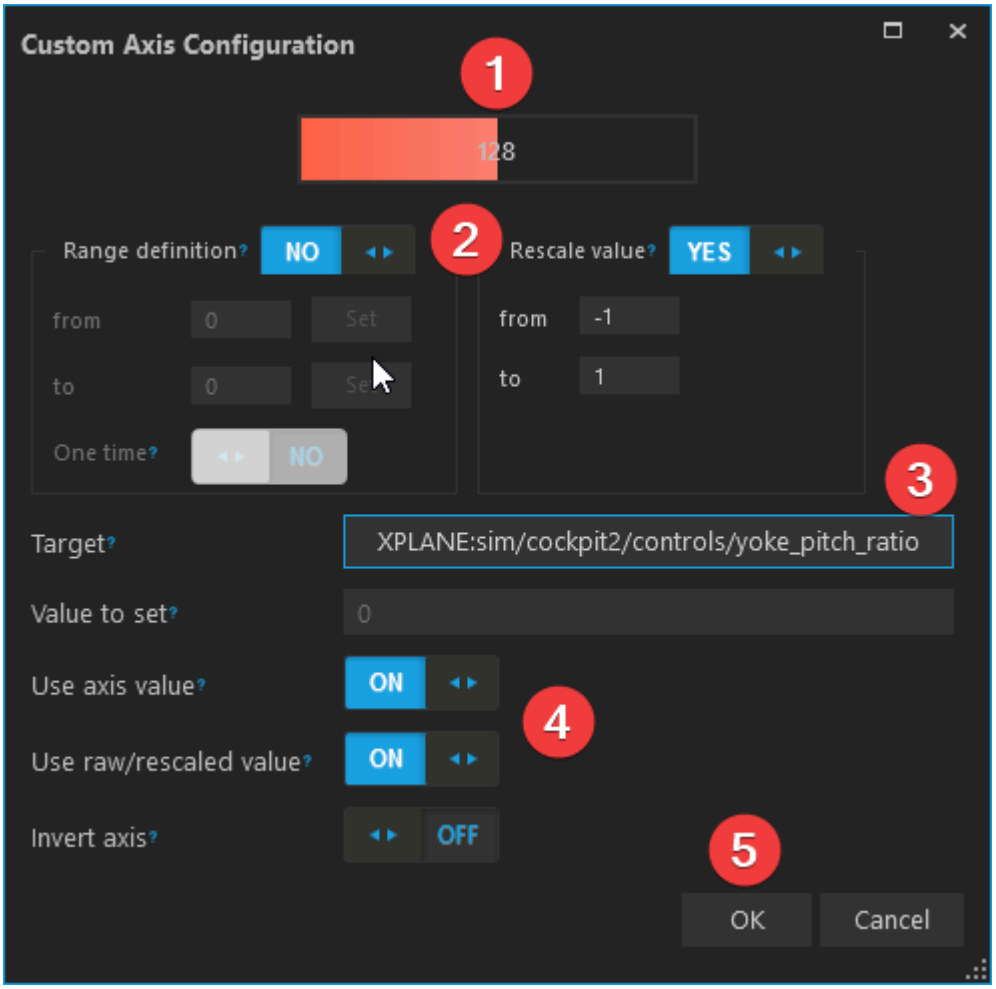
  Next:

  Click the 🎯 (plus) icon
  Select "Custom Axis Event"



✈ Let’s Configure Pitch (Elevator)

For this example, we’re wiring up the **Pitch axis**—the one that decides whether you climb majestically… or inspect the runway at high speed.
At first glance, it might look like the flight computer from a space shuttle. Don’t panic—we’ll break it down.
 What All This Means (Without the Headache)



1. **Axis Current RAW Value**
   This is your raw input—the unfiltered, straight-from-the-hardware signal.
   Think of it as your control saying:
   “I’m moving! I just don’t know what it means yet!”

2. **Range Definition** ,Ignore this for now. Seriously. Taxi right past it.

   It’s for more advanced setups—like when you start building logic that would make real avionics engineers.
   Rescale Value → YES
   This tells SPAD.neXt to translate your hardware input into something X-Plane actually understands.

   Because your joystick might speak “random USB nonsense,”
   but X-Plane speaks **nice, clean aviation numbers**.

   Understanding Axis Ranges

   Not all axes are created equal:

- ✈ **Pitch / Roll / Yaw (centered axes):**
  Range = `-1` to `+1`
  - `0` = centered
  - Negative = one direction
  - Positive = the other
- **Throttle / Prop / Mixture (non-centered axes):**
  Range = `0` to `1`
  - `0` = idle/off
  - `1` = full send

> *Refer back to your Axis Table if your brain starts buffering.*

3. **Target (The Important Bit)**
   This is where you tell SPAD.neXt what you actually want to control.

   For **Pitch**, use: sim/cockpit2/controls/yoke_pitch_ratio
   That's the elevator control—the thing keeping you from becoming a lawn dart.

4. **Use Axis Value**

   Select this option—this tells SPAD.neXt:

   "Take whatever the axis is doing… and use it."

5.

Final Approach

Click **OK**.

And just like that—your **Pitch (Elevator) axis is configured**.

Give it a test in-sim:

- Pull back → nose goes up
- Push forward → nose goes down
- No screaming or confusion → perfect

Rinse and Repeat

You can now repeat this process for:

- Roll

- Yaw
- Throttle
- Prop
- Mixture
- And anything else that slides, spins, or behaves suspiciously like an axis

Just use your **Axis Table** as your flight manual for picking the correct datarefs and value ranges.

Basic Flight Controls - Button Setup

(Because Clicking With a Mouse Is So Last Flight)

Step 1: Select Your Device

Head over to the **left panel** in SPAD.neXt and pick the device you want to configure:

- Yoke
- Button box
- Throttle quadrant
- Or your custom-built “looks like it belongs in a spaceship” panel

If it has buttons, it’s invited to the party.

Step 2: Activate the Lego Brick

Before you start pressing buttons like a caffeinated first officer, make sure the **Lego brick icon** is turned on.

This enables **input detection**, so SPAD.neXt can see what you’re pressing.

> *If this isn’t on, you’ll be pressing buttons into the void wondering why nothing is happening. The sim equivalent of yelling “Hello?” into a dead radio.*

Step 3: Press the Button

Now press the button you want to configure.

SPAD.neXt will highlight and select it instantly like:
“Ah, Button 5... a bold choice.”

Step 4: Create the Event

- Click **“Add Event”**
- Choose when it should trigger:

Options include:

- **Pressed** → when you push the button
- **Pressed (short)** → Press for less than the default time (1000 ms)
- **Pressed (long)** → Press for longer than the default time (1000 ms)
- **Button Held** → When you press and hold the button Options include:
  Threshold (time when spad considers the button being held) Frequency (how often it repeats the Action being executed.
- **Released** → when you let go
- **On / Off** → for toggle switches

> ✈ *Rule of thumb:*
>
> - Momentary button? → use **Pressed**
> - Toggle switch? → use **On/Off**

Step 5: Add an Action

Now click the **(plus) icon** to add what the button actually does.

You've got two main flight paths here:

Option A: Use a Command (Recommended for Most Cases)

- Select **"Send Simulation Event"**
- Search for your command (no more file gymnastics—SPAD.neXt's got this covered now)

Examples:

```
sim/lights/beacon_lights_toggle
sim/lights/beacon_lights_off
sim/lights/beacon_lights_on
```

Choose how it runs:

- **Execute Command Once** → quick tap (most common)
- **Begin Command / End Command** → for press-and-hold actions

> *Commands are perfect for things like toggles, autopilot buttons, gear, lights—basically anything you'd "press" in the cockpit. Most things in X-plane are tied to commands, only use Option B when you have run out of Option A.*
> *Note: Some Aircraft developers Do not tie cockpit buttons or switches to the*

> *default commands, nor do they tie them to Custom Commands. In this case Option B*

🔧 Option B: Use a Dataref (For Precision Work)

- Select **“Set Dataref Value”**
- Enter your dataref
- Define the value

Example:

```
sim/cockpit2/switches/landing_lights_on = 1
```

> *🧠 Use this when you want full control instead of “toggle and hope for the best.”*

🧪 Step 6: Test It

Fire up X-Plane and press your button:

- Light turns on? 💡
- Gear drops? 🛬
- Aircraft doesn’t explode? 💥

You’re good to go.

If not… well… welcome to troubleshooting. We’ve all been there.

🧠 Pro Tips from the Left Seat

- 🔁 Use **datarefs as conditions** to sync LEDs or states
- 🧩 Stack multiple actions if you want one button to do several things (because why not?)

✈️ Final Words from the Captain

You’ve now officially moved beyond “clicking things on-screen” and into the world of **proper cockpit interaction**.

Your buttons now:

- Do what you want
- When you want

- Without mouse gymnastics

P3d and FSX

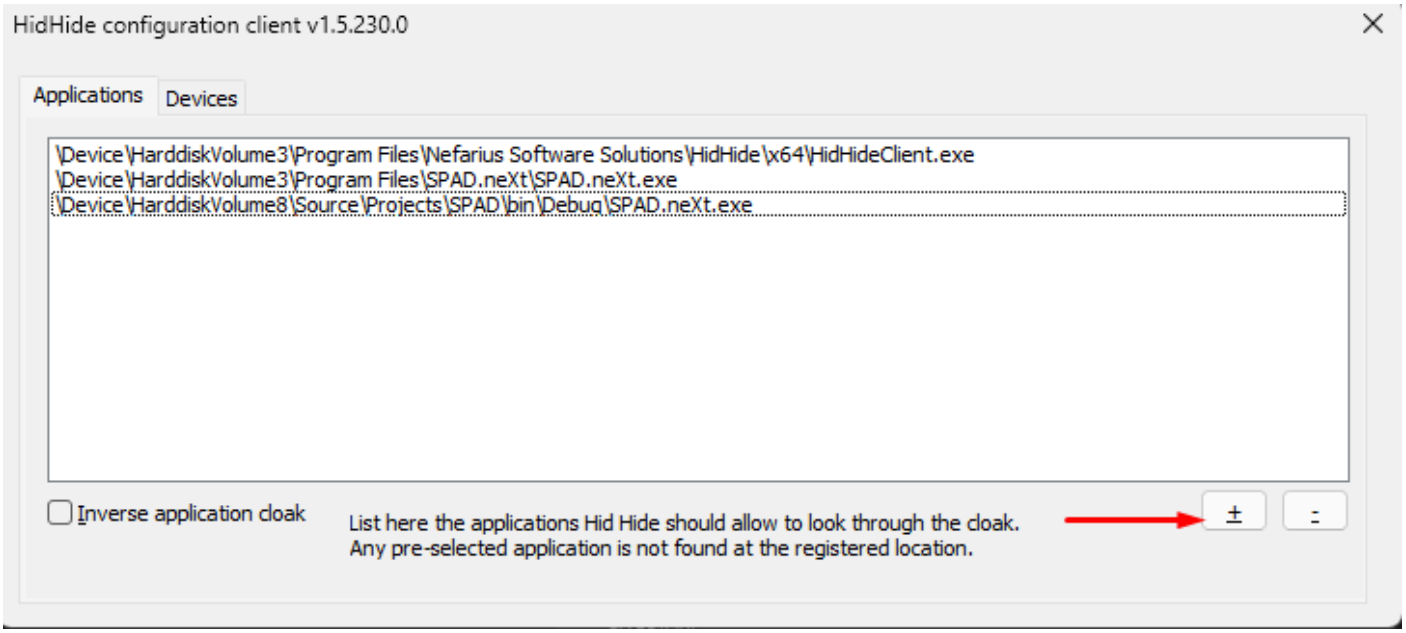
Other Simulations

Star Citizen / ETS2 / ATS and others

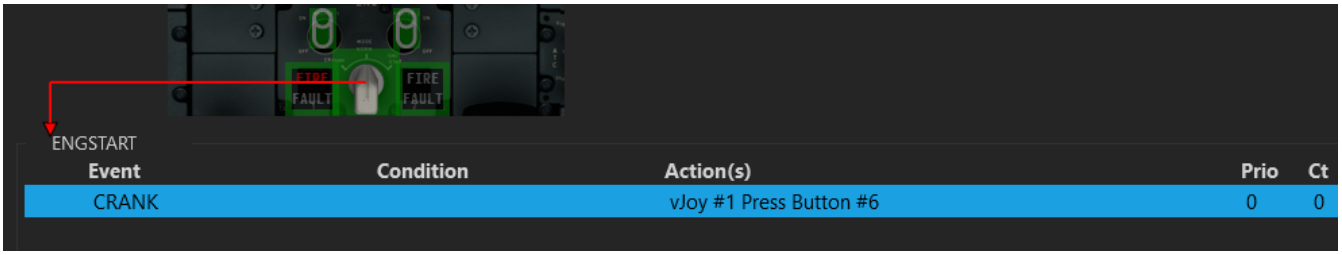
Basically you can drive any software using SPAD.neXt.
While a common way to control those software (e.g. Star Citizen) is using Keyboard events, this has some limitations due to windows security.

If the target software supports joysticks, follow these steps for the best experience:

- Install vJoy https://github.com/jshafer817/vJoy/releases
  - This will create a virtual joystick which you can control from within SPAD.neXt
  - Configure one or more virtual Joysticks to your likings
- Install HidHide https://github.com/nefarius/HidHide/releases
  - This software allows you to hide specific joystick devices to the system making it easier to bind functions.
  - Configure HidHide to hide all devices that you want to drive by SPAD.neXt
  - Configure HidHide to expose those devices to SPAD.neXt by adding the SPAD.neXt executable (C:\Program Files\SPAD.neXt\SPAD.neXt.exe usually) to the Applications tab



- Configure your devices in SPAD.neXt to forward buttons/axis to vJoy
  - Example



- In the target software now bind the vJoy buttons/axis by pressing the button/moving the axis on the device

USB HUB Requirements

1. Power in A needed = Devices connected * 0.5
2. Normal (cheap china) hub usually do **not** provide a power adapter to drive **all** it's ports at the same time. (They usually assume only 50% of the ports will be drawing power at the same time). You will usually see that they provide like 1.5-2A for a 6 port hub. That is not enough if you have 6 flight devices connected.
3. Every device that has some kind of display (Fip, radio panel , multipanel , even a Bravo (leds)) will *constantly* pull 500mA of the hub-psu
4. Rule of thumb is to have a psu that provides Devices*0.5A + 0.5A , to kinda ensure the psu is not falling down if max is pulled of it. (Again cheap power adapters tend not to be able to provide their max output value constantly, especially right after power on).
   It's like a car... it can only drive full power so long, and then it will slow down because everything is overheating and loosing effectiveness
5. If you connect a **unpowered** hub to your computer (shame on you!) it will be able to drive **exactly** one device. Not more.
6. Also connecting 5 fip to **one** usb 2.0 hub will most likely fail as well, due to usb 2.0 bandwidth limitations. All devices connected to the hub **share** the bandwidth.

> Make sure USB Power Management is turned off for all used USB Devices and the HUB/Root-Ports they are connected to.
> Also make sure USB Sleep mode is disabled in the Windows Energy Profile

CDU Devices

SPAD.neXt supports a wide range of hardware CDU devices, to display aircraft CDU/FMC on it. Since there is no standard defined every aircraft has to be implemented manually.

You can use snippet #13161 / #13165 for a reference implementation of CDU's

> All CDU devices are a complete edition feature

Currently supported aircraft

MSFS/MSFS24

- Aerosoft CRJ
- Fenix - All Aircraft
- FlyByWire - SimBridge must be running
- FsLabs
- iFLY 737
- Inibuilds A300 / A340 (MSFS 2024 Only)
- MaddogX
- PMDG - All aircraft. SDK must be enabled
- SAAB 340 (MSFS 2024 Only) LiM UNS-1 Plugin Required
- TFDI MD11/B717
- WT 707 (MSFS 2024 Only) LiM UNS-1 Plugin Required
- WT 747v2 (8i Pax & Cargo / Dreamlifter / Tanker) (MSFS 2024 Only) LiM 747 Plugin Required
- z-dementedmonkey Mod (v0.10.0 March 2026) SPAD.neXt v0.9.24 required
    - Download Link to DM Github
        - **MSFS 2020 Supported Aircraft**:
            - Citation CJ4 - Working Title (This is the Default CJ4 since 2020 AAU1)
            - FFX P180 v2.0.3 (Read dementedmonkey mod instructions about additional package installation process) - Remove any z-lim-ffx-cdu mods

        - FFX C750 v1.3.9 - Remove any z-lim-ffx-cdu mods
    - **MSFS 2024 Supported Aircraft:**
        - 737 Max 8 - Asobo (Default Sim Plane)
        - Citation CJ4 - Working Title (Default Sim Plane - WT21v2)
        - C90 King Air - MS/Carenado (Default Sim Plane - WT21v2)
        - FFX C750 v2.0.2 (WT21v2) - Remove any z-lim-ffx-cdu mods
        - FFX P180 v2.1.9 (WT21v2) - Remove any z-lim-ffx-cdu mods
        - FFX 800XP v1.x - Remove any z-lim-ffx-cdu mods

FSX/P3D

- PMDG - All aircraft. SDK must be enabled
- iFLY

X-Plane

- all aircraft that use the X-Plane standard cdu datarefs
- ToLiss aircraft
- ZiBO mod
- Rotate MD-11
- Rotate MD-80
    - The MD-80 has a 30 characters wide display, while CDU devices only have 24 character per line.
      Only the first 12 and last 12 characters will be shown on the device.
    - The horizontal/vertical lines are currently not supported
- CL650
- FF 767 / 757
- FF 777v2

CDU Devices with HDMI Interface

To drive CDU devices with a HDMI connector through SPAD.neXt to get access to all extended features without needing to popout a window in the simulator, SPAD.neXt provides a RemoteCDUDisplay-Tool which will popout a customizable window.

You can use snippet #13161 / #13165 for a reference implementation

The tool can be found in the SPAD.neXt installation directory in the subfolder Tools/cdu. It can be run on a remote computer if necessary

- to run display upon load of a profile create a `profile loaded` event and run the external command "<spad_install_dir>/tools/cdu/RemoteCDUDisplay.exe" or add it to a batch-file
- To run the display on a remote computer copy tools/cdu to target computer, and start "RemoteCDUDisplay.exe -h=ipOfSPAD -p=28001 -t=`targetid`". To get the targetid open the cdu tool once using SPAD.neXt.

WinCTRL CDU Devices

> WinCTRL CDU Devices must be configured in SimAppPro as CAPTAIN, you can select to show the FO-Side inside SPAD.neXt then.

Updating the CDU Font is currently not supported. Update the font using SimAppPro

Embedded Codes

If you program your own CDU pages in SPAD.neXt you can use embedded codes to control the formatting. Defaults: Background is black , Big font

Embedded codes start with `{tag}` and will apply to the following text until an `{end}` is found.

Embedded codes can be stacked.

`{tag}text{end}`

`{white}SPAD.{small}ne{end}{red}X{end}{small}t{end}{end}` SPAD.neXt

Colors: {white} {cyan} {green} {magenta} {amber} {red} {blue} {darkgreen} {black} {gray} {yellow}

Other:

{rev} - switch background and foreground

{small} - small font

{inop} - inoperable entry

CPFlight Devices

Support for CPFLight devices has been implemented "as it is". Any limitations by the firmware of the devices will apply.

Supported Devices (RS232/USB Versions only currently):
* MCP
* EFIS (chained)
* COM-Panel (chained)
* ATC-Panel (chained)

Configuration

RS232/USB Version

Add the device as serial device in Settings->Devices->Serial and select CPFlight as protocol

Ethernet Version

Add the device as serial device in Settings->Devices->Serial, select CPFlight (Ethernet) as protocol and configure ip and port according to the CPFlight manual. (not available yet)

MCP

Encoders

You can use the normal Counter-/Clockwise events, to use SPAD-acceleration like with all other devices. Additionally you can use a ENCODER_SET event which will use the device-builtin acceleration. In this case use a Send-Simulation-Event action and use "EVENT:VALUE" as the data value to send the value coming form the device to the simulation.
Due to the firmware handling encoders internally, it might be that the value displayed on device and simulation might differ.

A/T Led

The LED is controlled by the firmware and not controllable from SPAD.neXt, however you can control the magnetic lock (pro version) or make the led blink using the CPF_MCP_ATDICONNECT Variable.

COM-Panel

Standby Frequency cannot be addressed from outside.
Older devices do not support 8khz spacing.
Usage of device is limited to com frequencies only
Use VALUE_SET event with EVENT:VALUE parameter to react on Xfer-button

EFIS

If both CS and FO Efis are present , global buttons (FPV/MTRS/STD/RST) will only be executed on whichever device announces itself first

ATC Panel

- The Encoders do not send any events. When turning the encoders the VALUE_SET event on the display will be raised. Use EVENT:VALUE in the action to get access to the new Transponder Value.
- The ATC Fail LED is not documented and not controllable yet

Variables

Some Variables are provided to control behaviour of the devices:

Name	Purpose
CPF_MCP_BACKLIGHT	Control the backlight of all devices
CPF_MCP_ATDISCONNECT	Disconnect the A/T magnetic lock (MCP pro) or blink the A/T Led

Name	Purpose
CPF_COMx_SPACING	Switch between 8 (=0) and 25 (=1) kHz spacing. Defaults to 8khz if device supports it.

Saitek FIP (Flight Instrument Panel)

Special notes about Saitek FIP requirements and support

For SPAD.neXt to handle the Saitek/Logitech FIP correctly the following steps need to be performed:

- Install the Saitek FIP Drivers. (Download at their webpage)
- After installation, uninstall the "DirectOutput Service"
- MSFS: Uninstall the Logitech "Flight Simulator Plugin"

For best practices about FIP connection to your PC please see https://fipgauges.com/hardware.php

Also note that the USB HUB requirements apply!

> Please note, in contrast to the original Saitek implementation, which limits updates to one device at a time , SPAD.neXt will drive all FIP at full speed with concurrent updates, resulting in up to 16fps per device.
> However, this will draw the full 500mA per deviceport from the hub. Therefore it is **important** to follow the best practices.

SPAD.neXt supplies the standard saitek gauges that come with the FIP. For more sophisticated gauges it's recommend to have a look at fipgauges.com or Fsx times
Also there are some user created gauges available inside SPAD.neXt and on the Community Discord.

Error Rendering on Device

This is indicates the device is not getting enough usb power to keep up rendering.
See HUB requirements and the best practice above

The FIP with Serial XXX did not respond to the initialize request. This can happen if the FIP internally crashed. Please powercycle the FIP.

Recent Windows 11 changes can rapidly cycle USB device power during startup. Saitek FIP devices do not support this because they have a long boot cycle and may crash as a result.
Best practice is not to plugin the FIP devices before windows is fully booted and running

Saitek FIP: Driver & Function Check

For the Saitek FIP to work correctly with SPAD.neXt, you will need to install a windows driver for it.

- Download https://download01.logi.com/web/ftp/pub/techsupport/simulation/Flight_Instrument_Panel_x64_Drivers.exe
- Install it
- Reboot Computer
- The FiP should now show Saitek Advertisements
- Open "Apps & Features" (Programs & Apps)
- uninstall "Direct Output" (this conflicts with SPAD.neXt)

FIP Function-Check:

- unplug the FIP
- wait 15 seconds
- plug the FIP back in
- Now the Page-Up button (the triangle in the middle) should flash exactly eight times, then turn off.
- the backlight of the FIP display should be on and you should see it shimmering grayish.

Loupedeck Image Devices

Loupedeck Software 6.0.3 **required**

No later version of Loupedeck Software is supported, as Loupedeck broke dynamic image updates in all later versions.

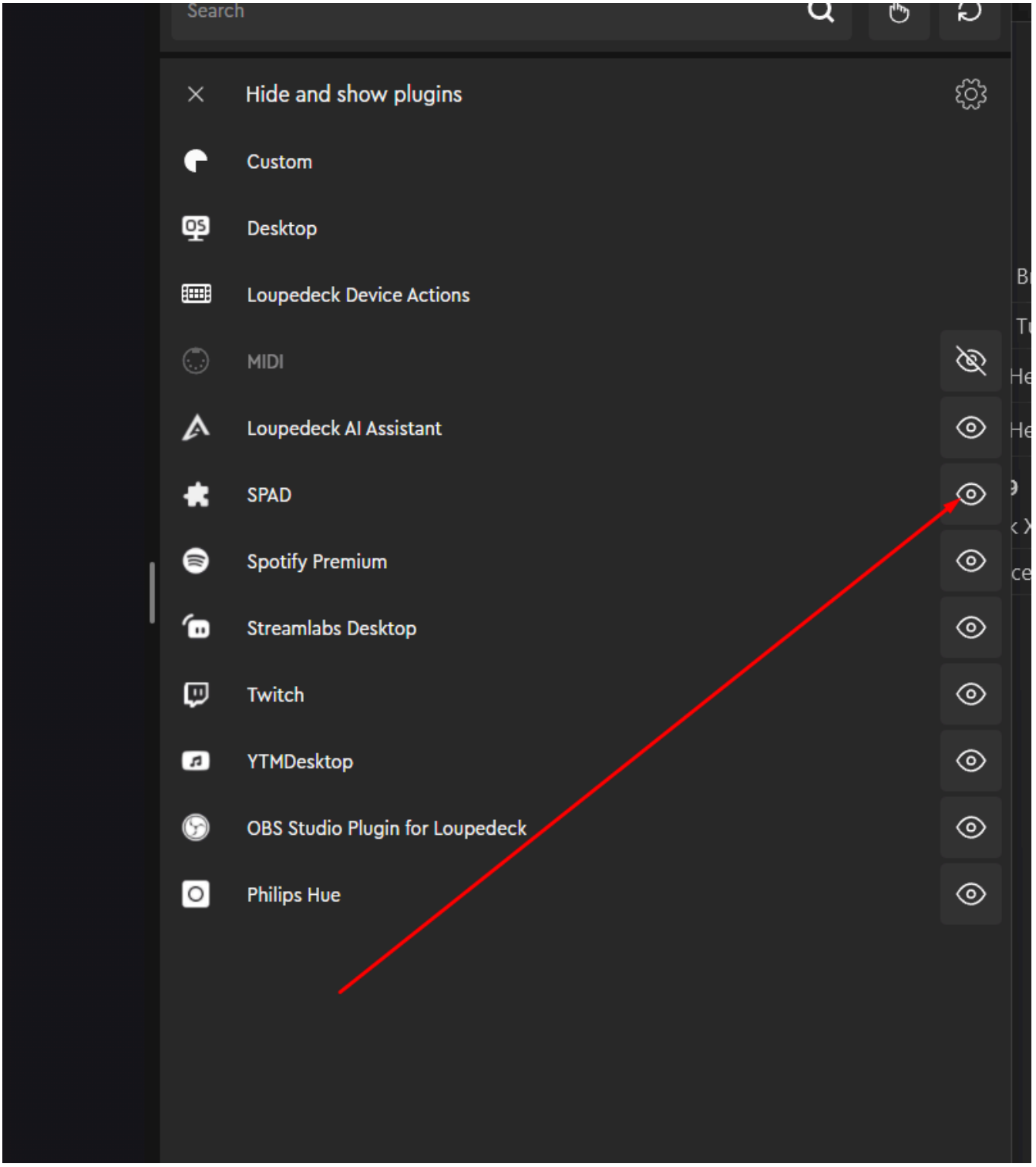
The Loupedeck Software 6.0.3 Installer can be found here

Logitech Options+ must not be installed

Loupedeck support is a complete edition feature

- Exit SPAD.neXt
- Install SPAD Loupedeck plugin. You will find it in the GamePlugins-folder of SPAD.neXt installation folder
- Restart Loupedeck app

- In Loupedeck Software unhide the SPAD plugin



- Create a new Loupedeck Profile and drag the Spad-Actions onto the buttons

- Buttons should show the Buttonnames on them now on device
- Start SPAD.neXt , make sure WebApi is enabled. (You can check by opening http://127.0.0.1:28001/ , it should show some api page)
- After about 15secs the Loupedeck should show in Panels , and all buttons on device should go blank. (You might also want to enable the "Wait" option in Settings->Devices->Remote Devices)
- if everything fails, open a ticket and attach

```
%LOCALAPPDATA%\Logi\LogiPluginService\Logs\plugin_logs\SPAD.log
```

The source code of the SPAD.neXt Loupedeck plugin is open source and can be found on the SPAD.neXt Github. Pullrequests are welcome if you update the code to a later Loupedeck version.

Midi Devices

Windows 11 24h2 Issues

Windows Midi driver will go into sleep mode and there is currently no way to prevent that, because you cannot configure usb power options for it. This results in the midi device not sending any input.
The only known solution is to unplug the device and replug it.

Windows 11 25H2 Issues

Important: This is only for W11 25H2!

Sometimes, the 25H2 update does not convert MIDI devices correctly. Microsoft has released a tool to fix this: MIDI Fix.

Also make sure the midisrv service is running

X-Touch Mini

New Mackie Mode

Since release 0.9.21.1 SPAD.neXt uses the Mackie-Mode for the X-Touch Mini which gives full control over led's and buttons.
For a guide how to use this new feature see the video section.

> **SPAD.neXt will attempt to switch the X-Touch Mini into Mackie-Mode automatically, however this might fail. Specially if you are in W11 24H2 Update, which has a known issue with Midi/audio devices.**

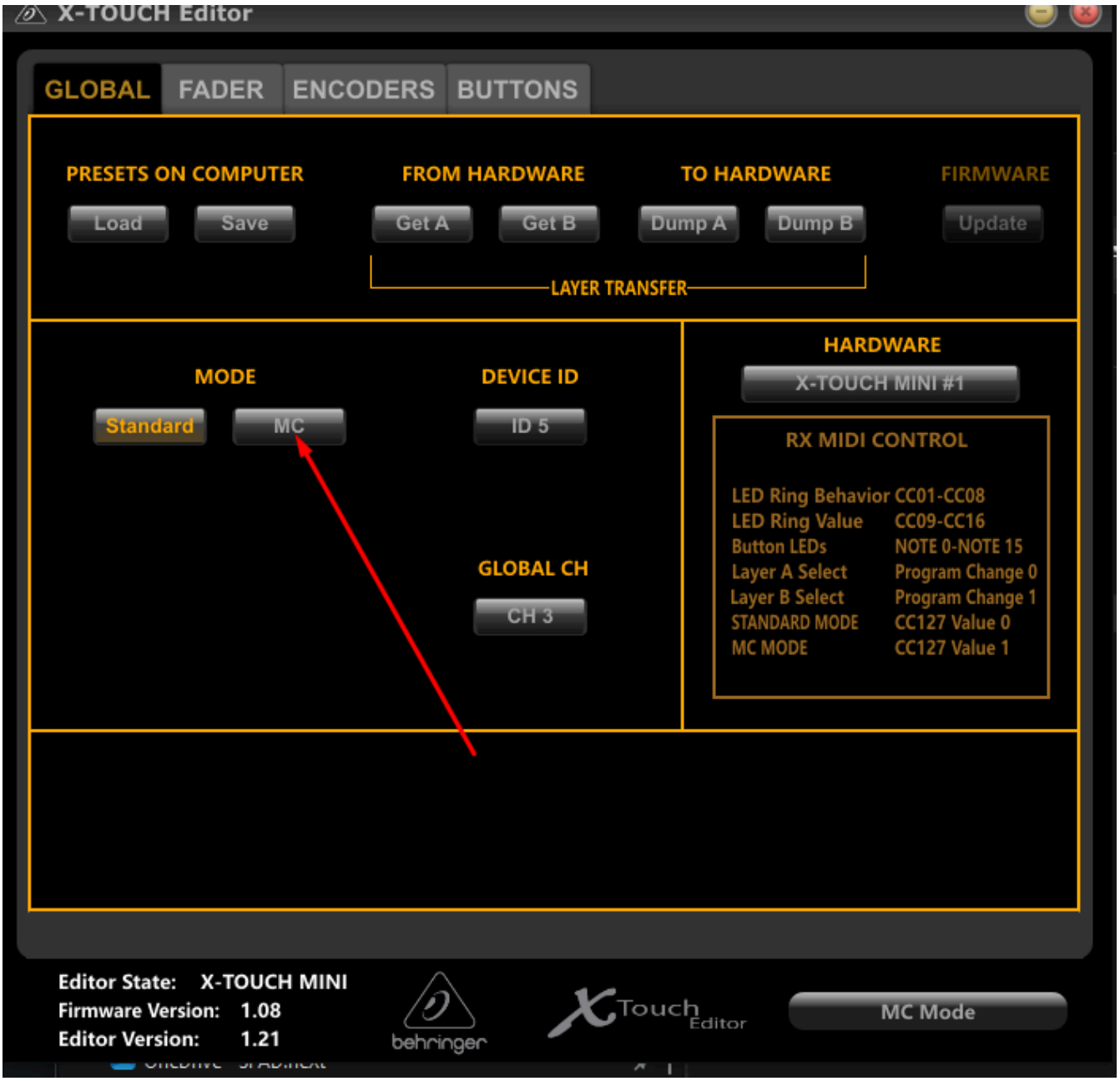
Switching did **not** work if:

- the Layer A led is on and the MC mode led above it is off
- Button presses are not recognized by SPAD.neXt

First, unplug and reconnect the X-Touch. Then restart SPAD.neXt and check whether the device switches automatically. If it does, the issue is resolved for now, but you may need to reconnect the device occasionally because of this Windows 11 24H2 bug. (As of January 12, 2025, a fix was in preview at Microsoft and was expected in a future Patch Tuesday update.)

If that did not fix it, the X-Touch Mini need to be switched into Mackie-Mode manually:

- Exit SPAD.neXt
- Download (if not already present) https://mediadl.musictribe.com/download/software/behringer/X-TOUCH/X-TOUCH-EDITORv1.21.zip
- Unzip container and start the x-touch.exe
- An ui like this should appear

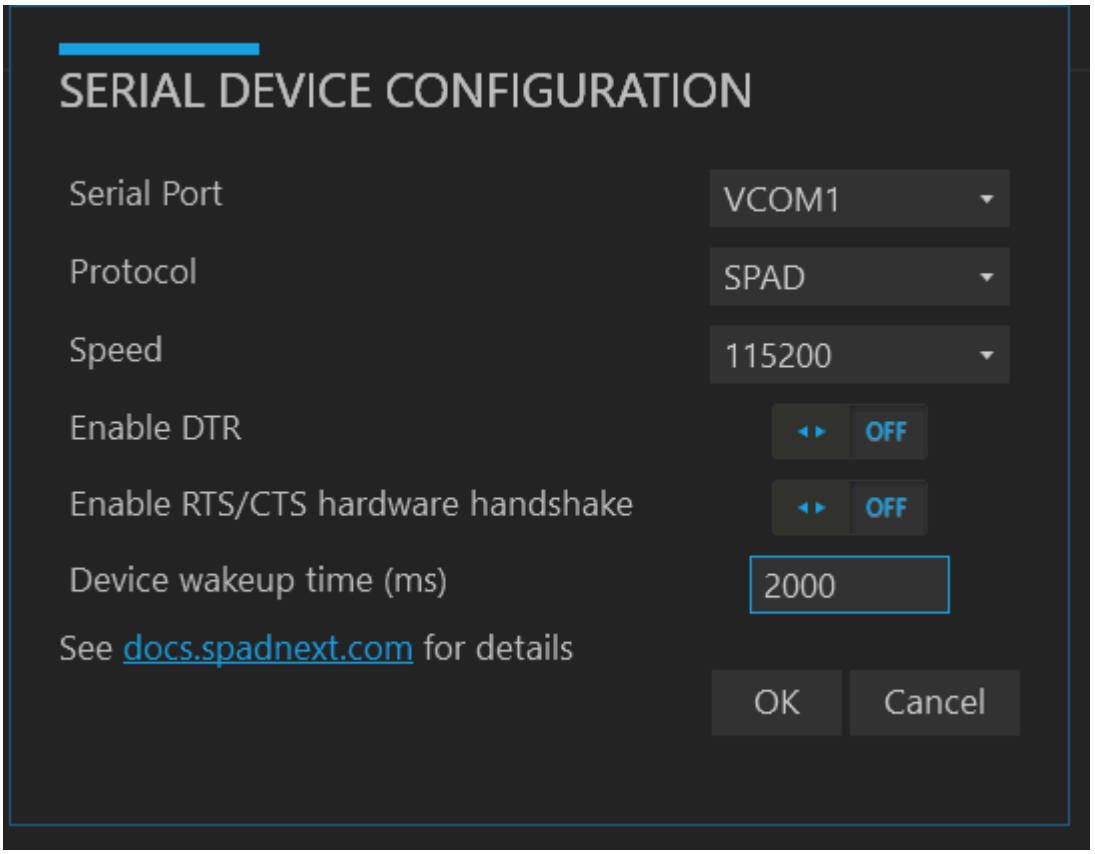


- Click on the MC Button
- The Layer led should go off and the MC MODE led above it should go on.
- Close the x-touch editor
- Start SPAD.neXt.
- All Buttons should be working now.

Serial (COM) Devices

SPAD.neXt supports a wide range if serial (COM) devices, however there are many different serial chipsets around, specially for USB2Serial devices.

Some of those chipset require a special treatment (notably the fake CH340 chipset), which you can configure in the serial device setup.



Select the appropriate Port and Protocol. For some protocols (e.g. VRInsight) no options will be available.

Speed

If you notice garbage being sent/received then lower the speed. For an USB2Serial device 115200 is usually a good choice, but if the device is too slow to handle this speed, lower it.

Enable DTR

Data Terminal Ready - It tells the device that something is connected to it's serial port. This should be ON in most of the cases.

> Exception: If the device is CH340 based, this has to be set to OFF, because the DTR signal will make the device reset.

Enable RTS/CTS

RequestToSend/ClearToSend Signaling
A hardware handshake protocol to ensure no data is lost in transmission because of a buffer overrun on the device. In most of the cases this should be set to ON. If you device does not seem to send/receive any data, set it to OFF

> For CH340 based devices this has to be turned off, unless it's configured in the device's firmware.

Device wakeup time

Time in ms before spad will try to send anything to the device after opening the port. For a real USB2Serial chipset this is not needed (set to 0).
However e.g. the CH340 fake serial chip has an odd bootloader which will go into device-programming mode if anything is received on the port while device is booting up and in result the device will not respond to any data sent to it.
If that happens, set this wakeup time. You might need to experiment with the settings but 2000ms is a good starting point. Devices with a lot of stuff in it (e.g. displays) might need a longer wakeup time (e.g. 3500ms).

Common Issues

Access to ComPort denied

If you get "Access to ComX is denied" error when starting SPAD.neXt after a fresh computer reboot with a serial device, this is most likely because you have Skype/Teams or the Arduino IDE running. Serial devices are sometimes recognized as audiodevices and locked by skype/teams for unknown reason.
(Same can apply to other software locking up the port)

Solution: Unplug serial usb and replug it, this will release the lock on it.

Skalarki

> All Skalarki devices as complete edition features

SPAD.neXt supports all Skalarki Ltd. devices, however you will need to have Skalarki's ProfilerIO running.

Current supported ProfilerIO version: 5.1.911.1000

Generally all inputs and outputs of all devices can be identified by

`SKALARKI_<device>_<type>_<sublocation>_<name>`

<type> will either be **I**nput (button/switch) **E**ncoder **D**isplay **L**ed
For the name, the naming provided by Skalarki is used, even if it might be a bit misleading sometimes.

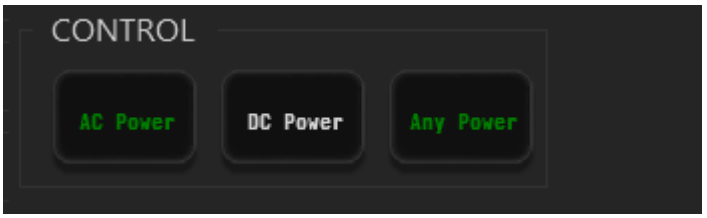
Throughout this guide several online snippets will be referenced. Those will work with the FlyByWire (FBW) A320 and are actively maintained.

General

All devices will provide variables for some settings:

Variable	Purpose	Description
..._DEVICE_POWER	Holds the virtual power state	defaults to 1 (on) if virtual power handling is disabled
..._BACKLIGHT	controls the device backlight	if supported by device

Skalarki Global



Since there are many different devices, setting up the basic power configuration for all of them is a lot of work. It is a repetitive task, and changes can require significant effort.

To help with this task SPAD.neXt has an additional virtual device called "Skalarki Global". It provides 3 variables that you can use to control general behavior of the panels at a central place.

As a reference refer to snippet #7635 which contains the programming for the FBW A320

Variable	Description	Purpose
LOCAL:SKALARKI_GLOBAL_AIRCRAFT_DC_POWER	Holds if the aircraft currently has any DC Bus powered	Use as condition for functions that only work if there is DC power
LOCAL:SKALARKI_GLOBAL_AIRCRAFT_AC_POWER	Hold if the aircraft currently has any AC Bus powered	Use as condition for functions that only work if there is AC power
LOCAL:SKALARKI_GLOBAL_AIRCRAFT_POWER	Holds if any of the AC/DC Bus is currently powered	Use as condition for functions that work if there is any

Variable	Description	Purpose
		kind of power

VRInsight Devices

Common Issues with VRInsight Devices

Do enable the support of a VRInsight device in SPAD.neXt add the serial port as new serial device (Settings->Devices->Serial Devices) and select **VRInsight** as protocol.

- Newer hardware versions of the overheads have **fake** knobs for e.g. Temp and Breakers. Those are **not** wired
- The firmware can behave unpredictably. On the first connection, it may sometimes overflow the serial port, preventing SPAD.neXt from recognizing the device. Click the Connect button in the SPAD.neXt UI again to connect.
- Sometimes the Firmware dies and is not responding to commands anymore. Solution: Unplug USB **and** power , replug power , wait 15 secs, replug usb

Gauges and Extensions

SPAD.neXt generally supports ESP compatible gauges.

A good first overview about gauges coding can be found here
Gauges loaded into SPAD.neXt have to match this schema: `Gauge.xsd`

RPN Expressions have to be ESP RPN conform!

SPAD.neXt is generally Saitek/Logitech compatible, but it has **NOT** implemented Saitek Bugs:

Saitek gauges require a Non-Space-Character before a closing brace "}" e.g. in if{ statements:
`if{ someexpression`` ``%} els{ someotherexpression`` ``%}`

SPAD.neXt does **not** support this. You will need to alter the gauge accordingly and remove the %

Extensions for (RPN) Expressions

To access any other Data than Simconnect (FSX / P3D / MSFS) use the Dataref-Tag as it is used by SPAD.neXt. Those Datarefs have always the format

`RPN Expressions: (PROVIDER:DATREFNAME) or (PROVIDER:DATREFNAME,unit)`
`SPAD Expressions: [PROVIDER:DATREFNAME] or [PROVIDER:DATREFNAME,unit]`

`If the requested unit does not match the datarefs default unit, SPAD.neXt will automatically convert it to the requested unit, if such a conversion is known`

To access LVAR use the prefix **LVAR: (FSX/P3D/MSFS only)**
`(LVAR:AB_MCP_AP_ON, Number)`

To access FSUIPC Offsets use the Prefix **FSUIPC:**
(FSUIPC:0000:0AB2, Number)

You can copy a dataref from the data browser to your clipboard by selecting it and pressing Ctrl+C or clicking the OK button.

Other examples:

PMDG (P3D):
`(`PMDG737:1000:010E) `= Glareshield.Mode Control Panel.Altitude`

`ETS2`
`(SCS:scs/truck/fuel)`

`X-Plane:`
`(XPLANE:sim/cockpit2/annunciators/afterburner)`

Local Variables access

LOCAL:VARNAME
Local var , will not be saved, will be initialized to 0 if it does not exist already

DEVICE:VARNAME
Local variable: This value is specific to the device and is initialized to its saved value when the gauge is loaded on that device. The same variable can have different values on different FIP devices.

P**ROFILE**:VARNAME
Local var , Value will be saved to the *PROFILE* , and will be initialized to the saved value again when the profile is loaded

Additional Time-Variables

All time variables as shown here can also be used with the "LOCAL:"-Prefix to retrieve the local computer time as UTC/Local.
To get the local computer hour you would use

```
(LOCAL:LOCAL TIME,Hours)
```

Dynamic Images

The Image-Tag supports the following extensions for dynamically changing images:

- Attribute "Dynamic"
- Sub-tag "Value"

Example:

```
<Element id="RevBar">
        <Position X="164" Y="144" />
        <Image Name="null1.png" Dynamic="rpm_{VALUE}.png"
ImageSizes="472,135,472,135">
           <Value Minimum="1" Maximum="32">(A:RPM) ABS 250 / %</Value>
       </Image>
</Element>
```

Dynamic Labels

In order to show numbers on the gauge, SPAD.neXt support an additional sub element: <Label>

See: Dynamic Labels

Extended Button Configuration

To assign a function to a FIP button other than a SimConnect Control Event, you can deliver an extended button configuration together with your gauge.
Just configure the button as desired within SPAD.neXt, and then export the button configuration (copy to clipboard and save to a file) to a file `BUTTONNAME.bml` in the gauge main directory.
The following buttons are supported:
`FIP_BUTTON_S1 - FIP_BUTTON_S6 , FIP_RIGHTDIAL, FIP_LEFTDIAL, FIP_BUTTON_PAGEUP, FIP_BUTTON_PAGEDOWN`

GaugeSettings

SPAD.neXt also defines an additional tag for providing settings in a gauge:

```
<Gauge>
..
..
..
  <GaugeSettings .....>
  </GaugeSettings>
</Gauge>
```

-
- SPAD.neXt GaugeSettings Sample

SPAD Gauges: GaugeSettings

Each SPAD.neXt compatible gauges should contain a GaugeSettings-node to identify the gauge, it's author and some general settings.

Minimal Example:

```
<GaugeSettings GaugeCompatibility="SPAD" GaugeID="!!! INSERT NEW GAUGEID
HERE !!!!" PowerMode="AlwaysOn">
        <Author>Your Name here</Author>
        <AuthorId>!! INSERT AUTHORID HERE!!!</AuthorId>
        <Description>General Airliner Gauge</Description>
        <Simulation>MSFS</Simulation>
        <!-- Only if thumbnail provided, else spad will generate one,
filename of thumb in 1024 directory
        <Thumbnail>myThumb.png</Thumbnail>
        -->
</GaugeSettings>
```

GaugeId

Unique Identifier identifying the gauge. It should follow the GUID-Registryformat e.g. "{A8AA15C5-7BB6-4AC6-A558-A88CAFB78729}". To create a GUID you can use this site: https://www.guidgenerator.com/online-guid-generator.aspx (enable at least hyphens)

AuthorId

Your AuthorId. You can get it by sending the command `!deviceinfo` to the SPAD.neXt Bot on the community discord.

C# Scripting Interface

The scripting interface is C#6.0 compatible.

You can check the DemoScript provided.

Basically you create Classes in a .cs-File for the needed Purposes.
To ease development it's recommended to use a C#-IDE to develop the Scripts. (e.g. VS 2015 Community edition (free)). Just create a new C#-Class-Library Project and Reference the Assembly "SPAD.Interfaces.dll" from the SPAD.neXt Binaries or Include the Interface-Project from github. (You will need to create your own signing key) This way you will get Intellisense and can test-compile your script.

All classes that derive from IScript somehow will be considered by SPAD.neXt.
Classes that inherit IScriptPrivate will never be called by SPAD.neXt

The following BaseClass-Interfaces (from SPAD.neXt.Interfaces.Scripting) are supported:

IScriptAction
Class provides a custom action which can be bound to a Button/Switch.
When the Event is raised the method "void Execute(IApplication app, ISPADEventArgs eventArgs)" will be called.

Additional Interfaces that you can implement:

IScriptCreation
Class supports Initialization/Deinitialization. E.g. if you want to initialize/store some stuff Initialize will be called when the object is instantiated , Deinitialize before the object is Disposed.

IScriptActivation (only suitable for Objects that also implement IScriptAction)
Class supports activation/deactivation.
Activate will be called when an event using this Action is activated in a profile
Deactivate will be called when an event using this Action is deactivated in a profile

To provide an object that is no ValueProvider or Action, derive it from IScript and add the additional necessary Interfaces (e.g. IScriptCreation)

The IApplication-Interface given as Parameter is a gateway to some global functions of SPAD.neXt like creating a logger to write to the logfile.
Comes in handy for debugging:

```
var myLogger = app.GetLogger("myCoolLoggerName");
myLogger.Error("This will be shown in Logfile");
```

For checking/viewing the Logfile I recommend a NLog-Compatible LogViewer like e.g. Log4View.
The LogFormat-String is "${longdate}|${level}|${threadid}|${logger}|${message}"

To work with Data from the Sim/SPAD.neXt the global Eventsystem-Object is the key.

using "EventSystem.GetDataDefinition" you grab a IDataDefinition from SPAD.neXt and can read or manipulate the Values. This also will makes sure that your script will be called when the data changes.

It's a bit try and error right now, but all interfaces exposed in the SPAD.Interfaces Assembly are usable and working.
(SPAD.neXt uses them itself for all Panels etc)
Inline-Documentation for the Interfaces will be added over time, as it is needed

Just play around a bit. If you need some help getting started, open an issue, describe what you want to do and I'll try to help you :-D

C# Scripting: ScriptStub

Starting with 0.9.18 SPAD.neXt.Interfaces contains an abstract ScriptStub which can be used for easier implementation:

```
/// <summary>
/// A basic Script Stub for free use
/// </summary>
public abstract class ScriptStub : IScript, IScriptCreation
{
    /// <summary>
    /// String to prefix all datat with. Data will be available as
LOCAL:[PREFIX]_[name]
    /// </summary>
    protected abstract string ScriptDataPrefix { get; }
    protected abstract void InitializeScript();
    protected abstract void DeinitializeScript();

    /// <summary>
    /// Interface to Application for this Script
    /// </summary>
    protected IApplication Application { get; private set; }
    /// <summary>
    /// Our logger for the script.
    /// </summary>
    protected ILogger ScriptLogger { get; private set; }
    protected CancellationTokenSource CancelTokenSource { get; private
set; } = new CancellationTokenSource();
    protected CancellationToken CancelToken => CancelTokenSource.Token;

    /// <summary>
    /// The loaded SPAD.neXtprofile has changed
    /// </summary>
    protected EventHandler<IProfile, string> ProfileChanged;
    /// <summary>
    /// The active aircraft/vehicle has changed
    /// </summary>
    protected EventHandler<IAircraft, string> AircraftChanged;
    /// <summary>
    /// A flight has been started
    /// </summary>
    protected EventHandler<ISimulationInterface, IAircraft>
FlightStarted;
```

```
    /// <summary>
    /// The flight ended
    /// </summary>
    protected EventHandler<ISimulationInterface, SimulationGamestate>
FlightEnded;
    /// <summary>
    /// The Simulation gamestate has changed (NewState,OldState)
    /// </summary>
    protected EventHandler<ISimulationInterface,SimulationGamestate,
SimulationGamestate> GameStateChanged;

    private Dictionary<string, IDataDefinition> scriptDataValues = new
Dictionary<string, IDataDefinition>
(StringComparer.InvariantCultureIgnoreCase);
    public void Initialize(IApplication app)
    {
        Application = app;
        ScriptLogger = app.GetLogger("Script." + ScriptDataPrefix);
        scriptDataValues = new Dictionary<string, IDataDefinition>
(StringComparer.InvariantCultureIgnoreCase);
        ScriptLogger.Debug("Initialize");
        Application.RegisterSimulationConnected(simulationConnected);
        Application.ActiveProfileChanged += (s, e) =>
ProfileChanged?.Invoke(Application.ActiveProfile, e.PropertyName);
        Application.AircraftChanged += (s, e) =>
AircraftChanged?.Invoke(Application.CurrentAircraft, e.PropertyName);
        SafeCall(() => InitializeScript());
    }

    private void simulationConnected(SimulationConfiguration sender,
IValueProvider e)
    {
        if (sender != null && sender.SimulationInterface is
ISimulationInterface simulationInterface)
        {
            simulationInterface.SimulationGamestateChanged -=
SimulationInterface_SimulationGamestateChanged;
            simulationInterface.SimulationGamestateChanged +=
SimulationInterface_SimulationGamestateChanged;
        }
    }

    private void
SimulationInterface_SimulationGamestateChanged(ISimulationInterface
sender, SimulationGamestate newState, SimulationGamestate oldState)
    {
        SafeCall(() =>
        {
```

```
            if (newState != oldState)
            {
                if (newState == SimulationGamestate.Flying)
                {
                    FlightStarted?.Invoke(sender,
Application.CurrentAircraft);
                }
                else
                {
                    FlightEnded?.Invoke(sender, newState);
                }
            }
            GameStateChanged?.Invoke(sender, oldState, newState);
        });
    }

    public void Deinitialize()
    {
        ScriptLogger.Debug("Deinitialize");
        CancelTokenSource.Cancel();
        SafeCall(() => DeinitializeScript());
        scriptDataValues?.Clear();
        scriptDataValues = null;
    }

    protected IDataDefinition GetOrCreateScriptDataValue(string name)
    {
        if (name.Contains(":"))
        {
            throw new InvalidCastException("Symbol ':' is not allowed in
datanames");
        }
        var realName = "LOCAL:" + ScriptDataPrefix + "_" + name;
        if (scriptDataValues.TryGetValue(name, out var data))
            return data;
        data = EventSystem.GetDataDefinition(realName);
        scriptDataValues[name] = data;
        return data;
    }
    protected T GetScriptDataValue<T>(string name, T defaultValue =
default)
    {
        var data = GetOrCreateScriptDataValue(name);
        return data.GetValueAs<T>(defaultValue);
    }
    protected void SetScriptDataValue(string name, object value)
    {
        var data = GetOrCreateScriptDataValue(name);
```

```
        if (data != null) { data.SetRawValue(value); }
    }

    private void SafeCall(Action _call)
    {
        try
        {
            _call();
        }
        catch (Exception ex)
        {
            ScriptLogger.Error(ex.ToString());
        }
    }
}
```

C# Scripting: Provide Values

Scripting reference implementation for providing Values to SPAD.neXt

```
    /// <summary>
    /// A Demo for a script that provides some value(s) to SPAD.neXt
    /// reference Implementation.
    /// </summary>
    public class ProvideValueDemoScript : ScriptStub
    {
        protected override string ScriptDataPrefix => "DEMO";

        protected override void InitializeScript()
        {
            ScriptLogger.Info("In Init");
            FlightStarted += OnFlightStarted;
            FlightEnded += OnFlightEnded;
            RegisterData();
            workerTask = Task.Run(workerThreadAction, CancelToken);
        }



        protected override void DeinitializeScript()
        {
            // Signal to end worker Task is set by base class via the
CancelToken already
            // Wait for worker task to gracefully end
            Task.WaitAll(workerTask);
        }

        private bool isFlightStarted = false;
        private Task workerTask = null;


        private void RegisterData()
        {
            // Register all the Data's this script provides
            GetOrCreateScriptDataValue("CYCLE"); // => LOCAL:DEMO_CYCLE
        }
```

```
        /// <summary>
        /// Increment LOCAL:DEMO_CYCLE by 1 every 1 second
        /// </summary>
        private async Task workerThreadAction()
        {
            while (!CancelToken.IsCancellationRequested)
            {
                if (isFlightStarted)
                {
                    SetScriptDataValue("CYCLE", GetScriptDataValue<int>
("CYCLE") + 1);
                }
                await Task.Delay(1000);
            }
        }

        private void OnFlightStarted(ISimulationInterface sender,
IAircraft aircraft)
        {
            ScriptLogger.Info($"FlightStarted: {aircraft?.Name}
({sender.SimulationGamestate})");
            isFlightStarted = true;
        }

        private void OnFlightEnded(ISimulationInterface sender,
SimulationGamestate newState)
        {
            if (isFlightStarted && newState !=
SimulationGamestate.Flying)
            {
                ScriptLogger.Info($"Flight end. ({newState})");
                isFlightStarted = false;

                SetScriptDataValue("CYCLE", 0);
            }
        }
    }
```

C# Scripting: PreCompile

Using precompiled Scripting Assemblies

The internal on-demand scripting interface is limited to several OS-Interactions for security reasons.

You can provide a precompiled C#-Script which has no restrictions, e.g. if you need to access System.Net.

Prerequisites:

- .Net Framework 4.8 Assembly
- SPAD.Interfaces.dll must be referenced (Copy to local = false!)
- If the Version-Number of SPAD.Interfaces changes, the assembly must be recompiled

Installation

- Create a directory "Addons" in Documents/SPAD.neXt
- Copy the assembly dll and all it's (non-system) dependencies to that directory
- The assembly will be loaded when SPAD.neXt starts, **before** any devices are initialized
-

Device Interface

SPAD.neXt device interface V2

SPAD.neXt can communicate with external devices like e.g. Arduino/Teensy via a very simple line based communication.

Request and reply on both sides will follow this format:

`<ChannelId>,<Param1>[,...,<ParamN>];`

Throughout the description of the device interface the following syntax will be used:

, is the separator for parameters

; is the terminator of a command

/ is used as escape character

[...] defines optional parameter

<Name> defines an actual value. This can be a number or a string (without quotes!)

Anything not enclosed in <> defines the exact string (without quotes!)

If `, / ;` are used within a parameter, they have to be escaped by a leading /.
E.g.
`15,Hello/, from Arduino;`

For a SPAD.neXt compatible Arduino library import CmdMessenger (4.0) into the Arduino IDE (recommended) or clone the github repository at https://github.com/thijse/Arduino-CmdMessenger

There is also a customized version available at the SPAD.neXt GitHub (TODO LINK) which includes SPAD.neXt specific functions to make device programming more easy.

The same protocol can be used to communicate with SPAD.neXt e.g. via a Websocket (TODO WEBSOCKET) and procide e.g. a custom webinterfaced device

Channel-ID's

ChannelID	Purpose	Direction
0	General Commands	SPAD <--> Device
1	Data Commands	Device -> SPAD
2	SPAD.neXt Events	SPAD -> Device
3	Debug	Device -> SPAD
4	Simulation Event	Device -> SPAD
5	Data Channel	SPAD -> Device
6	Device LED updates	SPAD -> Device
7	Device Display updates	SPAD -> Device
8	Device Input	Device -> SPAD
9	(reserved for future use)	
10 .. 49	Dedicated Data Channels	SPAD <-> Device

By default cmdMessenger is limited to a max of 50 channels to preserve device memory. You can change that by altering the line

`#define MAXCALLBACKS 50`
in cmdMessenger.h, however this is **not** recommended.

For the specs of the Serial V1 protocol (still supported but deprecated) please see here.

All <tag> or <targettag> are limited to the characters `A-Z,0-9,_`

Device Communication Flow

Initialization Phase

SPAD.neXt sends	Device sends	Comment
0,INIT...		
	0,SPAD,...	
0,CONFIG[,Parameters]		
	0,INPUT/OUTPUT/OPTIONS...	Only if device needs to configure anything
	0,CONFIG;	Device Setup completed **Needs to be sent always**
0,STATESCAN;		
	0,STATESCAN,1;	see STATESCAN
	8,....;	Any scanstate of stateful inputs
	0,STATESCAN,2;	
(any queued events and data)		
2,START;		Device is ready to start normal operation and SPAD.neXt will accept

SPAD.neXt sends	Device sends	Comment
		inputs now. Note this is on EVENT-Channel

Device INIT-Phase

When starting SPAD.neXt will send an INIT Request to the device and expects a response within 5 seconds, else the device will be marked as offline/unavailable.

Request from SPAD.neXt:
`0,INIT,<SerialVersion>,<AppVersion>,<AuthToken>`

`Parameters`

<SerialVersion>	**Version of the serial protocol supported 2 means SPAD.neXt supports version 1 and version 2 The current version is 2**
<AppVersion>	SPAD.neXt version (e.g. 0.9.12.0)
<AuthToken>	A unique ID identifying the SPAD.neXt instance used as authentication tag in AuthenticationCipher (See http://rweather.github.io/arduinolibs/classAuthenticatedCipher.h for details)

Reply from device:
`0,SPAD,<Guid>,<Name>,<SerialVersion>,<DeviceVersion>[,<opt1>,...,<optN>;`

`<Guid>`	**Unique Identifier identifying the device. It should follow the GUID-Registryformat e.g. "{A8AA15C5-7BB6-4AC6-A558-A88CAFB78729}". To create a GUID you can use this site: https://www.guidgenerator.com/online-guid-generator.aspx (enable braces and hyphens)**
<Name>	the human readable name of the device. This will be used throughout the SPAD.neXt UI for everything related to that device
<SerialVersion>	Serialversion the device is supporting (Current Version is 2)

`<Guid>`	**Unique Identifier identifying the device. It should follow the GUID-Registryformat e.g. "{A8AA15C5-7BB6-4AC6-A558-A88CAFB78729}". To create a GUID you can use this site: https://www.guidgenerator.com/online-guid-generator.aspx (enable braces and hyphens)**
<DeviceVersion>	Version of the device (Major.Minor[.Build.Patch])
<opt1>,..., <optN>	optional device option Key=Value pairs (see OPTIONS)

INIT-Options

available only in INIT Command.
(all additional options can be added as well)

Key	Value	
AUTHOR	<your authtorkey>	
ALLOWLOCAL	0|1|2	where SPAD.neXt will look for device configuration files (default: 0)
PID	string	A unique **short** product id for the device.
SERIAL	string	optional serial number of device

> If PID option is provided, the AUTHOR option must be provided as well

AUTHOR

To receive your unique authorkey, issue the command "!deviceinfo" on the SPAD.neXt discord.
This key will identify the device author and enable author-only functions like e.g. editing the device UI or device configuration database, if the current SPAD.neXt user is the device author.

> The authorkey is case-sensitive!

ALLOWLOCAL

The ALLOWLOCAL option controls where SPAD.neXt will search for device configuration files/images etc. before checking the database.
&#xNAN;*Documents* refers to the Documents-Directory from SPAD.neXt configuration.
Default: "Documents/SPAD.neXt"-Folder of Windows user

SPAD.neXt will search for files/images in the following order

Value = 0 (default)

SPAD.neXt will not search for any local files and will always use

1. Internal database
2. Device provided data

Value = 1

1. *Documents*/devices/local/<VID>/<PID>/
2. Internal database
3. Device provided data

Value = 2

if the current SPAD.neXt user is the author of the device, else it will behave like 0

1. *Documents*/devices/<AUTHOR>/<VID>/<PID>/
2. *Documents*/devices/local/<VID>/<PID>/
3. Internal database
4. Device provided data

> if no local configuration exists (or is not allowed), the device sent-configuration will be used always.
> if a local configuration exists (and is allowed) it will always overwrite any device sent configuration!

VID

The VID (or Vendor ID) defines the **author** of a device uniquely in a similar manor to USB devices.

The VID cannot be chosen freely, but has to be requested by opening a ticket, to prevent duplicates. The VID is provided and set automatically by SPAD.neXt once it has been assigned.

PID

The PID (or Product ID) defines a **device** uniquely in a similar manor to USB devices.

The PID can be chosen freely and should be unique per device(type) as it will be also used to find the corresponding device profile within a SPAD.neXt profile.

> The PID will be used in SPAD.neXt all over the place (e.g. Variable names), so **keep it short and simpl**e. The only allowed characters for the PID are A-Z,0-9 (no spaces, no special chars)

SERIAL

If provided the serial number of the device (unique per device connected!) to identify device variables (ADD Command). If not provided or a duplicate is found, SPAD.neXt will use an internal unique identifier

Device variables

if PID is defined for a device, all its inputs will be available as LOCAL variables within SPAD.neXt as
`LOCAL:<VID>_<PID>_<TAG>[suffix]`
[suffix] will only be present if more than one identical device is connected

Examples

`LOCAL:SHAKEPRINT_ECHO_BUTTON1`
`LOCAL:C0NNEX_FCU_I_AP1`
`LOCAL:C0NNEX_FCU_I_AP1_2` (second FCU device)

INIT Examples

Request from SPAD.neXt:
`0,INIT,2,0.9.10.0,73993732632;`
Reply from device:
`0,SPAD,{A8AA15C5-7BB6-4AC6-A558-A88CAFB78729},Altimeter Display,2,1.0;`
(No one will be able to locally modify the device configuration)

Request from SPAD.neXt:
`0,INIT,2,0.9.10.0,73993732632;`
Reply from device:
`0,SPAD,{A8AA15C5-7BB6-4AC6-A558-A88CAFB78729},Demo,2,1.0,AUTHOR=a3dhfc4323dca,PID=Echo`
The user associated with authorkey a3dhfc4323dca will be able to edit the device configuration. SPAD.neXT will first search in `Documents/SPAD.neXt/devices/a3dhfc4323dca/shakeprint/echo` for local configurations

Device Configuration-Phase

After the device initialisation SPAD.neXt will send the command

`0,CONFIG[,Parameters];`

to the device and start the configuration Phase.

The device can now issue commands to configure the device.

The configuration phase will end when the device acknowledges with

`0,CONFIG;`

> The device will not show up in SPAD.neXt UI before configuration phase is completed

Device options

Set device specific options

Syntax: `0,OPTION,<Key>=<Value>[,<Key2>=<Value2>,...];`

Simple Device options

Syntax: `Optionkey=Value`

Key	ValueType	Description
OUT_COOLDOWN	number	Time in milliseconds SPAD will wait after each sent command/data to give the device time to process it (Default: 25ms)
PROFILE_RO	0|1	If the device profile cannot be edited by the user
PROFILE_ENC	0|1	the profile part of the device will be encrypted
ISGENERIC	0|1	Device will not provide any UI, SPAD wil generate a generic UI
PAGESUPPORT	0|1	If the device supports profile pages
NO_DISPLAY_CLEAR	0|1	Disable clearing of displays on page changes
NO_LED_CLEAR	0|1	Disable turnoff of led on page changes
NO_LABEL_CLEAR	0|1	Disable Label reset on page change
UI_TYPE	0|1|2|3	0 = SPAD Generic UI 1 = Joystick UI 2 = MCP/FCU Style UI

Key	ValueType	Description
		3 = Custom UI (can be overwritten per INPUT)
DEFAULT_PANEL	string	Name of the Default UI Panel if UI is a split panel
TARGET_PANEL	string	name of the target ui panel of an in-/output
NOECHO	0|1	Disable echoing back changes of Device Data (ADD Command)
DISPLAY_STATEUPDATE	0|1	Resend all displayvalues if display is turned on (default: false)

Complex Device Options

Device COLORSET configuration

Defining a custom colorset for LED

Syntax: `0,COLORSET,<Key>,<Name>=<Value>[,....,<Name>=<Value>]`

`Key` contains the identifier for this colorset.

The `Name=Value` pairs contain

`Name` being the displayname of the color in the SPAD.neXt UI

`Value` being a valid color value. A valid color is either a named color (as defined in KnownColor) or a #RRGGBB hexadecimal value (e.g. AliceBlue , #FF0000)

> Each color set must not contain a definition for OFF. It will be generated automatically and is internally fixed.

Examples

`0,COLORSET,1,Red=Red,Green=Green,Yellow=Yellow,Blue=#0000FF`

`0,COLORSET,MyCoolSetName,Red=Red,Green=Green,Yellow=Yellow,Blue=#0000FF`

Device PROFILE configuration

Device default profile configuration

Defines a device profile that shall be offered/loaded as default for the device, if no device profile exists in the current SPAD.neXt profile (Device not configured dialog)

Syntax: 0,PROFILE,<SourceType>,<id>

SOURCETYPE

0	<ID> will contain snippet-id(s) of a complete device snippet to load from SPAD.neXt online database
1	<ID> will contain one snippet-id of a device snippet to load from SPAD.neXt online database, but the user will not be asked for confirmation.
2	<ID> will contain one snippet-id of a device snippet to load from SPAD.neXt online database. Any existing device profile in currently loaded SPAD.neXt profile will be **overwritten**.
3	<ID> will contain an url that will be downloaded. The url must return a valid SPAD.neXt device profile. If the url is not reachable or invalid, no default profile will be loaded/offered.

To provide more than one snippet-id for the user to choose from, separate the id's by `#`
`e.g. 123#456#12345`

> This command is only valid during the configuration phase of the device or from within the device configuration xml

Examples

`0,PROFILE,0,4567`

`0,PROFILE,3,http:////pastbin.com//a4JhSw`

> `Note the escaped '/' in the URL!`

Device OUTPUT Configuration

Define a device output descriptor (SPAD -> Device)

For defining an output descriptor the following command syntax is used:

```
0,OUTPUT,<DeviceIndex>,<Tag>,<Type>,<Inherits>[,<Option>=<Value>,...];
```

<DeviceIndex> is the numerical index the output should be used by SPAD when sending updates to the device (See LED / DISPLAY update). For Led commonly the pin on the device can be used as <DeviceIndex>. This way no more internal processing needs to be done on the device.
The <DeviceIndex> should be unique per <Type>

<Tag> being the internal name SPAD.neXt should use for events bound to that output descriptor. <Tag>'s must be unique device wide. It's generally a good idea to give the <TAG> a prefix (e.g. `L_` for led and `D_` for Displays)

<Type> the type of the output descriptor
Currently there are two types of output descriptors supported:

`LED` an led that has either only a ON/OFF state or a multi color LED

`DISPLAY` any kind of display

<Inherits> is a list of base configurations the output shall use within SPAD.neXt. This decides about the events available for the output and the behaviour.

Device output LED

A led is a simple output descriptor that always has at least the state ON or OFF

Available <Inherits> for led

`SPAD_LED` a simple ON/OFF led

`SPAD_LED_3COL` a 3 color led predefined as Red,Green,Yellow

`SPAD_LED_C` a led with custom defined colors. The option `COLORSET=<Colorset_Index>` must be added to the command to inform SPAD.neXt about the available colors.

Available specific options

DYNLABEL=1	Enabled updates of the led-labeltext (UI_FACE=2) via events	
UI_FACE=0	A round led that changed fillcolor depending on it's state	
UI_FACE=1	A rectangular led that changes foreground-color depending on it's state	`FOREGROUND=Red`
UI_FACE=2	A rectangular led that can change fore/background depending on it's state	`FOREGROUND.ON=Red` `FOREGROUND.OFF=LightGray` `BACKGROUND.ON=Transparent` `BACKGROUND.OFF=Transparent`
UI_FACE=3	Use custom provided images	
NOIMG=1	No background image	
COL_0	Color of OFF in UI (UI_FACE != 3)	LightGray
COL_1	Color of ON in UI (UI_FACE != 3)	Yellow or #FFFF00
IMG_OFF	Image for OFF (UI_FACE = 3)	_PanelImages/LED_OFF.png
IMG_ON	Image for ON (UI_FACE = 3)	_PanelImages/LED_ON.png

Examples

`0,OUTPUT,1,L_LED1,LED,SPAD_LED;` defines a ON/OFF led with tag `L_LED1` and index 1

`0,OUTPUT,2,L_LED2,LED,SPAD_LED_3COL;` defines a 3-color led with tag `L_LED2` and index 2

`0,OUTPUT,3,L_LED3,LED,SPAD_LED_C,COLORSET=1;` defines a custom led with tag `L_LED3` and index 3 using the defined colorset 1 (See COLORSET)

Device output DISPLAY

A display is any kind of alphanumerical output. It can e.g. be a 7-Segment lcd or a complex display with 5 rows of 20 characters each

The only available <Inherits> for a display is

`SPAD_DISPLAY`

All configuration is done via the options:

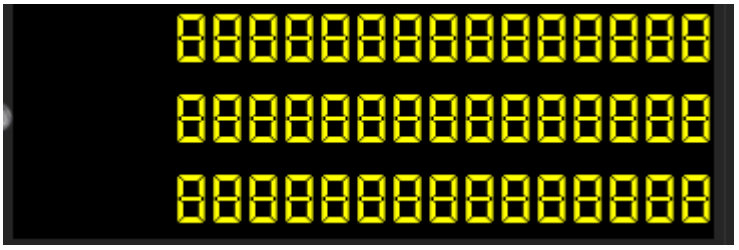
Option	Value	Description
LENGTH	integer	Length of each row in characters
ROWS	integer	number of rows
SEGMENTS	integer	number of segments per row. Each segments will be LENGTH/SEGMENTS width (Default: 0)
DEFAULT	any	Default value to display
NOPADDING	0|1	Disable padding and alignment of output values

Option	Value	Description
NOSEGMENTROWEVENTS	0|1	SPAD will only send updates per row, not per segment
SEGMENTALIGN	LEFT|RIGHT (... per segment)	Alignment of segment value (Default: Right)
TEXTALIGN	LEFT|RIGHT	alignment of row (Default: right)
`FONT`	string	Name of the font to use (Default: Builtin LCD Font)
FOREGROUND	#RRGGBB	Foreground color (Default: Blue #FFF0F8)
BACKGROUND	#RRGGBB	Background color (Default: Black #000000)
FONTSIZE	number	Size of the UI font (Default: Height of one display row)
HEIGHT	number	Height of display in pixels (UI)
WIDTH	number	Width of display in pixels (UI)

Examples

`0,OUTPUT,1,D_1,DISPLAY,SPAD_DISPLAY,LENGTH=6,ROWS=1,WIDTH=133,HEIGHT=40`

`0,OUTPUT,1,D_1,DISPLAY,SPAD_DISPLAY,LENGTH=15,ROWS=3,WIDTH=350,HEIGHT=120`



`0,OUTPUT,1,D_1,DISPLAY,SPAD_DISPLAY,LENGTH=24,ROWS=12,WIDTH=400,HEIGHT=300,FONTSIZE=16`

Other outputs

For any other kind of output (e.g. a servo) no output needs to be defined. Just provide a RW-data for the output using the ADD command and act on the device accordingly to changes to that data

Device INPUT configuration

Define a device input (Device -> SPAD)

`0,INPUT,<Index>,<Tag>,<Type>[,<Inherit>[,<OPTIONS>]];`

All input definition follow the same definition.

<Index> being a numeric index that is used for sending input updates to SPAD.neXt

<Tag> being the internal name of the input. This will be used for identifying actions bound to this input

<type> being the internal type of the input

<inherit> the default events to include in this input. if it's not provided the definition must provide the custom events.

<options> optional options to alter the behavior/ui of an input

Options supported for all Inputs and Outputs

NO_LABEL=1	Disable the state-value label above the input in SPAD.neXt standard UI	
ORDER=<number>	Defines the ordering in SPAD UI. Default 0 = All elements are shown in order as they arrive	

Sending INPUT updates

For all inputs the devices uses the channel 8 to send updates to SPAd.neXt:

`8,<index>,<value>;`

<index> being the index given in the input configuration

<value> being the new value

Device input AXIS

Define a device axis

`0,INPUT,<index>,<tag>,AXIS,<Inherit>[,<options>];`

Available Inherits

`SPAD_AXIS` defines a SPAD.neXt standard axis with calibration and responsecurve support

Available Options

AXIS_MIN	Axis minimum value	default: 0
AXIS_MAX	Axis maximum value	default: 1024
DIRECTIONFUNC	Value expression to convert normalized value to 0-100% UI value	default: value*100.0

`AXIS_MIN` and `AXIS_MAX` are used to calculate the normalized value (0 .. 1.0) of the axis.

> MIN can be negative if device has a auto-center axis. E.g (-512 .. 512) for a rudder

The 8-Input commands must send absolute raw values (Min ... Max) and not a normalized value!

Device input ENCODER

Define a stateless encoder

`0,INPUT,<index>,<tag>,ENCODER,<Inherit>[,<options>];`

Available Inherits

`SPAD_ENCODER_NOACC`	a single encoder with Clockwise/Counterclockwise events	no acceleration
`SPAD_ENCODER`	a single encoder with Clockwise/Counterclockwise events and acceleration	
`SPAD_DOUBLE_ENCODER_NOACC`	a double encoder with CW/CCW events for inner and outer knob	no acceleration
`SPAD_DOUBLE_ENCODER`	a double encoder with CW/CCW events for inner and outer knob and acceleration	

Sending encoder input updates

`8,<index>,<directionvalue>;`

Single Encoder

send -1 for counterclockwise turn

send 1 for clockwise turn

Double Encoder

Inner Encoder `-1 / 1`

Outer Encoder `-100 / 100`

Device input PUSHBUTTON

Stateless push button

`0,INPUT,<index>,<type>[,<inherit>[,<options]];`

Pushbutton with Press / Press short / Press long / Release events

`0,INPUT,<Index>,<TAG>,PUSHBUTTON,SPAD_PUSHBUTTON[,<OPTIONS>]];`

Simple Pushbutton with Press and Release events

`0,INPUT,<Index>,<TAG>,PUSHBUTTON,SPAD_SIMPLEBUTTON[,<OPTIONS>]];`

Available inherits

SPAD_PUSHBUTTON	a standard SPAD.neXt push button with `PRESS / PRESSSHORT / PRESSLONG / RELEASE / HELD` events	
`SPAD_SIMPLEBUTTON`	a simple pushbutton with PRESS / RELEASE Events	

Available options

No specific options

Sending button input updates

`8,<index>,<statevalue>;`

statevalue = 0 button released

statevalue = 1 button pressed

Device input ROTARY

Define a stateful rotary switch

`0,INPUT,<index>,<tag>,ROTARY,<inherit>,<options>;`

For a stateful rotary the states have to be defined in the options.

In SPAD.neXt standard UI the rotary is visualized as following (uivalue = 0):

`IMG=svg/rotary.svg`

Available Inherits

`SPAD_ENCODER_NOACC`	provide additional Clockwise/Counterclockwise events	Acceleration not supported
`<EMPTY/OMITTED>`	No tuner events	

Options

POS_NAMES=statename1#statename2#....	a list of the statenames of the rotary separated by "#" (hash)	POS_NAMES=OFF#LEFT
POS_VALUES=val1#val2#val3....	a list of the statevalues the device	POS_VALUES=0#1#2#3

	will use in the input events of the rotary separated by "#" (hash)	
POS_UIVALUES=val1#val2#val3....	the rotation ui values for the states (real rotation = 15*value degrees)	POS_UIVALUES=-3#0#3

Sending rotary input updates

`8,<index>,<statevalue>;`

e.g to tell SPAD.neXt that the rotary (index 5 tag E_MAGNETO) has been changed to BOTH-Position the device will send a

`8,5,3;`

command.
SPAD.neXt will execute the events

`E_MAGENTO.BOTH`

`E_MAGNETO.TUNER_CLOCKWISE` if previous position was RIGHT and encoder inherit was present

and rotate the UI-representation absolute by 6 * 15 = 90 degrees

Device input SWITCH

Define a stateful device switch

`0,INPUT,<Index>,<TAG>,SWITCH[,<Inherit>[,<OPTIONS>]]`

<Inherit>

The following inherits are available for switches to control the behaviour and available events:

SPAD_SWITCH	an ON/OFF switch	

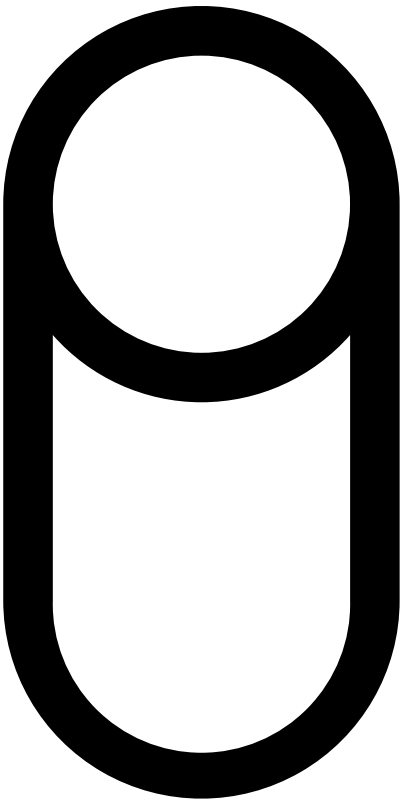
If no inherit is present, the switch will not provide any standard events. The events have to be defined separately in the options

<Options>

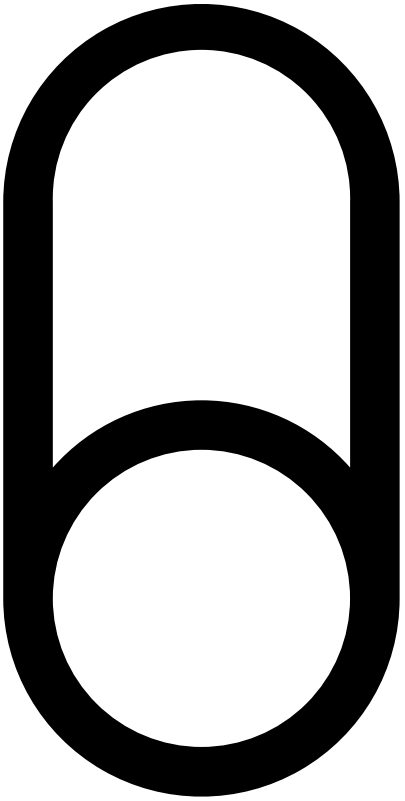
For customizing the switch UI see (CUSTOM UI).

Default images used:

`IMG_ON`` = *svg/switch_1.svg*`



`IMG_OFF` = *svg/switch_0.svg*



For the SPAD.neXt standard UI of a switch the following options are supported:

ROTATE= <angle>	rotate switch-ui	ROTATE=90 (create left to right switch) ROTATE=180 (create switch with OFF-Position is up instead of down)

Sending switch input updates

`8,<index>,<statevalue>;`

statevalue = 0 switched to OFF

statevalue = 1 switched to ON

Device input SWITCH3

Define a stateful 3-Position (On1-Mid2-On0) switch

`0,INPUT,<index>,<tag>,SWITCH3[,<inherit>[,<options]];`

Mandatory options

POS_NAMES=name0#name1#name2	Event names of the 0/1/2 states	POS_NAMES=DOWN#UP#OFF

UI customization

`IMG_0= svg/switch_0.svg`

`IMG_1= svg/switch_1.svg`

`IMG_2= svg/switch_2.svg`

Sending switch3 input updates

`8,<index>,<statevalue>;`

statevalue = 0 switched to ON0 position

statevalue = 1 switched to ON1 Position

statevalue = 2 switched to MID2 Position

Device LABEL

`0,LABEL,<index>,<tag>[,options];`

Creates a label on the UI

> By default labels are page aware and will store their current value if the page is changed and restore it again, when the page is switched back.

Available options

DYNLABEL=0|1	1 enables dynamic labeltext updates by device or events 0 adds a static labeltext	
READONLY=1	labeltext can only be changed by device	
LABEL=<text>	default labeltext	
[Generic UI Options]		

Sending label updates

`8,<index>,0,<newtext>;`

Updates the labeltext of the label

Device config: Advanced topics

Add functionality to inputs and outputs

Input Routing

Basically every device input is a separate control for SPAD.neXt, however you can re-route inputs in the SPAD.neXt UI to add functionality to one input in the UI.

An additional input is created, and the following options are added to it:

HIDDEN=1	Input will not be exposed in the UI	
ROUTETO=<targettag>	Events from this input will not be handled by the input, but routed to the input associated with <targettag>	ROUTETO=E_ENCODER

Example: An Encoder with a Pushbutton

First the Encoder is configured

`0,INPUT,2000,E_HDG,ENCODER,SPAD_ENCODER;`

Then the pushbutton is configured, telling SPAD.neXt to route it to the encoder

`0,INPUT,7,I_HDG_PUSH,PUSHBUTTON,SPAD_PUSHBUTTON,HIDDEN=1,ROUTETO=E_HDG;`

SPAD.neXt will now add the pushbutton-events to the encoder in the UI, and not expose the pushbutton I_HDG_PUSH in the UI at all. Whenever the device sends now an input event (`8,7,1;`) for that pushbutton, the according events on the encoder will be executed.

Example: An encoder that can be pulled out as a switch

Again, just configure the encoder and the additional switch

`0,INPUT,2000,E_HDG,ENCODER,SPAD_ENCODER;`
`0,INPUT,7,S_HDG_PULL,SWITCH,SPAD_SWITCH,HIDDEN=1,ROUTETO=E_HDG;`

Complex Examples

Example: Encoder that can be pushed and pulled as a pushbutton

Now things get a bit more complicated.

First, as before the encoder is configured

`0,INPUT,2000,E_HDG,ENCODER,SPAD_ENCODER;`

Now 2 additional inputs shall be attached to the encoder that will use the same events. SPAD.neXt needs to know what to do and how to separate the events. To achieve this, additional options can be added to the input(s) to create different events for each of them:

Option	Default	
`ROUTETO.PRESS`	PRESS	
ROUTETO.PRESSSHORT	PRESSSHORT	
ROUTETO.PRESSLONG	PRESSLONG	
ROUTETO.HELD	HELD	
ROUTETO.RELEASE	RELEASE	

Literally every event from an input can be rerouted this way. For a switch the options would be `ROUTETO.ON` , `ROUTETO.OFF` , for stateful input (rotary/3-way) those would be `ROUTETO.<positionname>`

> The option-value will be used as event name in SPAD.neXt. If it does not exist yet, it will be automatically created.

But back to the example, now create the 2 additional input buttons (simple ones to keep this short and simple):

```
0,INPUT,7,I_HDG_PUSH,PUSHBUTTON,SPAD_SIMPLEBUTTON,HIDDEN=1,ROUTETO=E_HDG,ROUTETO.PRESS=PUSH_PRESS,ROUTETO.RELEASE=PUSH_RELEASE;
```

```
0,INPUT,8,I_HDG_PULL,PUSHBUTTON,SPAD_SIMPLEBUTTON,HIDDEN=1,ROUTETO=E_HDG,ROUTETO.PRESS=PULL_PRESS,ROUTETO.RELEASE=IGNORE;
```

> Note that `INHERIT` is omitted for the button to prevent SPAD.neXt from adding default events to it.

For the pull-button SPAD.neXt is instructed to ignore the release event and only support the push event. The value "IGNORE" can also be omitted which will have the same effect.

In the SPAD.neXt UI the E_HDG encoder will now have the events `CW / CCW / PUSH_PRESS / PUSH_RELEASE / PULL_PRESS`

> Note: This can become quite long configuration commands and should be done in the device editor/ device xml configuration to preserve device space

> Important!
> Routed inputs still have their own variables to hold their current value!

Example: Airbus FCU Altitude encoder

The Airbus FCU altitude encoder has following additional functions:
Push , Pull and a ring-switch for 100 / 1000 steps

```
0,INPUT,1000,E_ALT,ENCODER,SPAD_ENCODER;
```

```
0,INPUT,1,I_ALT_PUSH,PUSHBUTTON,SPAD_SIMPLEBUTTON,HIDDEN=1,ROUTETO=E_ALT,ROUTETO.PRESS=PUSH,ROUTETO.RELEASE=;
```

`0,INPUT,2,I_ALT_PULL,PUSHBUTTON,SPAD_SIMPLEBUTTON,HIDDEN=1,ROUTETO=E_ALT,ROUTETO.PRESS=PULL,ROUTETO.RELEASE=;`

`0,INPUT,3,S_ALT_RING,SWITCH,SPAD_SWITCH,HIDDEN=1,ROUTETO=E_ALT,ROUTETO.ON=ALT100,ROUTETO.OFF=ALT1000;`

The E_ALT encoder will have the events

`CW / CCW / PUSH / PULL / ALT100 / ALT1000`

> Note that the INHERIT for the buttons `/switch` are omitted, to prevent SPAD.neXt to add any default events to the buttons.

Output routing

The other way around is an output that physically located on a button (e.g. MCP / FCU). Here SPAD.neXt needs to know not to expose the led in the UI, but add it to another input, while keeping the output events separated. This is currently only supported for LEDs.

Again the led will need to have the `HIDDEN=1` option so it will not be exposed in the UI, but now the button needs an option to associate it with the led. This is done by adding the option `LED=<ledtargettag>` to it.

Example: FCU AP1 button

`0,INPUT,8,I_AP1,PUSHBUTTON,SPAD_PUSHBUTTON,LED=L_AP1;`

`0,OUTPUT,5,L_AP1,LED,SPAD_LED,HIDDEN=1;`

This will add the led-control actions to the button, but when executed they will be routed to the `L_AP1` led using the led-update channel (6)

Device STATESCAN-Phase

`0,SCANSTATE;`

this command tells the device that SPAD.neXt is now ready to receive the current state of stateful inputs (switches/rotaries/axis)

if the device does **not** support scanning of the states reply with `0,STATESCAN;`

if the device supports scanning of stateful inputs reply with

`0,STATESCAN,1;`

After the STATESCAN reply, issue any Device Input commands (8)
if any events will be executed in statescan phase is defined in the SPD.neXt profile

and finally send a

`0,STATESCAN,2;`

to notify SPAD.neXt that the state-scanning has completed

Device General Commands

General Command (SPAD.neXt <--> Device) on Channel 0

PING

Every 10 seconds SPAD.neXt will send a PING to the device and expects a PONG in response containing the received TICKET as payload.

> If no PONG is received for longer than 30 seconds, SPAD.neXt assumes the device is offline.

Request: `0,PING,<TICKET>;`
Reply: `0,PONG,<TICKET>;`

END

When SPAD.neXt exits it will send an `0,END;` command to the device, so the device knows SPAD.neXt is exiting.

> There is no guarantee that this command is sent, and SPAD.neXt will not wait for a reply and close the communication immediately

Request: `0,END;`
Reply: none

Device commands

The 1-Command is used by the device to send commands to SPAD.neXt.

Request: `1,<SUBCOMMAND>,...;`
Reply: none or 2,ERROR

Provide Data to SPAD.neXt

The subcommand `ADD` informs SPAD.neXt about a data value the device will provide. This data can be used in events or expressions like any simulation data in the SPAD.neXt UI.

`1,ADD,<CHANNELID>,<path>,<valuetype>,<access>,<name>[[,description][,options]];`

Parameter	Description	Limits
`<CHANNELID>`	the Channel the device/SPAD.neXt will use to send updates for that data	>= 10
<path>	name of the data value. To place it in a tree "//" can be used to separate the nodes. e.g. `buttons//button1` (**note the escaped "/" !!** If using CmdMessenger-Lib the escaping will be done automatically for you)	name must not start with /
<valuetype>	type of the data (**S**igned/**U**nsigned)	`S8` `S16` `S32` `S64` `U8` `U16` `U32` `U64` `FLT32` `FLT64` `ASCIIZ`

Parameter	Description	Limits
<access>	**RO** for readonly, **RW** for readwrite	
<name>	human readable name of the data for the UI	
<description>	optional description for the UI	

Supported options

PERSIST	0|1	0 create LOCAL variable (default) 1 Create a persistent DEVICE variable

PERSIST=1

Within SPAD.neXt the data will be available in Device->Devicename.
Internally the Data is referenced as `DEVICE:{DeviceId}/<path>`
e.g. DEVICE:`A8AA15C5-7BB6-4AC6-A558-A88CAFB78729/buttons/button1`

Device Data is only available within the profile of the device.

Getting and setting

SPAD.neXt will send data updates to the device and expect data using the given CHANNELID channel using the 5-Channel: 5,`<CHANNELID>,<value>;`

Examples

Persistent Device Variable

`1,ADD,11,pages//activepage,U8,RW,Active device page,My cool description,PERSIST=1;`

creates a variable `DEVICE:VID/PID/pages/activepage` available only within the deviceperofile of the device. outside of the deviceprofile this variable will always have the value 0

Normal non-persistent local variable

`1,ADD,11,pages//activepage,U8,RW,Active device page;`

creates a variable `LOCAL:VID/PID/SERIAL/pages/activepage` available to all deviceprofiles within SPAD.neXt

> DEVICE variables persist for each connected device. Their values are loaded from and saved to the profile.
>
> LOCAL variables will not be saved and default to 0

Subscribe to a data

the subcommand `SUBSCRIBE` with subscribe the device to a data value from SPAD.neXt or the simulation

`1,SUBSCRIBE,<index>,<path>[,<unit>[,<epsilon>]];`

`<index>`	the index associated with the data
<path>	internal name of the data value to subscribe to. E.g. `SIMCONNECT:AUTOPILOT HEADING LOCK DIR` `XPLANE:SIM/COCKPIT/AUTOPILOT/HEADING_MAG`
<unit>	desired unit, if not default (e.g. 'degrees','radians','Hz','kHz','MHz')
<epsilon>	minimum change that must happen before an update will be sent to the device. If it's omitted the default value for the data will be used

> SPAD.neXt will send the subscribed data on the 5-Channel in the format
>
> `5,<index>,<value>;`

Example
1,SUBSCRIBE,12,SIMCONNECT:AUTOPILOT HEADING LOCK DIR
1,SUBSCRIBE,12,SIMCONNECT:AUTOPILOT HEADING LOCK DIR,radians
1,SUBSCRIBE,12,XPLANE:sim//cockpit//autopilot//heading_mag,,5

> You can get the internal name of a data value, by selecting it in the data browser and pressing `CTRL-C` or clicking on "book"-icon in the data monitor. It will be copied to the clipboard

Unsubscribe from a data

The subcommand `UNSUBSCRIBE` will unsubscribe from a previously subscribed data value
Syntax: `1,UNSUBSCRIBE,<index>;`

Emulate an event on a different device

The subcommand `EMULATE` will emulate an event from another device. (E.g. a dial turned on radio panel)

`1,EMULATE,<TARGET_DEVICE>,<TARGET_SWITCH>,<TARGET_EVENT>;`

`<TARGET_DEVICE>` is either the internal id `VID:PID:COUNTER` or `VID:PID:SERIAL`
`<TARGET_SWITCH>` is the internal name of the switch/dial/led etc (e.g. FIP_LEFTDIAL)
`<TARGET_EVENT>` is name of the event to emulate (e.g. TUNER_CLOCKWISE)

To get the TARGET_SWITCH and TARGET_EVENT you can program the event in the UI and then copy the complete event to the clipboard, pasting to a texteditor.
The TARGET_SWITCH will be the "BOUNDTO" and the TARGET_EVENT the "Trigger"

Alternatively you can get a list of all devices and events at Settings->Application->Expert->Available Remote Events

Update the labeltext of an input/output or label

`1,LABEL,<index>,<newlabel>;`

Updates the labeltext of a input/led (DYNLABEL=1 !) or a label pseudo input

Ask SPAD.neXt to resend all subscribed data

`1,REFRESHDATA;`

SPAD.neXt will send all currently subscribed data

Page-Control

Serial: Configuration (1) Page-Control

If the device supports paging (see OPTIONS), the current page being active can be controlled from the device

SWITCHPAGE: Switching to a specific Page

Syntax: `1,SWITCHPAGE,<Guid>[,<create>,<name>];`

<Guid>	GUID of the target page
<create>	0 | 1 , create page if it doe snot exist yet
<name>	name of the new page

SPAD.neXT will either respond with PAGE-Events or a ERROR

GOTOPAGE: Navigate absolute or relative

Syntax: `1,GOTOPAGE,<number>,<isrel>;`

Navigate (absolute) (<isrel> = 0)

<number> contains the absolute pagenumber (1..n)

Navigate (relative) <istrel> = 1

<number> contains the number of pages to navigate left (< 0) or right (> 0)

a rollover to first/last page will be done automatically

Device SPAD.neXt Events (2)

SPAD.neXt raises internal events on Channel 2

The channel 2 is used by SPAD.neXt to inform the device about events. There are some events that the device might ignore, and some that it should react to.

`2,START;` -> The device can send data updates now , if any
`2,STOP;`-> The device should stop sending data updates
`2,AIRCRAFTCHANGED,<new name>;` -> the loaded aircraft changed

`2,PROFILECHANGING,<new profile name>;` -> the profile is about to change
`2,PROFILECHANGED,<new profile name>;` -> the loaded SPAD.neXt profile changed and is active
`2,CONNECT,<simulation name>;` -> a simulation connected to SPAD.neXt
`2,DISCONNECT,<simulation name>;` -> a simulation disconnected from SPAD.neXt

2,PAGE,<GUID>,<completed>,<name>; -> Switch to page

2,VIRTUALPOWER,<onoff>

2,LABEL,<index>,<newlabel>;

2,PROVIDER,<name>,<status>; -> The status of a provider has changed

When the profile is switched in SPAD.neXt the following events will be sent to the device:

1. PROFILECHANGING
2. PAGE (completed = 0)
3. PAGE (completed = 1)
4. PROFILECHANGED

(2 and 3 only if device supported pages)

Device Simulation Events

The channel 4 is used by the device to send commands/events to the connected simulation.
Request: `4,<SIMNAME:EVENTNAME>[,<Value Default=0>[,<Value2>];`
Reply: none

`<SIMNAME:EVENTNAME>` is the event to send to the simulation. e.g. SIMCONNECT:FLAPS_UP. The "SIMNAME:" part can be omitted, then SIMCONNECT: is assumed.
`<Value>` can be an int value to send with the command

SIMCONNECT: Events/Commands will be automatically transcribed to the according connected Simulation (e.g. XPLANE)

MSFS / H:Events: e.g. MSFS:AS1000_MFD_DIRECTTO

Device Data (Channel 5)

`5,<index>,<value>;`

Device LED Update (6)

SPAD tells the device to change the status of a LED

`6,<LEDINDEX>,<ONOFF>,<TAG>[,<COLORNAME>,<COLORVALUE>];`

<LEDINDEX> is the index the device defined in configuration phase

<ONOFF> will be 0 or 1 to turn the led off or on

<TAG> is the tag the device defined in configuration phase

<COLORNAME> and <COLORVALUE> are only present for colored led's (3-State or ColorSets)

Device Display Update (7)

Update device display

`7,<DISPLAYINDEX>,<ROW>,<COMMAND>,<VALUE>;`

<DISPLAYINDEX> is the index the device defined in configuration phase

<ROW> is the row-index that should be updated

<COMMAND>

0 Turn display off

1 Turn display on

2 Update display value

<VALUE> the actual value to display

Device Input Updates (Channel 8)

See the according INPUT definitions

Device Virtual Power

Virtual Power Support for serial devices

To enable Virtual Power support for the device, the following steps have to be completed:

- [ ] Add the Option VPSUPPORT=1
- [ ] Define an INPUT with
  - [ ] DeviceIndex 0
  - [ ] Type `SYSTEM`
  - [ ] Tag `CONFIGURE_PANEL_STATUS`
  - [ ] Inherit `SPAD_VIRTUAL_POWER`
  - [ ] OPTION `UI_TYPE=3 CUSTOM_TYPE=POWER`
  - [ ] (in Alpha < 0.9.12.38) `WIDTH=20 HEIGHT=20`

This will add the SPAD.neXt standard power-configuration button to the device UI, and the device will be treated like any other device that supports virtual power.

Example

```
0,INPUT,0,CONFIGURE_PANEL_STATUS,SYSTEM,SPAD_VIRTUAL_POWER,UI_TYPE=3,CUSTOM_TYPE=POWER
```

Device Custom UI

Providing a Custom UI for a device

All custom UI is defined as SVG or PNG images. Transparent PNG are preferred.
Limitations:
* Maximum size is 640x480 pixel
* Images must have a DPI of 96. Else windows automatically will upscale them when rendering. E.g. Images has 72dpi => upscaling to 96dpi will cause the image dimensions to grow by 33% => Positions will not match what you expect them to be

SVG

You can use your favorite editor to create SVG images.

As a free online editor https://boxy-svg.com/ is a good choice

Also the repository at https://www.svgrepo.com/ is a good first stop if you are looking for free SVG icons and graphics, or ideas

All common SVG features can be used, but keep in mind that users may use a different theme. Make sure the UI looks good in both the light and dark themes of SPAD.neXt.
If a SVG does not define a fill-color the button text color of the chosen SPAD.neXt theme will be used.

PNG

PNG images should be transparent

USAGE

For development, make sure you set the ALLOWLOCAL option and store the svg/png-files in Documents/devices/<AuthorKey>/<VID>/<PID>/

- Subdirectories are supported
- For centralized images you can travel up two directories max. (../../gfx/button.svg)
- File-/Directorynames must not start with an underscore

- SPAD.neXt has some builtin images that can be used (TODO REF)

To have a custom UI for an input being displayed, the following options for an input must be set:

- [ ] WIDTH and HEIGHT should be set. (SVG/PNG will be rescaled to this size automatically, default: 32x32)
- [ ] TOP and LEFT can be set to define the position of the UI element relativ to the UI panel (default 0,0)
- [ ] UI_TYPE=3
- [ ] BEHAVIOR is set, if it differs from input-default
- [ ] IMG=<relativePathAndFilename> (See the BEHAVIOR for any special requirements)
- [ ] The type of the file will be determined by the extension. Only .svg and .png are supported

BEHAVIOR

The BEHAVIOR option defines how an UI element will behave in the UI if it's activated by a device event

Supported values

ROTATE

UI element will rotate around its center point, 15 degrees clockwise/counterclockwise per event

ROTATEABSOLUTE

UI element will rotate around its center point, to an angle of 15 * eventvalue degrees

ONOFF

Most commonly used for switches. Two images have to provided:

`IMG_ON` (shown when input is considered ON/PRESSED)

`IMG_OFF` (shown when input is considered OFF/RELEASED)

> When loading/initializing all inputs are treated as OFF

HIGHLIGHT

default behavior. The UI element will not be modified when activated, but it will be highlighted.

> If the input is a double encoder/rotary the options
> `IMG_INNER`
> `IMG_OUTER`
> have to be provided

Device HTML UI

Providing a custom HTML UI for the device

(TODO) not exposed yet

Device preconfigured definition

To take load from the device and save device rom/flash space the complete configuration of a device (see Configuration Phase) can be preconfigured.

The device.xml can either be created by hand (see device-xml-schema) or using the SPAD.neXt built-in device editor.

When using the device editor, the editor will automatically add all defined input/outputs to the ui and they "just" need to be moved around or new inputs/outputs can be added and configured.
To simplify the process of creating a basic device.xml without needing the device sending it's config at least once, the device-csv-importer can be used.

Once a `device.xml` is present it's save to remove all 0,INPUT/OUTPUT/OPTION-Commands from the device and just reply with a `0,CONFIG;` in the config phase.

Serial V2

Command 1,RAISE

1,RAISE,<InputName>,<Eventname>[,<EventValue>]

Command: 0,AUTH

Request 3rd Party Authentication of the Device

Contact us for further information about device authentication

demo-ino-snippet

serialdemo.ino see
https://github.com/c0nnex/SPAD.neXt/blob/master/Scripts/serialdemo_spad.ino

X-Plane Datarefs & Commands

> Since version 0.9.24 SPAD.neXt will scan for datarefs and commands automatically for X-Plane 12, if X-Plane and SPAD.neXt are running on the same computer and it is not necessary anymore to create datarefs.txt/commands.txt files.

Unfortunately there is no standard way in X-Plane to add data and commands to an aircraft, so every aircraft developer is doing it differently, and there is no way for SPAD.neXt to query those for X-Plane versions < 12

Scanning for Datarefs

DataRef Tool

Download DataRefTool .
Install it as a plugin.
Then start X-Plane, load the aircraft.
After exiting X-Plane there will be a X-Plane/preferences/output/drt_last_run_datarefs.txt and one for commands which should list all the aircraft ones.

The files have to be moved to the aircraft folder (see below) and be renamed.

Adding the datarefs and commands locally.

Those are only available for the local installation

Commands

Create a file "commands.txt" in the directory of the ACF-File of the targeted aircraft. List all commands you wan to expose to SPAD.neXt in this file. The Format of the File is the same as the standard X-Plane commands.txt you can find in "X-Plane 11\Resources\plugins"

Format:

```
command[TAB]description (optional)
```

Datarefs

Datarefs can be announced by placing them into a file "datarefs.txt" in the directory of the ACF-File of the targeted aircraft. The Format of the File is the same as the standard X-Plane datarefs.txt you can find in "X-Plane 11\Resources\plugins"
the First 2 lines of that file will be ignored (X-Plane Header)

Format:

```
dataref[TAB]type[TAB]writeable[TAB]unit[TAB]Description(optional)
sim/cockpit2/radios/actuators/com1_power        int     y       boolean
Com radio 1 off or on, 0 or 1.
```

dataref = name of the dataref e.g. sim/cockpit2/radios/actuators/com1_power
type = Type of the dataref typically "float" or "int". For dataref arrays add the size of the array e.g. "int[12]"
writable = "y" or "n" if the dataref is writable or readonly
unit = the unit that dataref is in. This is used as a hint for SPAD.neXt to offer conversions to the user

Limits:
* arrays are limited to 140 for performance reasons.
* if a dataref is not yet known to X-Plane when the plane is loaded, the SPAD.neXt plugin will retry to find it regularly and announce it to SPAD.neXt

Both files will be read by the X-Plane plugin when the aircraft is loaded.

Contributing Datarefs to SPAD.neXt

You can publish the collected datrefs and commands to SPAD.neXt for global availability via the online services menu.

FSUIPC: Custom Offsets

Adding custom offset to FSUIPC Data

FSUIPC Custom Offsets Sample XML

Please make sure you keep Version at 1

Each entry consists of one OffsetDefinition.
Attribute Selectable: If false, the offset will not be offered in the UI.
Name = Displayname of the offset in UI
ValueType + Size = One of

- S8 = Signed Byte , Size = 1
- U8 = Unsigned bytes, Size = 1
- U16 = Unsigned SHort, Size = 2
- S16 = Signed Short , Size = 2
- U32/S32 = Un-/signed Int , Size = 4
- U64/S64 = Un-/Signed long , size = 8
- FLT32 = Float , size = 4
- FLT64 = Double , size = 8

Category = Main Category in Definition Browser (first tree level)
SubCategory = Sub Category in Definition Browser (second tree level). If more levels needed, separate by "##"
Access = R -> read-only , RW -> Read/write , R will not be offered in e.g. Change Data value
Usage = Information shown in browser/mouseover. BBCode tags can be used here.
Key = FSUIPC Offset in HEXADECIMAL as long HI-DWord:LO-Dword , usually HI-DWord should be 0000. Must always be formatted 000:0000 . Special: if the Offset represent just a BIT in a Value the Key is 0000:0000:BITNUMBER , an offsetentry for the main offset with Selectable="false" must exist then.

optional Tag:
LinkedEntry = Offset that will trigger value change for this offset as well. See example. If 000:0022 changes SPAD.neXt will also trigger a value changed for 0000:0021. Justleave it away if not needed.

There some additional Tags which should not be necessary normally, e.g. for converting values on the fly (feet to meter etc). If you need those,let me know.

Save the XML to Folder "%APDDATA%\SPAD.neXt\conf\data_config\data" as "FSUIPC Local.xml"
It will be read ONCE when starting up. if you change it you have to restart SPAD.neXt.
If a problem occurs while reading the file, check the log file.

RestAPI

SPAD.neXt exposes a public ServiceStack Rest / web api which can be opened at http://127.0.0.1:28001

HTML Rendering API

For custom rendering (e.g. stream overlays or CDU Web-Interface)

not exposed to public yet

Other API's

Other API's available:

FIP Remote management

Allows complete remote management and image rendering onto FIP devices just like the Saitek DirectOutput Interface, but without the limitations (queuing of device updates) with up to 17 fps.
For specs please contact support@spadnext.com

General FAQ

Can I use my license on another computer as well?

If you have a Complete Edition license, you can request a free secondary license by submitting a support ticket.

If you have a Standard Edition license, deactivate it on the computer you are not using before activating it on the second computer. Otherwise, the license server may eventually blacklist it for license sharing.

All my profiles are suddenly empty

Most likely your Documents folder got migrated to OneDrive by a Windows update. Possible solutions:

a) In SPAD.neXt head to Settings->Directories and alter the Documents-Folder to the new OneDrive folder

b) Move the SPAD.neXt-Folder back from OneDrive to the old location and keep profile local

Does SPAD.neXt support device X?

If the device is listed in Supported Hardware, it is supported. If it appears as a joystick in Windows' USB Game Controllers, it is supported as a generic joystick. You can enable or disable joystick support in Settings → Devices → Joysticks.

Why does my Streamdeck show "not licensed" although it worked fine in trial?

Stream Deck support is a Complete Edition feature.
You need to upgrade your license to the Complete Edition (Settings → Registration) or disable Stream Deck support (Settings → Devices → Elgato). The free trial includes the same features as a Complete Edition license.

MSFS: Lost profile aircraft assignments

- exit SPAD.neXt
- open Windows File Explorer and navigate to %APPDATA%/SPAD.neXt/conf
- delete AircraftAssignments_MSFS.xml
- copy AircraftAssignments_FSX.xml to AircraftAssignments_MSFS.xml
- edit AircraftAssignments_MSFS.xml in notepad
- change Simulation="FSX" to Simulation="MSFS" in first line
- save and exit notepad, start spad.

For MSFS2024 the same procedure applies, just change Simulation="FSX" to Simulation="MSFS24" in first line and the file is called AircraftAssignments_MSFS24.xml

Running release and beta side by side

#Running release and beta side by side release and beta are config-compatible, no need to duplicate profiles, however it is strongly recommended to run Beta and Release separate from each other because a beta is a beta and bugs can and will happen. It will contain bugs, and depending on current source state it might take a week or two until a game-breaking bug is is fixed.

Preparations

- Create a subdirectory somewhere where the beta-version will live (e.g. D:\SPAD-Beta)
- Create a "bin" subdirectory (D:\SPAD-Beta\bin)
- Create a "logs" subdirectory (D:\SPAD-Beta\logs)
- Create a "conf" subdirectory (D:\SPAD-Beta\conf)
- Optional: If you want separate beta profiles:
  - Create a "profiles" subdirectory (D:\SPAD-Beta\profiles)

Content copy

- Copy all contents from %APPDATA%/SPAD.neXt/conf to D:\SPAD-Beta\conf
- Copy the complete SPAD.neXt installation directory contents including subfolders (C:\Program Files\SPAD.neXt) to D:\SPAD-Beta\bin
- copy "%APPDATA%/SPAD.neXt/SPAD.neXt.conf" to D:\SPAD-Beta
- Optional: If you want separate alpha profiles:
  - Copy Documents/SPAD.neXt/profiles to D:\SPAD-Beta\profiles

Configuration Step 1

- Open Windows File Explorer, navigate to "%LOCALAPPDATA%/SPAD.neXt"
- Make a copy of the "RELEASE"-Folder and name it "BETA"
- In BETA-Folder edit the file "directories.json" in you favorite text editor.
- Change "Data" entry to D:/SPAD-Beta
- Change "Logs" entry to D:/SPAD-Beta/logs
- Change "Configuration" entry to D:/SPAD-Beta/conf

- Optional: If you want separate beta profiles:
  - Change "profiles" entry to D:/SPAD-Beta/profiles
- **Important: you must either escape "\" (backslash) by "\\" , or use "/" to separate path-fragments**

Configuration Step 2

- Edit "D:\SPAD-Beta\SPAD.neXt.conf" in your favorite text editor
- Find the `<Option Key="Application.UpdateChannel" Value="Release" />` line and change it to `<Option Key="Application.UpdateChannel" Value="Beta" />`

Profit

- Start SPAD.next.exe from D:\SPAD-Beta\bin. It will automatically update
- Always 3 Green

26.7.0

not released to public yet

Devices

- Add MobiFlight devices support
- Add WinCTRL TCAS support
- Add WinCTRL RMP support
- Add WingFlex RMP Cube support
- Add WingFlex DAP-500 support
- WinCtrl AGP: Fix ambient sensor
- WinCTRL CDU: Add setting for disabling lower case character support
- X-Touch Mini set device to midi mode upon exit
- Rotary-Switches (e.g efis / tcas) will raise tuner events additionally
- Add ToLiss Airbus A346 CDU support

Important changes/additions

- Add Support for L:1:XXXX (local aircraft) sim lvars
- Add an option to create a custom LVAR (Definition Browser: select an LVAR category and press "Create LVAR")
- CDU: All aircraft cdu pages are enforced to uppercase, as defined in ARINC 424
- CDU: Scratchpad row no longer uses smallfont by default

Event-Programming

- Add new EVENT:-Data: DEVICE_ID, DEVICE_PAGE_ID, DEVICE_PAGE_NUM, DEVICE_PAGE_NAME
- Add "Set String" operation to EventChangeData
- Play sound: Add option to wait for finish of audio
- Conditions: Add IsEmpty (string) , IsNotEmpty(string)

Functions

- ButtonGauges: Add Drop&Drop swapping
- ButtonGauges: Add Templates
- ButtonGauges: Add ScratchPad
- Enable Paging for all Joysticks and all devices that already migrated to new device concept
- Add testmode button to applicable devices

Expressions

- Expressions: Add IndexOf(stringWhat,stringWhere[,bool])
- Expressions: Add SendDeviceCommand(deviceId,eventId,parameter). Currently only supported eventid: PAGE_SWITCH
- Expressions: Add Fract(value) to get decimal fraction part of a number (opposite to Trunc(value))
- Expressions: Add IsEmpty (string) , IsNotEmpty(string)
- Expressions: Add IndexOf(stringWhat,stringWhere[,bool])

UI

- UI: Fix spad not reacting on taskbar close if any dialog is open
- UI: Fix dialogs not always reacting to ESC to close
- UI: Rework of Response-Curve editing
- Fix profile rename might delete profile
- UI: Show position in expression edit
- ImageCache: Switch to SQLite db

0.9.24

2026-03-28

- X-Plane: Update Plugin to v6.12
- Devices WinCtrl: Add AGP Support
- Devices WinCtrl: Add ECAM Support
- Devices WinCtrl: Add PAC-32 Support
- Devices WinCtrl: Add PDC 3N/3M support
- Devices WinCtrl: Add support for URSA MINOR Joystick backlight and vibration control
- Devices Virtual-Fly: Add Compass support
- Devices Virtual-Fly: Add Switcho Trims led support
- Devices Wingflex: Add A320 FCU CUBE support
- Devices CPFlight: Add new MCP-Version support
- Devices Honeycomb: Add Sierra Gear led control
- Devices Elgato: Add Stream Deck XL+ support
- Devices VRInsight: Add uProPt support
- Devices Cockpit-Simulator: Add chained devices support (e.g. MCP/FCU + EFIS)
    - If you plan to use chained devices, do not chain them yet.
    - Start SPAD.neXt load the profiles where you have the "in future chained devices" programmed and save the profile again. (This will make sure you do not need to reprogram after chaining)
    - Exit SPAD.neXt, chain the devices (don't forget the 20V power supply!).
    - the FCU/MCP has to be Chainmaster (usb connected)
    - Start SPAD.neXt , have fun

- Devices Skalarki: Add DCDU Support
- GoFlight/OpenCockpits: relax interface stress
- MSFS24: Add Inibuilds A300/A340 CDU Support
- Expose LOCAL:SIM_PAUSE (0/1), LOCAL:SIM_PAUSE_EX1 , LOCAL:SIM_PAUSED (0/1)
- SPAD Serial: Add UI.INACTIVE=1 to make an element non-interactable in spad ui
- SPAD Serial: Add VP.IMMUNE=1 to make an element immune to virtual power
- Add AudioControl action (External/Output->Audiocontrol)
- IScriptAction: Fix eventargs have wrong data. IScriptAction is now deprecated. use IScriptAction2 instead
- vFIP: Add button to reset vFIP window in case screen resolution/configuration changed
- UI: Data/Event search adds ";" as a multi-search delimiter
- UI: Data Monitor search adds ";" as a multi-search delimiter
- Fix aircraft assignments for P3d not loading
- Event UI: Add ordering of conditions
- RemoteCDUDisplay has been changed and requires an additional commandline parameter. To get it start it once from within spad, rightclick the display "copy cmdline" to copy it to the clipboard
- UI: Data Monitor add unit column
- UI: Device Rename is active immediately
- UI: Allow creation of subfolders when creating a new profile (use '/' in name)
- UI: Remember last selected device
- UI: Settings->General->UI Add option to automatically open edit dialog if only one action configured (Same as CTRL-E)
- UI: Settings->General->UI Add option to default new led actions to ON / AutoOff

- UI: Delay-Action: Add "SPAD common" static/data/expression for delay-time
- UI: Add button to Collapse/Expand device ui to gain more space for events
- UI: Visualize application shutdown
- UI: Enable Expression-dialog resizing
- UI: Enable online snippets without context
- X-Plane: Fix additional indexed datarefs not registered when already in use
- X-Plane: Dynamically discover all commands (XP12 only)
- X-Plane: Event monitor is working again (XP12 only)
- Axis Variable Unit can be configured per axis/device (Device Settings->Global)
    - The default value is "Percent" which has been the current value. Possible values are

      Percent (0..100)
      Raw value from device
      Normalized (-16383 .. 16383)
      AxisValue (0..1)
      Inverted Raw value from device
      Inverted Normalized (-16383 .. 16383)

    - For virtual devices (e.g. vJoy) it is recommended to set it to Raw value

You can switch rendering for ButtonGauges to the "future" device independent Skia renderer to prepare for Linux/Mac releases of SPAD.neXt in Settings->Application->Expert (Globally) or via a slider on the device (temporary) (No restart required,active instantly) Please report any major issues where buttons are rendered incorrectly

Some slight render differences for textlayers **are expected**, specially if you defined a textbox that is too small to fit the text.

Please note that the vDeck base files have been changed, and if you get a "Unsupported connect" when opening the vDeck as webpage, please delete your customized html/css files (as well as the .disableUpdates) to have SPAD.neXt recreate them.

0.9.23

2025-09-14

- Expressions: To reference device-specific data in an expression, use "[DEVICE:VARNAME:SERIAL]" (e.g., "[DEVICE:BRIGHTNESS:CL28J1A00419]"). When adding device data to an expression, use the context menu's "Use concrete id" option to copy its concrete name. Use this **only** when accessing data from a different device. By default, [DEVICE:VARNAME] (e.g., [DEVICE:BRIGHTNESS]) is used, which returns the device-specific variable for the device evaluating the expression. [DEVICE:VARNAME##SERIAL] is not a valid reference and is evaluated as [DEVICE:VARNAME] (see above).
- Profiles: Add option to mark a profile as Default Profile (will be loaded if "Auto load last used" is turned off)
- Button gauge: Fix Disabled-layers are rendered anyway
- UI: Event-Configuration add checkbox if comment shall be used as label or not
- Introduce "Scheduled Event" to automatically schedule an event every x milliseconds.
- Skalarki: Add support for Desktopline overhead Calls ALL button
- Add noise collection to event monitor. While "Remove Noise" is active all incoming events will be ignored.
- Events: Enable acceleration of "Send Simulation Event"
- X-Touch Mini: Fix access to led 11 of controller rings
- Events: Add ProfileSwitch command
- Expressions: SubStr will return empty string is len <= 0
- X-Plane: Add FlightFactor Right cdu support
- X-Plane: Add Rotate MD80 cdu Support
- X-Plane: Add FlightFactor 777v2 CDU support
- Add WinWing PFP7 Support (737 ui yet),
- WinWing PAP:
  - New config ReplaceTrailingZeros to replace trailing "0" by "o" (1200 -> 12oo 0500 -> 05oo),
  - New config ShowPositiveRateIndicator to show leading "+" if rate is positive,
  - Add control for V/S (L_VS_VS) and FPA (L_VS_FPA) Display-Tags,
  - Fix negative V/S sign is not visible if rate <-999

- UI: CDU: Show warning if display is darkened
- Settings: Add setting to disable Systemsounds
- UI: Add filter (name and/or value) to Data Monitor
- UI: Rework aircraft assignment dialog (Phase 1)
- SDK: Add Parameters to IScriptAction
- SDK: Add UpdSender-Sample to https://github.com/c0nnex/SPAD.neXt/scripts
- MSFS DM CJ4 Mod: Update support to 0.8.0
- Expose LOCAL:SIM_VIEW containing current simView (FSX/P3D/MSFS/MSFS24 only)
- Settings: Add setting to use internal clipboard instead of windows builtin
- Keyboard Action: Add option to add special keys via menu
- Expressions: Add Replace() , IsMatch()
  https://docs.spadnext.com/features/expressions/functions
- Conditions (string): Add "Matches" comparator
- GoFlight/VA: Update to new led logic
- WinWing: Add MCP/PAP support.
  - Not all display options (e.g decimal points) supported yet
  - Global MCP snippets are compatible
- Skalarki: MIP expose KORRY_BRIGHTNESS variable to change korry brightness on MIP
- Skalarki: Update SDK to ProfilerIO 5.1.911.1000
- X-Plane: Add FlightFactor 257/256 CDU support (requires plugin update to 6.10)
- Skalarki: Expose ABRK/MIP
- miniFCU: Show update firmware button always if outdated fw detected
- CDU: Add "Detach Display" for cdu devices with hdmi rendering. You can use snippet #13161 / #13165 for a reference implementation
  - to run display upon load of a profile create a profile loaded event and run external command "<spad_install_dir>/tools/cdu/RemoteCDUDisplay.exe" or add it to a batch-file
  - To run the display on a remote computer copy tools/cdu to target computer, and start "RemoteCDUDisplay.exe -h= -p=28001"
- Add miniCockput EFIS support.
- X-Plane: Add CL650 CDU support
- Add Short/long-Press to hatswitches
- Add vDeck support:
  - Add a vDeck in Settings->Devices->Virtual Devices
  - It behaves exactly like a streamdeck

    - Click on the detach button to open it as a web device
    - You can customize the web interface by editing Documents/SPAD.neXt/html/VDCK0001.html.
    - if you customize html or css and want to prevent spad overwriting your changes create a file "vdeck001.html.disableUpdates"/"vdck.css.disableUpdates"
    - your must not change the title in the html, as it is used to identify the vdeck to connect to
    - For remote access open "http://<ipofspad>:28001/VDCK0001.html" in a browser
- WW FCU add variables for altitudechange/baro state
- Delete no longer persisted variables from profiles automatically
- ButtonGauge-Editor: Fix Axis Center point at wrong design location
- UI: Fix Airbus Korry-Styles
- CDU: Add right CDU support for X-Plane
- CDU: Fix PMDG777 errors
- RadioPanel: Fix left/right display not updating correctly
- CDU: Added SendKey-Event Support for Fenix,IFly737 Max,TFDI MD11,FLabs
- CDU: Expose LOCAL:Aircraft/CDU/[number]/LED/* with automatic updates for all supported aircraft
- CDU: Expose LOCAL:Aircraft/CDU/[number]/MODE to determine the cdu aircraft mode (IFLY/PMDGXXX ... X-ZIBO)
- WinWing: Add support for WinWing PFP3N
- WinWing: Add support for WinWing MCDU (see https://discord.com/channels/229058021651644416/1290860853930885150/1339288051092033597)
    - Reference Snippet: #13161
- ButtonGauges: Add configuration for DefaultButtonGauge Device Settings->Settings (Global) which will be automatically applied to new ButtonGauges or can be applied manually
- ButtonGauges: Add delete and reset options
- HC Bravo: Add Virtual Power support
- CDU Devices: Add Device-Settings (Profile) for remapping colors
- Optimize Startup/shutdown performance
- UI ButtonGauges: Add a LayerGroup property to group and sort layers in the UI. It does not affect rendering; layers are rendered on top of each other according to

the "Layer" property.

- UI ButtonGauges: Add Visibility-Property as expression to control visibility of a layer without layer mode actions.

0.9.21

2025-01-11

- X-Touch Mini: Add Mackie-Mode support.
  - Existing profiles/snippets will be upgraded
  - Full control over all Led is provided. Drawback: Mackiemode only supports 11 led per control, so 2 led on the encoders are lost.
  - Layer A/B Buttons are fully controllable (e.g. for Page switching)
  - Add Page support
  - When upgrading profiles, events for Layer B will be moved to "Page 2"
  - Sending custom midi commands to the device is no longer supported
  - If profile conversion fails or some stuff is missing, open a ticket and attach the profile backup from **_ProfileBackup_0.9.20.5.zip**
  - If buttons do not work, the device may not have switched to Mackie mode automatically. See how to switch it manually.
- VelocityOne Flight: Add Led Support for Yoke
- RSG G1000XFD: Add support for PAN-Held events
- CockpitSimulator CDU **Breaking Change**
  - For being flexible across different device types all specific keys (e.g. KEY_MENU) are renamed to BTN_Bx (e.g. KEY_MENU = BTN_B5 on 737 cdu).
  - Also the according variables are renamed
  - All events and variable usages are automatically migrated
  - New setting: KeyThreshold (in ms) for debouncing keys. default 100ms
  - Device display updates should be noticeably faster now, depending on the aircraft SDK.
  - Add Support for v2 CDU Devices.
- CS MCP: Full Ascii-Charset now supported for all displays
- Add CockpitMaster MCP-LITE support
- LoupeDeck: Update PLugin to v6.6, address image update issues
- Add support for PU Air Korea CDU v2
- Skalarki: Update SDK (SkalarkiIO Profiler 5.1.907.1000)
- Skalarki Overhead: Fix external light switches not working, add Navlight-State 1 support

- Skalarki MCDU: Add support for Blank1/Blank2 buttons
- Skalarki OVHD: Add variable KORRY_BRIGHTNESS to dim korry buttons (0..15)
- Skalarki ECAM: Add "Door unlock"-Button (ECAM-Switches panel)
- Added 394 missing datarefs (Simconnect->MSFS24)
- Added 293 missing SimConnect Events (Parameter count might be wrong for some still)
- Add FSLabs A32X Support (MSFS + P3D)
  - CDU Left+Right are fully supported
  - All FSLabs Rotor Brake events are available as FSLABS:
- Profile assignments for FSX/P3D/MSFS20/MSFS24 are separated now
  - If you encounter problems with your profile assignments being lost: how to fix lost profile assignments
- MSFS24: Add complete aircraft indexing
- SpadSerial: Fix data subscriptions get lost on profile switch
- MSFS: Fix custom B-Input-Events are not handled correctly
- UI: Make sure all clickspots are actually clickable
- UI: Add DataPoint editing to Axis curves
- UI: Change "remove device" to "reset device"
- UI: Pressing Left ALT-Key while holding CTRL will highlight only clickspots that have events assigned
- UI: When holding CTRL all clickspots for rich ui-devices are highlighted in UI. (Disable via Settings->Application->General)
- UI: Simconnect Data: Add context menu for data index and unit
- UI: Performance: Show performance of costy events/scripts
- UI: Simconnect Data: unlock all data indexes
- UI: Add context menu to B-Input-Events

0.9.14.0

2023-10-18

0.9.14.0 Release

- Profiles: Exclude VIM-Backup files
- Conditions: Fix IsBitNotSet
- Add plugin Version to Status screen
- AddonDevices: Fix Panel size
- VRInsight 737OH: Add Sync-Skip button
- Remove all remaining "uppercase" datareferences (Lvar)
- XPlane: Fix problems with SAM-plugin
- XPlane: Add deprecated datarefs to exclude list
- XPlane: Update plugin to 6.5.0.0
- Expressions: Add "Select" Function
- Bridge-Installers: Separate FSX and P3D-Installers
- P3D: Add Prepare3d v6 LVAR Bridge support
- Variables: Fix Variables are initialized with wrong valuetype
- Elgato: Move options to global settings instead of profile
- SimpleGauge Editor: Add "Duplicate layer"-Action

0.9.13.39

2023-10-08

- X-Plane: Update required XPlane Plugin to 6.3.0.0
- X-Plane: Do not announce model refs
- Axis: Add Axis relative Direction-Events (up/down)
- Expressions: Fix Format/ToString
- Online-Snippets: Fix searching
- MSFSS Bridge: Fix delayed lvar updates on indexing
- X-Plane: Add option to disable/enable builtin commands
- X-Plane: Fix command events not activating if the aircraft is not loaded yet
- X-Plane: Fix datarefs defined in datarefs.txt are made lower case
- X-Plane: Extend datarefs array limit to 240
- X-Plane: XP12 only: Implement automatic registering of all custom aircraft datarefs (datarefs.txt still supported as well)
- X-Plane: Add option to select the X-Plane Master Computer (more than one sim with role master in network, defaults to this computer)
- X-Plane: Add option to disable SPAD.neXt builtin Datarefs (recommended to disable for XP 12) and commands
- MSFS: Update required LVAR Plugin to 4.1
- Stream Deck Plus: Fix EVENT:X/Y not being evaluated in expressions or ChangeDataValue actions
- VRInsight OVH 737 Sync: Sync is completed by pressing the "HIGH ALT LDG" button
- CJ4 AAU1: Fix SendCDUKey sends to wrong CDU
- Expressions: Fix incorrect conversion error
- CJ4 AAU1 CDU fixes
  - Row 13 will show fmc message if scratchpad is empty, else scratchpad
  - MSG led will light up if FMC Message is not empty
  - EXEC led will light up if FMC Message signals "EXEC"
- Action RestCall: Do not show any errors. To debug problems consult the logfile.
- Added P3d V6 Support (no lvar bridge installer yet)
- VrInsight 737H Overhead: Added init synchronization to reliably sync switch states

- iRacing: Update SDK
- Cockpit Simulator: Add support for v2 MCP hardware
- Cockpit Simulator MCP: Fix non-working brightness/backlight variables
- Conditions: make string-comparators (equal,unequal,contains) case insensitive
- Streamdeck: Fix config dialog buttons
- Streamdeck: Fix Device-Rotation is lost after configuring buttons
- Axis Deadzones: Adjust deadzone slider range to calibrated axis range
- Axis Events: Add condition support
- Axis Events: Add target value configuration for standard axis
- Elgato: Add support for EVENT:X/Y-Parameters for tap events

Glossary

SimConnect

SimConnect is a synonym for a protocol Microsoft published together with the FSX Acceleration Pack/SP2. All simulations that are based on FSX (this includes Prepare3D and MS Flight Simulator 2020) support this protocol. For a complete description of SimConnect and its feature please see Microsoft's ESP Documentation.

Have you had a chance to answer the previous question?

Yes, after a few months we finally found the answer. Sadly, Mike is on vacations right now so I'm afraid we are not able to provide the answer at this point.

Old-Getting-Started-Guide

Getting Started

Contents

1. What is SPAD.neXt and will I need it? 3
2. Requirements and features 5
3. Where to get it and full vs. free vs. trial version 7
4. Installation 8
5. First Start: The Configuration Wizard 12
6. A quick glance through the main interface 19
7. Registration / Trial Key 21
8. Profile selection, aircraft binding and takeoff 23
9. Panels, Controls and FIPs 29
10. Go flying! 34
11. Simulator integration 36
12. General Configuration Settings 38
13. Working with profiles 42
14. Once more on panels, controls, and fips - how to assign a button yourself 44
15. Gauges, gauges, gauges 50

16. SimConnect and networked configurations 53

17. Where can I get further help? 58

18. Author and Version 59

What is SPAD.neXt and will I need it?

SPAD.neXt set out as a replacement for Saitek Flight Simulation Panels (http://www.saitek.com/) drivers, but supports much more Hardware today. While Saitek provides drivers for their panels, these have several shortcomings, notably lack of configurability. SPAD.neXt closes this gap and provides almost unlimited options to configure your panels not only for use with flight simulation software.

SPAD stands for Saitek Panels Advanced Driver. There was an earlier attempt to develop substitute drivers by Massimo de Nadal (http://fstools.weebly.com). Unfortunately, development stopped several years ago. Thus, Ulrich Strauss (http://www.fsgs.com/) made a "neXt" approach. Despite similar names, SPAD.neXt is not based on the former SPAD drivers but has been written from scratch.

Saitek has been recently acquired by Logitech and all devices will be renamed. This guide uses the old common names for the devices.

SPAD.neXt had the reputation of being a bit hard to master initially. However, the latest versions enjoy a redesigned user interface as well as several helpful enhancements like online profiles and code snippets which should make it easier to come to terms.

This Guide is divided into two parts. **Part A** tries to give you a hand to get up into the air as soon as possible taking an A2A C172 and some basic Saitek hardware as examples. Essentially, these are the steps to get things going:

1. Download and install SPAD.neXt Sections 3, 4
2. Run the Configuration Wizard Section 5
3. Register SPAD.neXt (optionally, for the registered version) Section 7
4. Select and bind a proper profile for your plane Section 8
5. Select gauges for your Flight Instrument Panels, if you own any (optionally, requires the registered version) Section 9

6. Go flying! Section 10

Part B surveys a number of further features of SPAD.neXt which go beyond a first start, but should be helpful even for the average user to get the most out of SPAD.neXt.

Attention!

There are a few critical places were things tend to go South (and were, possibly, the author of this guide failed before). These are marked in red.

Registered only

SPAD.neXt comes in two flavors, a free one and a registered one (cf. Section 3 for more detail). Most of this guide is based on the assumption you are using the registered (or trial) version of SPAD.neXt. To avoid confusion, some aspects not working in the free version are marked in green.

Part A : How to get into the air

Requirements and features

- Operating Systems: Windows 7, Windows 8 / 8.1, Windows 10 (all 64 bit)
- .Net Framework 4.5.1 or later (https://www.microsoft.com/de-de/download/details.aspx?id=30653, pre-installed under Windows 10)
- Supported Flight Simulators:
  - FS9
  - FS2000
  - FSX
  - FSX:SE
  - Prepar3D v2.3
  - Prepar3D v3
  - (X-Plane)
- Supported Simulation-Connection Methods:
  - Microsoft's SimConnect interface
  - Pete Dowson's FSUIPC, available from http://www.schiratti.com/dowson.html (freeware is sufficient),
  - FSconnect (http://www.dirks-software.ca/), for WILCO planes only.

SPAD.neXt supports the following devices/features as of the time of this writing:

- Saitek Switch Panel
- Saitek Multi Panel
- Saitek Radio Panel
- Saitek Backlit Information Panel

Registered only from here on

- Saitek Flight Information Panels
- Saitek Pro Flight Yoke
- Saitek Quadrant
- Saitek Trim Wheel
- Saitek Rudder Pedals
- SimConnect local/remote connection
- General Joysticks/Gamepads
- Access to LVARS (e.g. for A2A aircraft)
- Aircraft Profile-Database
- Code Snippets
- PMDG 737 NGX
- PMDG 777
- Aerosoft Airbus
- Wilco Airbus
- Ultimarc PACLED64/PACDRIVE (http://www.ultimarc.com/pacled64.html)
- Pololu Servo Controller (https://www.pololu.com/category/12/rc-servo-controllers)
- Gauge-Market
- Send Keyboard-Keys to any program
- Keyboard macros
- Custom C# Scripting for BIP etc. (https://github.com/c0nnex/SPAD.neXt/wiki)

Attention!

As a general rule, it is not required that any Saitek drivers are installed on the system for SPAD.neXt to function correctly. However, there may be cases where you can't get Windows to properly detect some of your devices. In this case, there is no other way than installing the Saitek driver for the corresponding device. However, in any case stay off any Saitek additional "Software".

To make SPAD.neXt drive your Saitek Flight Instrument Panels (FIP) it is important to either disable or uninstall the Saitek DirectOutput Service.

Where to get it and full vs. free vs. trial version

The most current installer of SPAD.neXt can always be found at http://www.spadnext.com.

To enable the "registered only" features of SPAD.neXt, you will need to purchase an activation key turning the unregistered version into the registered one (cf. Section 7). This activation key is available from SimMarket. Just follow the link on the r.h.s. of the SPAD.neXt portal or the in-program link (see again Section 7).

In addition, there are a freeware and a trial version available. The freeware version has limited functionality (essentially Radio/Switch/Multi/BIP only, no FIPs support, no networked configuration etc.) while the trial version has full functionality, but runs for 5 days only turning to the freeware version afterwards.

A clever strategy might be to install the free version first, to see if you will be able to get it running and can master the interface. Next, you can activate the trial for 5 days to check if the additional features work. If you like it – you certainly will – you can buy a key for the full version either via SimMarket or directly from within SPAD.neXt later (see section 7).

In case of questions, you will find a very helpful and friendly user forum under http://www.spadnext.com/forum/. All announcements concerning SPAD.neXt will be done in this user forum. There is no mailing list.

Additional sources of information about SPAD.neXt are:

- YouTube Channel https://www.youtube.com/channel/UCsgyEhnCPKZiTvXV2gBj5dA
- Facebook http://facebook.com/spad.next
- Discord: https://discord.gg/sRwcWkB
- Twitter: https://www.twitter.com/spadnext

In case of issues, there are several ways to handle them, see Section 17.

Installation

Please note, the SPAD.neXt Installer will always automatically download the most recent release from the webservers. A working internet connection is required for installation.

Double-clicking the downloaded installation file "SPAD.neXt.Setup.exe" starts the installation. In case a security question pops up, just confirm "Run".

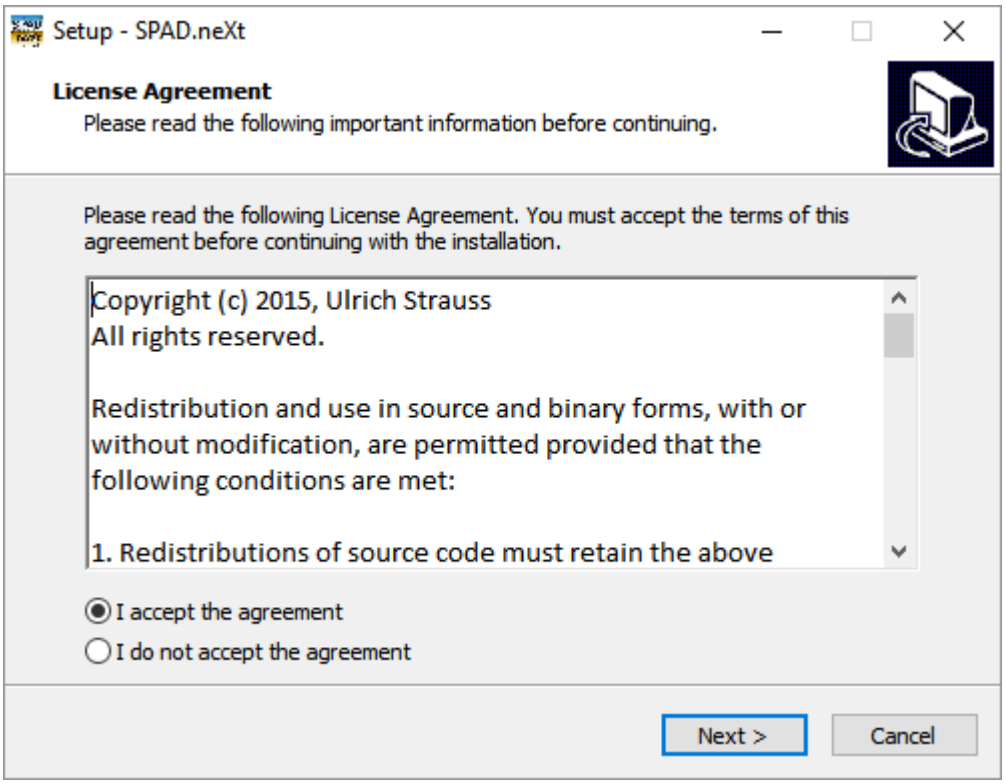


Fig. 1 License agreement

Click on Next to accept the license conditions and continue the installation.

The following screen lets you choose the location of the installation.

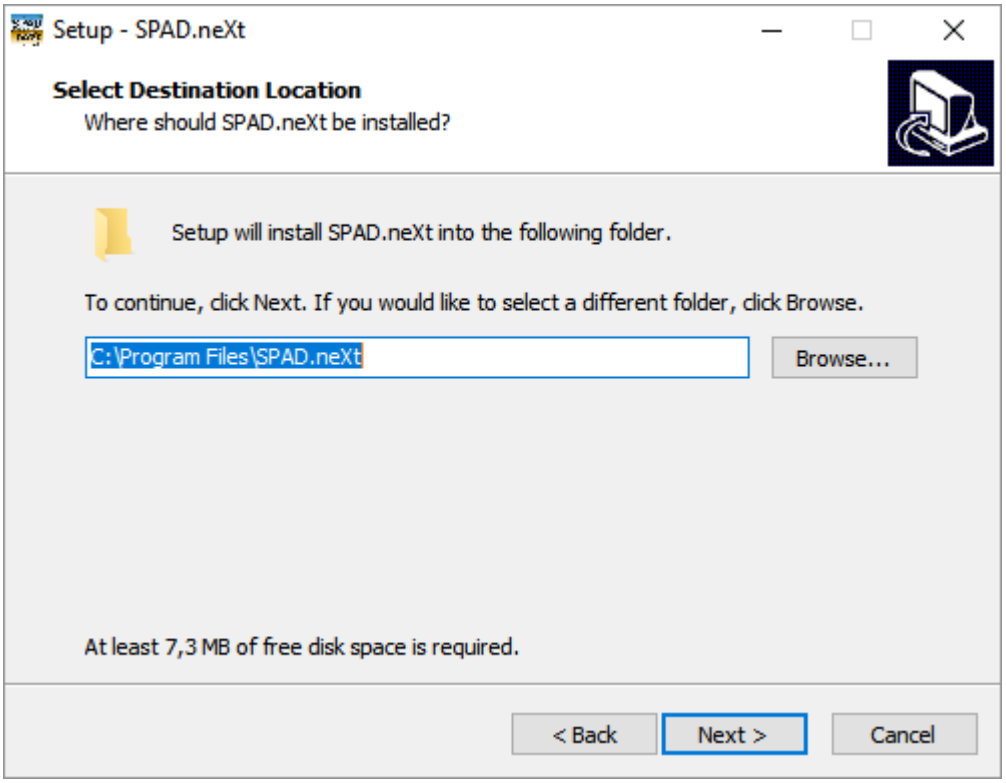


Fig. 2 Install Location

For normal operation, it is recommended to install SPAD.neXt at the default location, however if you would like to install SPAD.neXt somewhere else, you can change the location here.

Due to Windows security restrictions, do not install SPAD.neXt to the Root-Folder of your system drive (e.g. C:\SPAD.neXt)! The software will not be able to access its necessary files. If in doubt, just click next.

For review the selected location will be shown. And clicking on next will start the download process.

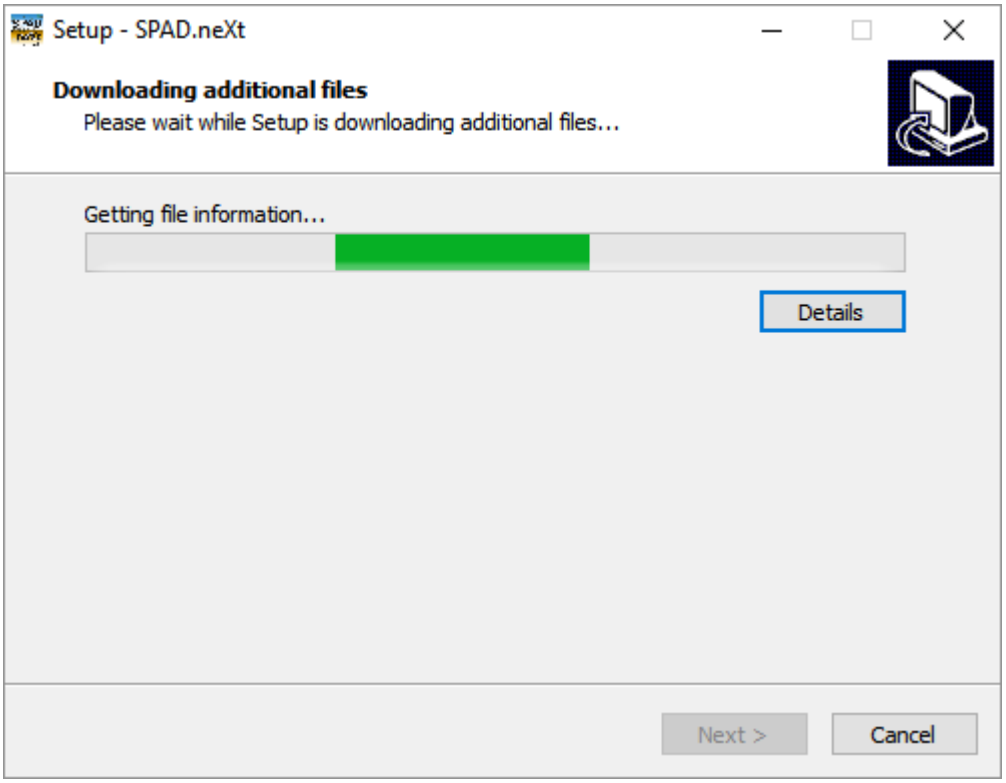


Fig. 3 Downloading installation files

After all files have been downloaded and installed the installation is complete

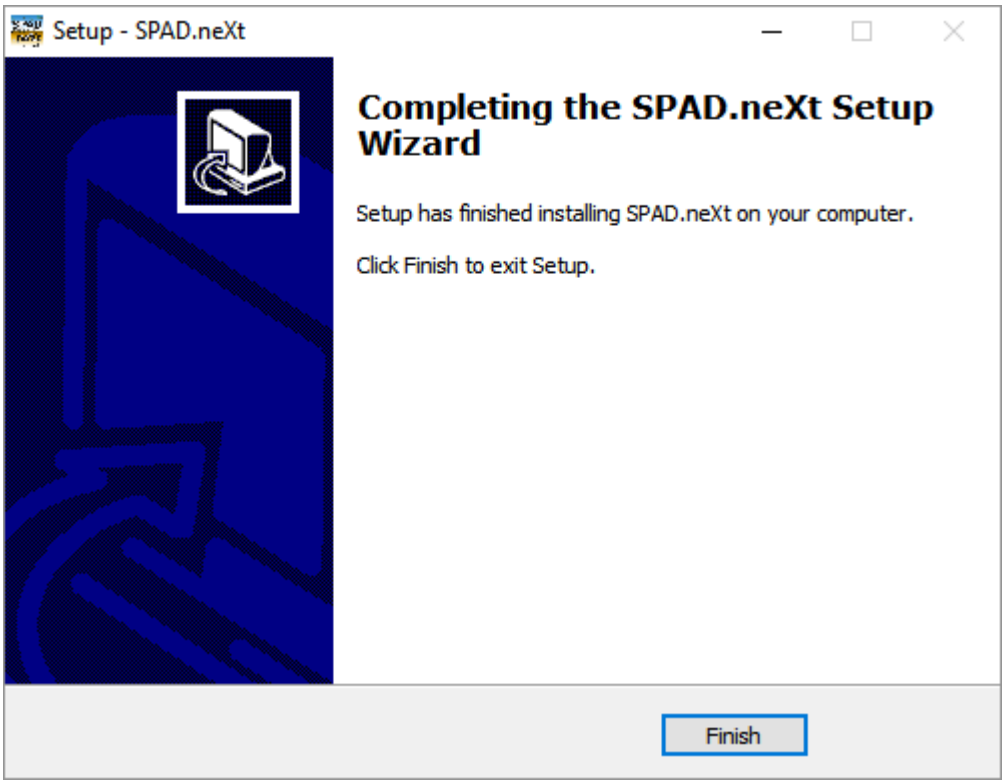


Fig. 4 Installation completed

and an Icon will be placed on your Desktop automatically.

First Start: The Configuration Wizard

When starting SPAD.neXt the first time, you are offered a Configuration Wizard. While you can skip the wizard and either (i) configure SPAD.neXt manually, or (ii) call the Wizard at a later point of time, it is highly suggested to accept its help. All settings can be modified later within the program.

We will guide you through the various Wizard windows now.

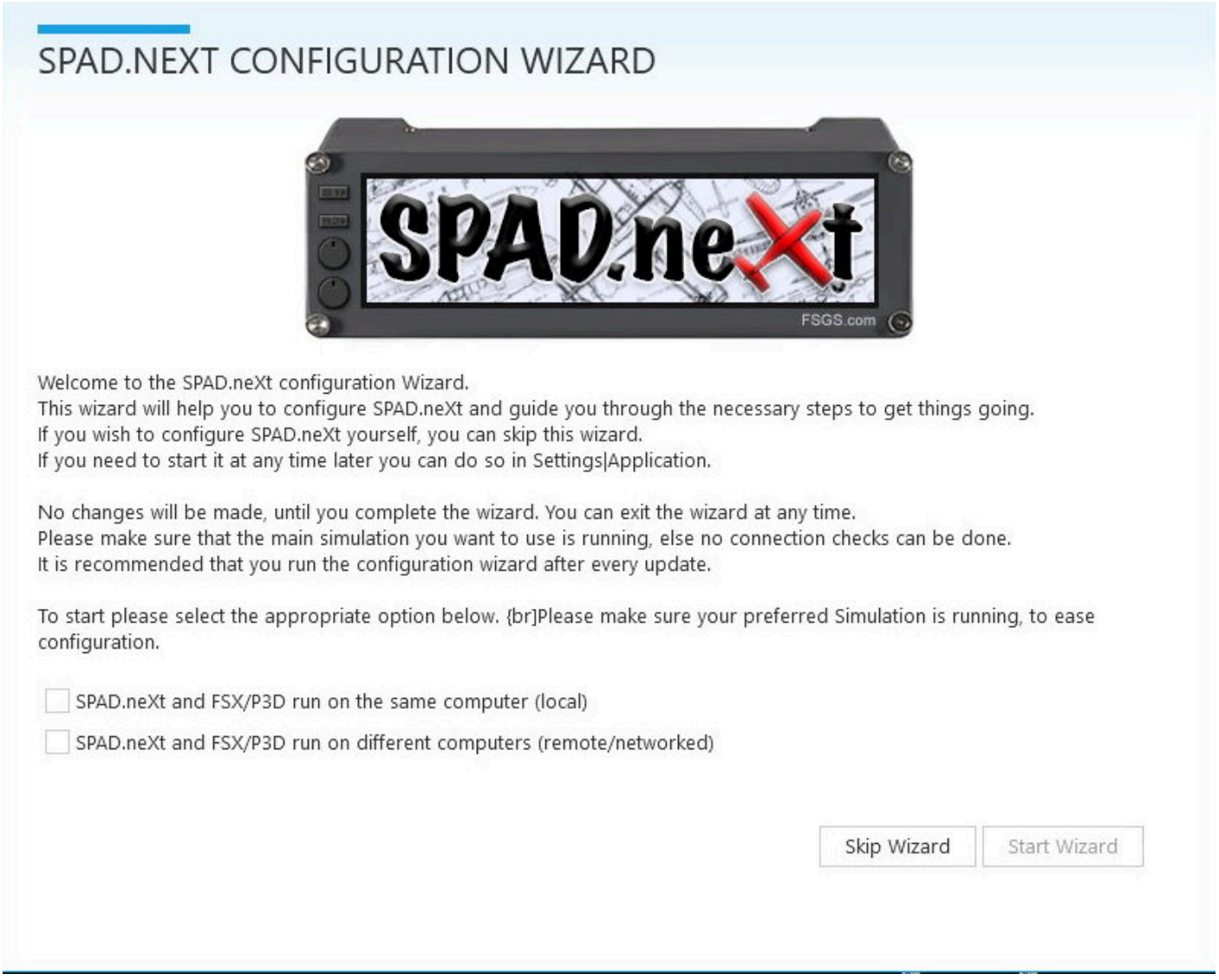


Fig. 5 Configuration Wizard: Welcome screen

Local or Client-Server: First, you have to select between a local or Client-Server installation. We assume a local installation (i. e. SPAD.neXt running on the same machine as your simulator) for now.

Registered only

Client-Server installations are dealt with in Section 16.

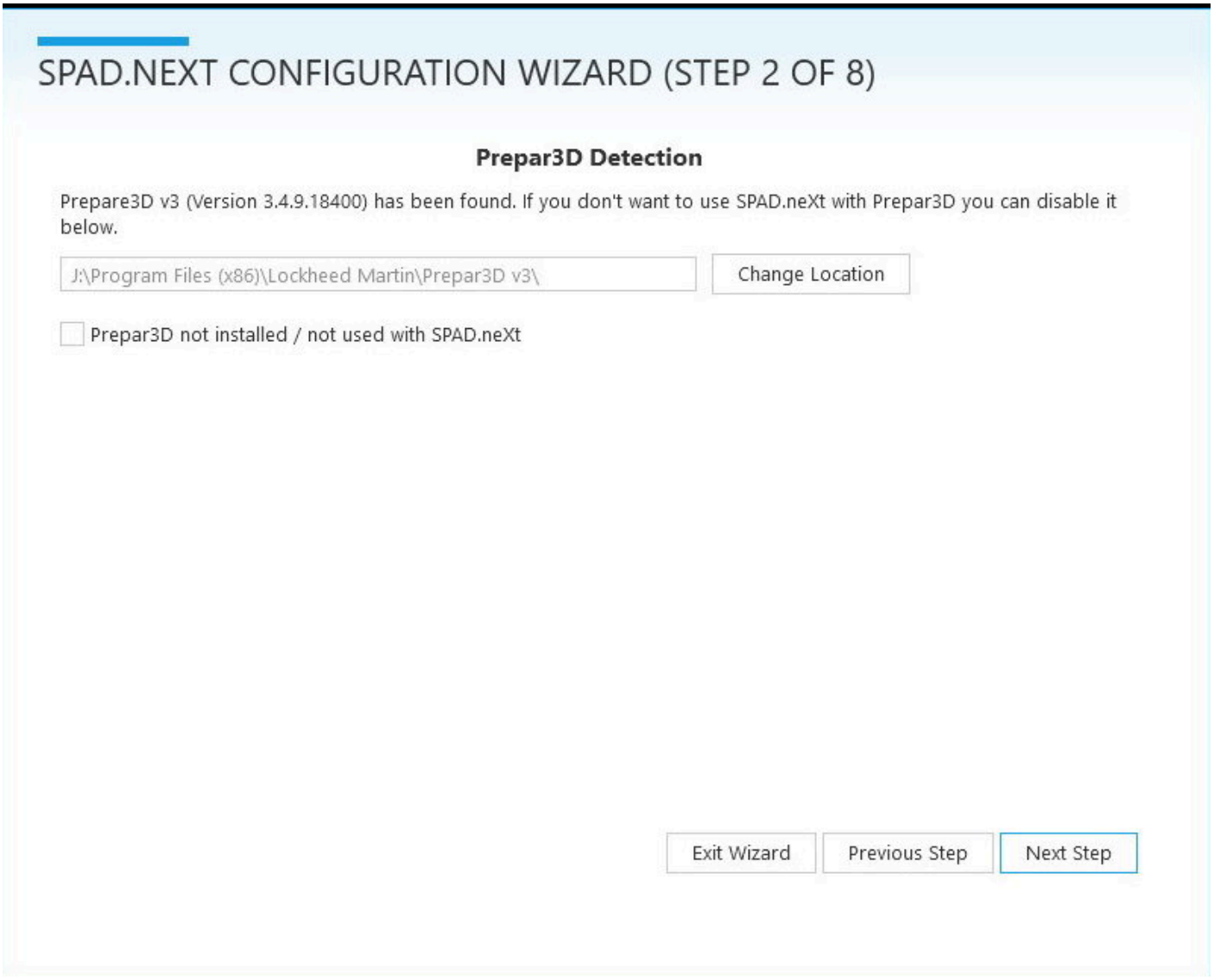


Fig. 6 Configuration Wizard: Simulator detection.

Simulator Detection: The next screen allows you to select the simulator you are going to use with SPAD.neXt. If the appropriate simulator is found, its path is given in the field, which can be changed, between, e. g., FSX boxed and FSX:SE. There will be two screens, one for FSX and another one for Prepar3d.

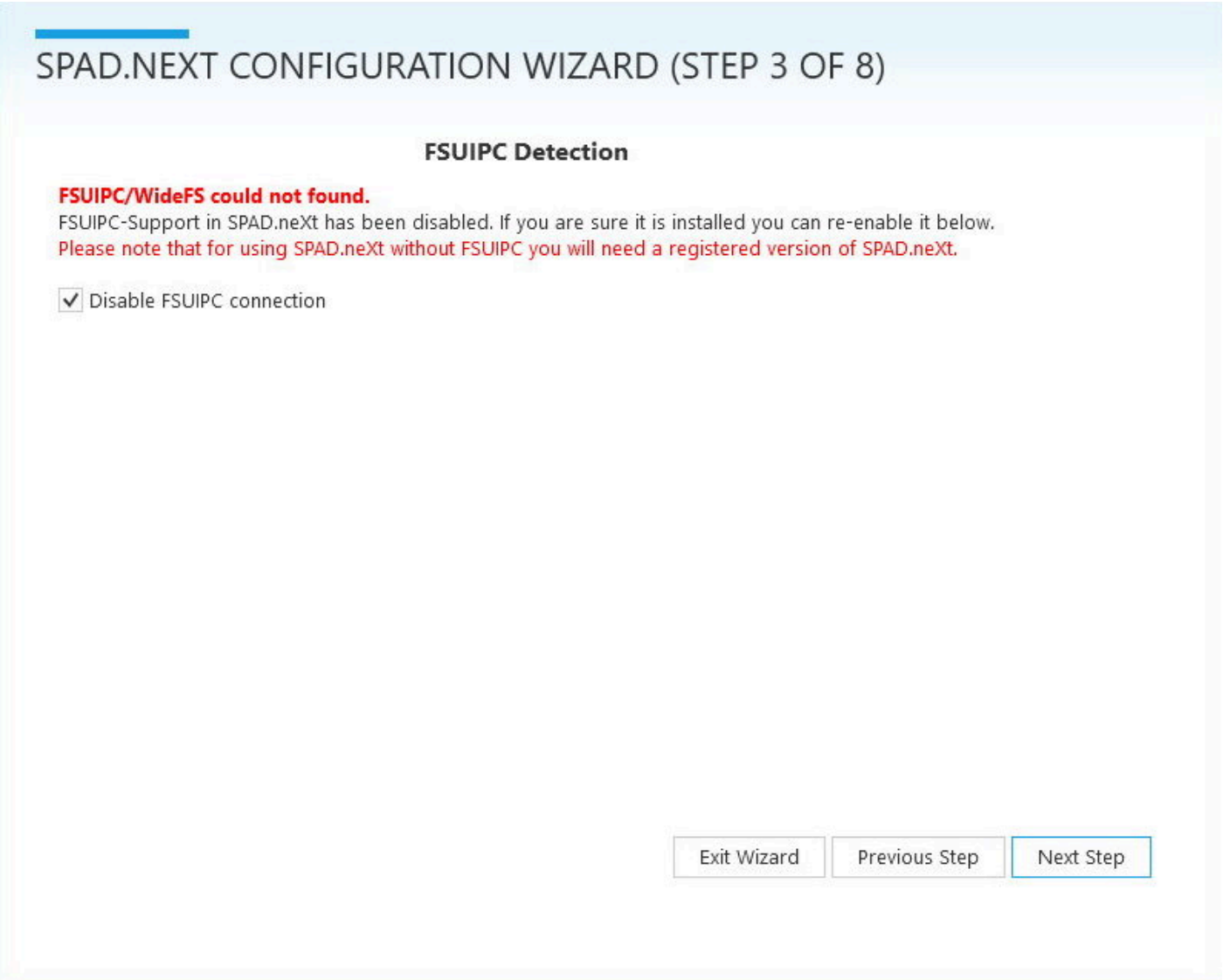


Fig. 7 Configuration Wizard: FSUIPC Detection

FSUIPC Detection: SPAD.neXt supports connection to the simulation via FSUIPC (free version is sufficient). It is suggested to leave this field disabled to save resources and use SimConnect (see below) unless you are common with FSUIPC.

You must enable this option in case you are going to use a Wilco plane.

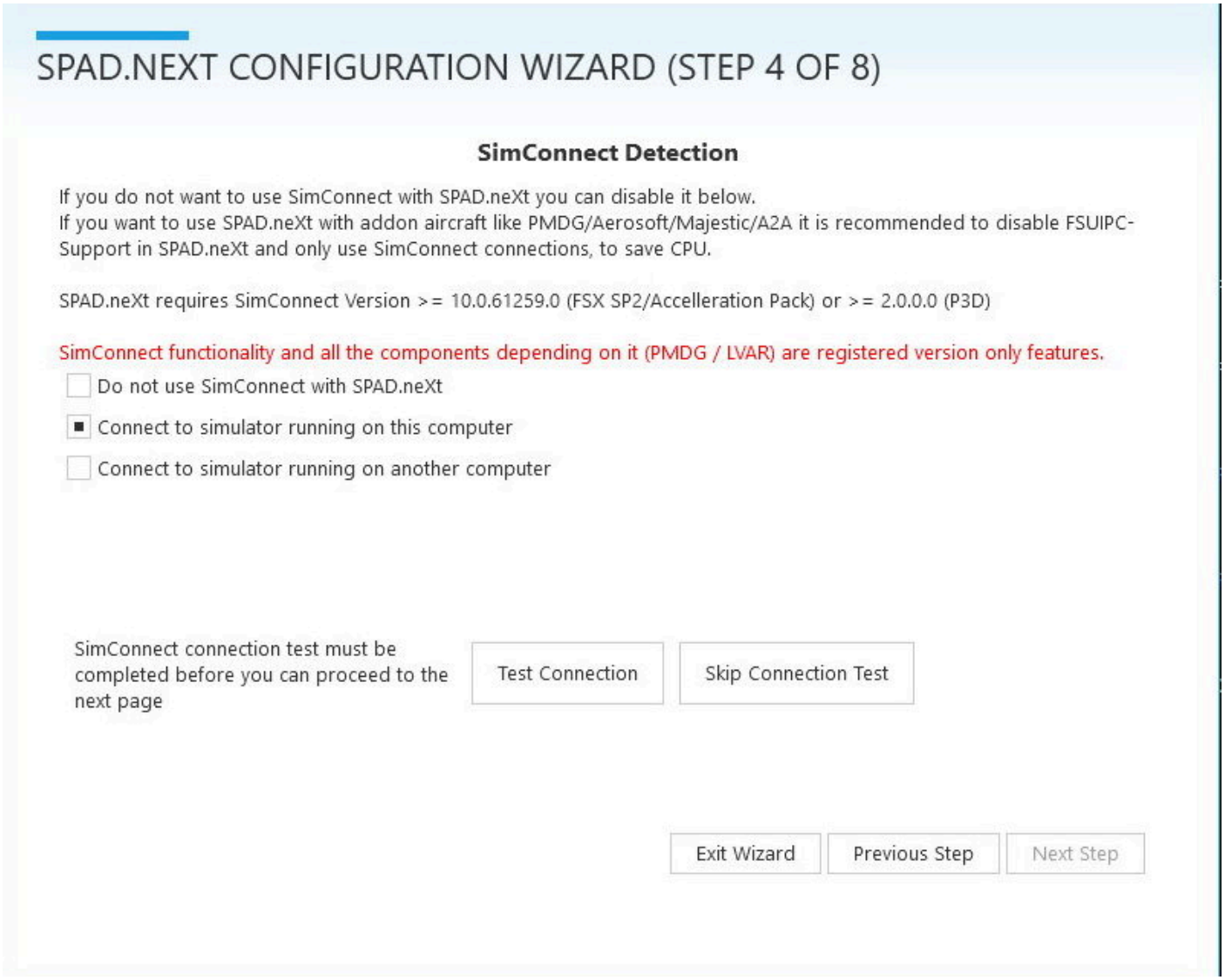


Fig. 8 Configuration Wizard: SimConnect Detection

SimConnect Detection: SPAD.neXt mainly communicates with the simulator via the SimConnect interface, thus the Wizard checks if a proper SimConnect version is installed. SimConnect is an interface developed by Microsoft to allow exchange of data between FSX (and now Prepar3D) and external programs running either on the same or on a networked computer. In case of trouble or if you are uncertain about SimConnect being installed, please refer to Section 16 for more detail.

Next, you have to select if the simulator is running on the same machine (Server) or on another one (Client). For "Test Connection" to work the simulator must be up and running on the respective machine, otherwise it will fail. The test returns version numbers of the simulator and SimConnect.

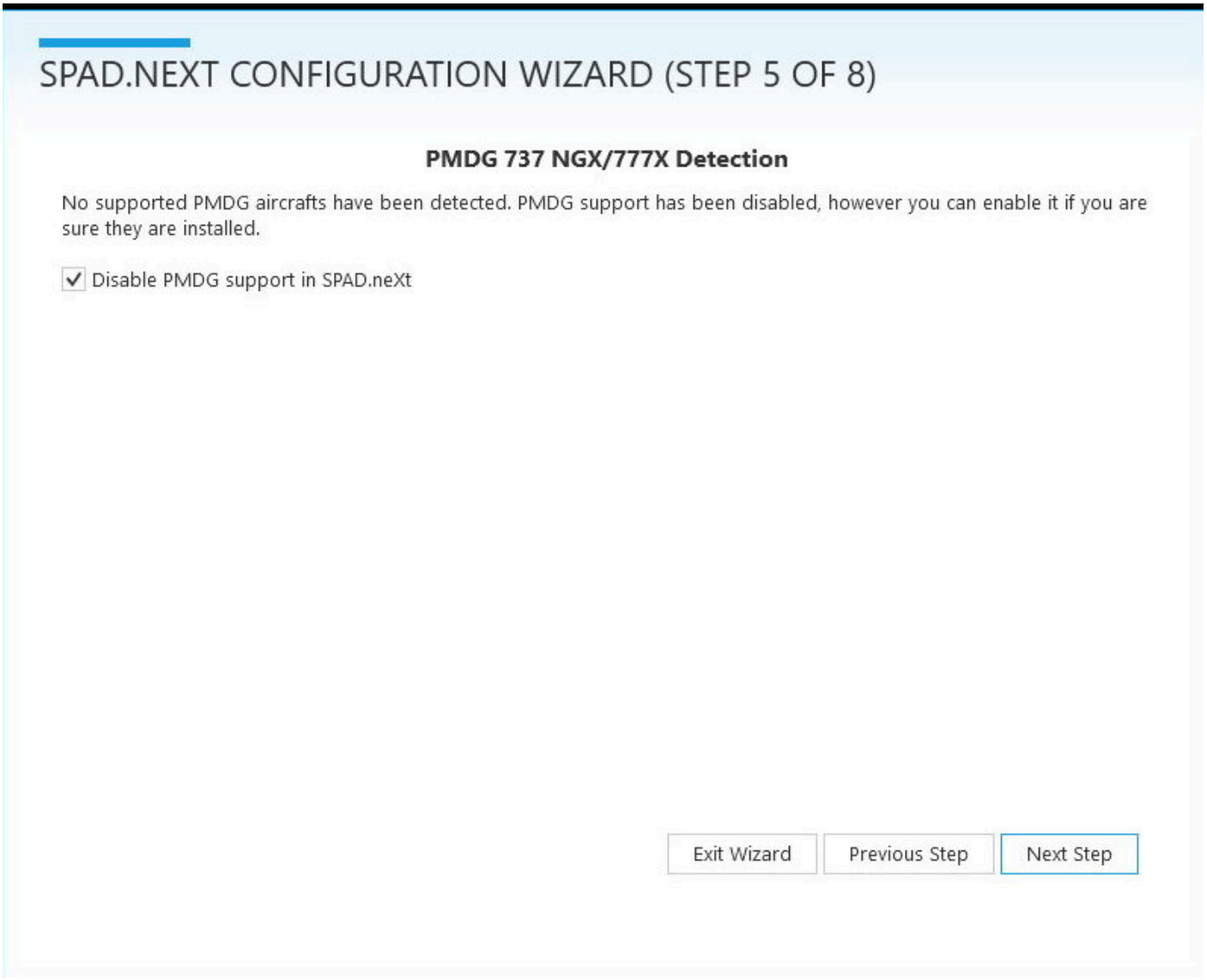


Fig. 9 Configuration Wizard: PMDG 737 NGX/777X Detection

PMDG 737 NGX/777X Detection: Determines if one of the PMDG planes is installed to be supported by SPAD.neXt. (Registered only)

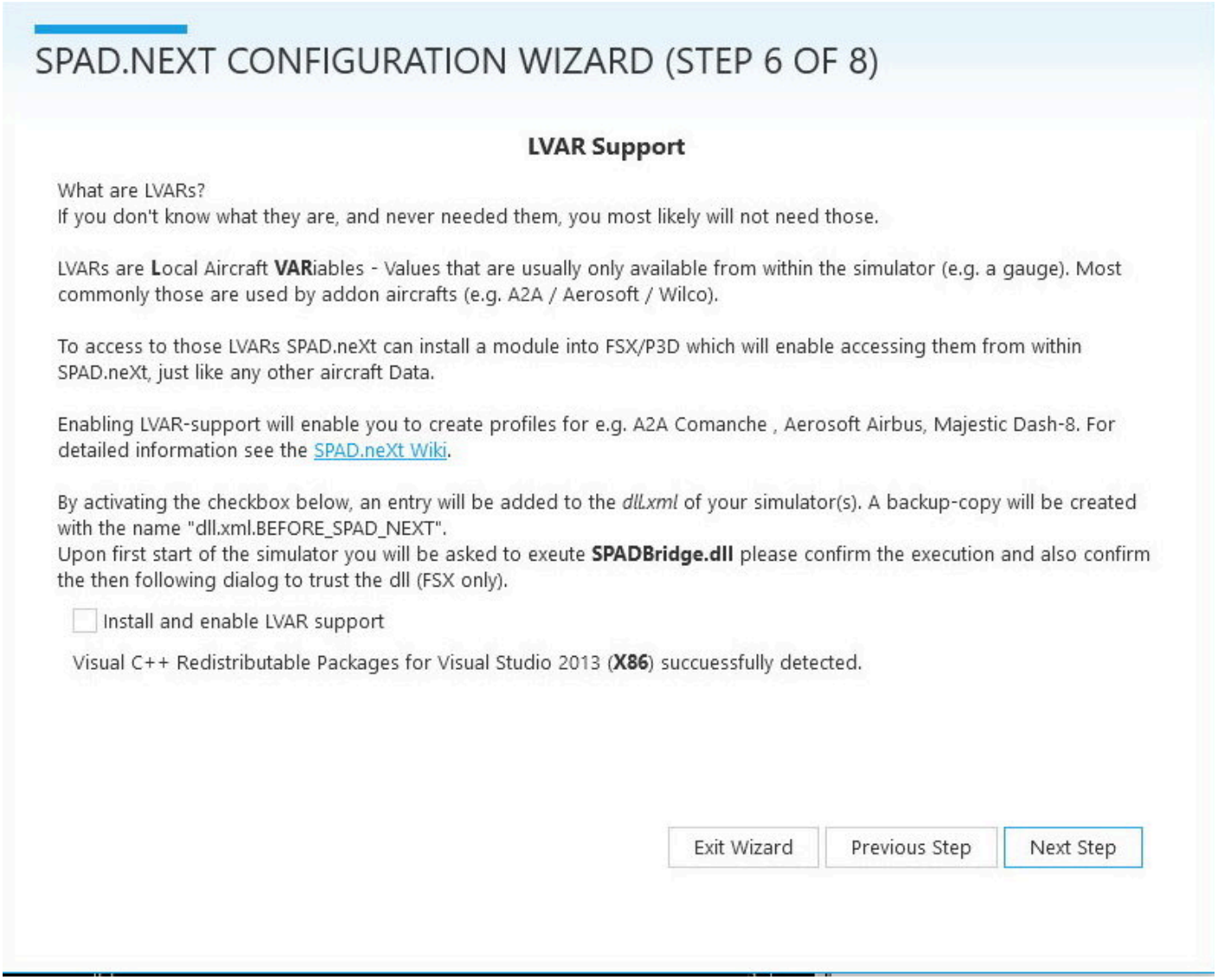


Fig. 10 Configuration Wizard: LVAR Support

LVAR Support: See the screen for more information what LVARs are. You should "Install and enable LVAR support" in case you use any addon aircraft like A2A, Aerosoft, Majestic Dash-8 Q400. In this case a specific module "SPDbridge.dll" will be installed into your simulator (precisely into the dll.xml file). Otherwise leave this unchecked. (Registered only)

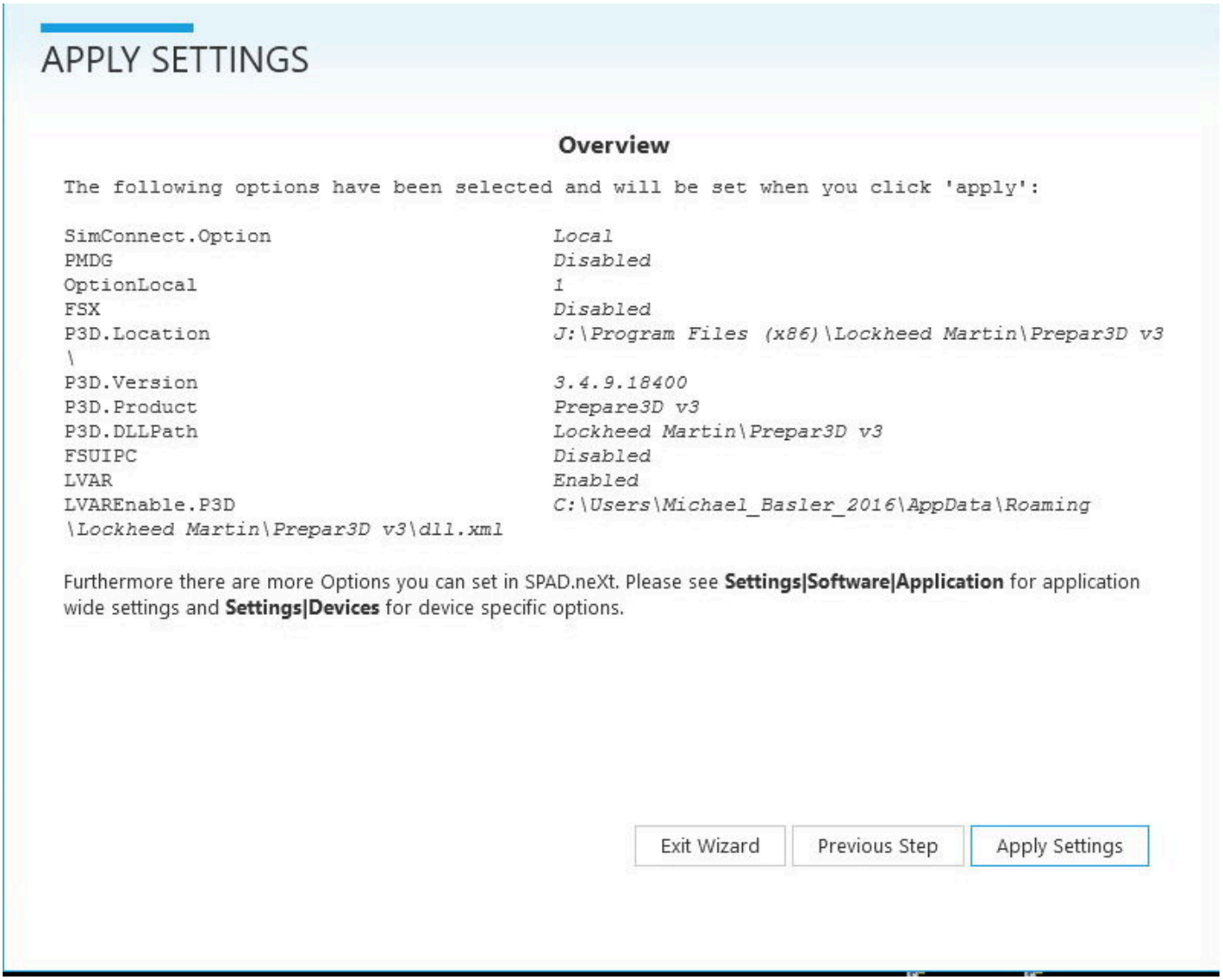


Fig. 11 Configuration Wizard: Overview

Overview: This screen gives a summary of the options selected. There has been no change made on your configuration or simulator so far. You can go back in case you want to revise them. Otherwise press "Apply Settings". The following screen just confirms that your options have been applied.

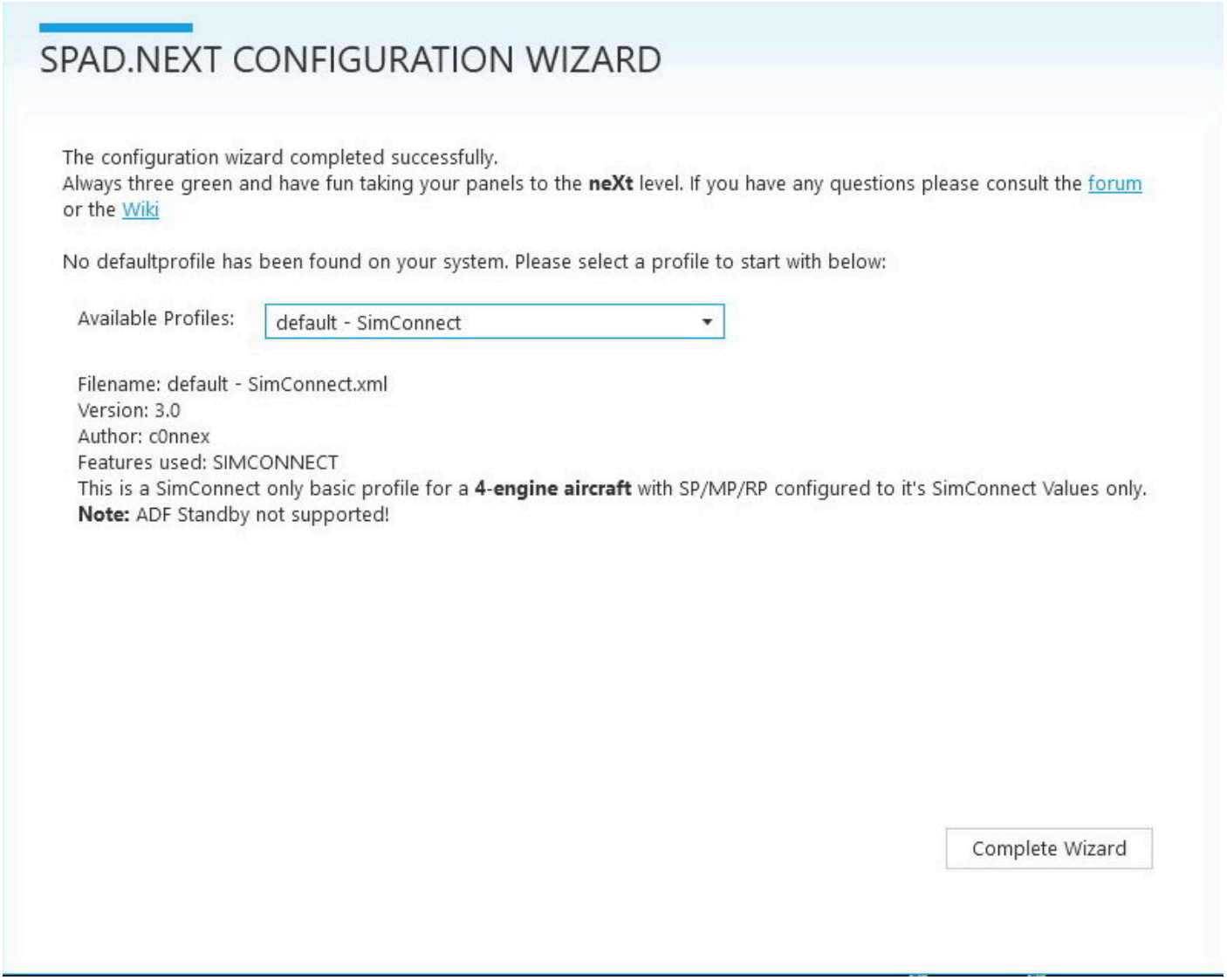


Fig. 12 Configuration Wizard: Final Step

The final selection concerns the profile selection. Profiles describe the relation between your Hardware (i. e. Panels, FIPs, Controls) and their functions. At this stage, just select "default - SimConnect" and "Complete Wizard". You will be given a possibility to select others profiles later.

A quick glance through the main interface

Once you completed the Wizard, you will be greeted by a WELCOME screen (Fig. 9).

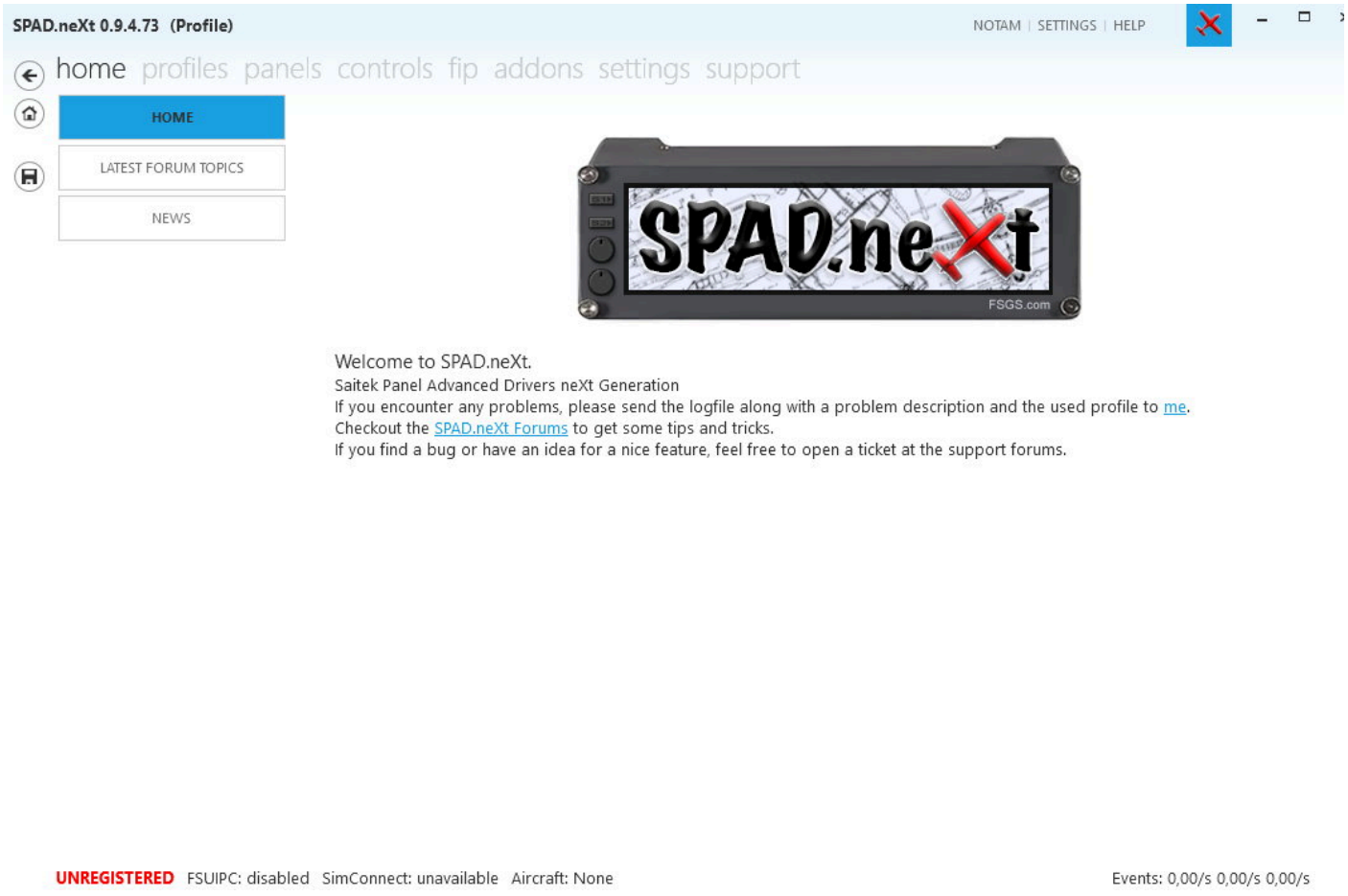


Fig. 13 The main interface: home

This will be your entry point every time you start SPAD.neXt. SPAD.neXt will check for possible updates at any restart. It is highly suggested to "Apply" any updates available. The updater runs in a separate window. After applying updates and clicking on "Close" at the bottom, SPAD.neXt has to be restarted.

There are some areas of the screen layout that will stay the same throughout the application:

1. Quick-Access buttons (on the l.h.s., top to down)

- **Back Icon**: will return to the page you are coming from, e.g. from settings directly back to the Switch Panel.
- **Home Icon**: will return to the WELCOME screen
- **Save Icon**: will save the active profile

1. Main-Menu (on top)
    - **home** this page
    - **profiles:** Specify the devices-functions-airplane connection
    - **panels:** Saitek Radio etc. panels, as far as installed and recognized by SPAD.neXt

    - **controls:** Yoke etc., as far as installed and recognized
    - **fips:** Saitek Flight Instrument Panels, as far as installed and recognized
    - **addons**: Special functions like data monitor and other advanced features
    - **settings**: Registration, Application Settings, and more
    - **support**: connection to the SPAD.neXt support ticket system
2. Sub-Menu (left pane)
    - Here you will see the menu-options available for the selected area in the main menu, e.g., the devices recognized on your system.
3. Status-Line (at the bottom)
    - Will show if you are using the unregistered (free) version vs. the registered one, the connection status of SimConnect and optionally FSUIPC as well as the currently loaded aircraft.

We will not go into detail with all the menu entries, as some of them go way beyond just "getting started". Instead, we will guide you now to get into the air as soon as possible.

Registration / Trial Key

The following description will be based on the registered version; we will mention some features not working in the free version in some places, though. If you don't own a registered copy yet, you either can apply for a 5-days trial key or buy a register code. You will have to go to the "settings" screen and select "REGISTRATION" from the submenu on the l.h.s. in both cases.

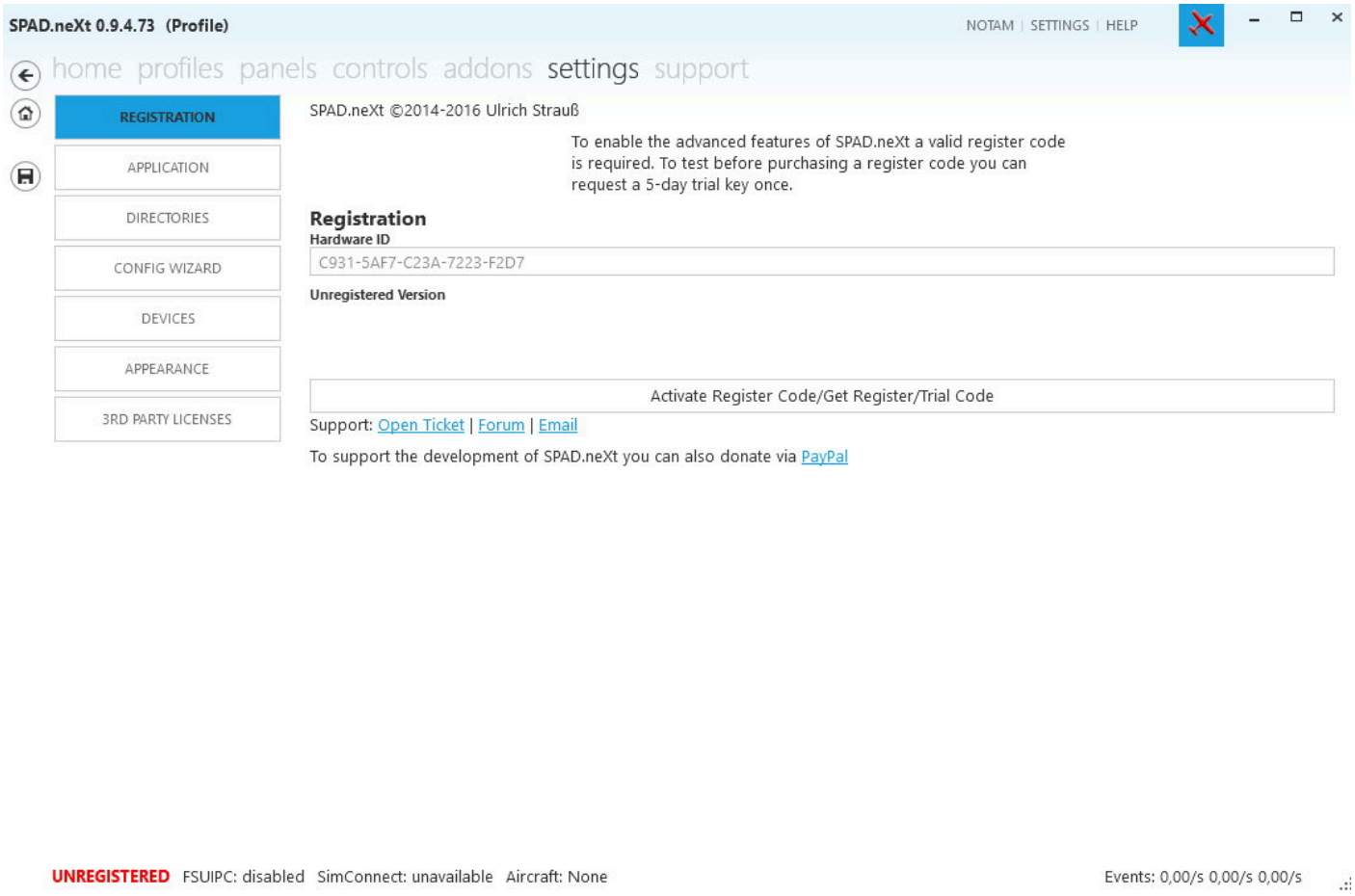


Fig. 14 The registration submenu

Press "Activate register Code/Get register/Trial Code". This opens the following activation window:

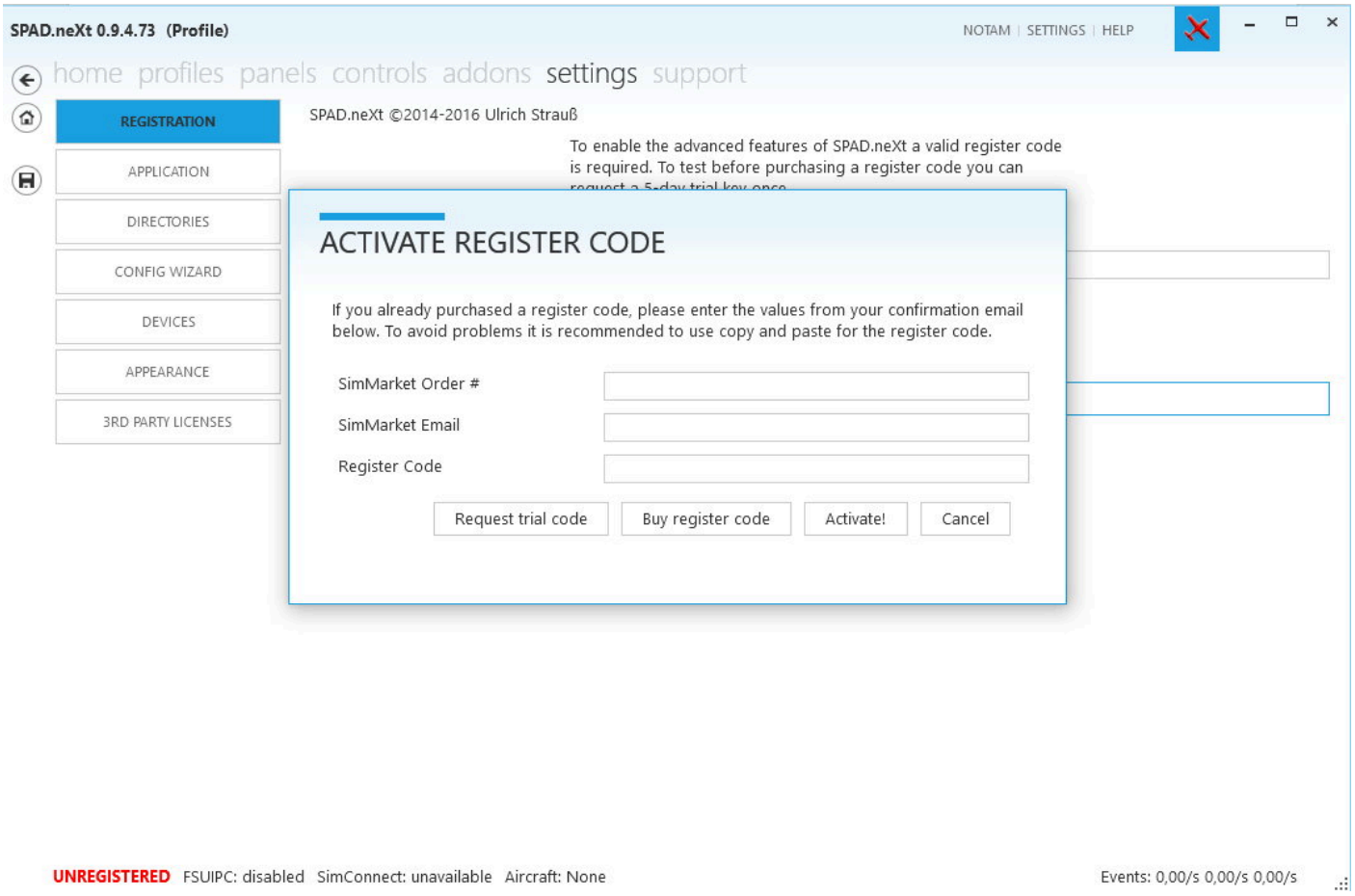


Fig. 15 Buying / Activation / Trial

There are three possibilities:

- If you didn't buy SPAD.neXt so far, you can press "Buy register code" which leads to SimMarket to buy it now.
- After buying from SimMarket you will get a confirmation email your registration code. Now you are ready to fill in your details (order #, mail and code) and press "Activate".
- You can apply for a 5-days trial. Keep in mind this cannot be extended, thus plan in advance to have time for testing it.

The request for a trial code is granted immediately (and anonymously, no details required) and leads to the following result:

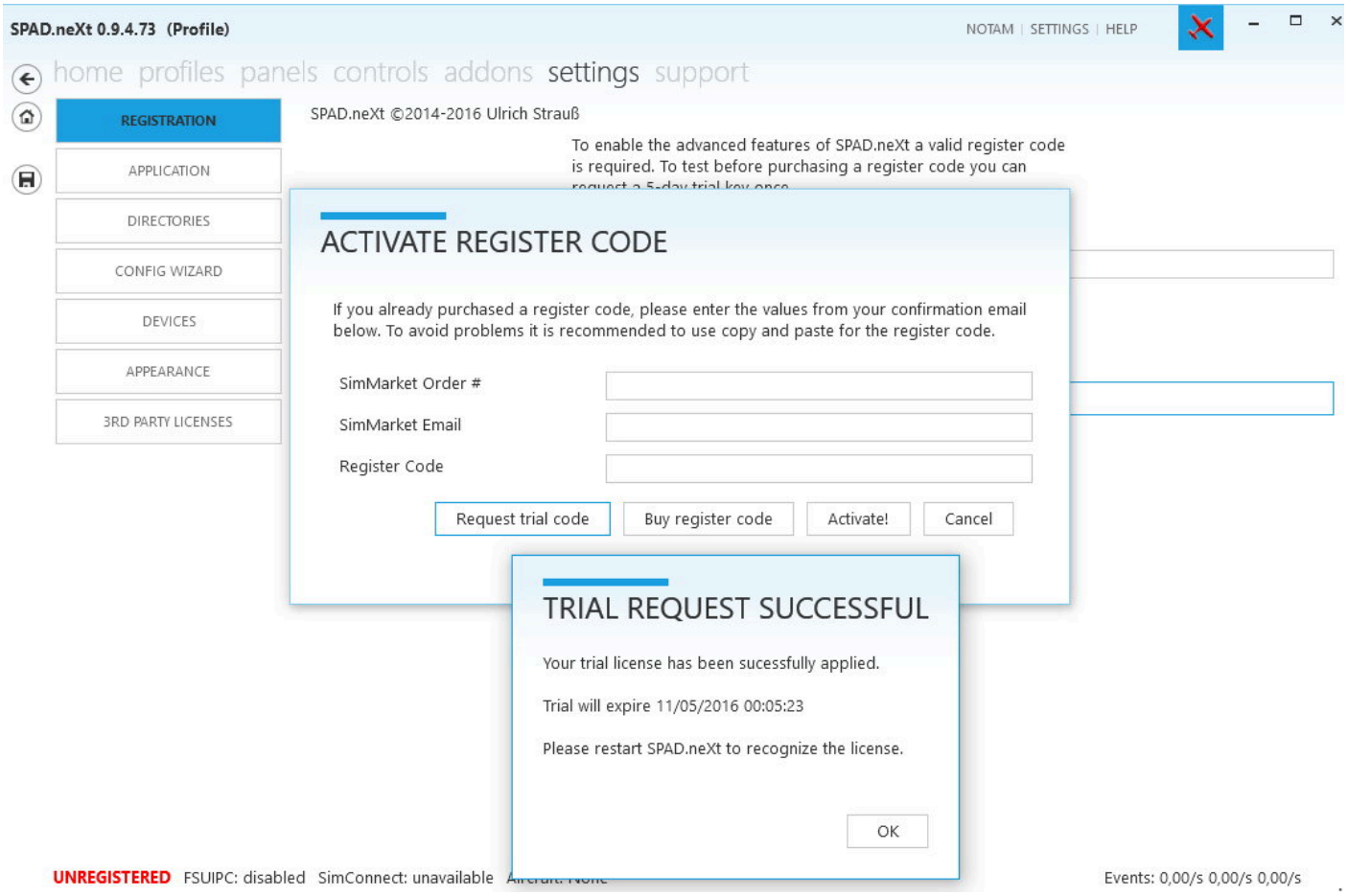


Fig. 16 Buying / Activation / Trial request

You can only request a 5-day trial key once per computer.

After inputting the registration details and restarting SPAD.neXt, the red "UNREGISTERED" in the Status Line should turn into a black "Registered".

Profile selection, aircraft binding and takeoff

At this point we have to decide which aircraft we are going to fly first. It's certainly a good idea to begin with a simple GA plane like a Cessna C172 or C182. One of the more popular planes is the A2A Cessna 172, which we will take as an example. However, the following general guidelines can be followed to get SPAD.neXt to support any other plane, including even tube liners and choppers.

Registered only

A2A planes require LVAR support being limited to the registered version.

To proceed, we switch to the profiles menu entry.

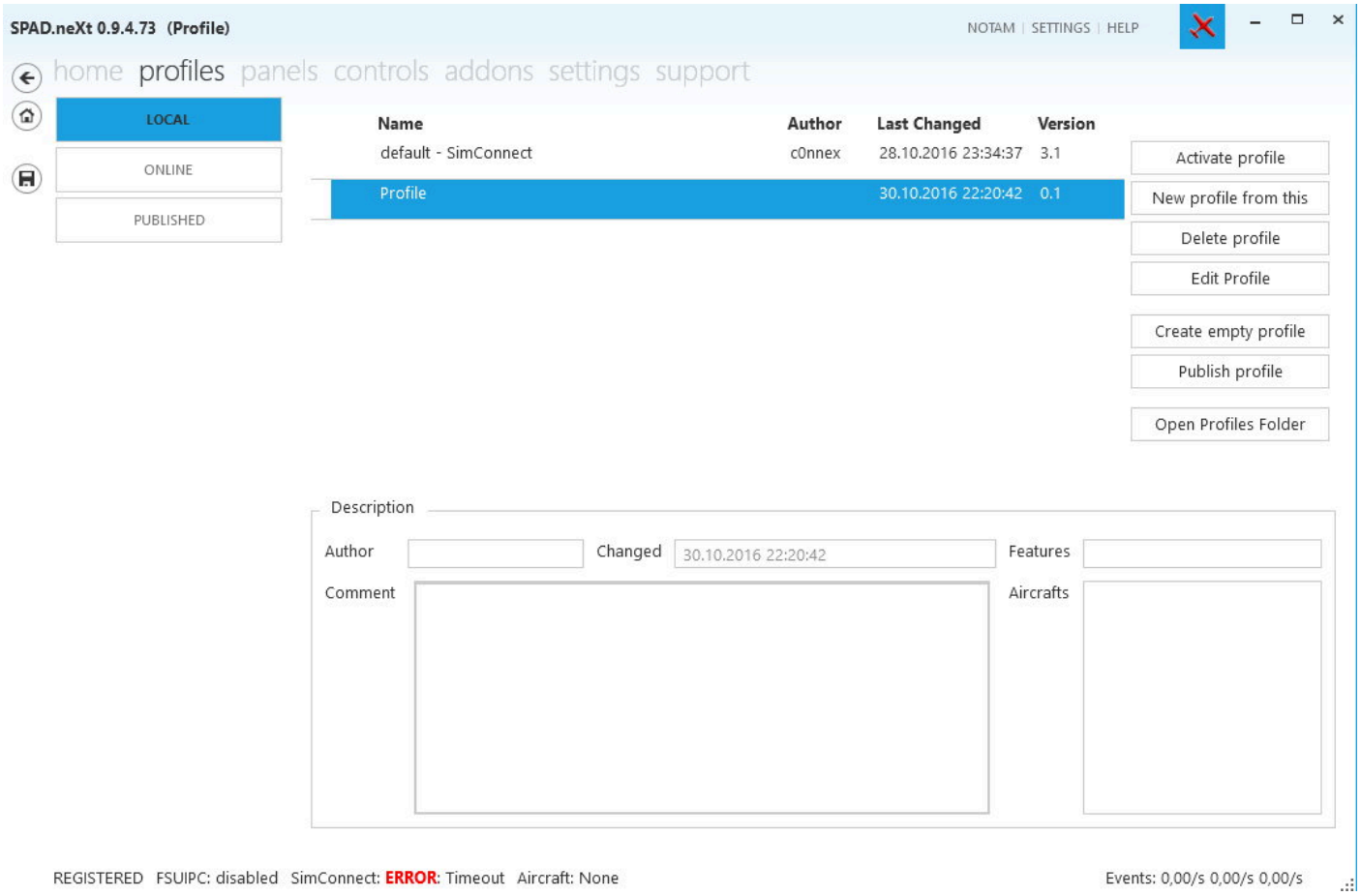


Fig. 17 The profiles menu

There are three panes. The middle one shows all locally installed profiles ("LOCAL"). At present, there is only the one "default - Simconnect" which we choose when running the Wizard (cf. Fig. 7). Generally, you can build your profile (i.e. the functions of all knobs, levers etc.) from scratch, but this is a tedious process which moreover requires some knowledge of SimConnect variables.

Fortunately, a number of users as well as the author of SPAD.neXt have worked out and provided a number of working profiles already. You can use them "as is" or modify them according to your needs after getting a bit familiar with SPAD.neXt. To access these profiles (being stored online) press "ONLINE" on the left pane. Scrolling down provides a – quite impressive – list of all the online profiles available. Every profile is characterized by a unique name, its Author who submitted it, the Date of Submission and the Provider, i.e. the features required to get it working.

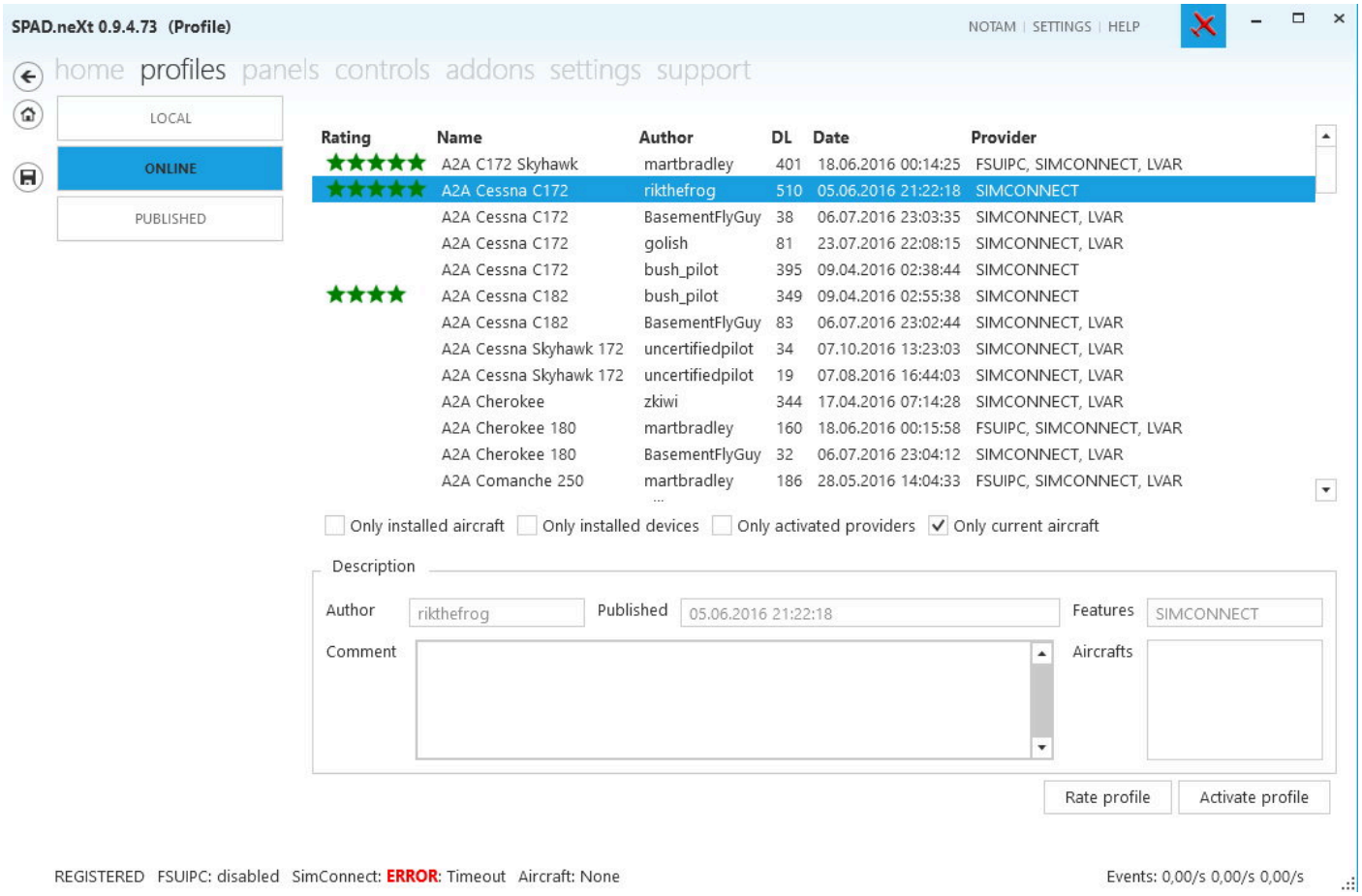


Fig. 18 Online profiles selection

At first glance this may be a bit overwhelming, thus here are some hints for selecting a specific profile:

- a. The leftmost column contains green stars as a result of user ratings. A five-star-rating should be a good indication the profile is at least working properly.
  b. The number of downloads may give another indication.
  c. Some (unfortunately only a few) profiles include Comments elucidating which functions are working and which don't or are limited in this profile.
  d. The SPAD.neXt website provides a Profiles subforum, where you will find discussions on several of the profiles submitted.

To see all online profile available, untick the checkbox next “Only current aircraft”, which will remove the filter to only show profiles for the current aircraft (or profiles that have no aircraft set).

Returning to the chosen A2A C172 we are in a quite comfortable situation in view of several choices. Now I put my former experience in and select the profile "A2A Cessna Skyhawk C172" submitted by "uncertifiedpilot" which I know to work at least basically. It includes a description with some of the special functions assigned.

To select it, put the blue bar upon the profile and press "Activate profile".

This results in the quick access disk icon on the l.h.s. to run red meaning there are unsaved changes. Press the icon to save your selection.

Attention!

Whenever the disk symbol is red there are unsaved changes in a profile which you should save to keep them for further use. Nothing will be saved automatically.

Changing to the LOCAL profiles the selected one should be in your list now.

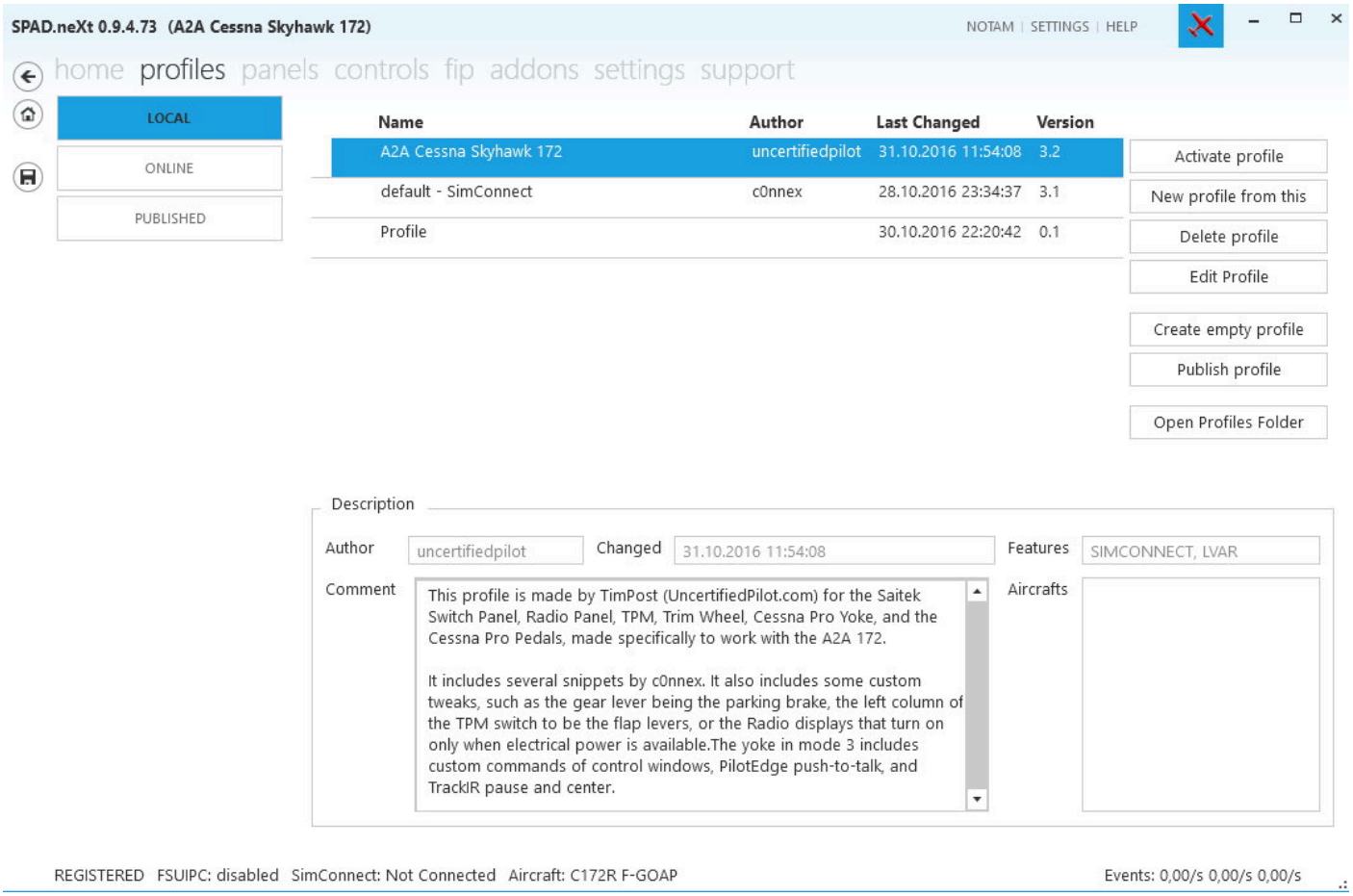


Fig. 19 The added A2A Cessna profile in the profiles list

There is a second potential trap. You might have selected a profile requiring a feature ("provider") which you did not activate during the Configuration Wizard run. In this case the profile will not be selected and you will get a corresponding error message. If you, e.g., did not activate LVAR support, which is required for our profile, you can go to SETTINGS|APPLICATION|SIMULATIONS where you find switches for activating/deactivating these features in hindsight.

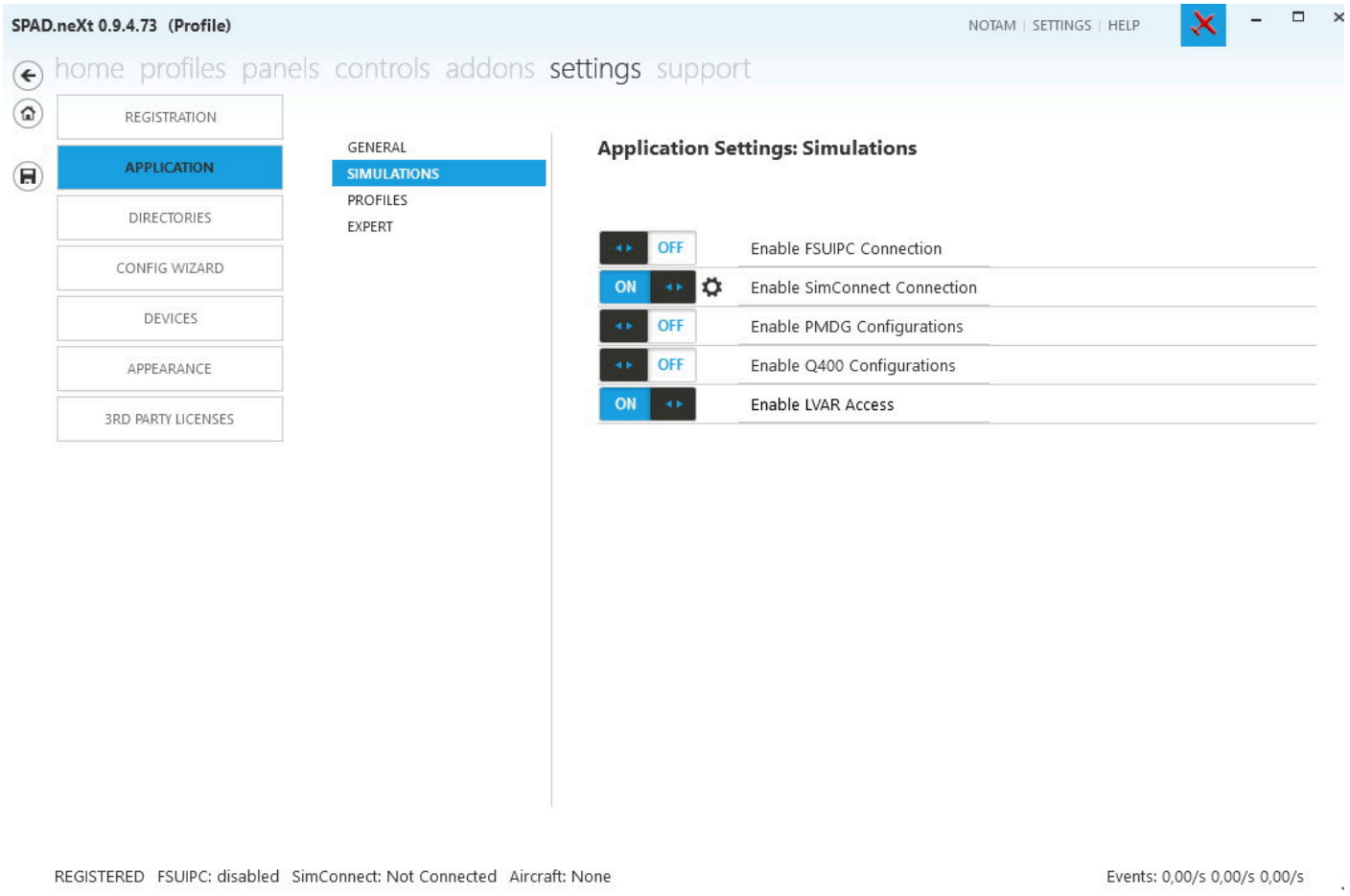


Fig. 20 Enabling/disabling features under Applications Settings

There is even a third potential trap. Who tells you your Hardware will fit to the profile being based on its author's Hardware? SPAD.neXt has a clever substitution mechanism pairing e.g. the Saitek Pro Flight Rudders Pedals with the Saitek Pro Flight Cessna Pedals. However, it certainly needs some hints in case the author used a yoke while you own a joystick. We assume your hardware is fitting for now and return to this case in Section X.

At this point we are not yet done, as we have to tell SPAD.neXt which aircraft(s) is/are going to make use of this profile. For this purpose, we select the profile via the blue line and press "Edit Profile" on the r.h.s. This opens a window for editing the profile.

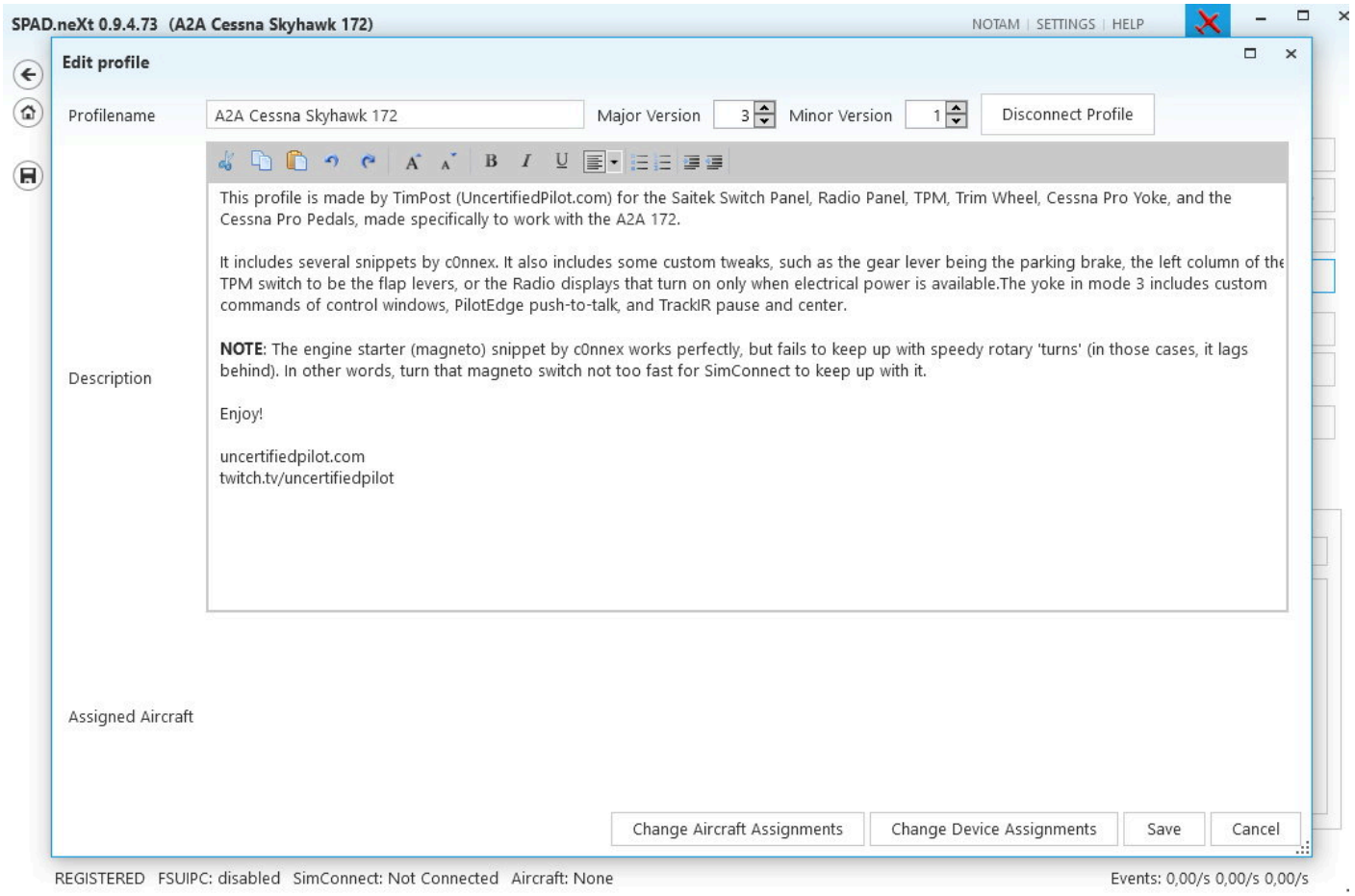


Fig. 21 Editing a profile

We are not going to edit the profile at this point, but choose "Change Aircraft Assignments" at the bottom.

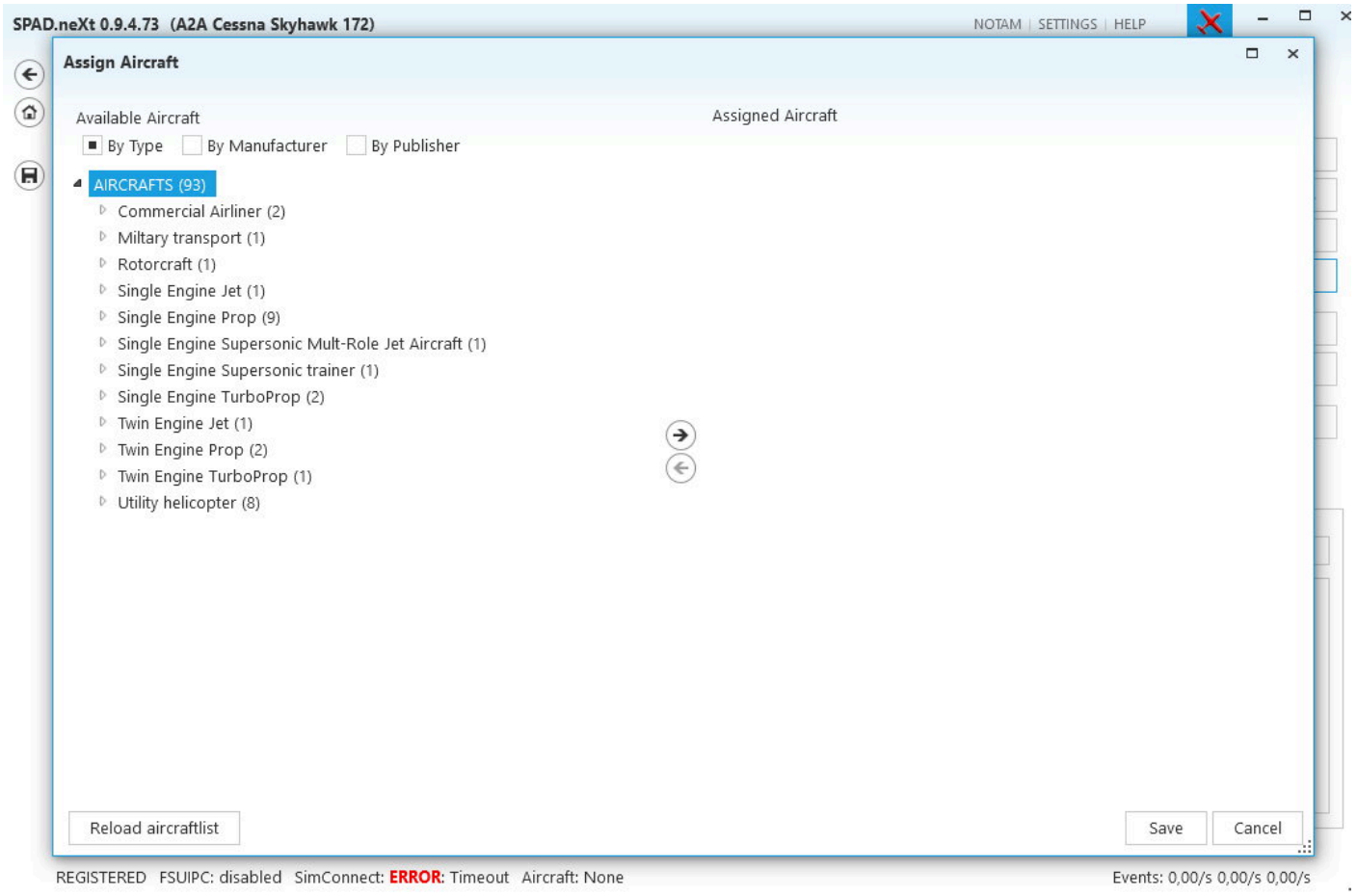


Fig. 22 Assigning an aircraft to a profile

The further procedure is quite straightforward.

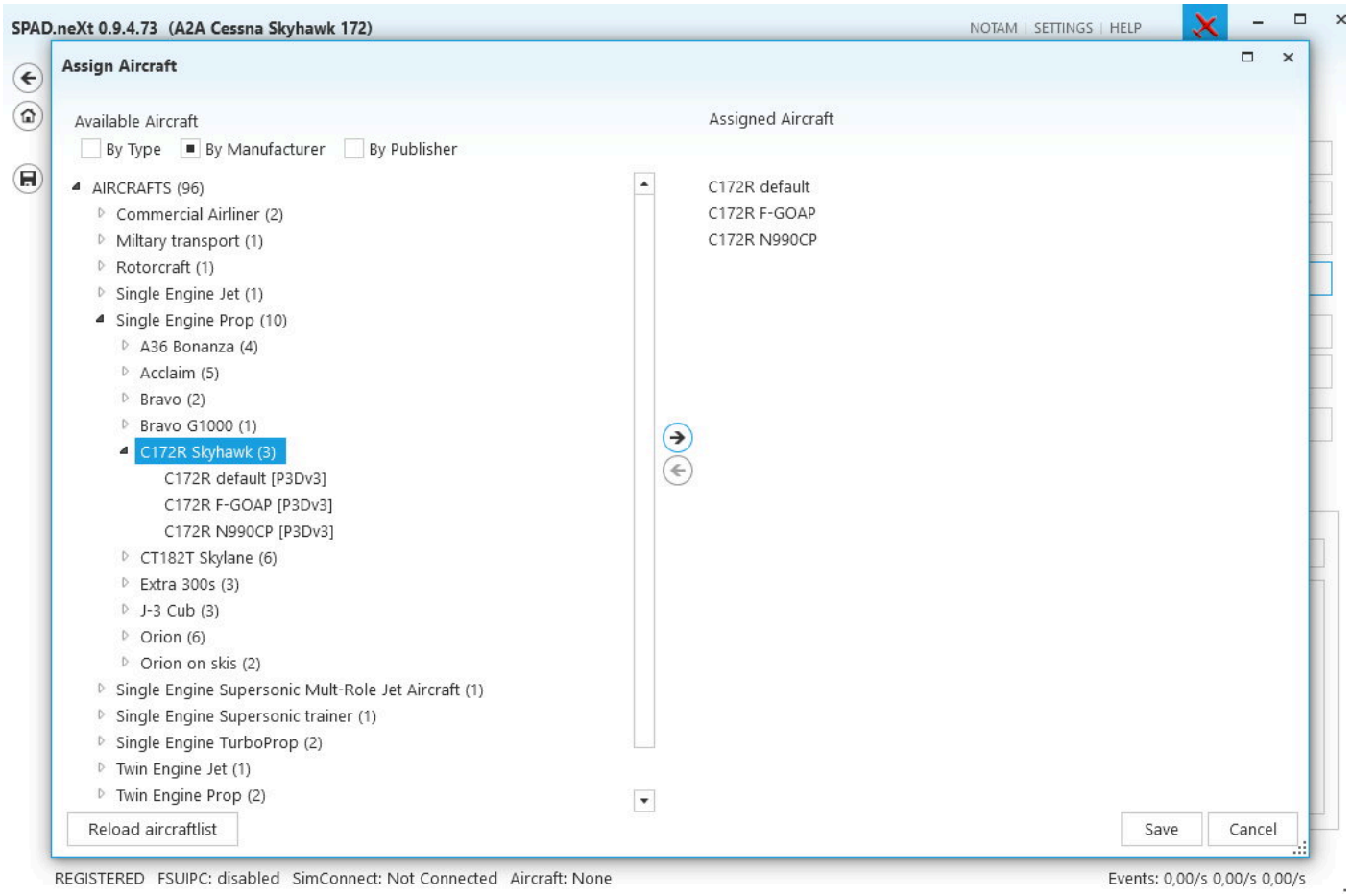


Fig. 23 Assignment of the A2A C172 to the profile "A2A Cessna Skyhawk 172"

After two times "Save" we are in the "profiles" menu again.

Panels, Controls and FIPs

While we want to get into the air as soon as possible, we should at least take a short glance onto our assets. Select "panels" under the main menu and, depending which one(s) you own, one of the panels. The multi panel should look as follows:

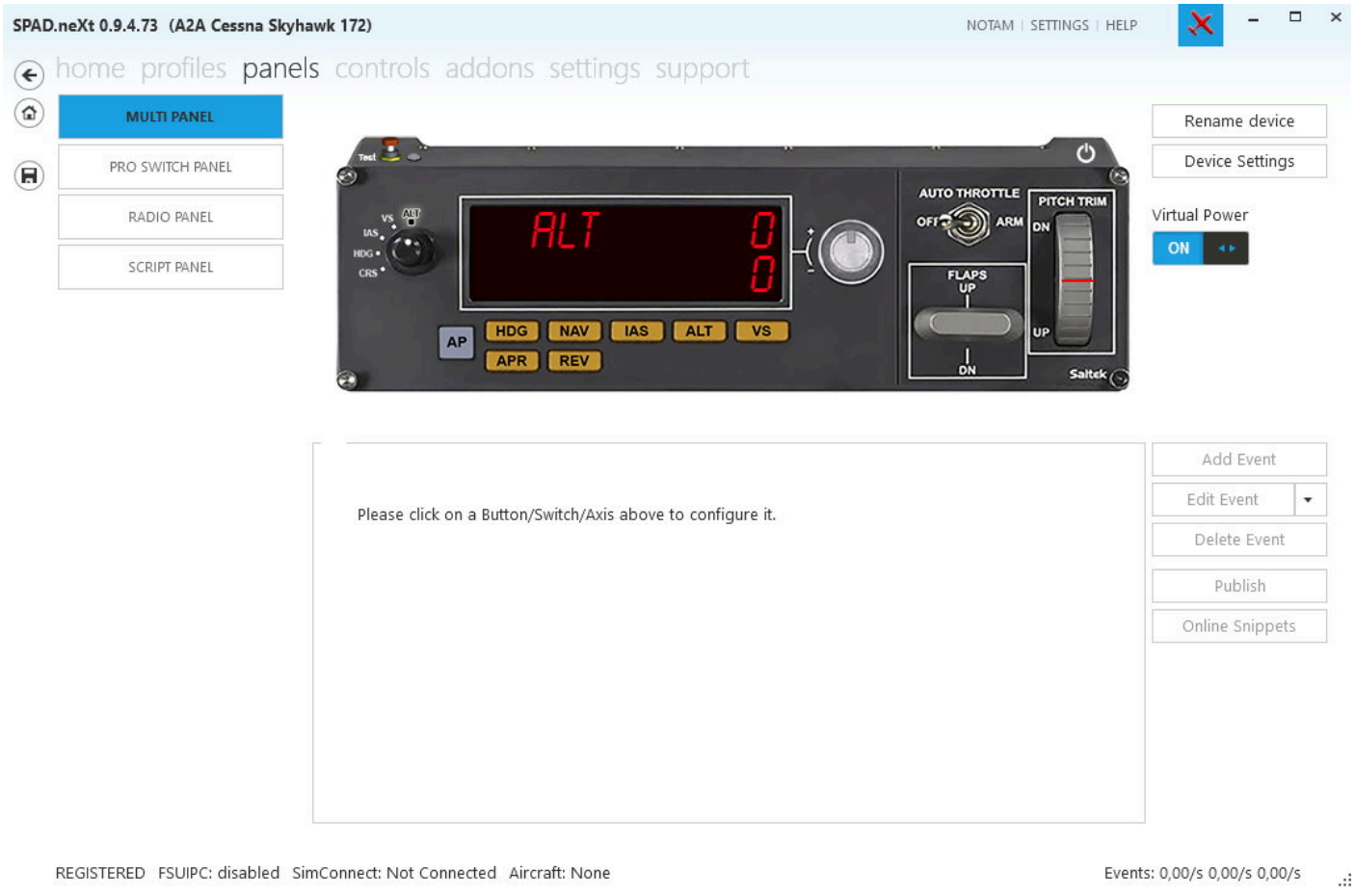


Fig. 24 The interface for the Saitek Multi Panel

You may want to press some of the autopilot buttons on the physical device to make sure the corresponding counterparts in SPAD.neXt will react properly (e.g. being lit). Actually, this screen provides a powerful interface for programming all sorts of functions, which, however, goes beyond just Getting Started.

Next, we switch to the "controls" main menu tab. Let's have a look upon the Saitek Cessna yoke as an example.

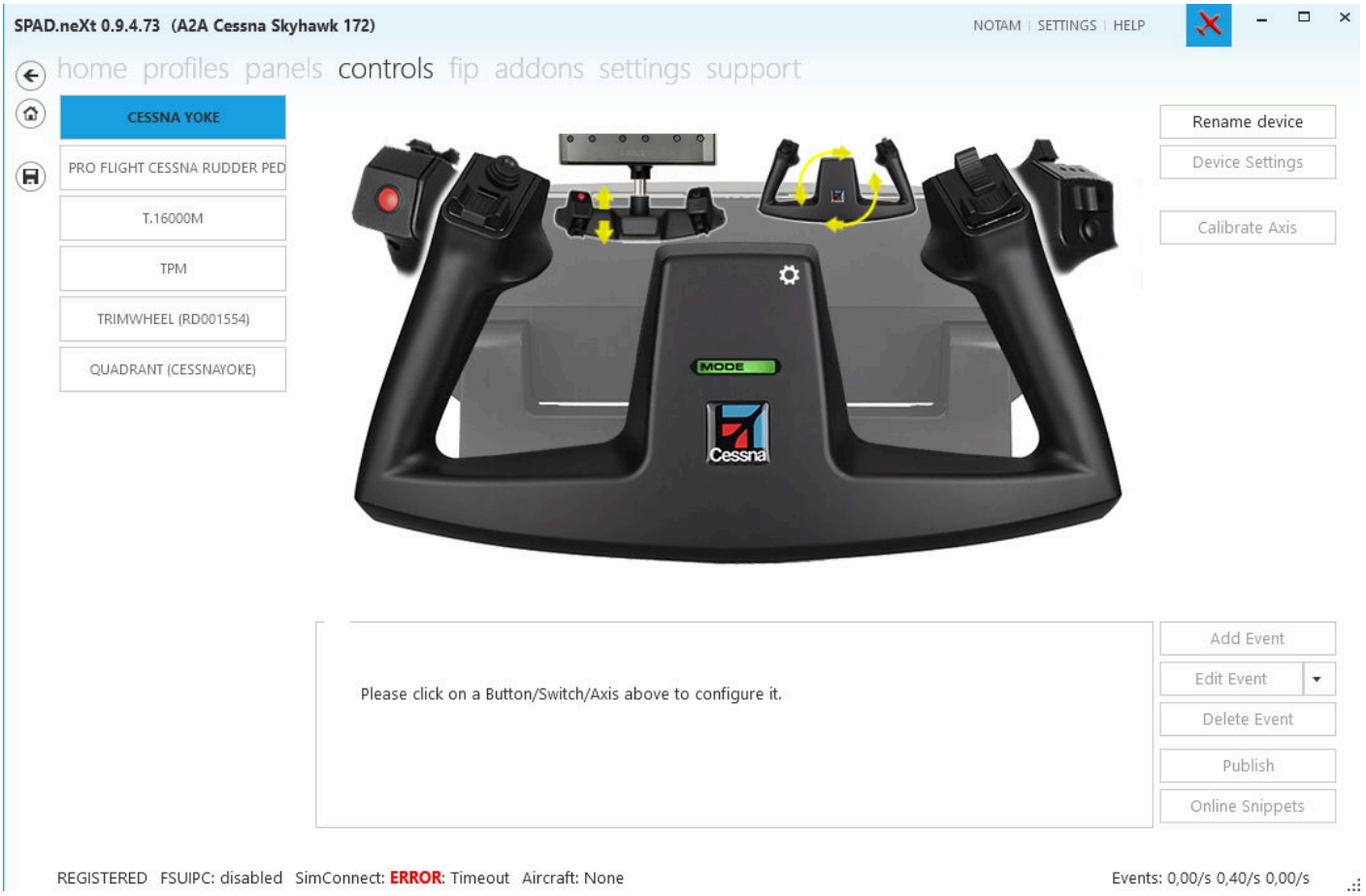


Fig. 25 The interface for the Saitek Cessna Yoke

Again, you should turn the Yoke a few degrees to make sure, the SPAD.neXt representation to react properly.

Registered only

Controls are only supported in the registered version.

The next main menu entry is devoted to "fip". We will assume you own at least one of the FIPs (Saitek Flight Instrument Panel), otherwise this menu entry is just missing.

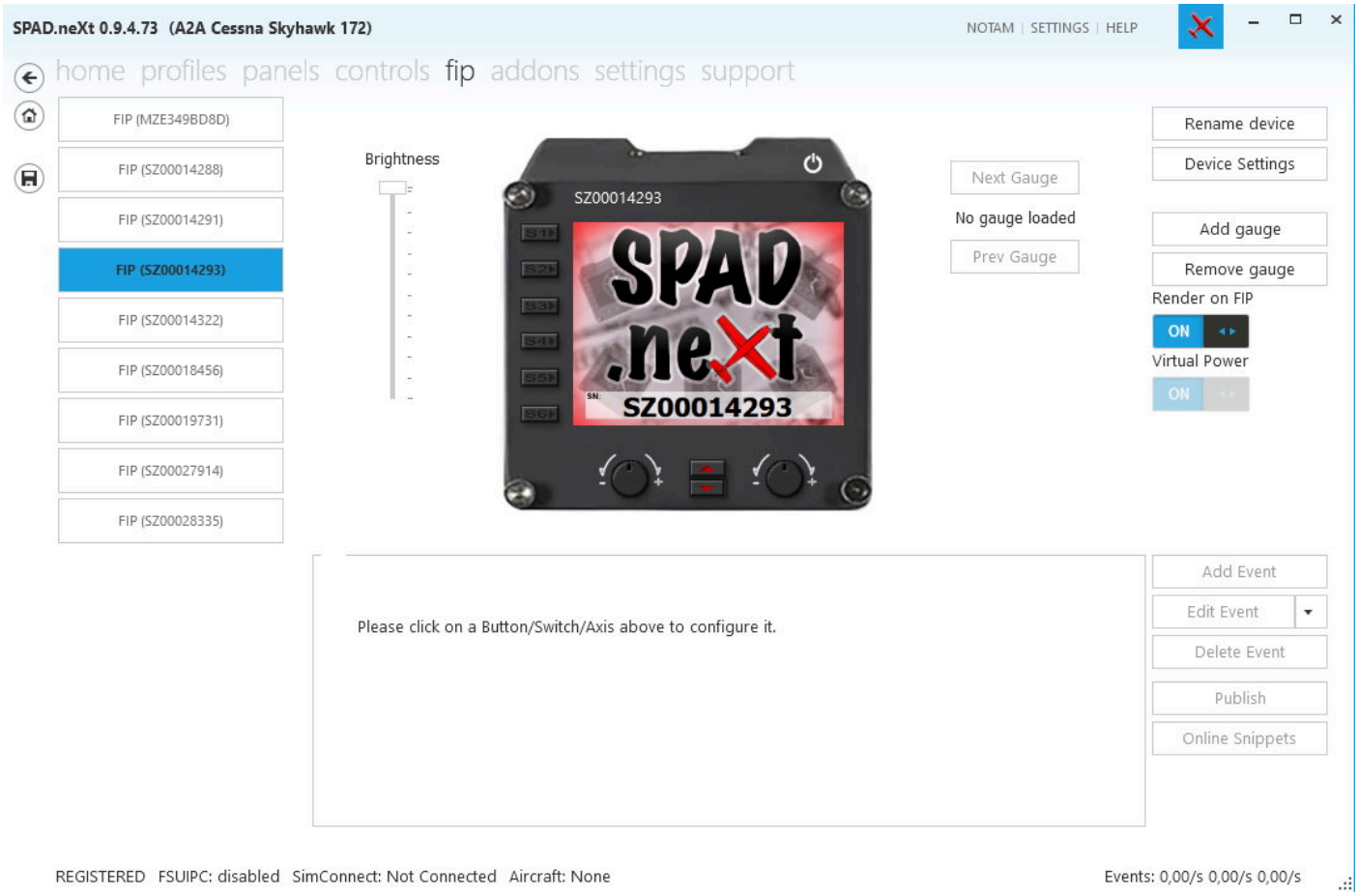


Fig. 26 One of the Saitek Flight Instrument Panels

If you own more than one of the FIPs they are listed in the column on the l.h.s. with their corresponding serial numbers. A similar image as above should be on the display of the physical FIP device. The serial number provides an easy correspondence between the physical device and the one displayed in SPAD.neXt.

Registered only

While FIPs are always displayed in SPAD.neXt, if connected, actual display of gauges on the physical device is only possible using the registered version.

Now that image isn't really helpful for piloting a plane, is it? You need gauges for making use of them. Again, we begin with a very simple solution relying on the Saitek gauges being included in SPAD.neXt for convenience. (Recall: You MUST not have installed the Saitek FIPS software!)

Click onto "Add gauge" on the r.h.s. to get to the gauge selection which should look like in Fig. 23.

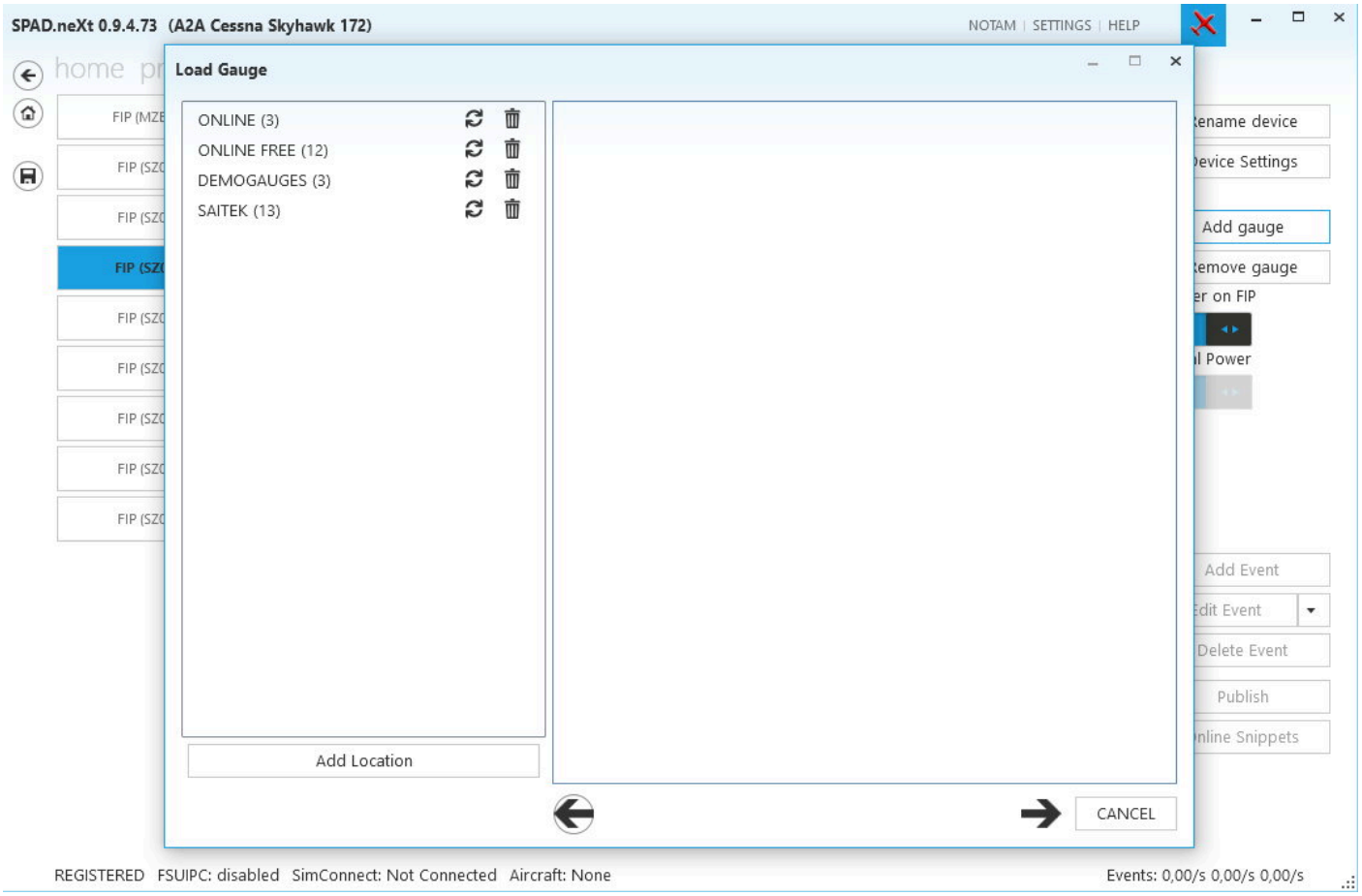


Fig. 27 Main gauge selection

Next, go to the SAITEK entry.

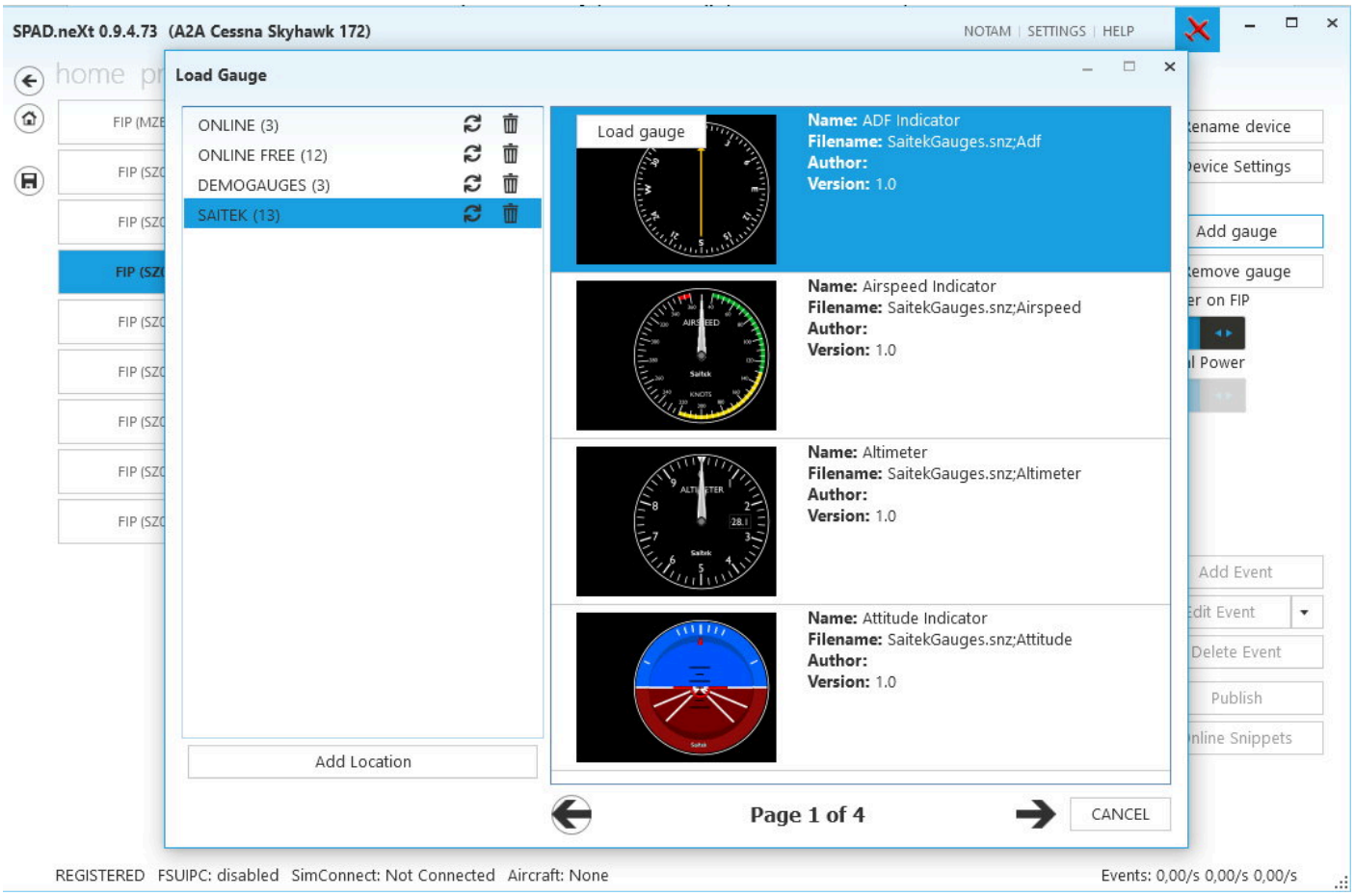


Fig. 28 Gauge selection

A natural choice for the first (or only) gauge is the Airspeed Indicator which can be activated just by double-clicking onto its image. Now your FIP should look like this one:

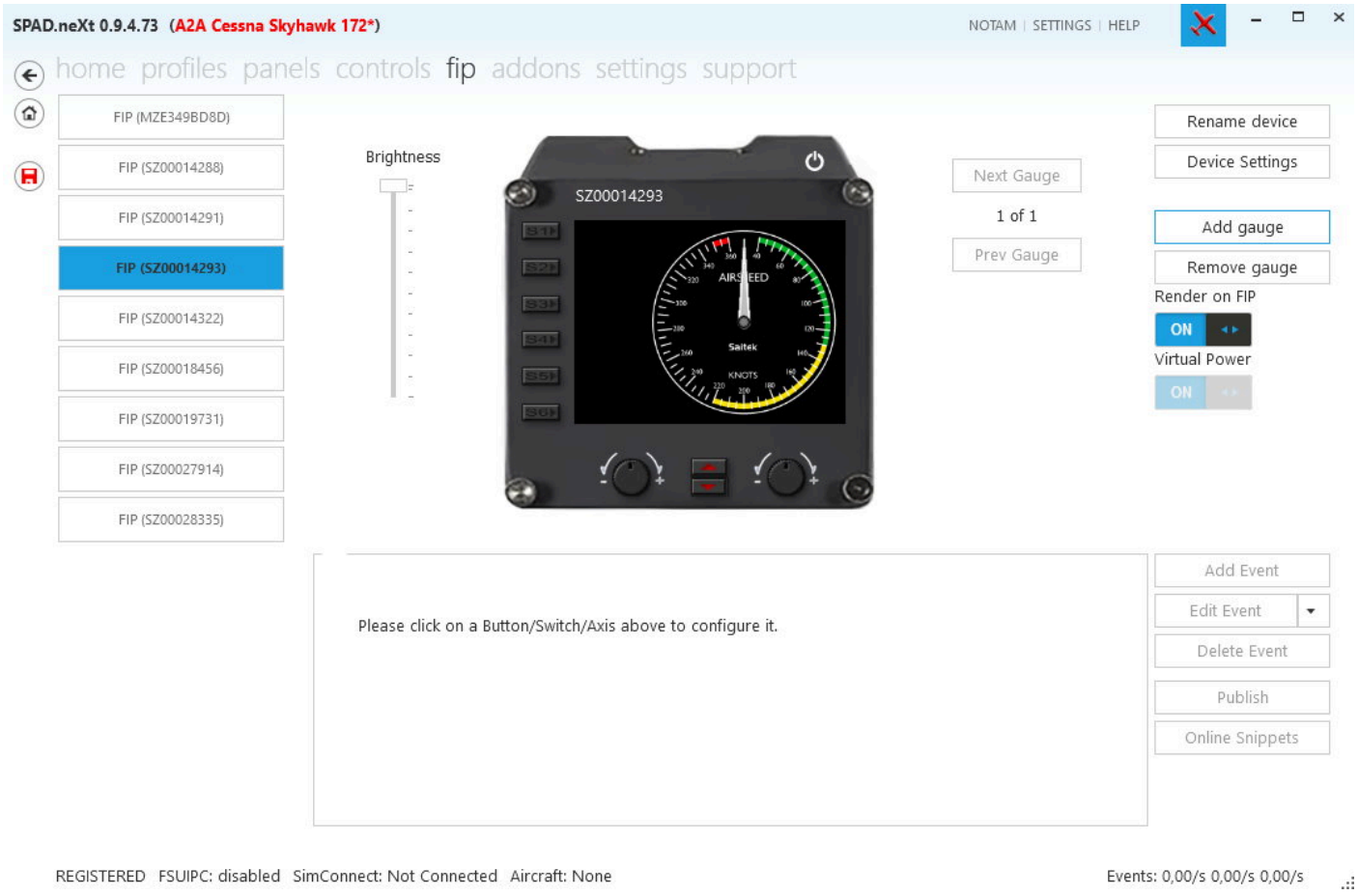


Fig. 29 FIP selection completed - the Saitek Airspeed gauge

You will see the red disk symbol close to the left border meaning you did not yet save your selection in the selected profile. Press it now, to have the gauge selection written to the profile.

Confirm that "Render on FIP" is set to "ON" which is the default setting. Now your physical FIP should display the Saitek Airspeed gauge. As understood, you can assign more gauges either to the same physical FIP (switching between them with "Next Gauge" and "Previous gauge") or assign other gauges to others of your physical gauges in the same way.

Go flying!

At this point you should be ready to go flying. Generally, SPAD.neXt can be started before, during or after the start of the simulator. I made it a habit to start it before the

simulator providing a possibility to see if all the controls do their job. As always, there may be a little Gremlin introducing a loose connection between the yoke and the hub etc. No need to have the simulator running to catch him.

Don't forget to select the A2A C172 (or whatever plane you selected the profile for) in the Scenario Setup Screen. Otherwise, SPAD.neXt does not know which profile to use for the present plane and presents you a selection box (which you are probably unable to handle at the present state).

If all goes well, your C172 is ready to be started on the selected runway or ramp. You can even loose the parking brake using the gear lever - do you recall that assignment when reading the profile properties in Section 8? Given you own the proper hardware, all the basic functions like pitch, roll, throttle. etc. should work as assumed.

Part B : A closer look upon helpful features

Simulator integration

As already mentioned in Section 10, SPAD.neXt can be started before, during or after the start of the simulator. To make SPAD.neXt run automatically whenever you start FSX/P3D you can make use of the EXE.XML file which can be found under

C:\Users\(your profile name)\AppData\Roaming\Microsoft\FSX (FSX)

or

C:\Users\(your profile name)\AppData\Roaming\Lockheed Martin\Prepar3D v3

(or similarly for other simulator configurations). It can be edited, e.g., using notepad (please do NOT use Word or any other Text Processor).

Please, insert the following lines into EXE.XML:

<Launch.Addon>

<Name>SPAD Next</Name>

<Disabled>False</Disabled>

<Path>(Installpath of SPAD.neXt)\SPAD.neXt.exe</Path>

<CommandLine>-m</CommandLine>

</Launch.Addon>

In the (improbable) case you don't already have an EXE.XML file, you can create one, again using notepad, as follows:

<?xml version="1.0" encoding="windows-1252"?>

<SimBase.Document Type="Launch" version="1,0">

<Descr>Launch</Descr>

<Filename>exe.xml</Filename>

<Disabled>False</Disabled>

<Launch.Addon>

<Name>SPAD Next</Name>

<Disabled>False</Disabled>

<Path>(Installpath of SPAD.neXt)\SPAD.neXt.exe</Path>

<CommandLine>-m</CommandLine>

</Launch.Addon>

</SimBase.Document>

and save it into the corresponding directory.

The “-m” in CommandLine will make SPAD.neXt to start minimized to the system tray.

In both cases, it is suggested, to just copy/paste the code from the present document. Needless to say, you will have to adapt the path to the driver according to your installation.

To make SPAD.neXt exit automatically when the connection to the simulator is lost, you can enable this option under SETTINGS|APPLICATION.

General Configuration Settings

After you got SPAD.neXt up and running, it might be a good idea to have a closer look upon some general program settings. For this purpose, we return to the settings menu:

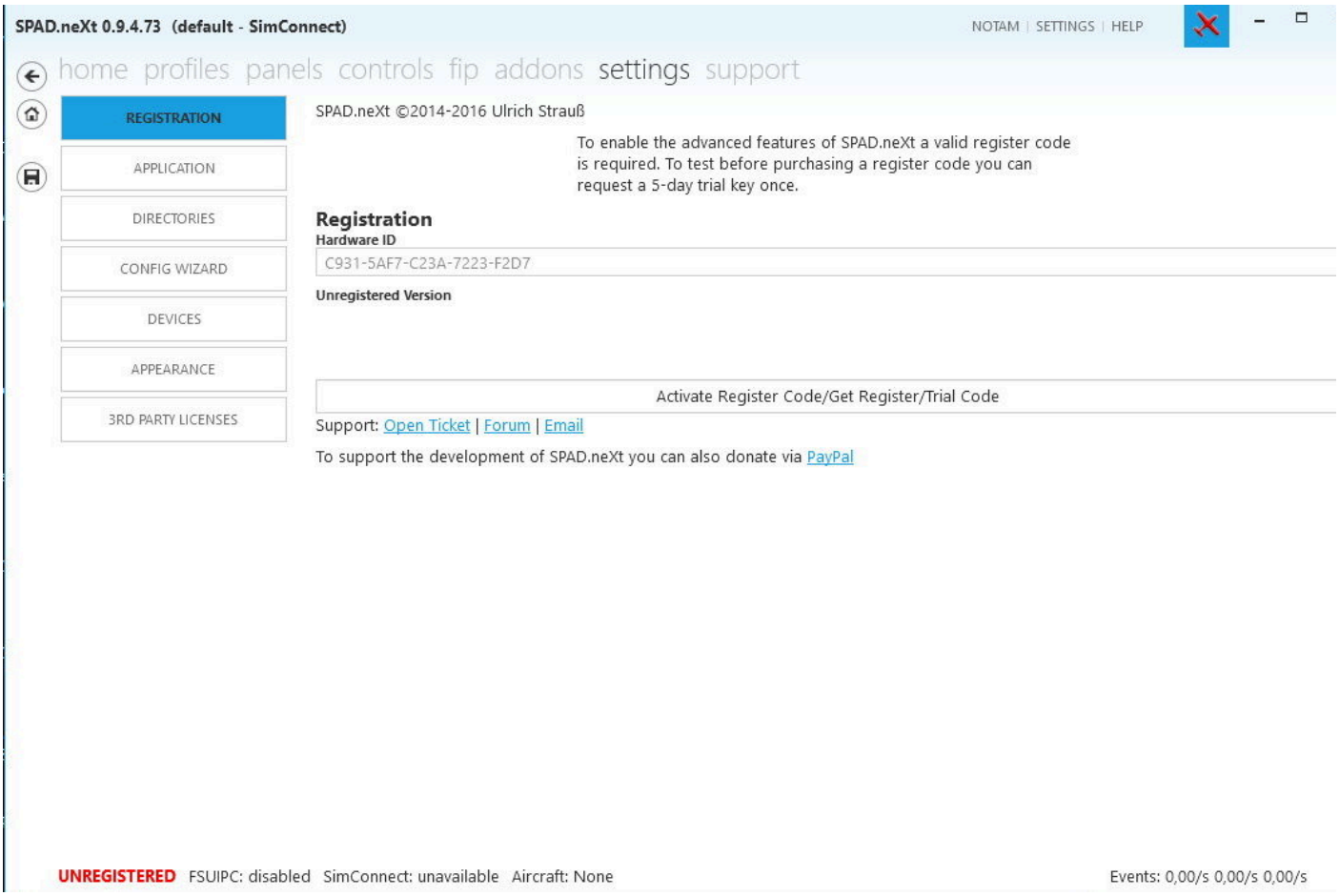


Fig. 30 The settings menu revised

We already dealt with the REGISTRATION submenu. So, let's go on to APPPLICATION, offering four submenus.

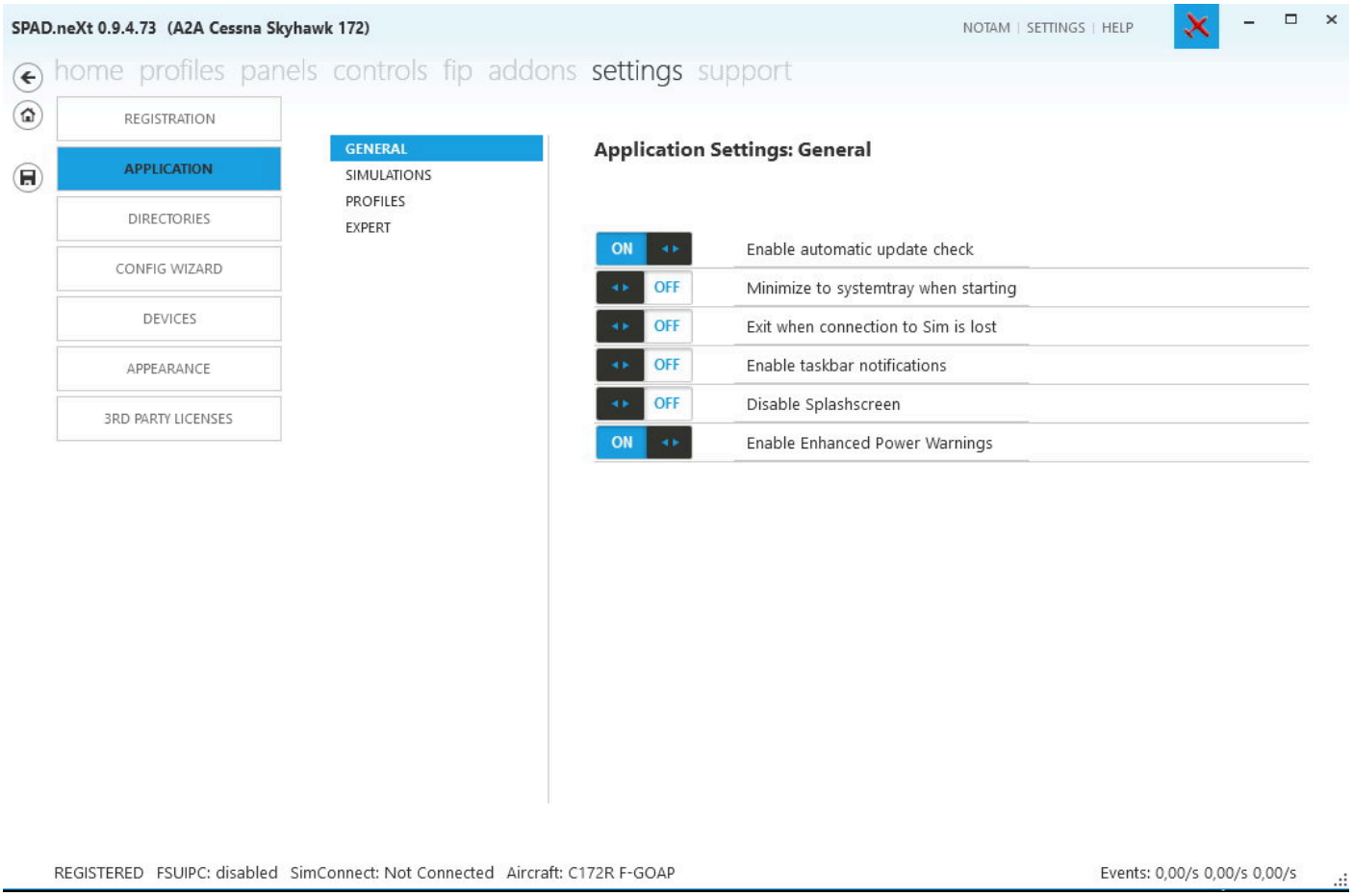


Fig. 31 Application Settings: General

Most of them are self-explanatory. As already mentioned, it's suggested to leave the automatic update check on. There is an active beta team, thus it's quite rare an error creeping into a release version. I'd suggest leaving taskbar notifications off, at least under Windows 8 or 10 - otherwise you'll have quite often that ugly Info Center from the right border popping in telling you something you don't want to know.

The last entry is a bit more cumbersome. It refers to the Windows USB port energy saving settings. These settings may cause devices to power down even at times when they shouldn't. If the switch is set to "ON" SPAD.neXt will detect this (Windows default) setting and produce a warning including instructions how to proceed. According to them, you should close SPAD.neXt once and restart it as an administrator which will modify those settings. Thus, it is suggested to leave this setting always "ON".

The next screen refers to various features required by certain profiles (in other works, for certain types pf planes) to work.

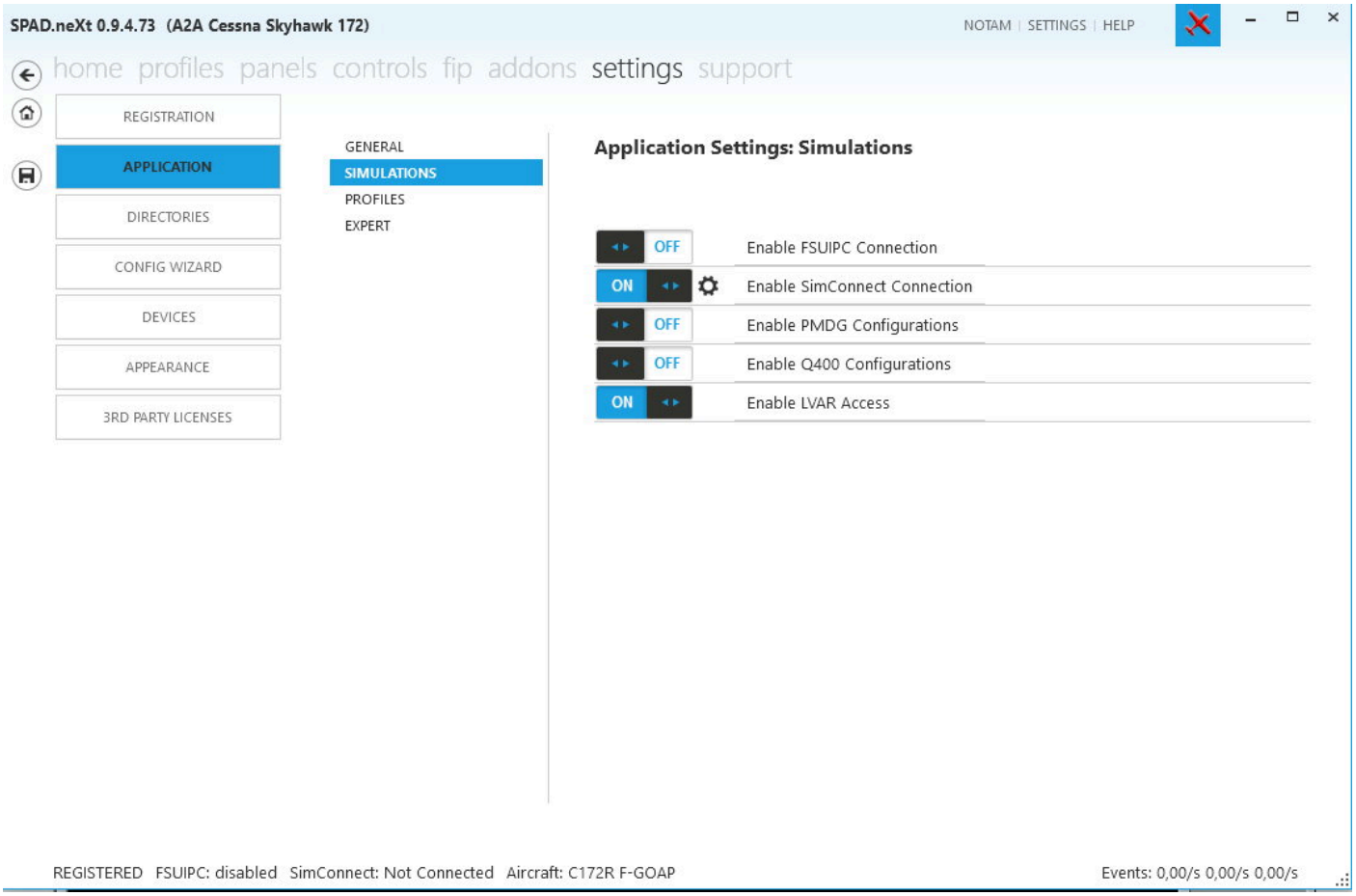


Fig. 32 Application Settings: Simulations

You can read off the requirements under "Provider" when importing online profiles. As already mentioned, A2A planes require LVAR support, and so on.

Registered only

All of these additional support features can only be enabled in the registered version.

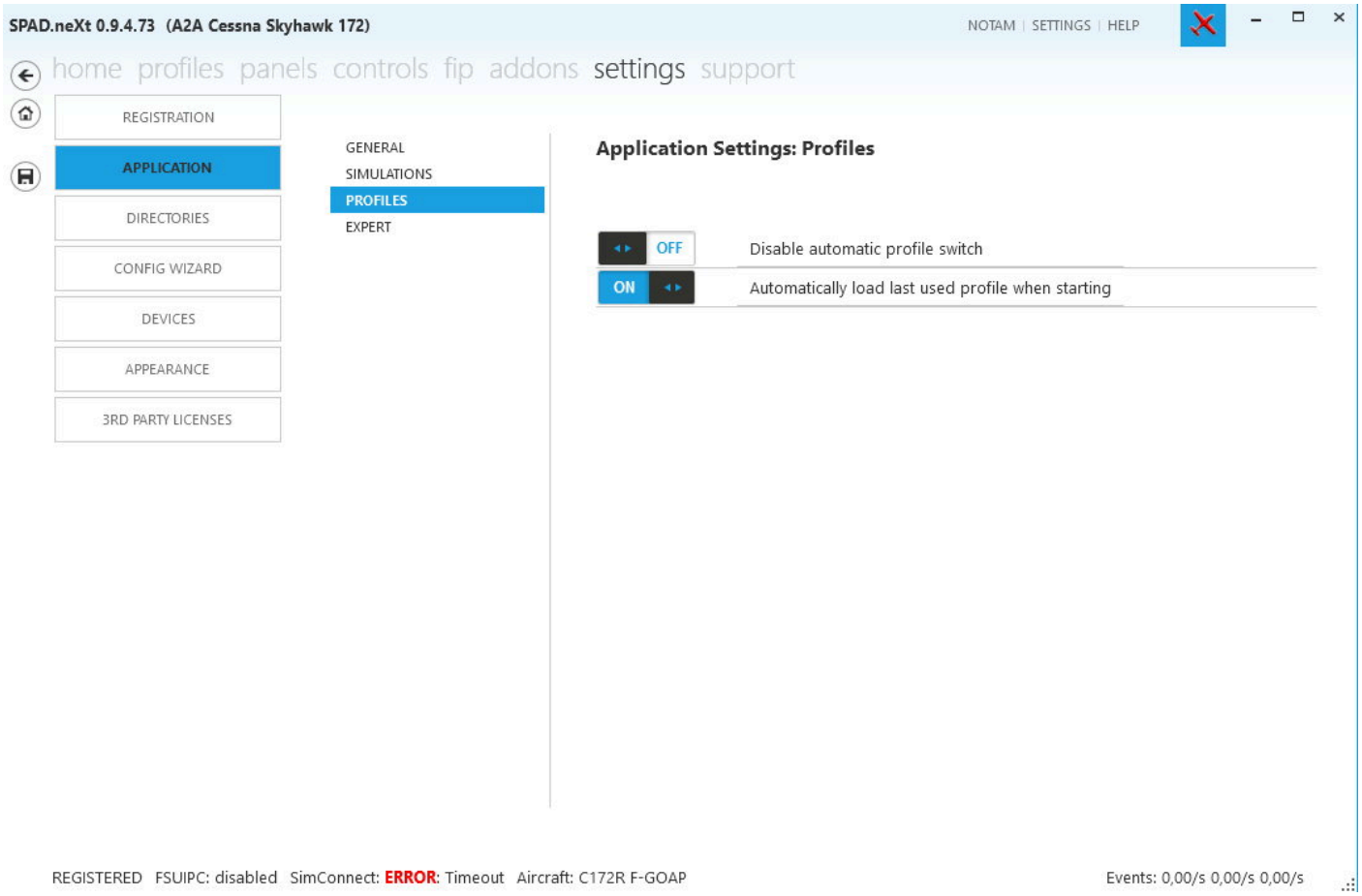


Fig. 33 Application Settings: Profiles

The 3rd submenu provides two settings for using the profiles. The first one is useful as soon as you gather quite a number of planes and corresponding profiles. SPAD.neXt will detect you assigned that "PMDG 777" to the Profile "My excellent 777 profile" and use it automatically as soon as you call the PMDG777 either at the start or while the simulator is running. If the switch is turned off, SPAD.neXt will not switch profiles automatically and just keep the currently loaded one, when switching aircraft in the simulation.

The last entry should be left as is as well. If you start SPAD.neXt before the simulator or just for maintenance purposes, it can't detect an active plane, thus it opts for the last active one - makes sense, doesn't it?

We will leave the Experts menu to the experts, skip DIRECTORIES to the left and switch to Config Wizard which provides an opportunity to repeat the Config Wizard we recall from Section 5, just in case you feel something went wrong. Most of the settings don't require the complete Wizard to be rerun as they can be changed via the Settings menu. Besides, this menu entry provides a possibility to enter the LVAR Bridge entry into you DLL.XML file, should it been missing (e.g., in case you started with the freeware version and registered it later providing the possibility to use LVAR support).

The DEVICES submenu gives a possibility to make some global (i.e. profile independent) settings for your devices. Most of them (e.g. switching them on/off) should be self-explanatory. As understood you can switch off support for devices you don't own.

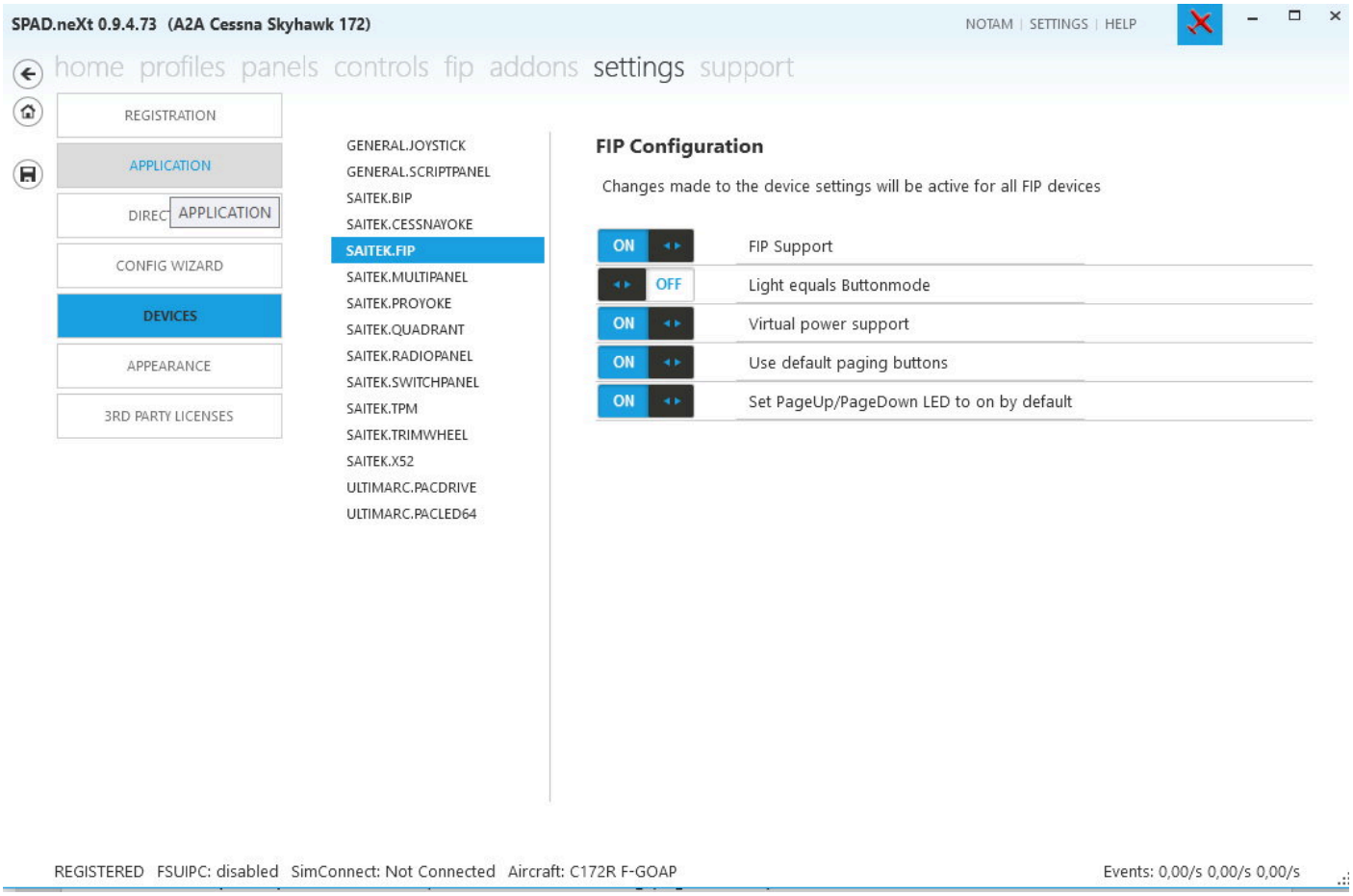


Fig. 34 Devices: Global FIPs Settings

The FIPs submenu has a few useful entries.

"Virtual power support": This will enable or disable the support for virtual power, to turn the FIP off based on conditions in the aircraft (e.g. if the avionics switch is off)-

This may be a matter of taste, but does the airspeed gauge in a real Cessna turn black as soon as the battery is switched off?

Besides, do you hate those unrealistically reed-glowing triangles below the SAITEK FIPs? The last option gives you an opportunity to turn off their LEDs altogether.

The submenu entry APPEARANCE allows you to adapt SPAD.neXt to your liking (and environment, e.g. bright or dark) and should be self-explanatory.

Don't forget to "Save Changes" after modifying any configuration setting.

Working with profiles

After our rather sketchy treatment in Part A we will have a closer look upon working with profiles. We already described in Section 8 how to load an online profile, but there are more options.

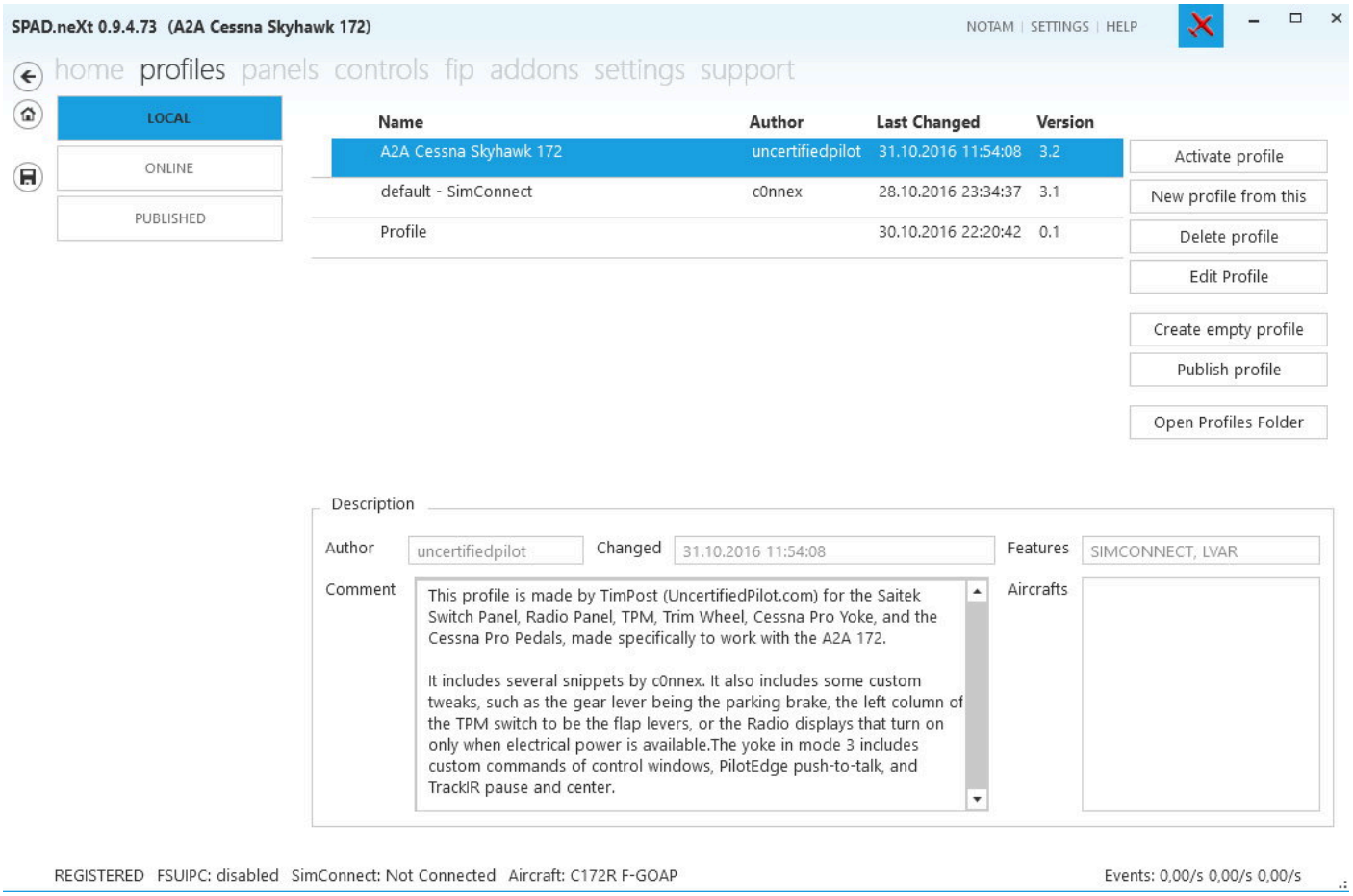


Fig. 35 Once more on profiles

As you can see, the "A2A Cessna Skyhawk 172" we imported is listed under "LOCAL" now. According to the button selection on the r.h.s. you can do several manipulations with your local profiles.

Activating a profile might come handy, e.g., in case the simulator isn't running and you want to do some manipulations on the profile (e.g. exchange FIPs or modify button settings).

"New profile from this" creates a double of the active profile. Did you note the "A2A Cessna Skyhawk 172" on top still being assigned to "uncertifiedpilot"? This isn't you, is it? Thus, if you start adding or modifying stuff (e.g. adding FIPs), it's a better habit to make your personal copy like this:

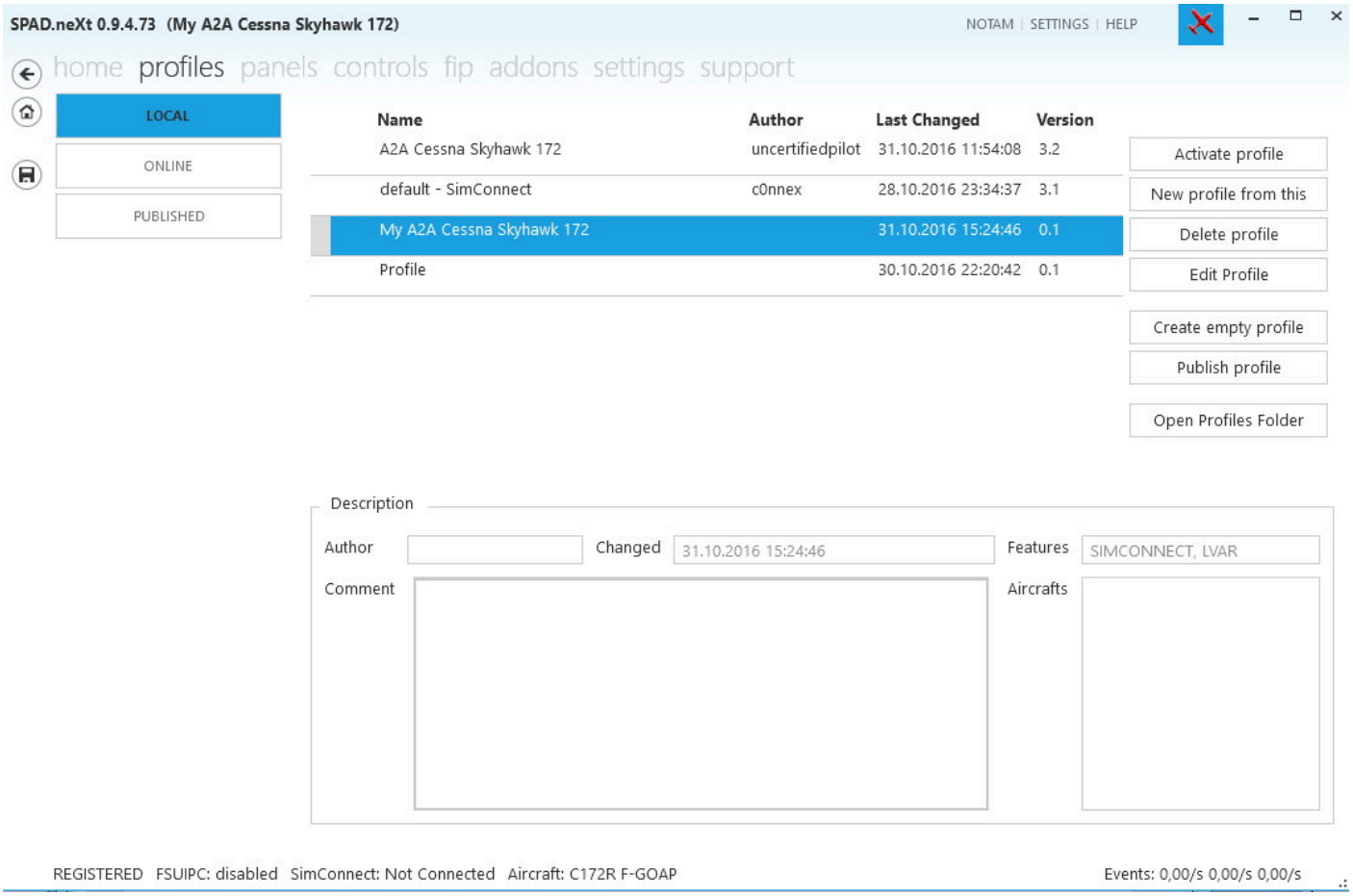


Fig. 36 Copying a profile

Moreover, this allows to provide your own comments (use "Edit profile" for that purpose). "Delete profile" and "Create empty profile" are self-explanatory. The last button leads you to the folder where your profiles are saved (usually under *C:\Users\(your profile)\Documents\SPAD.neXt\Profiles*). It is certainly a good idea to make a backup of this folder now and then, even more, if it contains your own valuable creations.

Where do all those Online profiles come from? Actually, they are published by users. Publishing your profiles can be done via the "Publish profile" button. It requires a forum account and, even more important, some background on programming your devices and thus is beyond just Getting Started.

Once more on panels, controls, and fips - how to assign a button yourself

SPAD.neXt is a very powerful and highly flexible tool which allows pretty much any conceivable assignments to buttons, switches, or display functions. This, however,

requires some knowledge not only of the assignment mechanisms but of SimConnect, LVAR variables etc. as well.

While a complete programmer's guide is way beyond this introduction (and not required by the average user either), we will show you at least two ways how to proceed in case you really want to add or modify functions. The following will be described taking a Saitek Cessna Yoke as an example, but application to other control devices should be more or less straightforward.

Registered only

This description will be given for modifying Controls settings, which most users might be interested in. Controls are only supported in the registered version. However, the general idea does also apply if you want to modify, e.g., a switch on the Saitek Switch Panel.

14.1 Using Online-Snippets

We start by activating our personal profile "My A2A Cessna Skyhawk 172". Next we change to the main controls menu entry and activate CESSNA YOKE.

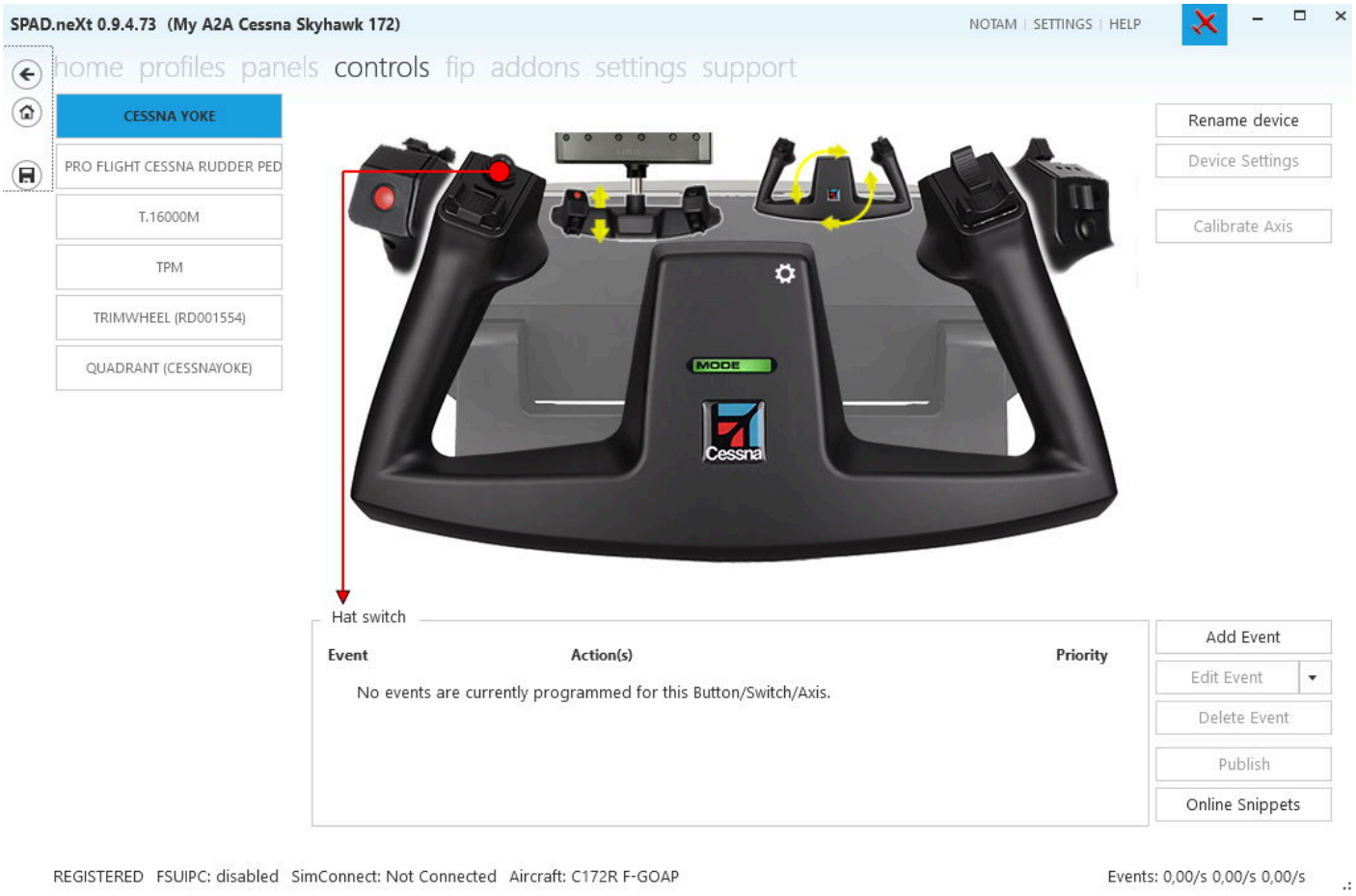


Fig. 37 : Controls: The Saitek Cessna Yoke

Clicking on the hat switch you will note there has been no action (given in the pane below the yoke) assigned to it. Thus, it does not sport the convenient function of looking around by tilting the hat switch.

Now we could start assigning these functions by hand, but there is a smarter method, called Online Snippets. Leave the hat switch active (i.e. red line visible) and press the Online Snippets button on the r.h.s.

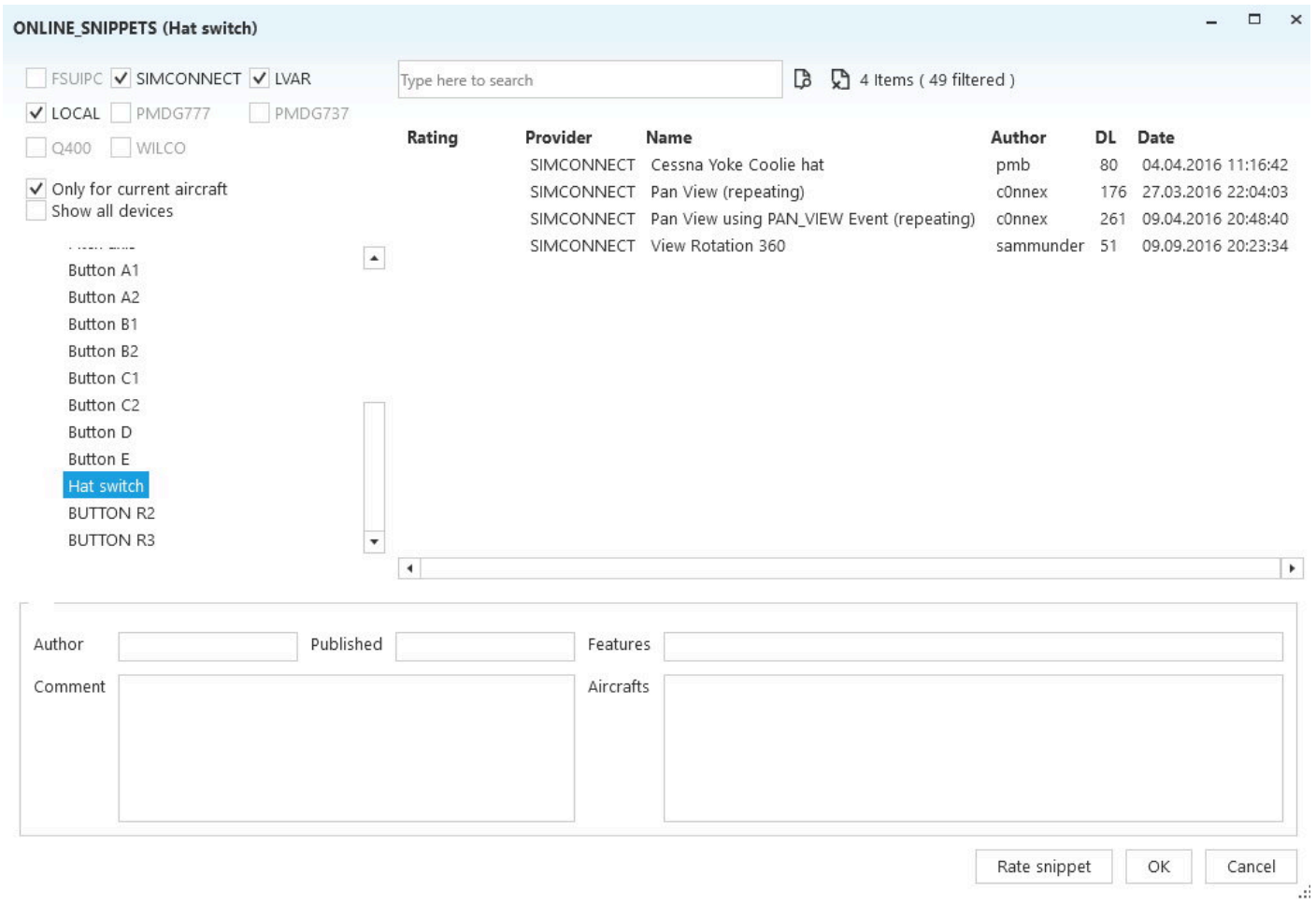


Fig. 38 Importing an Online Snippet

As you can see, there were 4 Online Snippets submitted for the coolie hat so far. In absence of any detailed description it is up to you to try which one you like best (probably there is not much of a difference). In this case the selection is easy: The first one "Cessna Yoke Coolie hat" was submitted by the author of this guide, thus I will give it preference. Put the blue bar onto it and press OK.

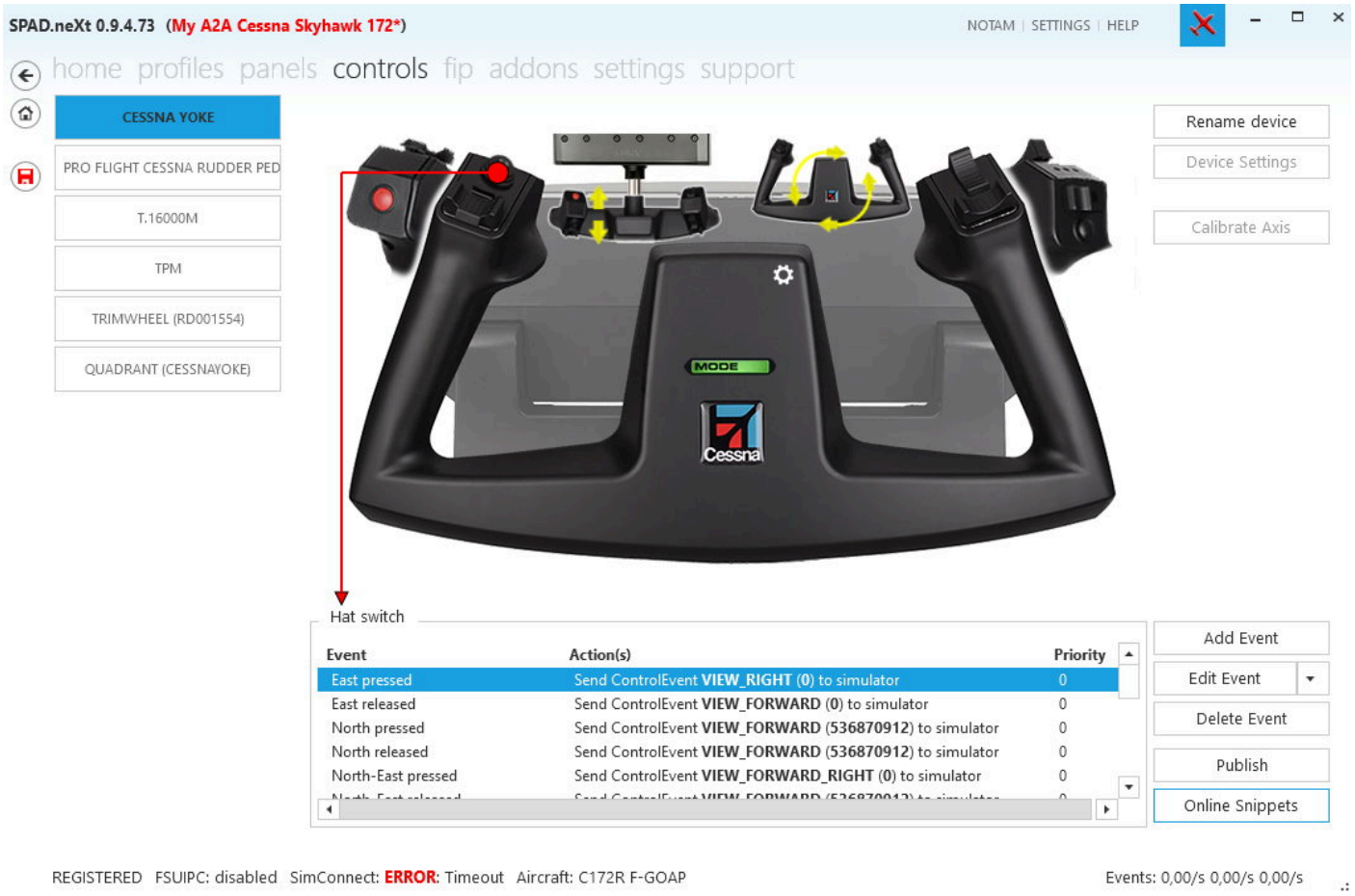


Fig. 39 Coolie hat functions have been assigned

You will note the functions of the several view directions in the pane below the graphics. Fortunately, you don't have to care about the entries at this point but just enjoy the functions.

Again, don't forget to "Save Changes" (red disk icon) after modifying the configuration.

14.2 How to assign a function to a button

This is probably at the borderline between Getting Started and Advanced, but I want to give anyone a hand being brave enough to try it himself or herself. After saving our present configuration, we are going to assign another useful function: It is often required to pause the simulator for whatever reason, thus we want to assign the Pause function to another button. Clicking through the various choices shows that the right rocker button (the one in the North-South direction) has not yet been assigned a function for its backwards movement.

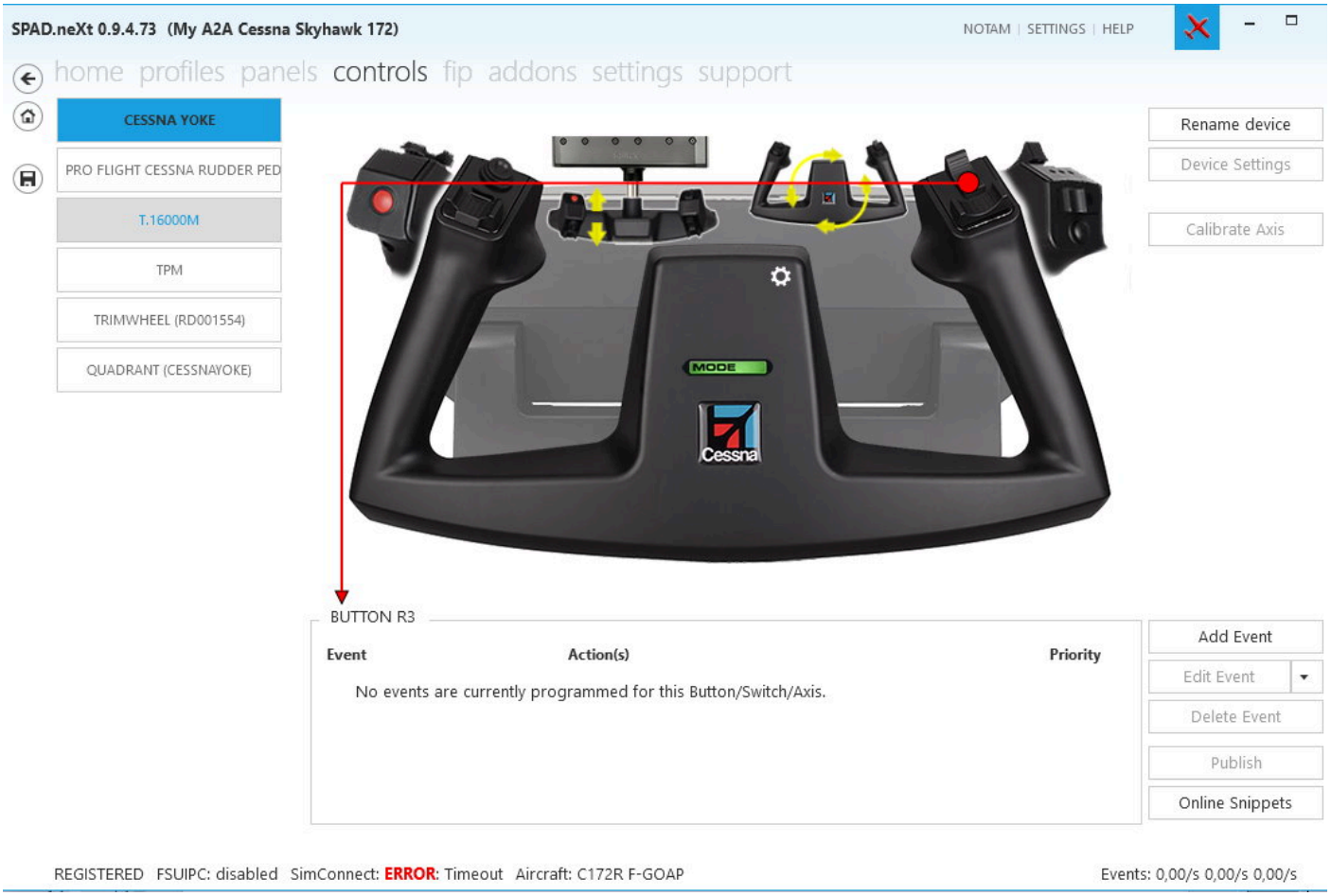


Fig. 40 The backwards movement of the r.h.s. N-S rocker switch is still unassigned

This will become our pause button. Press "Add Event" on the r.h.s. and select "Pressed (Mode 1)".

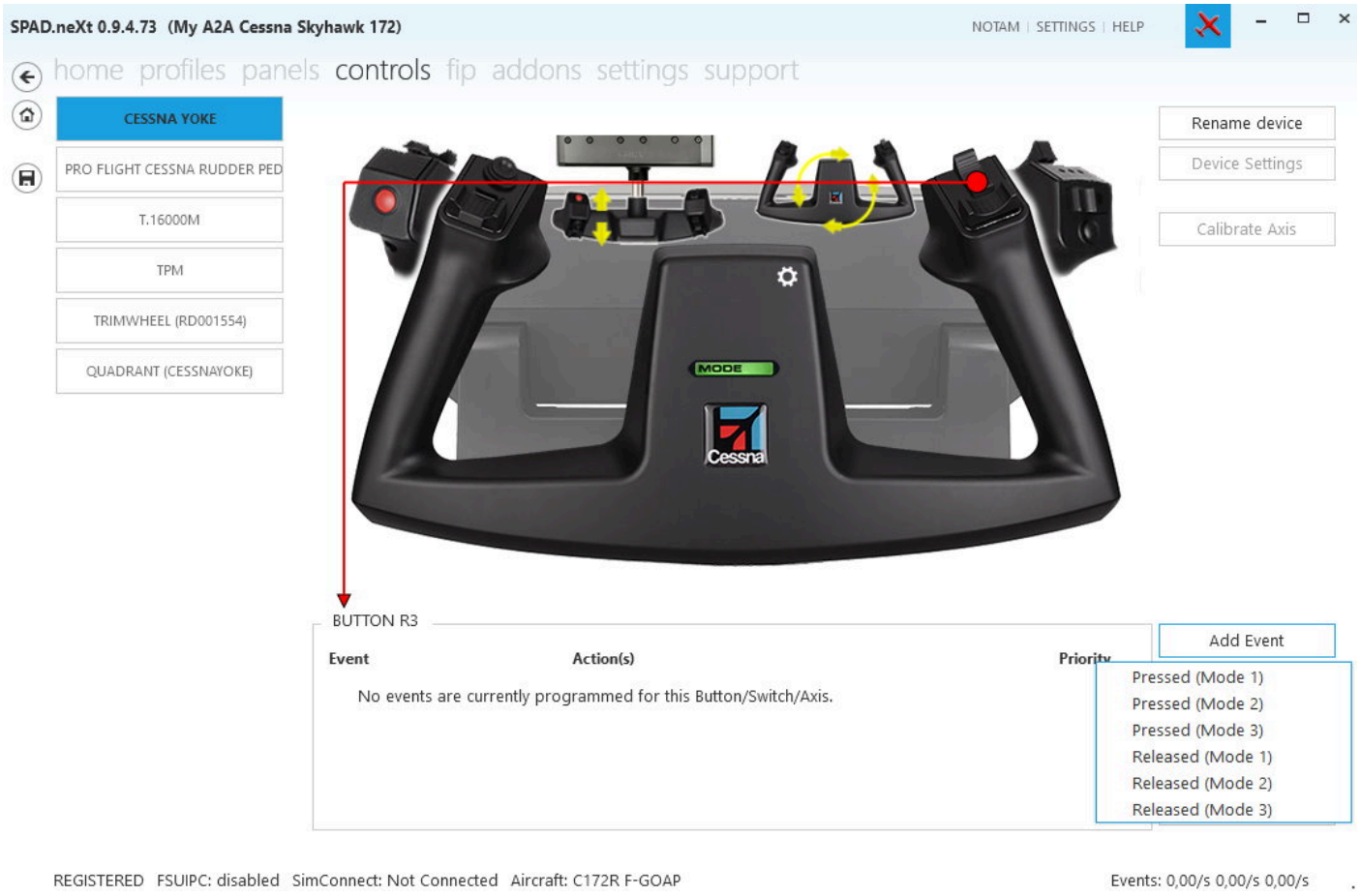


Fig. 41 Select Pressed (Mode 1)

Next you will be asked to "Add Action". Select "Send Simulation Event" here. You may note there are more interesting options like assigning a keyboard key to a button.

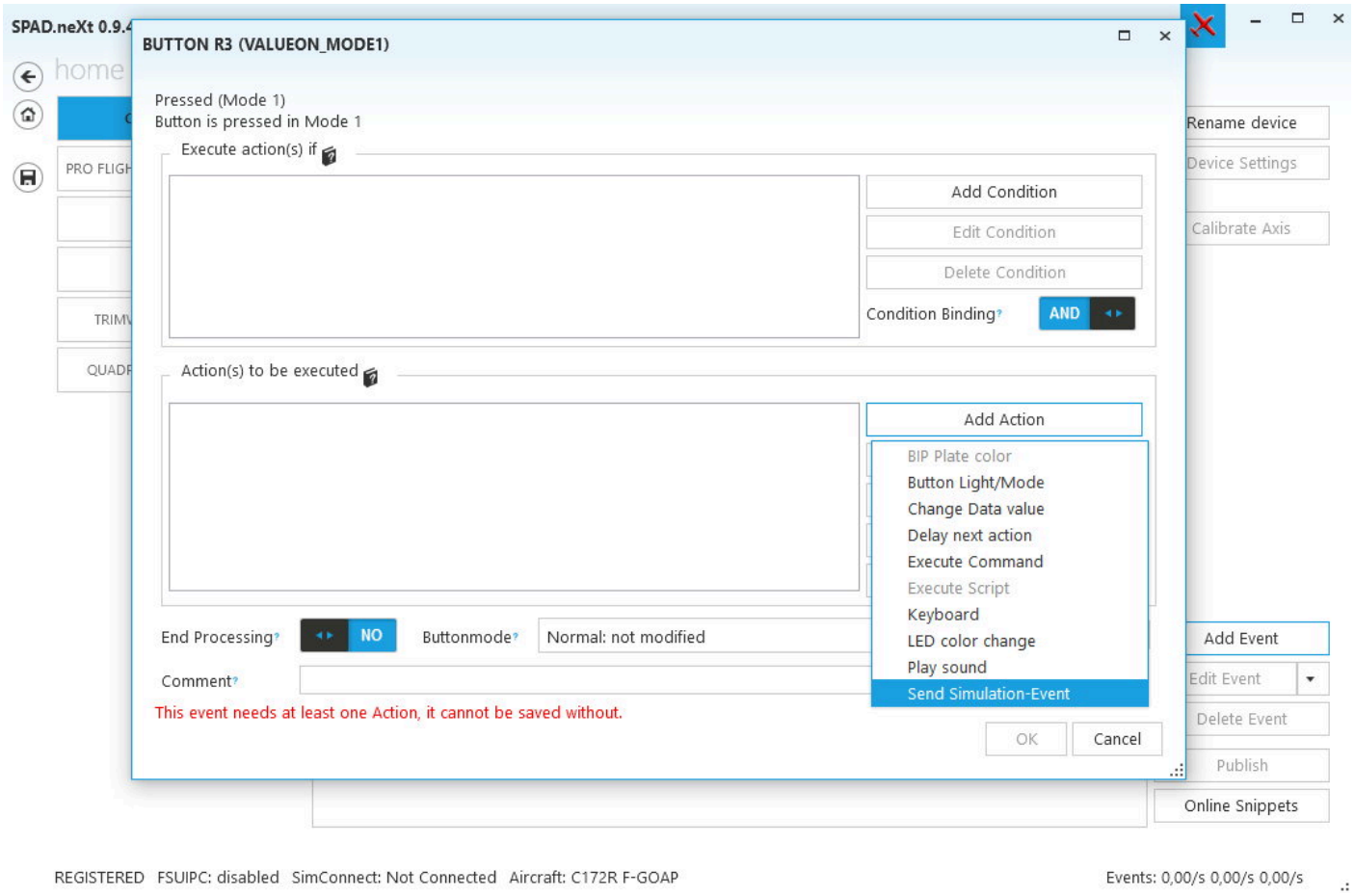


Fig. 42 Select Send Simulation Event

Now you are presented a long list of events to select from. Fortunately, there is a search function to input "pause". This will provide five results, where obviously the last one fits our intentions: Tilting the rocker switch backward once toggles pause on while the next tilting will switch it off.

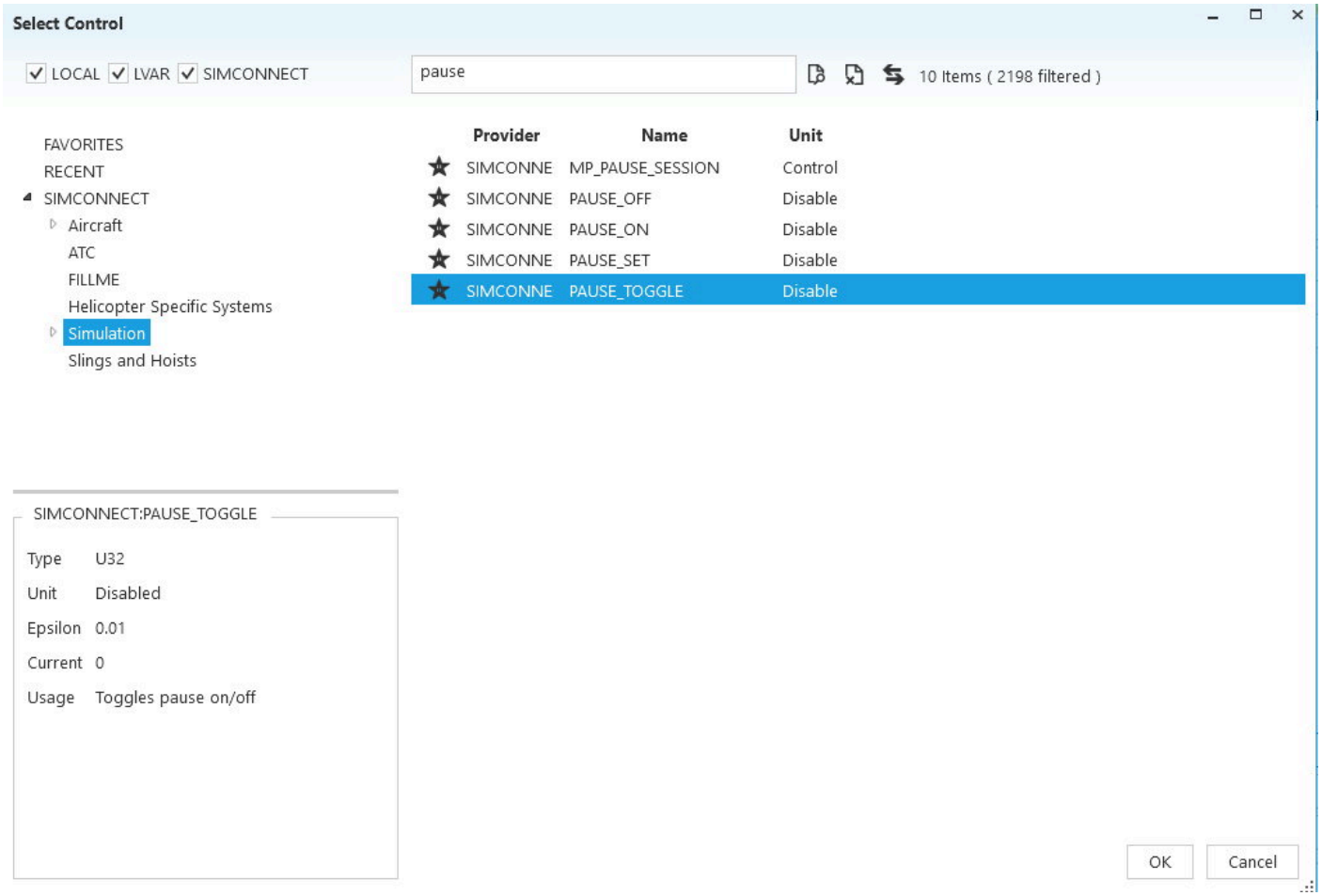


Fig. 43 Select PAUSE_TOGGLE

After making your selection confirm "OK". Leave the next window "Define ControlEvent Action" as is and confirm another time "OK"

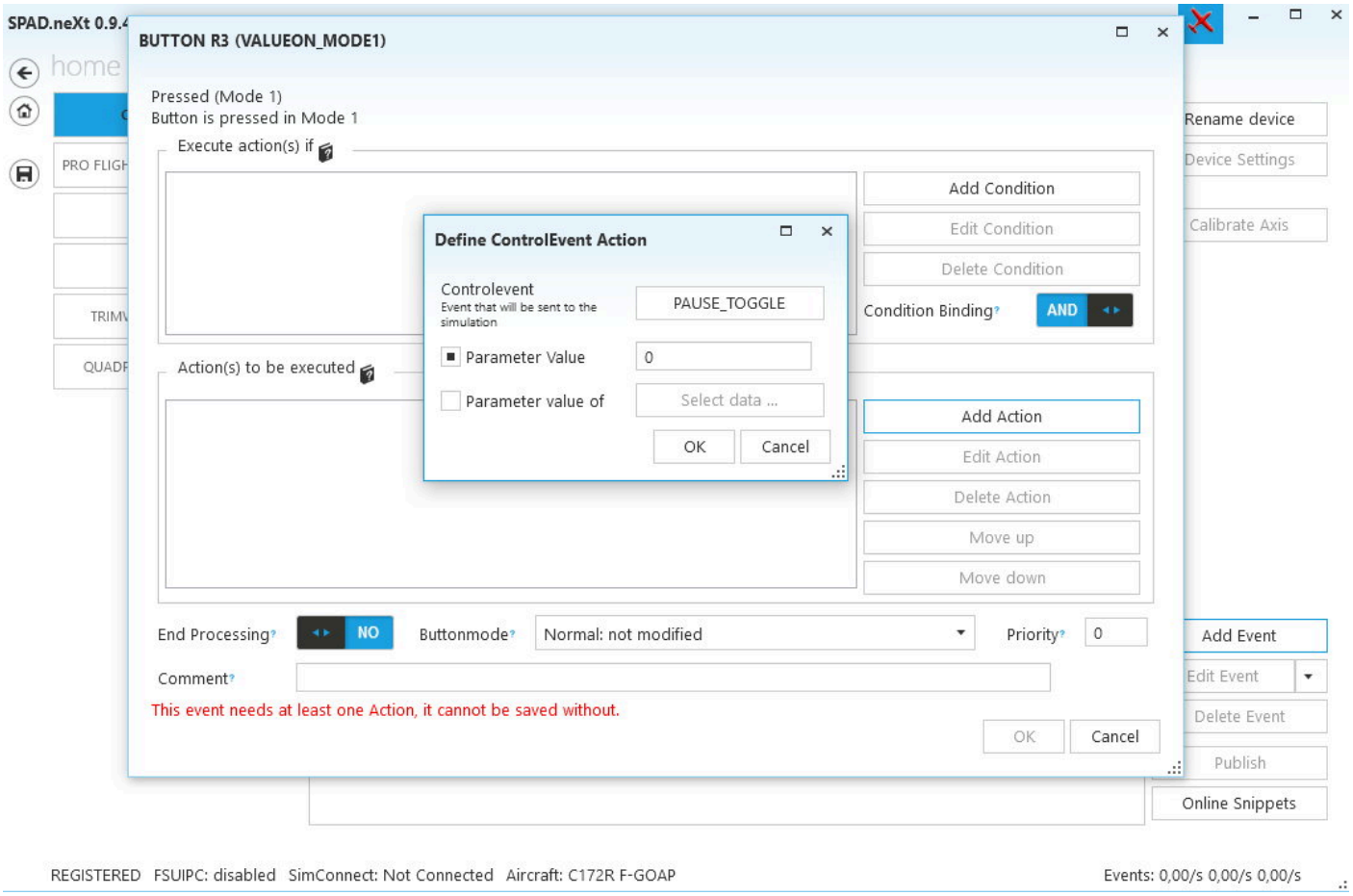


Fig. 44 Confirm the "Define ControlEvent Action" dialog as is with "OK".

This completes our assignments which should look like this:

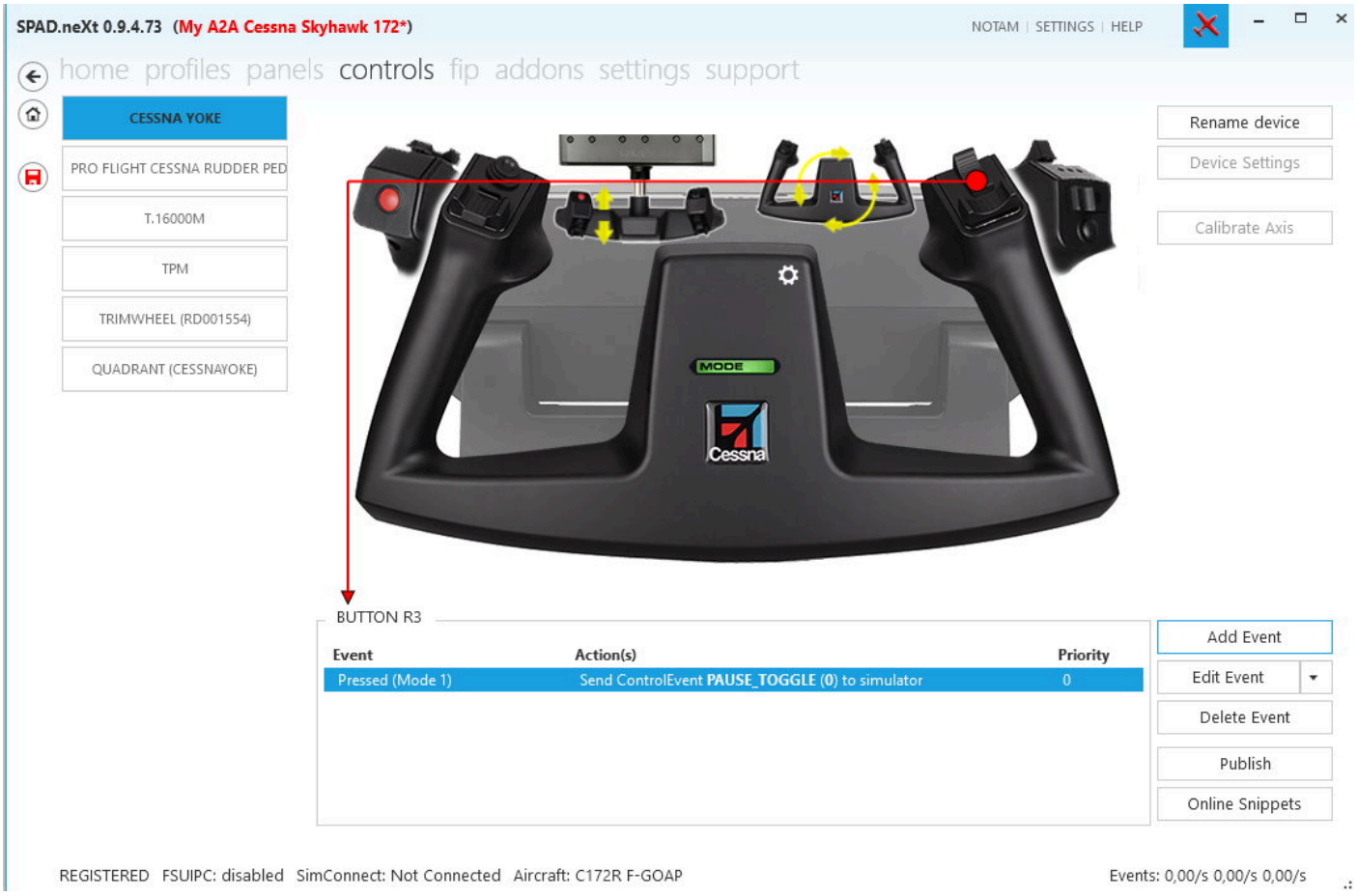


Fig. 45 Assignment done!

Again, don't forget to save your assignment and happy flying to try it!

Gauges, gauges, gauges

We have seen in section 9 how to add a simple Saitek default gauge to a FIP. Granted, the One-for-all-style-Saitek panels work as such, but don't they look a bit, well, plain? Notably, if you own quite a number of those FIPs or even have them integrated into a bought or home-made panel, you may deserve something better.

Fortunately, SPAD.neXt supports a number of far more sophisticated and beautiful gauges, partially even developed in parallel with it. To access them for installation, recall how to assign a gauge to a Flight Instrument Panel (cf. Fig. 24). Instead of the last entry "SAITEK" we try the first one "ONLINE".

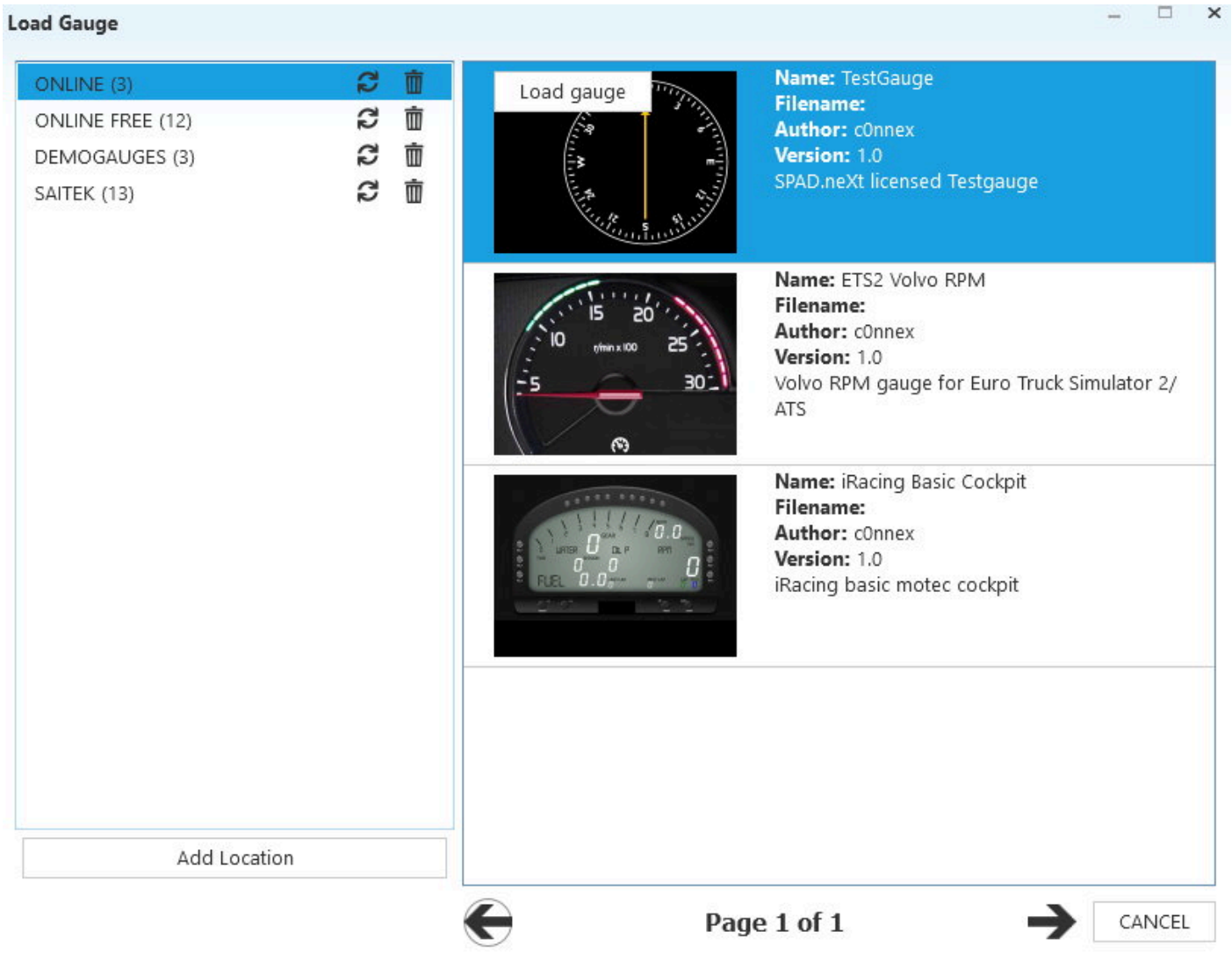


Fig. 46 The future gauge market

That's not yet impressive, but a preview for the future online gauge market where you will be able to buy gauges directly from within the program. The gauges in Fig. 41 indicate though, that SPAD.neXt is aimed to support much than just flight simulations.

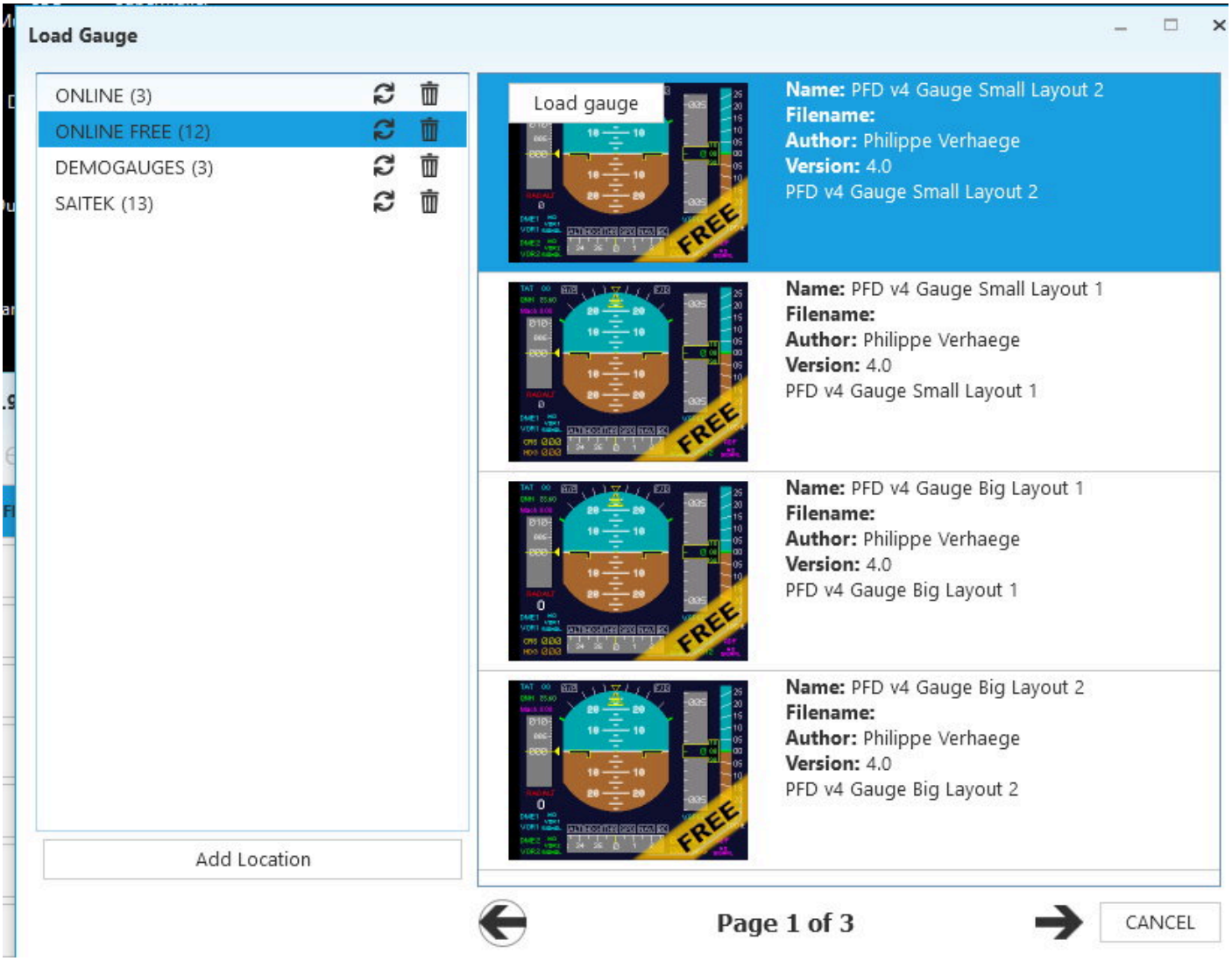


Fig. 47 Freeware gauges

The next entry provides a preview for a number of freeware gauges. You can download and install them, just by clicking onto them.

Finally, the 3rd entry showcases creations by two commercial providers providing their own gauge shops under fipgauges and FSXTimes, resp.

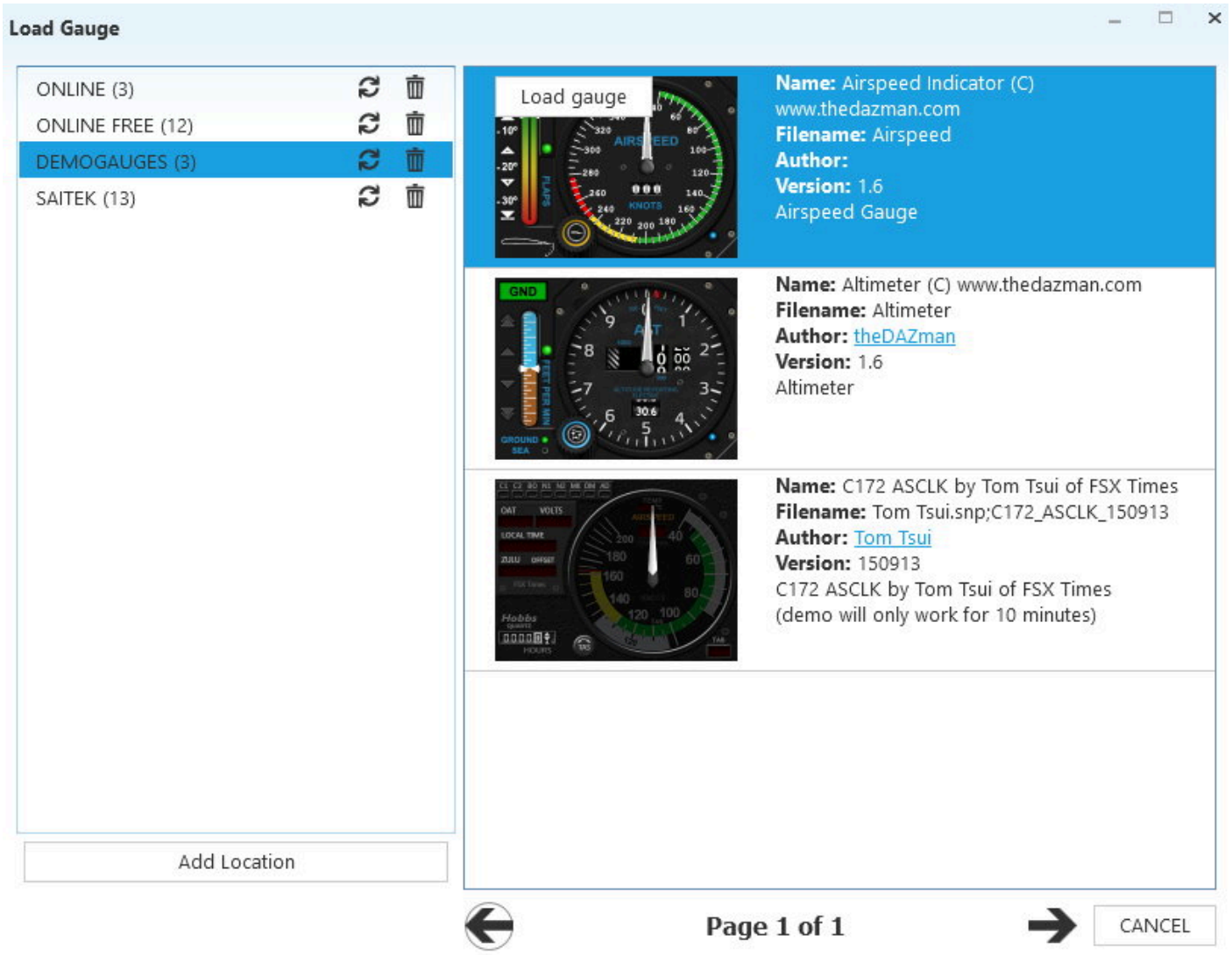


Fig. 48 Demo versions of payware gauges

You can install the demo gauges, the only limitation is they only will work for 15 minutes. Both providers' creations are two classes above the Saitek gauges and sport a number of additional functions like flaps display, time, and more, which the Saitek gauges lack. Having used their creations for years now, I will never return to the line-graphics style Saitek gauges.

Granted these professional-grade gauges are not "cheap" if you are going to buy a larger number of them, but they are worth every cent. Moreover, look for bundles if you buy several of them.

Installation of any addon gauges is trivial: Just make a new folder under the "Gauges" in *C:\Users\(your profile)\Documents\SPAD.neXt\Gauges*, place them there and the next time SPAD.neXt will be started they will be found.

SimConnect and networked configurations

As already mentioned in several places, SPAD.neXt makes use of the SimConnect interface developed by Microsoft and now by Lockheed Martin.

SPAD.neXt presupposes a working SimConnect interface in any case, independent on if it runs on the same computer as your simulator or networked on a Client.

15.1 SimConnect installation on the Flightsim PC

To begin with: SimConnect installation is a bit tricky and can even be a hassle. The following is a guideline to get you started. We will refer to some further reading at the end of this Section.

Before going to install SimConnect it is advised to check if it is already installed on your flightsim machine. This is highly probable as a lot of programs like Active Sky, FSGlobal Real Weather, Plan-G, or others use it. Check under

C:\Windows\WinSxS

If there are any SimConnect versions available you are done (in case there are several of them, do NOT uninstall them). Skip all further steps in Section 15.1 in case you already have a working SimConnect connection!

If this is not the case you have to install SimConnect. FSX:SE and Prepar3D have SimConnect installed by default, while you have to install it yourself for FSX (boxed). A detailed guide can be found under

http://fsdeveloper.com/wiki/index.php?title=SDK_Installation_(FSX)

Next, you have to configure SimConnect. Look for a file "SimConnect.xml" under the simulator's configuration directory, either, for FSX,

C:\Users\(your profile)\AppData\Roaming\Microsoft\FSX

or, for Prepar3D3,

C:\Users\(your profile)\AppData\Roaming\Lockheed Martin\Prepar3D v3.

This might look as follows:

<?xml version="1.0" encoding="Windows-1252"?>

<SimBase.Document Type="SimConnect" version="1,0">

<Descr>SimConnect</Descr>

<Filename>SimConnect.xml</Filename>

<SimConnect.Comm>

<Disabled>False</Disabled>

<Protocol>IPv4</Protocol>

<Scope>global</Scope>

<Address>192.168.2.110</Address>

<MaxClients>64</MaxClients>

<Port>4506</Port>

<MaxRecvSize>4096</MaxRecvSize>

<DisableNagle>False</DisableNagle>

</SimConnect.Comm>

<SimConnect.Comm>

<Disabled>False</Disabled>

<Protocol>Auto</Protocol>

<Scope>local</Scope>

</SimConnect.Comm>

</SimBase.Document>

Actually, there may be further entries, and they may look different. If this file is missing, create it using notepad and save it into the appropriate directory. Important entries are the "Address" and "Port" setting. "Address" is the IP address of your

Flightsim computer. You can read it off using, e.g., the command line utility "ipconfig" to be started on a Command Shell, as "IPv4 Address".

In case you intend to work in a networked configuration, it is highly advised to allocate a fixed IP address to the Server; otherwise (when relying on the DHCP protocol in your router) you would have to modify configuration files every time you reboot your machines. This can be done as follows: Open the Control Panel, select "Network and Sharing Center". In the menu on the l.h.s. select "Change adapter settings". Right-click on "LAN Connection" (or another connection you use) and "Properties". Now scroll to "Internet Protocol Version 4". Next, go to "Properties" and switch from "Obtain an IP address automatically" to "Use the following IP address". Select an address within the range of 192.168.2.000 to 192.168.2.100. In case of a networked configuration do the same (albeit with another IP number) on the Client machine.

If you have more machines (including, maybe, printers) connected make sure addresses don't match to theirs.

For the Port setting start with 4506, however, this may require some experimenting in case the port is already in use by some other program.

Excellent descriptions on how to configure SimConnect can be found in the following manuals:

Plan-G: http://www.tasoftware.co.uk/plan-g/files/Plan_G%20Manual.pdf

SimLauncherX: http://www.avsim.com/topic/453567-simlauncherx-the-freeware-tool-everyone-has-been-waiting-for/ (Manual installed with the Software)

15.2 Installing SPAD.neXt on a networked machine

In my eyes, one of the biggest advantages of SPAD.neXt is its ability to support a networked configuration. I have observed this to free quite an amount of resources on the flightsim machine, notably if you run a higher number of FIPs. In what follows the machine running the simulator will be called the Server, while an optional networked machine will be named a Client.

When installing SPAD.neXt on a Client, select the lower checkbox in the opening window of the Configuration Wizard (Fig. 38 cf. the former Fig. 1).

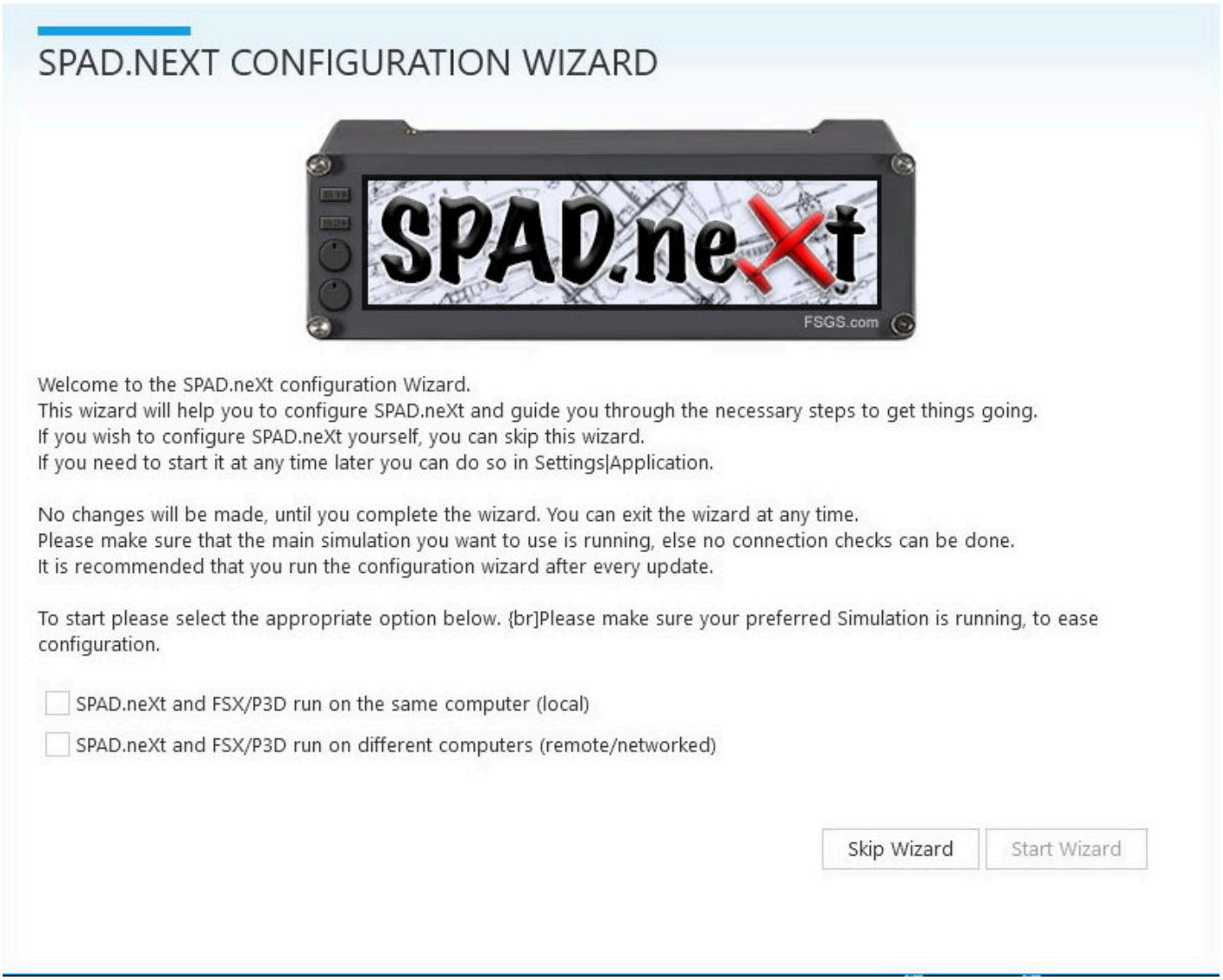


Fig. 49 The Configuration Wizard initial screen

In the following windows disable FSUIPC and skip to the SimConnect selection Window. Fill in the corresponding figures you selected either under Section 15.1 or had already preconfigured and can read off from your "SimConnect.xml" file.

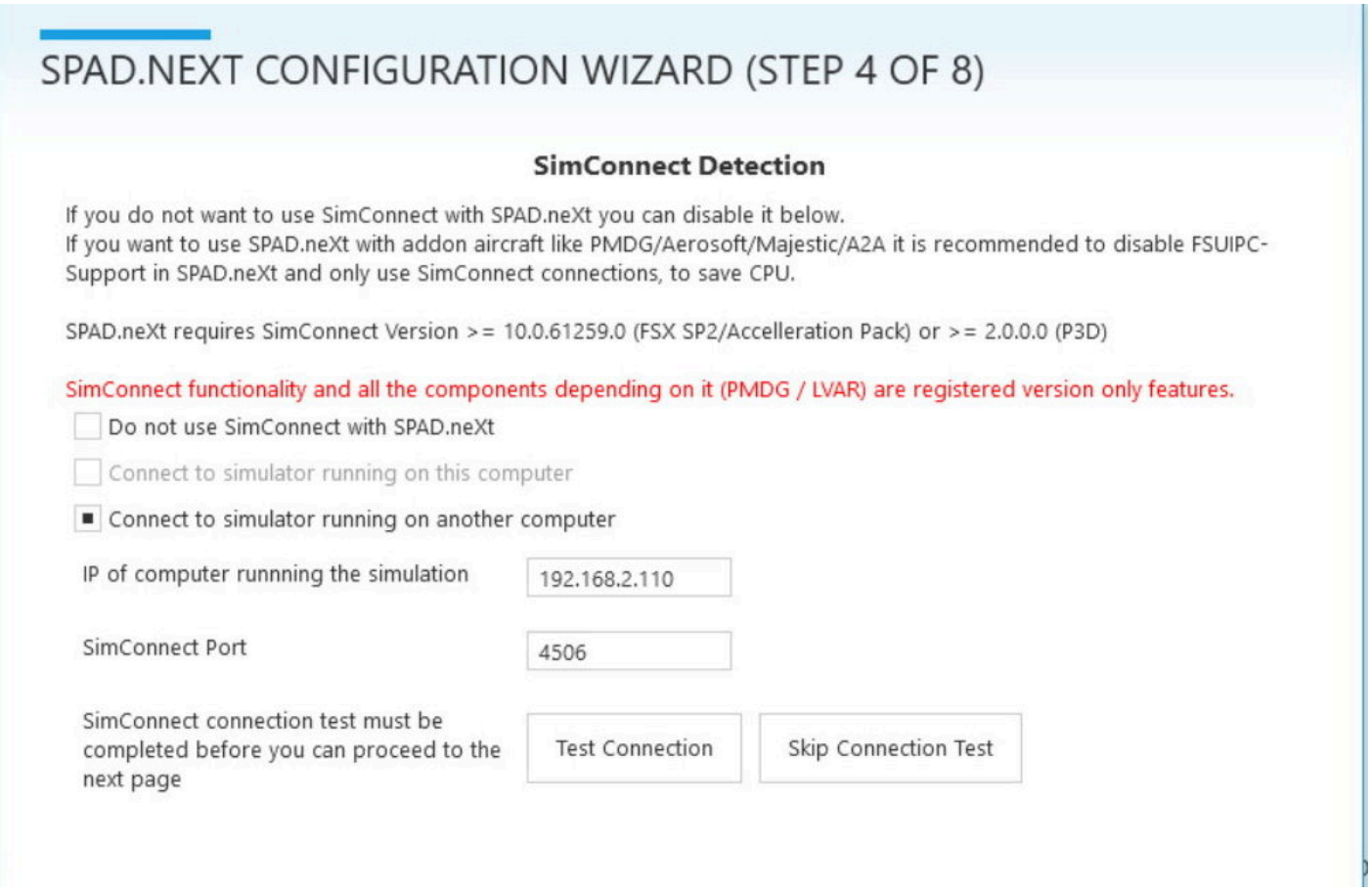


Fig. 50 Fill in the details from your "SimConnect.xml" file

Given the numerous potential error sources and continuous complains in forums, testing the connection should be a MUST in case of a networked configuration.

If all goes well the result should look as follows:

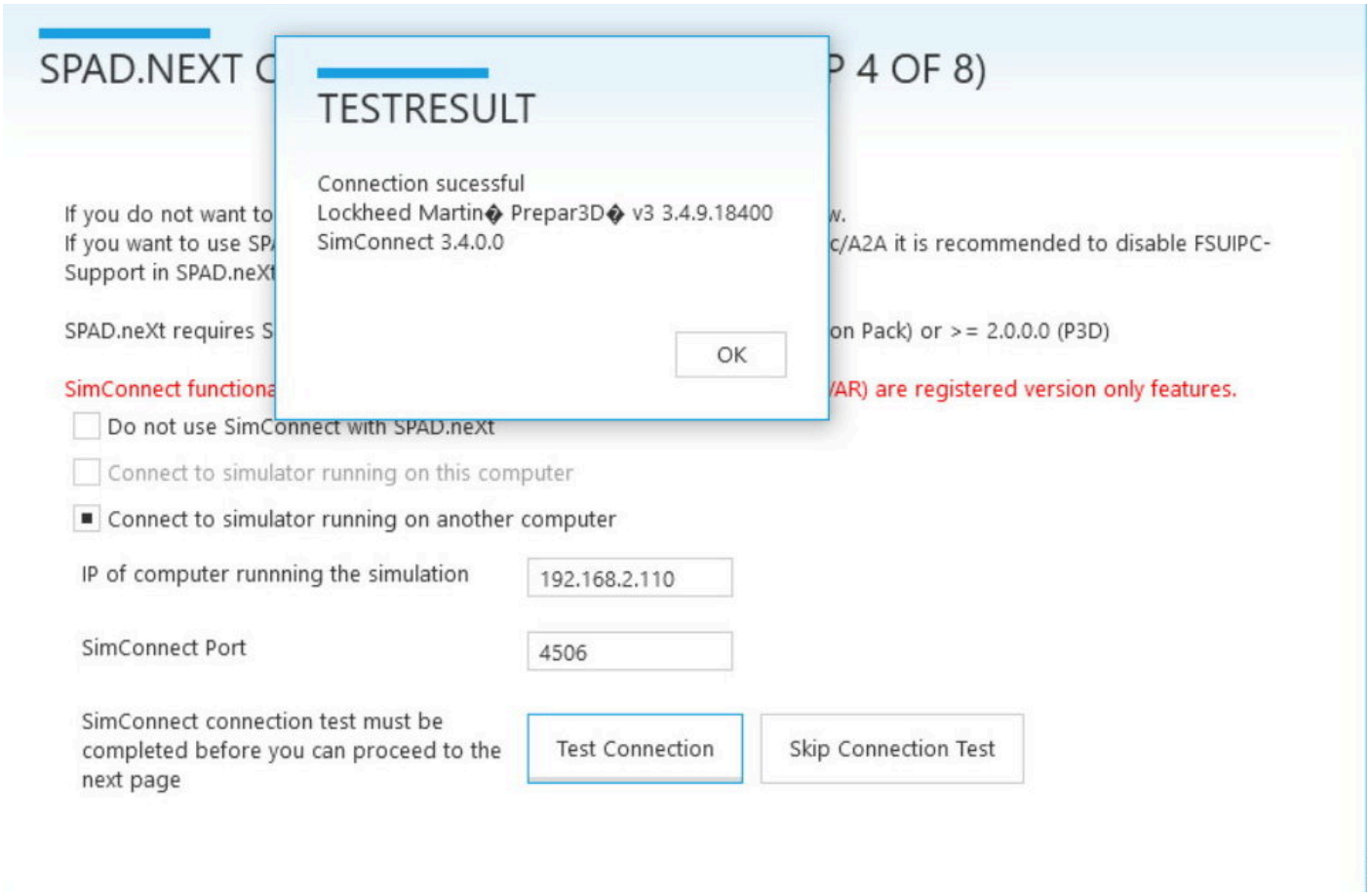


Fig. 51 SimConnect connection test passed

Now you can proceed as in Section 5. There is one more loophole, though. In case you will need LVAR support, e.g., for an A2A plane, SPAD.neXt needs "SPDbridge.dll" to run on the Server machine which the installer does not have access to. Thus, you have to include it into dll.xml yourself.

Registered only

Keep in mind LVAR support requires the registered version.

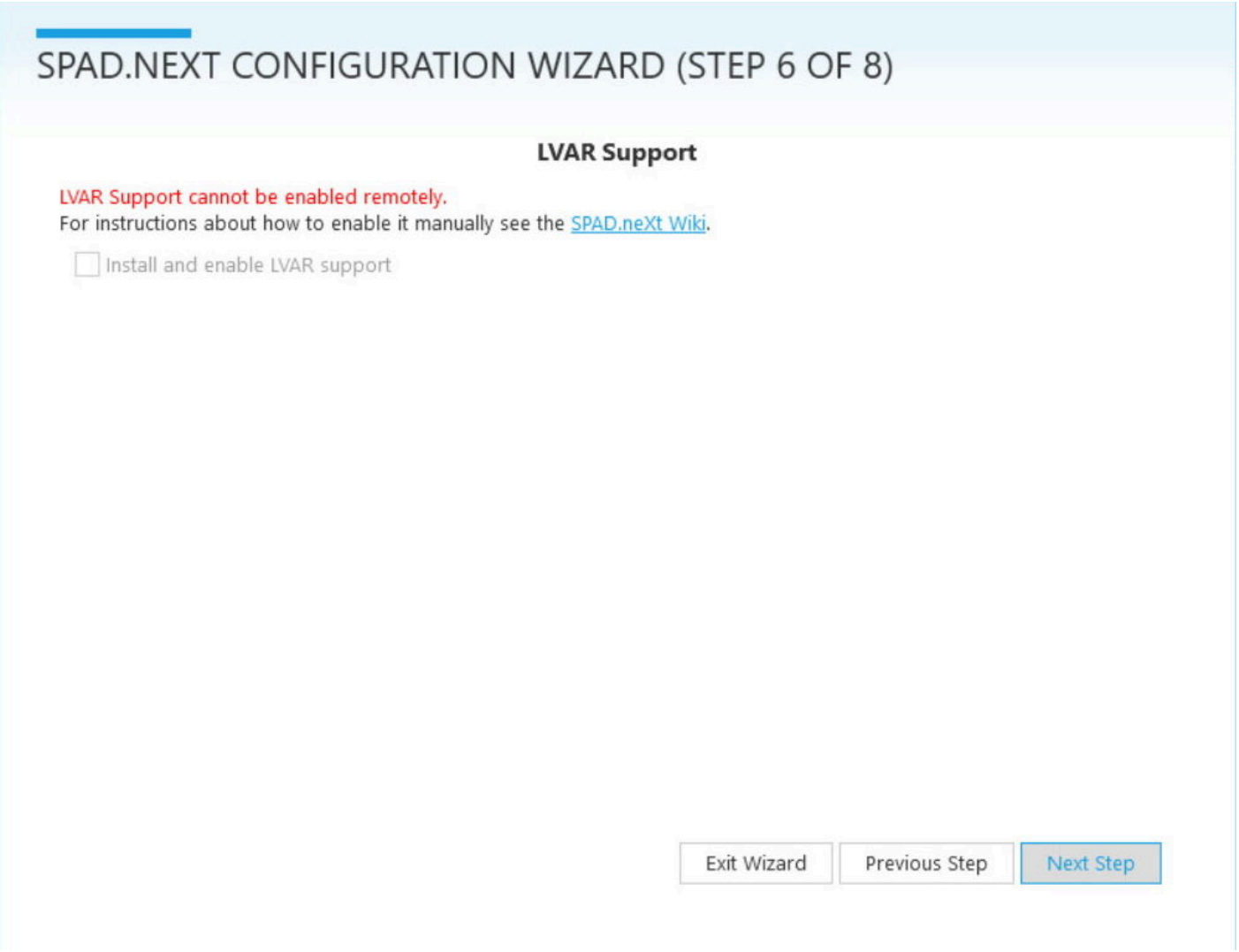


Fig. 52 You have to install the SPAD.neXt bridge manually on the Server

The procedure is simple and described in the link given in Fig. 40. Essentially, you have to copy the file "*SPDBridge.dll*" from the Client to the Server, e.g. to "Simulator maindirectory\Modules", plus add a corresponding dll.xml entry pointing to it.

After these steps, you can complete the Configuration Wizard as in Section 5.

While SPAD.neXt is running, the SimConnect status can always be watched on the Status Line at the bottom. Needless to say, don't forget to plug your panels/controls/FIPs or whatever into an USB port of your Client in this case. Again, you can start SPAD.neXt on the Client before, with or after the simulator.

Attention

In a networked configuration, there is no aircraft selection for the profile like in Fig. 17 and 18. Instead, the following box will pop up after starting the simulator:

Fig. 53 The no profile assigned box

This allows you to select a proper profile among the installed ones. This should happen only once, though, as SPAD.neXt will recall your choice for later.

Where can I get further help?

There certainly will be issues. Either, the user can fail, or the program can fail, or this guide can fail. There are several possibilities to look for help.

Passive forum search

Usually someone else had a similar issue before. You can navigate to the forum https://www.spadnext.com/forum/ and use the search function. I did this numerous times and did quite often find a suggestion or even solution.

Ask at the forum

It your search is without any useful result, you can place your own question in the forum. Posting requires establishing a forum account first. Please, use the following guidelines for reporting an issue:

1. Attach a logfile (SPAD.log from %APPDATA%\SPAD.neXt\logs) [Zip if >300 Kb or multiple files]
2. Attach the profile used or name the default profile if the problem is profile specific
3. Describe your problem as exact as possible. Maybe attach screenshots

Refrain from hijacking other peoples' threads, until you have exactly the same problem.

Open a ticket on the website

You can navigate to the website http://www.spadnext.com/support/ and open a ticket there. (This link is also in the red bar on top of the forum!) Choose the OPEN A NEW TICKET tab and proceed according to the guidance. You can check the status of your ticket in the same place as well.

To login to the support system, use your Forum credentials (username / password)

Open a ticket from within SPAD.neXt (recommended)

SPAD.neXt itself includes an interface to the ticket system allowing to report issues. To access it, navigate to the "support" main menu entry and select "Create new ticket" on the r.h.s. This opens an editor window to report your issue. It is highly suggested to add the files at the bottom (as applicable). You can check the status of your ticket from within SPAD.neXt as well.

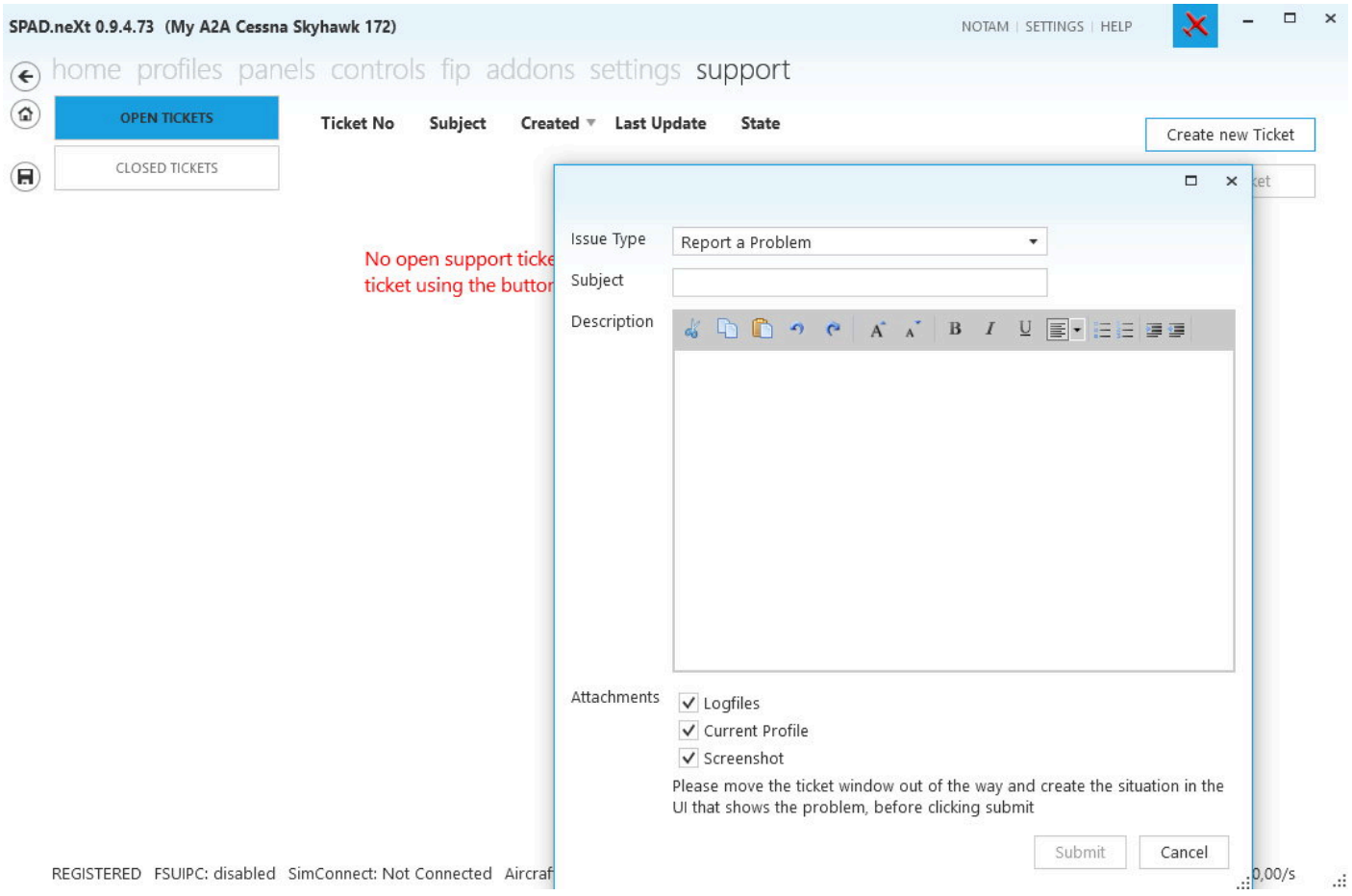


Fig. 50: Sending a support ticket

Author and Version

This is Getting Started (formerly Quickstart) Version 0.5 written by Michael Basler (http://michael-basler.net). It is valid for SPAD.neXt, Version 0.9.4.73, by Ulrich Strauß (http://fsgs.com/).

Changelog

06/24/2015 Initial release [MB]

06/25/2015 Graphical improvements, added UAC hint, several improvements throughout [US]

06/29/2015 Start via FSUIPC.ini, changelog added [MB]

06/30/2015 Reformatting [US]

07/06/2015 Added paragraph on General Configuration Settings [MB]

07/31/2015 Added section 4 Unregistered vs. registered version [MB]

09/15/2015 Updated all Sections for 0.6.5735, added material on registered version and registering, added Section on SimConnect [MB]

10/18/2015 Adapted to Version 7, Configurations Wizard Section added, new Version 7 Figures, Minor corrections [MB]

10/31/2016 Completely rewritten and extended version reflecting new features like FIPs support, Online Profiles, Code Snippets, Ticketing system, and much more. Renamed Getting Started. [MB]

Fig. 1 License agreement 8

Fig. 2 Install Location 9

Fig. 3 Downloading installation files 10

Fig. 4 Installation completed 10

Fig. 5 Configuration Wizard: Welcome screen 12

Fig. 6 Configuration Wizard: Simulator detection. 13

Fig. 7 Configuration Wizard: FSUIPC Detection 14

Fig. 8 Configuration Wizard: SimConnect Detection 15

Fig. 9 Configuration Wizard: PMDG 737 NGX/777X Detection 16

Fig. 10 Configuration Wizard: LVAR Support 17

Fig. 11 Configuration Wizard: Overview 18

Fig. 12 Configuration Wizard: Final Step 18

Fig. 13 The main interface: home 19

Fig. 14 The registration submenu 21

Fig. 15 Buying / Activation / Trial 21

Fig. 16 Buying / Activation / Trial request 22

Fig. 17 The profiles menu 23

Fig. 18 Online profiles selection 24

Fig. 19 The added A2A Cessna profile in the profiles list 25

Fig. 20 Enabling/disabling features under Applications Settings 26

Fig. 21 Editing a profile 27

Fig. 22 Assigning an aircraft to a profile 27

Fig. 23 Assignment of the A2A C172 to the profile "A2A Cessna Skyhawk 172" 28

Fig. 24 The interface for the Saitek Multi Panel 29

Fig. 25 The interface for the Saitek Cessna Yoke 30

Fig. 26 One of the Saitek Flight Instrument Panels 30

Fig. 27 Main gauge selection 31

Fig. 28 Gauge selection 32

Fig. 29 FIP selection completed - the Saitek Airspeed gauge 32

Fig. 30 The settings menu revised 38

Fig. 31 Application Settings: General 38

Fig. 32 Application Settings: Simulations 39

Fig. 33 Application Settings: Profiles 40

Fig. 34 Devices: Global FIPs Settings 41

Fig. 35 Once more on profiles 42

Fig. 36 Copying a profile 43

Fig. 37 : Controls: The Saitek Cessna Yoke 44

Fig. 38 Importing an Online Snippet 45

Fig. 39 Coolie hat functions have been assigned 45

Fig. 40 The backwards movement of the r.h.s. N-S rocker switch is still unassigned 46

Fig. 41 Select Pressed (Mode 1) 47

Fig. 42 Select Send Simulation Event 47

Fig. 43 Select PAUSE_TOGGLE 48

Fig. 44 Confirm the "Define ControlEvent Action" dialog as is with "OK". 48

Fig. 45 Assignment done! 49

Fig. 46 The future gauge market 50

Fig. 47 Freeware gauges 51

Fig. 48 Demo versions of payware gauges 51

Fig. 49 The Configuration Wizard initial screen 55

Fig. 50 Fill in the details from your "SimConnect.xml" file 55

Fig. 51 SimConnect connection test passed 56

Fig. 52 You have to install the SPAD.neXt bridge manually on the Server 57

Fig. 53 The no profile assigned box 57